



PH201

Physics I



PH201 PHYSICS I

ABET eCOURSEBOOK

2021-2026

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SECTION I

COURSE ASSESSMENTS



TAB A AY2026 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 26 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY26 Annual Guidance), 27 June 2025.
- d. Core Physics Program AY 25 Executive Summary, September 2025.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2026 (AY 26), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 26. For AY 26, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 26. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 26 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment								
	AY 19	AY 20		AY 21	AY 22	AY 23	AY 24	AY 25	AY 26
Cadets who successfully complete the Core Physics program can:		AY 19 Rubric	AY 20 Rubric						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Good
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good	Good

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 26 was **Acceptable**. This rating is consistent with rating from AY 25.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 26 was **Good**. This rating is an increase from the Acceptable rating from AY 25.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 26 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 25.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 26 was **Good**. There is no statistically significant change from AY 24 to AY 25.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 26 was **Good**. This rating is consistent with the Good rating from AY 25.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY26. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 26 is located in Appendix C. 251, and 252 were assessed with 201 and 202.

6. Significant Planned or Continuing Curricular Changes for AY 26. There are no significant changes planned for the Core Physics Program for AY 26. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence. Minor changes include continuing AI assisted grading in 201 and creating a test bank in both 2x1 and 2x2. A Core Physics book review was completed and attached in Appendix D. We are drafting an update to the SLAM and lab procedures in 2X1/2X2 to increase continuity between 2X1/2X2.

Enclosure 1 – Core Physics Program AY 26 Executive Summary

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 27.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment
- D. Core Physics textbook review

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Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 26. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 26, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase ($\nearrow$), decrease ($\searrow$), or equivalency ($\rightarrow$) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease ($\searrow$) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent ($\rightarrow$) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment								
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25	AY26
		AY 19 Rubric	AY 20 Rubric						
Cadets who successfully complete the Core Physics program can:									
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Good
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.

2. For AY 26, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

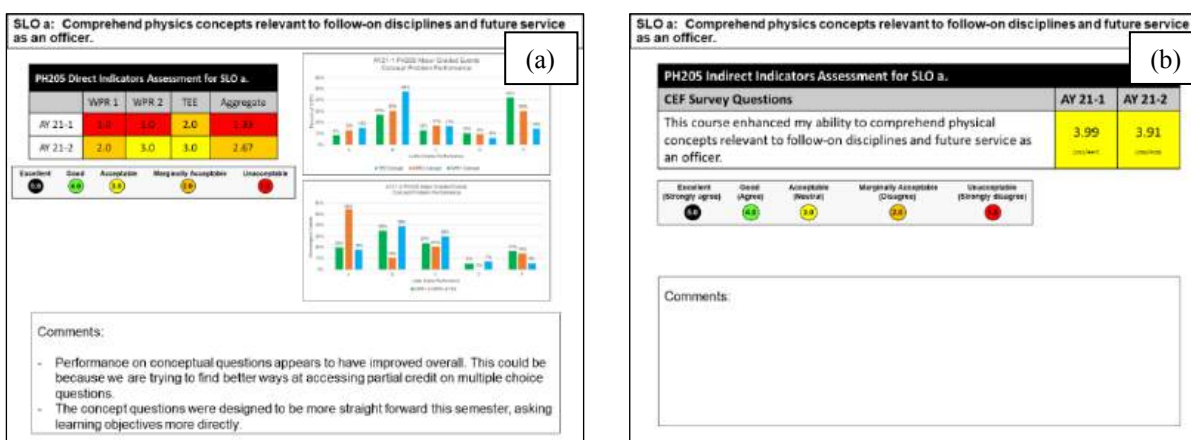


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

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PH205
SLO Assessment AY 21

SLO	AY	Indicators		CD	Comments	
		Direct	Indirect			
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31	
	AY 21-2	2.67	3.91			
b	AY 21-1	4.00	4.13	3.5		
	AY 21-2	2.00	4.00			
c	AY 21-1	3.33	N/A	2.5		
	AY 21-2	1.33	4.01			
d	AY 21-1	N/A	4.20	4.0		
	AY 21-2	N/A	4.01			
e	AY 21-1	4.50	N/A	4.0		
	AY 21-2	4.50	3.93			

Direct:

Excellent (5.0)	Good (4.0)	Acceptable (3.0)	Marginally Acceptable (2.0)	Unacceptable (1.0)
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Indirect:

Excellent (Strongly agrees)	Good (Agree)	Acceptable (Neutral)	Marginally Acceptable (Disagree)	Unacceptable (Strongly disagree)
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Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					Program Director
	PH2X1 (AY 26)		PH2X2 (AY 26)		Weighted Average	
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5		1.8		2.6	3
b. Apply physical principles and mathematical competencies with logical processes.	4.0		3.5		3.6	4
c. Analyze and solve problems of increasing complexity.	3.5		2.7		3.3	3.5
d. Develop the ability to learn independently.	4.0		4.0		4.0	4
e. Examine physical phenomena, analyze data, and communicate results.	3.8		4.0		4.0	4
Direct: Excellent (4.0) Good (3.5) Acceptable (3.0) Marginally Acceptable (2.5) Unacceptable (2.0) Indirect: Excellent (Strongly agree) (4.0) Good (Agree) (3.5) Acceptable (Neutral) (3.0) Marginally Acceptable (Disagree) (2.5) Unacceptable (Strongly disagree) (2.0)						

Table 6. Weighting Factors

Course	AY 26 Enrollment	Weighting Factor
PH2X1	1218	0.69
PH2X2	553	0.21

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 25 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, if the assessment of WGCD statement 2.2 is based on the assessment of SLO a (i.e. Marginally Acceptable) and SLO c (i.e. Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 26

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206/202 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205/201 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205/201 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205/201 (I), PH206/202 (R)	d. develop the ability to learn independently.	Good
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206/202 (R)	d. develop the ability to learn independently.	Good
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205/201 (R), PH206/202 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205/201 (R), PH206/202 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205/201 and PH206/202	e. examine physical phenomena, analyze data, and communicate results.	Good

Appendix D – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program Results Dashboard was used to assess the current E-textbook and possibility of changing textbooks is in Table 8. The 5 assessment buckets were A-Pedagogy & Content, B-Platform & Technology, C-Instructor Workflow, and D-Cost, Reks & Flexibility. The final recommendation was to retain ExpertTA/OpenStax, but explore new text for possible implementation in AY 28-1.

Table 8. Core Physics Program new textbook Assessment, AY 26

RESULTS DASHBOARD					
<i>All scores pull automatically from Scoring Matrix.</i>					
Metric	Pearson	Expert TA	Cengage-Openstax	Cengage - New Text	Kudu
★ OVERALL WEIGHTED SCORE	3.45	4.20	3.59	3.63	3.70
RANK (1 = best)	5	1	4	3	2
Category Weights: Max 1.25					
Bucket A — Pedagogy & Content	1.0	1.0	0.7	0.9	1.2
Bucket B — Platform & Technology	1.1	1.0	1.0	1.0	1.0
Bucket C — Instructor Workflow	0.8	1.1	0.9	0.9	0.8
Bucket D — Cost, Risk & Flexibility	0.6	1.1	1.0	0.9	0.8
RECOMMENDATION & DECISION NOTES					
Recommendation:	Retain Expert TA, explore Kudu (Self-Hosted Content from OpenStax) or Cengage (Webassign with OpenStax) in AY 27-1 with next target of course change occurring summer of 2027 implementation AY 28-1. The learning outcome benefits from changing books does not exceed the cost to change the course. Instead we should focus on platforms which maximize learning outcomes with Cadet cost and long-term content accessibility for STEM majors in mind. Expert TA and Kudu are the best options to achieve that. Cengage minimizes cost but does not improve material engagement and regresses our course.				
Primary Rationale:	Expert TA is at a good state of maturity with flexibility to implement our own content at a comparable price point for what it offers. They can offer us the flexibility to deliver our course the way we want to with one of the better online homework platforms. Kudu is a close second but not in a mature enough state for a confident transition.				
Key Risks / Concerns:	The cost of Expert TA has risen and with Cengage as an already paid for option. The question is if the \$50 for Expert TA is worth it for hosting our custom course content and its usefulness as a homework platform. Expert TA is slow to deliver on new features (similar to Kudu based on size of company and trends in last 2 years). Expert TA is behind other vendors in terms of interactive content integration and AI / LLM integration, though they have indicated it is on their roadmap.				
Next Steps:	Option 1 (if we want more self-developed interactive content leaning into leveraging LLMs): AY 27-1 adopts and builds a Kudu course over the 2026 summer (1:1 from our Expert TA content, est ~20 work hours to build). and run Kudu for 6 sections and assess faculty workflow, content development, and homework platform feasibility across split sections. Course end feedback measures cadet engagement with the material and achieving course outcomes. If we think the learning outcome benefits are worth it then in AY27-2 we decide to adopt Kudu in AY28-1 (semester). PH202 transitions in Summer 2027 for an AY28-1 concurrent adoption and NIT is trained on Kudu. Option 2 (if we want to lean into the single source/cost optimized solution and away from our self-developed material integrated in the book): Run a similar pilot course as above testing Webassign with OpenStax. Concurrently conduct textbook review in AY 27-1 and AY 27-2 for content migration in Summer 2027 for both PH201 and PH202 and adoption during s second semester.				



TAB B AY2025 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 25 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY25 Annual Guidance), 28 June 2024.
- d. Core Physics Program AY 24 Executive Summary, September 2024.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2025 (AY 25), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 25. For AY 25, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 25. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 25 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment							
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25
Cadets who successfully complete the Core Physics program can:		AY 19 Rubric	AY 20 Rubric					
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable	Good

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 25 was **Acceptable**. This rating is consistent with rating from AY 24.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 25 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 24.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 25 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 24.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 25 was **Good**. There is no statistically significant change from AY 24 to AY 25.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 25 was **Good**. This rating is consistent with the Acceptable rating from AY 24 with a slight improvement from AY25.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY25. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 25 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 26. There are no significant changes planned for the Core Physics Program for AY 26. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence. Minor changes include continuing AI assisted grading in 201 and creating a test bank in

Enclosure 1 – Core Physics Program AY 25 Executive Summary

both 2x1 and 2x2. A Core Physics Assessment Review will occur to provide further assessment of the curriculum.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 26.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

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Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 25. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 25, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment							
	AY 19	AY 20		AY 21	AY22	AY23	AY24	AY25
Cadets who successfully complete the Core Physics program can:		AY 19 Rubric	AY 20 Rubric					
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets →	Marginally Meets →	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds →	Meets →	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets →	Exceeds →	Good	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets →	Exceeds →	Good	Good	Acceptable	Good	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds →	Meets →	Acceptable	Good	Acceptable	Acceptable	Acceptable →	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 25, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.



Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes



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PH205

SLO Assessment AY 21



SLO	AY	Indicators		CD	Comments
		Direct	Indirect		
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31
	AY 21-2	2.67	3.91		
b	AY 21-1	4.00	4.13	3.5	
	AY 21-2	2.00	4.00		
c	AY 21-1	3.33	N/A	2.5	
	AY 21-2	1.33	4.01		
d	AY 21-1	N/A	4.20	4.0	
	AY 21-2	N/A	4.01		
e	AY 21-1	4.50	N/A	4.0	
	AY 21-2	4.50	3.93		

Direct:

Excellent (5.0)

Good (4.5)

Acceptable (3.5)

Marginally Acceptable (2.5)

Unacceptable (1.5)

Indirect:

Excellent (Strongly agrees) (5.0)

Good (Agrees) (4.5)

Acceptable (Neutral) (3.5)

Marginally Acceptable (Disagrees) (2.5)

Unacceptable (Strongly disagrees) (1.5)

1

Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					Weighted Average	Program Director
	PH201 (AY 25)	PH251 (AY 25)	PH202 (AY 25)	PH252 (AY 25)			
Cadets who successfully complete the Core Physics program can:							
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5	2.5	3.4	0.0	3.1	3	
b. Apply physical principles and mathematical competencies with logical processes.	4.0	4.0	3.9	0.0	3.7	3.5	
c. Analyze and solve problems of increasing complexity.	2.8	4.0	3.4	0.0	3.1	3.5	
d. Develop the ability to learn independently.	4.0	4.0	4.3	0.0	4.1	4	
e. Examine physical phenomena, analyze data, and communicate results.	3.8	4.0	4.5	0.0	3.7	4	
Direct: Excellent (4.0) Good (3.5) Acceptable (3.0) Marginally Acceptable (2.5) Unacceptable (1.0) Indirect: Excellent (throughly agree) (4.0) Good (agree) (3.5) Acceptable (neutral) (3.0) Marginally Acceptable (disagree) (2.5) Unacceptable (strongly disagree) (1.0)							

Table 6. Weighting Factors

Course	AY 25 Enrollment	Weighting Factor
PH201	973	0.60
PH251	203	0.13
PH202	433	0.27
PH252	0	0.00

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 25 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, if the assessment of WGCD statement 2.2 is based on the assessment of SLO a (i.e. Marginally Acceptable) and SLO c (i.e. Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 25

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206/202 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205/201 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205/201 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205/201 (I), PH206/202 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206/202 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205/201 (R), PH206/202 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205/201 (R), PH206/202 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205/201 and PH206/202	e. examine physical phenomena, analyze data, and communicate results.	Good



TAB C AY2024 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 24 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY24 Annual Guidance), 26 June 2023.
- d. Core Physics Program AY 23 Executive Summary, September 2023.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2024 (AY 24), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 24. For AY 24, the Core Physics Program implemented minor curriculum change to the PH201/PH251, PH202/PH252. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 23. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 24 Executive

Summary Table 1 - Summary of SLO Outcomes

SLOs	Assessment						
	AY 19	AY 20		AY 21	AY22	AY23	AY24
		AY 19 Rubric	AY 20 Rubric				
Cadets who successfully complete the Core Physics program can:							
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 24 was **Acceptable**. This rating is consistent with rating from AY 23.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 24 was **Good**. There is no statistically significant change from AY 23 to AY 24.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 24 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 23.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY24. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 24 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 25. There are no significant changes planned for the Core Physics Program for AY 25. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

Enclosure 1 – Core Physics Program AY 24 Executive Summary

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 25.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

JACOB W. CAPPS, PhD
LTC, AV (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 24. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 24, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment						
	AY 19	AY 20		AY 21	AY22	AY23	AY24
		AY 19 Rubric	AY 20 Rubric				
Cadets who successfully complete the Core Physics program can:							
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets →	Marginally Meets →	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds →	Meets →	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets →	Exceeds →	Good	Acceptable	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets →	Exceeds →	Good	Good	Acceptable	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds →	Meets →	Acceptable	Good	Acceptable	Acceptable	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 24, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.



Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

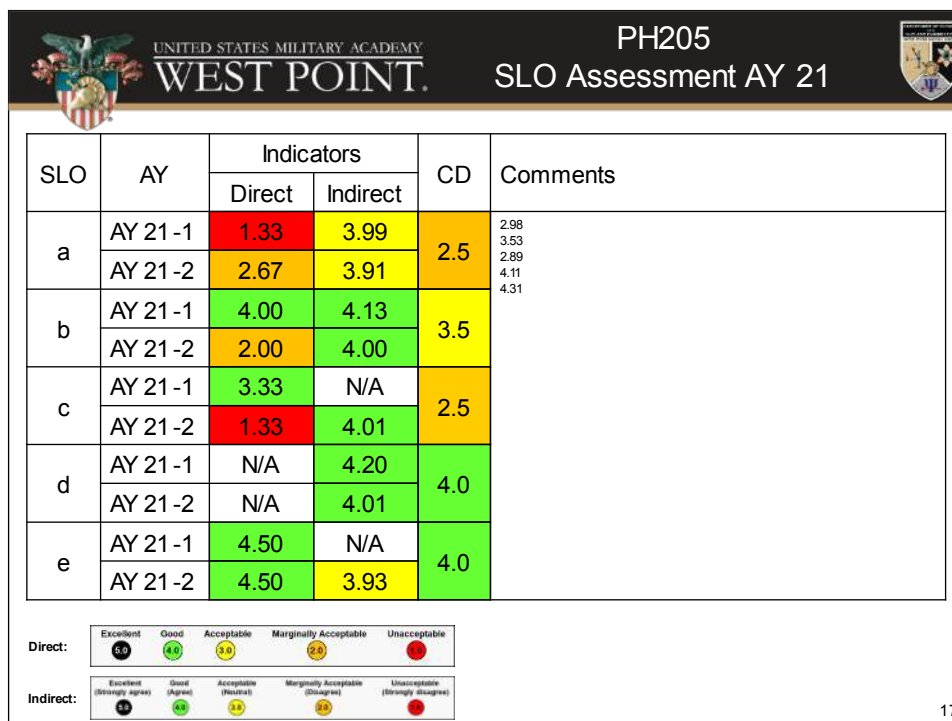


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					
	PH201	PH251	PH202	PH252	Weighted Average	Program Director
Cadets who successfully complete the Core Physics program can:	(AY 24)	(AY 24)	(AY 24)	(AY 24)		
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.8	4.0	3.4	4.0	3.0	3
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4.0	3.9	4.0	3.6	3.5
c. Analyze and solve problems of increasing complexity.	2.8	4.0	3.4	4.0	3.0	3.5
d. Develop the ability to learn independently.	4.0	4.0	4.3	4.3	4.1	4
e. Examine physical phenomena, analyze data, and communicate results.	3.3	4.0	4.5	4.5	3.7	3.5
Direct: Excellent (Strongly agree) Good (Agree) Acceptable (Neutral) Marginally Acceptable (Disagree) Unacceptable (Strongly disagree) Indirect: Excellent (Strongly agree) Good (Agree) Acceptable (Neutral) Marginally Acceptable (Disagree) Unacceptable (Strongly disagree)						

Table 6. Weighting Factors

Course	AY 24 Enrollment	Weighting Factor
PH201	1185	0.68
PH251	50	0.03
PH202	469	0.27
PH252	37	0.02

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 24

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable



TAB D AY2023 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 23 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
 - b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
 - c. Office of the Dean, MADN, memorandum (Dean's AY22 Annual Guidance), 02 July 2021.
 - d. Core Physics Program AY 22 Executive Summary, September 2022.
 - e. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH205 (Physics I) and PH255 (Advanced Physics I), 21 April 2022.
 - f. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH206 (Physics II) and PH256 (Advanced Physics II), 21 April 2022.
2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2023 (AY 23), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.
3. Status of Curricular Change Initiatives for AY 23. For AY 23, the Core Physics Program implemented curriculum change to the PH201/PH251, PH202/PH252 sequence approved by the General Committee of the Academic Board on 12 May 2022. There were no changes to the Student Learning Outcomes (SLOs).
4. Summary of Student Outcome Evaluation for AY 22. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Enclosure 1 – Core Physics Program AY 23 Executive Summary

Table 1 - Summary of SLO Outcomes

SLOs	Assessment					
	AY 19	AY 20		AY 21	AY22	AY23
		AY 19 Rubric	AY 20 Rubric			
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds	Meets	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Exceeds	Good	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Exceeds	Good	Good	Acceptable	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Meets	Acceptable	Good	Acceptable	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 23 was **Acceptable**. This rating is an improvement over the Marginally Acceptable rating from AY 22. The assessment is based on a weighted average across all courses of 2.98 out of 5, compared to 2.88 in AY 22. There is no statistically significant change from AY 22 to AY 23.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 23 was **Good**. This rating demonstrates an improvement from the Acceptable rating from AY 22. The AY 22 rating was based on weighted average of 3.98, whereas the AY 23 rating was based on a weighted average of 4.00 (see Table 5 in Appendix B). There is no statistically significant change from AY 22 to AY 23.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 23 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 22.

(1) The Core Physics Program continued the implementation of the two lab writs and the 10 labs that culminate in the written lab reports from AY22. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH2X2 and future STEM courses.

Enclosure 1 – Core Physics Program AY 23 Executive Summary

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 22 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 23. There are no significant changes planned for the Core Physics Program for AY 23. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 23.

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

For DAVID J. PHILLIPS, PhD
COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 23. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 23, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase ($\nearrow$), decrease ($\searrow$), or equivalency ($\rightarrow$) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease ($\searrow$) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent ($\rightarrow$) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 2 - Summary of SLO Outcomes

SLOs	Assessment					
	AY 19	AY 20		AY 21	AY22	AY23
		AY 19 Rubric	AY 20 Rubric			
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Meets →	Marginally Meets →	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable	Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Exceeds →	Meets →	Acceptable	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets →	Exceeds →	Good	Acceptable	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets →	Exceeds →	Good	Good	Acceptable	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds →	Meets →	Acceptable	Good	Acceptable	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.

2. For AY 23, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

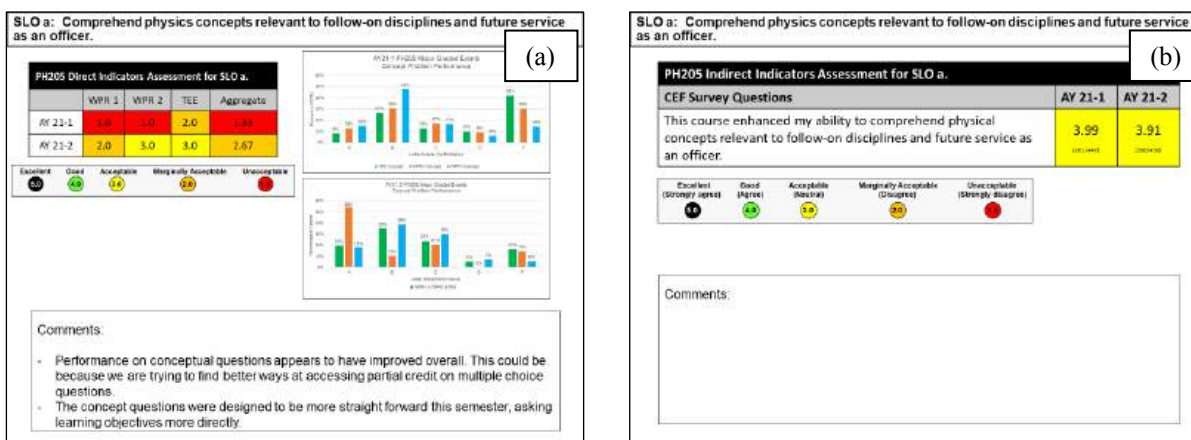


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

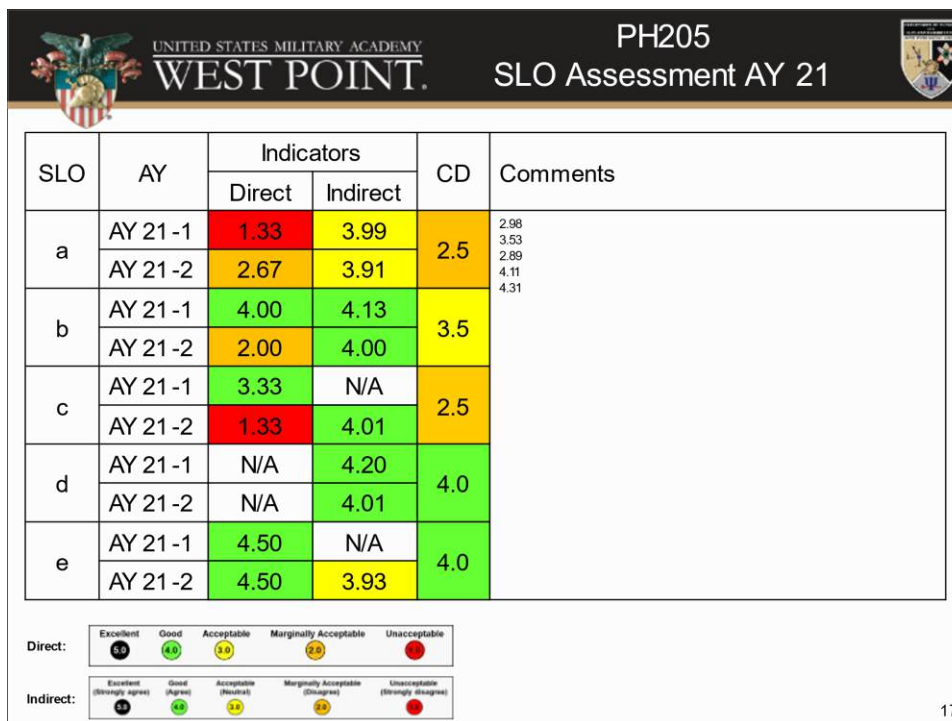


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 23 is shown in Table 5. The weighting factors for AY 23 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Program Director's Assessment of Student Learning Outcomes using Direct and Indirect Indicators.

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH201	PH251	PH202	PH252	Weighted Average	Program Director
	(AY 23)	(AY 23)	(AY 23)	(AY 23)		
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	3	4	2.5	3	2.98	3
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4	4	4	3.73	3.5
c. Analyze and solve problems of increasing complexity.	3.5	4	3.5	4	3.6	3.5
d. Develop the ability to learn independently.	4	4	4	4	4	4
e. Examine physical phenomena, analyze data, and communicate results.	3	4	4	4.5	3.5	3.5
Direct: Excellent 5.0 Good 4.0 Acceptable 3.0 Marginally Acceptable 2.0 Unacceptable 1.0 Indirect: Excellent (Strongly agree) 5.0 Good (Agree) 4.0 Acceptable (Neutral) 3.0 Marginally Acceptable (Disagree) 2.0 Unacceptable (Strongly disagree) 1.0						

Table 6. Weighting Factors

Course	AY 23 Enrollment	Weighting Factor
PH201	770	0.54
PH251	168	0.11
PH202	380	0.27
PH252	104	0.07

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 22

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable



TAB E AY2022 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 22 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Office of the Dean, MADN, memorandum (Dean's AY22 Annual Guidance), 02 July 2021.
- d. Core Physics Program AY 21 Executive Summary, September 2021.
- e. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH205 (Physics I) and PH255 (Advanced Physics I), 21 April 2022.
- f. Department of Physics and Nuclear Engineering, MADN-PNE, memorandum (Core Course Curricular Change Request for PH206 (Physics II) and PH256 (Advanced Physics II), 21 April 2022.

2. Purpose. This enclosure summarizes the Core Physics Program assessment, curriculum changes, and assessment process for academic year 2022 (AY 22), in compliance with the annual reporting requirements of references 1a, 1b, and 1c.

3. Status of Curricular Change Initiatives for AY 22. For AY 22, the Core Physics Program changed the sequence of content of the core course PH205/PH255 and core Science Depth Option course PH206/256. See references 1e and 1f. The Core Physics Program piloted PH201X in AY 22-1 and PH202X in AY 22-2. The General Committee of the Academic Board voted to approve the curriculum change to the PH201/PH251, PH202/PH252 sequence on 12 May 2022. There were no changes to the Student Learning Outcomes (SLOs).

4. Summary of Student Outcome Evaluation for AY 22. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated on the next page. The assessment methodology is given in Appendix B.

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 18	AY 19	AY 20		AY 21	AY 22
			AY 19 Rubric	AY 20 Rubric		
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Exceeds	Meets	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Exceeds	Good	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Exceeds	Good	Good	Acceptable
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Meets	Acceptable	Good	Acceptable

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, for AY 22 was **Marginally Acceptable**. This rating is consistent with the Marginally Acceptable rating from AY 21.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, for AY 22 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 21.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, for AY 22 was **Acceptable**. This rating is consistent with the Acceptable rating from AY 21.

d. Cadet performance on SLO d, *develop the ability to learn independently*, for AY 22 was **Acceptable**. This rating demonstrates a downturn from the Good rating from AY 21. The AY 21 rating was based on weighted average of 4.0, whereas the AY 22 rating was based on a weighted average of 3.98 (see Table 5 in Appendix B). The percent difference between the AY 21 and AY 22 rating was 0.5%. This percent difference is less than 1%, and no conclusions may reasonably drawn from this small value.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, for AY 22 was **Acceptable**. This rating represents a downturn from the Good rating from AY 21. The change is primarily due to changes to the laboratory program in PH205 and PH201X in AY 22-2.

(1) In AY 21, PH205 had a lab WPR, and the assessment of SLO e for the direct indicator of the lab WPR was Good. In AY 21, the PH205 lab program consisted of 8 labs that included a final design sequence, with lab 6 for design, lab 7 for execution, and lab 8 for presentation. The assessment of SLO e for the direct indicator of the Group

Enclosure 1 – Core Physics Program AY 22 Executive Summary

Lab Grade was Excellent. The aggregate assessment of SLO e for direct indicators for AY 21 was Good.

(2) In AY 22-1, both PH201X and PH205 had a lab WPR, and the assessment of SLO e for the direct indicator of the lab WPR for both courses was Good, consistent with AY 21. In AY 22-1, PH201X had 8 labs without the lab design sequence, whereas PH205 had 8 labs that included the lab design sequence. The assessment of SLO e for the direct indicator of the Group Lab Grade was Good for both courses. It is not known why PH205 had a downturn in performance on the Group Lab Grade. In AY 22-2, PH205 had two lab writs instead of a lab WPR. Cadet performance on lab writ #1 was Acceptable, and performance on lab writ #2 was Marginally Acceptable. The aggregate assessment of SLO e for direct indicators for AY 22 was Acceptable.

(3) The lab program was changed from the lab WPR and the lab design sequence in AY 21 to the lab writs and 10 labs in AY 22-2 to focus on technical writing of lab reports. In AY 21, cadets had limited exposure to writing a full laboratory report. In AY 22, the lab program culminated with the writing of two full lab reports. The change from a lab WPR in AY 21 to the two lab writs in AY 22-2 facilitated the spreading out of teaching and assessing lab material to more logically support the skills necessary to conduct a lab and write a lab report.

(4) The down turn in the assessment of SLO e from AY 21 to AY 22 likely has two causes. First, cadets likely assessed the lab WPR as a higher stakes event than the lab writs and therefore prepared more for the lab WPR, as reflected in the better performance on the lab WPR in AY 21 and AY 22-1 than on the lab writs in AY 22-2. Second, the lab design sequence (labs 6 through 8) was less rigorous than the full lab reports, and cadets scored better on the lab design sequence in AY 21 than on written lab reports in AY 22-2.

(5) The Core Physics Program will continue to implement the two lab writs and the 10 labs that culminate in the written lab reports. The lab reports improve the written technical communication skills of cadets and better prepare them for more advanced labs in PH202X and future STEM courses.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk for AY 22 is located in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 23. There are no significant changes planned for the Core Physics Program for AY 23. The Core Physics Program will continue to refine the PH201/PH251 and PH202/PH252 course sequence.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no planned changes to the SLOs for AY 23.

Enclosure 1 – Core Physics Program AY 22 Executive Summary

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

3 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

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COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 22. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 22, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase (↗), decrease (↘), or equivalency (→) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease (↘) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent (→) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 18	AY 19	AY 20		AY 21	AY 22
			AY 19 Rubric	AY 20 Rubric		
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Exceeds	Meets	Acceptable	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Exceeds	Good	Acceptable	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Exceeds	Good	Good	Acceptable
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Meets	Acceptable	Good	Acceptable

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.
2. For AY 22, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). In AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20).

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

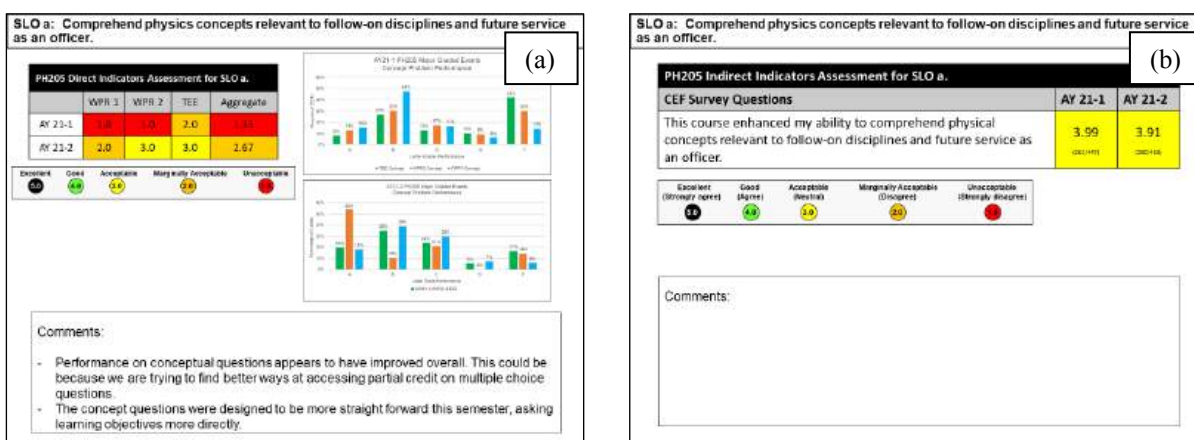


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

UNITED STATES MILITARY ACADEMY
WEST POINT.

PH205
SLO Assessment AY 21

SLO	AY	Indicators		CD	Comments	
		Direct	Indirect			
a	AY 21-1	1.33	3.99	2.5	2.98 3.53 2.89 4.11 4.31	
	AY 21-2	2.67	3.91			
b	AY 21-1	4.00	4.13	3.5		
	AY 21-2	2.00	4.00			
c	AY 21-1	3.33	N/A	2.5		
	AY 21-2	1.33	4.01			
d	AY 21-1	N/A	4.20	4.0		
	AY 21-2	N/A	4.01			
e	AY 21-1	4.50	N/A	4.0		
	AY 21-2	4.50	3.93			

Direct:

Excellent (5.0)	Good (4.0)	Acceptable (3.0)	Marginally Acceptable (2.0)	Unacceptable (1.0)
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Indirect:

Excellent (Strongly agree)	Good (Agree)	Acceptable (Neutral)	Marginally Acceptable (Disagree)	Unacceptable (Strongly disagree)
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Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH201X and PH205 was normalized to the enrollment for PH255, and the weighting factor for PH202X and PH206 was normalized to the enrollment of PH256.

5. The PD's assessment of the SLOs for AY 22 is shown in Table 5. The weighting factors for AY 22 are shown in Table 6.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 5. Core Physics SLOs Assessment Summary, AY 22, Direct and Indirect Indicators.

SLOs	Assessment							
Cadets who successfully complete the Core Physics program can:	PH201X (AY 22-1)	PH202X (AY 22-2)	PH205 (AY 22)	PH255 (AY 22-2)	PH206 (AY 22)	PH256 (AY 22-1)	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	3.0	2.5	3.0	4.0	2.5	2.5	2.88	2.5
b. Apply physical principles and mathematical competencies with logical processes.	4.0	4.0	3.5	4.5	4.0	4.0	3.74	3.5
c. Analyze and solve problems of increasing complexity.	4.0	4.0	3.5	4.5	3.5	4.0	3.62	3.5
d. Develop the ability to learn independently.	4.0	3.5	4.0	4.0	4.0	4.0	3.98	3.5
e. Examine physical phenomena, analyze data, and communicate results.	4.0	4.0	3.5	4.0	4.0	4.0	3.72	3.5
Direct: Excellent (5.0) Good (4.0) Acceptable (3.0) Marginally Acceptable (2.0) Unacceptable (1.0) Indirect: Excellent (Strongly agree) (5.0) Good (Agree) (4.0) Acceptable (Neutral) (3.0) Marginally Acceptable (Disagree) (2.0) Unacceptable (Strongly disagree) (1.0)								

Table 6. Weighting Factors for Core Physics SLOs, AY 22.

Course	Enrollment			Weighting Factor
	AY 22-1	AY 22-2	AY 22 Total	
PH201X	118	---	118	1.7
PH202X	---	84	84	1.2
PH205	505	427	932	13.7
PH255	---	68	68	1.0
PH206	200	230	430	5.9
PH256	73	---	73	1.0

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 22 is shown in Table 7. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 7. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 22

APG	WGCD Statement and Course Responsibility ¹	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Acceptable
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Acceptable
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Acceptable
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Acceptable

¹See Annex A: Academic Program Development Framework, DPOM 5-07.



TAB F AY2021 COURSE ASSESSMENT

Enclosure 1 – Core Physics Program AY 21 Executive Summary

1. References.

- a. DPOM 5-05, Managing Curricular Change, 08 April 2020.
- b. DPOM 5-07, Assessing and Improving Student Learning in the Academic Program, 24 April 2019.
- c. Dean's Annual Guidance, Academic Year 2021 (AY21), 10 July 2020.
- d. Core Physics Program AY 20 Executive Summary, September 2020.

2. Purpose. This memorandum summarizes the Core Physics program assessment, curriculum changes, and assessment process for AY 21, in compliance with the annual reporting requirements of references 1.a, 1.b, and 1.c.

3. Status of Curricular Change Initiatives for AY 21. For AY 21, no curricular changes requiring Academy-level approval were implemented. Internally, for AY 21 we maintained the change to the assessment mechanism for the Student Learning Outcomes (SLOs) that we first implemented in AY 20. The change has allowed us to combine assessment data from both direct and indirect indicators to assess all SLOs using a common 5-tiered system.

4. Summary of Student Outcome Evaluation for AY 21. Evidence for cadet achievement of the SLOs was obtained from direct indicators (e.g., WPRs, Lab reports, TEE, etc.) and indirect indicators (i.e., course end feedback surveys). A summary of the assessment results is given in Table 1 in Appendix A and repeated here. The assessment methodology is given in Appendix B.

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 17	AY 18	AY 19	AY 20		AY 21
				AY 19 Rubric	AY 20 Rubric	
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Meets	Exceeds	Meets	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Meets	Exceeds	Good	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Meets	Exceeds	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Exceeds	Meets	Acceptable	Good

Enclosure 1 – Core Physics Program AY 21 Executive Summary

a. Cadet performance on SLO a, *comprehend physical concepts relevant to follow-on disciplines and future service as an officer*, was **Marginally Acceptable**. This rating is consistent with the Marginally Acceptable rating from AY 20.

b. Cadet performance on SLO b, *apply physical principles and mathematical competencies with logical processes*, was **Acceptable**. This rating is consistent with the Acceptable rating from AY 20.

c. Cadet performance on SLO c, *analyze and solve problems of increasing complexity*, was **Acceptable**. This rating represents a downturn from the Good rating from AY 20. The reason for the downturn is unclear. The lack of clarity is due to the fact that the indicators used to assess this SLO are too broad; the SLO is assessed based on aggregate performance on both conceptual and calculation problems, in addition to indirect indicators from the CEF survey. For AY 22, the indicators need to be more narrowly focused. A best indicator, such as cadet performance on a specific problem on a WPR or TEE, will be considered to replace the current indicators for this SLO.

d. Cadet performance on SLO d, *develop the ability to learn independently*, was **Good**. This rating is consistent with the Good rating from AY 20.

e. Cadet performance on SLO e, *examine physical phenomena, analyze data, and communicate results*, was **Good**. This rating represents an upturn from the Acceptable rating from AY 20. The change is due to better cadet performance in AY 21 on the Signature Writing Event (SWE) and the group laboratory reports. The increased performance on these events can likely be attributed to changes in the lab program for AY 21 that focused on clarifying the requirements for cadets on the SWE and on laboratory reports.

5. Summary of Results of APG-WGCD Responsibility Evaluations Conducted During the Year. The summary of the Core Program APG-WGCD to SLO crosswalk is in Appendix C.

6. Significant Planned or Continuing Curricular Changes for AY 22. There is one planned curricular change for the Core Physics program for AY 22. The planned curricular change is to realign the course content for PH205 and PH206 to the traditional topical coverage of mechanics in PH205 and electricity and magnetism in PH206. The change is currently underway. For AY 22-1, the Core Physics program is piloting 8 sections of PH201X as a replacement for PH205.

7. Proposed Changes to Student Learning Outcomes / Significant Planned or Implemented Assessment Process Changes. There are no changes to the SLOs for the upcoming AY. The assessment process for AY 22 may change to include assessment data from cadet performance on national standardized tests, and best indicators will be considered for assessment of SLO c.

Enclosure 1 – Core Physics Program AY 21 Executive Summary

8. Updated Assessment Schedule and Time-line. Currently, the Core Physics program assess each of its SLOs every year. There are no plans to change this assessment timeline.

9. Point of Contact. The POC for this action is the undersigned at extension 845-938-5895.

4 Appendices

- A. Student Learning Outcomes and Assessment Matrix for Core Physics SLOs
- B. Assessment Methodology for Core Physics Program Student Learning Outcomes
- C. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment
- D. Planned Curricular Changes

DAVID J. PHILLIPS, PhD
COL, MS (FA 47)
Program Director, Core Physics

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

1. Core Physics SLOs for AY 21. Cadets who successfully complete the Core Physics program can:

- a. **comprehend** physical concepts relevant to follow-on disciplines and future service as an officer.
- b. **apply** physical principles and mathematical competencies with logical processes.
- c. **analyze** and **solve** problems of increasing complexity.
- d. develop the ability to **learn independently**.
- e. **examine** physical phenomena, **analyze** data, and **communicate** results.

2. The Assessment Matrix for Core Physics SLOs.

a. For AY 21, the Program Director (PD) continued to use the 5-tiered scale for direct indicators that the PD adopted from the Nuclear Engineering program in AY 20 and the associated assessment ratings and color codes. The change is discussed in more detail in Appendix B.

b. The Assessment Matrix for the Core Physics SLOs is shown in Table 1. Of note, the assessment for AY 20 shows the assessment for AY 20 using both the AY 19 rubric and the AY 20 rubric. The increase ($\nearrow$), decrease ($\searrow$), or equivalency ($\rightarrow$) in SLO achievement from AY 19 to AY 20 is indicated graphically with arrows. For example, for SLO b, the AY 19 assessment was Exceeds. The AY 20 Assessment was Meets using the AY 19 rubric, which represents a decrease ($\searrow$) in SLO performance compared to AY 19. The AY 20 Assessment was Acceptable using the AY 20 rubric, which is equivalent ($\rightarrow$) to Meets using the AY 19 rubric.

Appendix A – Student Learning Outcomes and Assessment Matrix for Core Physics SLOs

Table 1. Assessment Matrix for Core Physics SLOs.

SLOs	Assessment					
	AY 17	AY 18	AY 19	AY 20		AY 21
				AY 19 Rubric	AY 20 Rubric	
Cadets who successfully complete the Core Physics program can:						
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Fails to Meet	Fails to Meet	Marginally Meets	Marginally Meets	Marginally Acceptable	Marginally Acceptable
b. Apply physical principles and mathematical competencies with logical processes.	Meets	Meets	Exceeds	Meets	Acceptable	Acceptable
c. Analyze and solve problems of increasing complexity.	Meets	Meets	Meets	Exceeds	Good	Acceptable
d. Develop the ability to learn independently.	Meets	Meets	Meets	Exceeds	Good	Good
e. Examine physical phenomena, analyze data, and communicate results.	Exceeds	Exceeds	Exceeds	Meets	Acceptable	Good

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

1. Evaluation and assessment of each SLO is integrated into each individual core course offering, including the laboratory program. At the end of each semester, Course Directors (CDs) conduct a course assessment to gain situational understanding of immediate changes required for the next semester and to evaluate each SLO. CDs then provide this information to the PD in the form of a Course Assessment (CAT) brief. At the conclusion of each academic year, the PD completes a program review that focuses on assessing the SLOs and the assigned Academic Program Goals – What Graduates Can Do (APG-WGCD) statements.

2. For AY 21, the PD continued using the SLO Assessment Rubric adopted in AY 20. In AY 20, the PD changed the SLO assessment rubric for the program and for the direct indicators from a 4-tiered rating scale (see Table 2) to a 5-tiered rating scale (see Table 3). The PD made this change in order to combine data from both direct and indirect indicators into the SLO assessment for each SLO where possible. The main difference between the AY 19 rubric and the AY 20 rubric is that the AY 20 rubric added an additional tier of Excellent for direct indicators for when the most frequent grade is A and B, and there are no D or F grades. Additionally, the AY 19 rubric did not use a Likert rating for direct indicators (see Table 4). For AY 21-2, the PD also added two additional questions to the Course End Feedback (CEF) survey to provide indirect indicators to assess SLOs c and e.

Table 2. SLO Assessment Rubric, AY 19 and prior.

Assessment Rating	Color Code	Most Frequent Letter Grade	Low Performance Threshold
Exceeds Expectations		B or better	< 10% D or F
Meets Expectations		C or better	< 20% D or F
Marginally Meets Expectations		C or better	< 40% D or F
Fails to Meet Expectations		Any	> 40% D or F

Table 3. Core Physics Program SLO Assessment Rubric (Adopted AY 20)

Assessment Rating	Color Code	Likert Rating	Direct Indicators: Criteria	Indirect Indicators: Level of Agreement
Excellent		5	Most frequent grade is A and B, with no D or F	Strongly agree
Good		4	Most frequent grade is A and B, with some D or F	Agree
Acceptable		3	Most frequent grade is C or better, with < 20% D or F	Neutral
Marginally Acceptable		2	Most frequent grade is C or better, with > 20% D or F but < 40% D or F	Disagree
Unacceptable		1	Most frequent grade is D or F	Strongly disagree

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 4. SLO Assessment Rubric Comparison, AY 19 and AY 20.

AY 19		AY 20	
Assessment Rating	Color Code	Assessment Rating	Color Code
		Excellent	
Exceeds Expectations		Good	
Meets Expectations		Acceptable	
Marginally Meets Expectations		Marginally Acceptable	
Fails to Meet Expectations		Unacceptable	

3. Data from the AY 21-2 PH205 CAT brief will be used to explain the assessment process.

a. For SLO a, for example, the AY 21-2 PH205 CD assessed each direct indicator (WPR 1, WPR 2, and the TEE) and each indirect indicator (CEF Survey question) for AY 21-2 using the assessment rubric from Table 3. The CD's assessment of these direct and indirect indicators is shown in Figure 1. Note that Figure 1 includes assessment data from the previous term, AY 21-1. For the direct indicators, the CD computed a simple average to determine the aggregate assessment of the direct indicators for each semester.

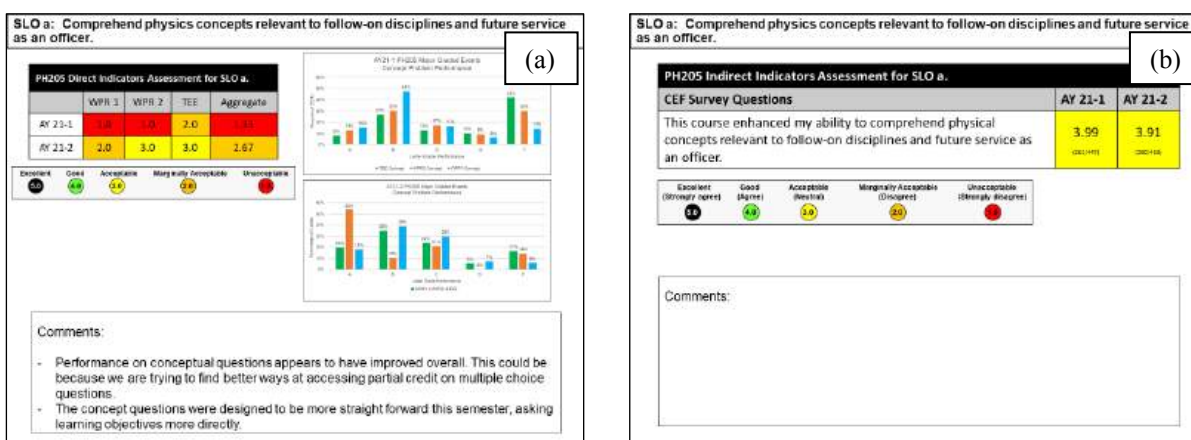


Figure 1. PH205 CD's assessment of (a) direct and (b) indirect indicators for SLO a.

b. The CDs summarize the assessment of the direct and indirect indicators for each SLO for their courses. To determine the overall assessment for each SLO, each CD computes a simple average of the direct and indirect indicator assessment data for each SLO, then rounds down to the nearest 0.5. Figure 2 provides an example from the AY 21-2 PH205 CAT brief.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

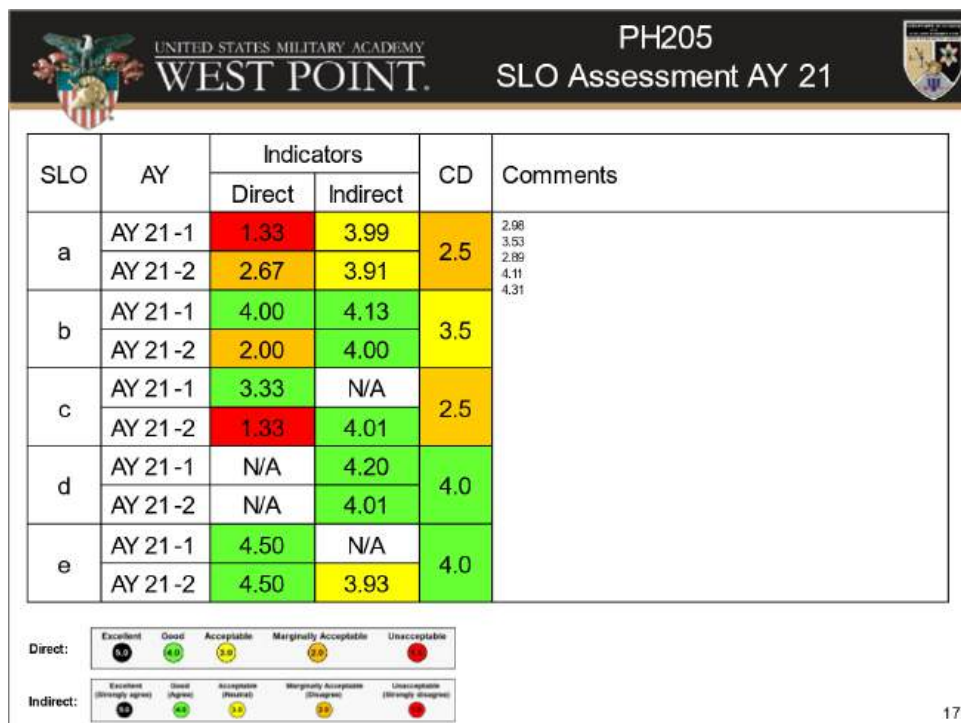


Figure 2. PH205 CD's assessment of each SLO, based on a simple average.

4. The PD aggregates the SLO assessments from each CD. To determine the overall program assessment, the PD performs a weighted average of the CD assessments for each SLO and rounds down to the nearest 0.5. The PD's assessment of the SLOs for AY 21 is shown in Table 5. The weighting factors are used to ensure that PH255 and PH256, which have lower enrollment and higher performance, do not unfairly skew the overall program assessment. The weighting factors are based on course enrollment. The weighting factor for PH205 is normalized to the enrollment for PH255, and the weighting factor for PH256 is normalized to the enrollment of PH256. The weighting factors for AY 21 are shown in Table 6.

Table 5. Core Physics SLOs Assessment Summary, AY 21, Direct and Indirect Indicators

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH205	PH255	PH206	PH256	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.5	4.0	2.5	3.0	2.77	2.5
b. Apply physical principles and mathematical competencies with logical processes.	3.5	4.0	4.0	4.5	3.85	3.5
c. Analyze and solve problems of increasing complexity.	2.5	4.5	4.0	4.5	3.48	3.0
d. Develop the ability to learn independently.	4.0	4.0	4.0	4.0	4.00	4.0
e. Examine physical phenomena, analyze data, and communicate results.	4.0	4.0	4.0	4.0	4.00	4.0

Direct: Excellent (5.0), Good (4.0), Acceptable (3.0), Marginally Acceptable (2.0), Unacceptable (1.0)

Indirect: Excellent (Strongly agree) (5.0), Good (Agree) (4.0), Acceptable (Neutral) (3.0), Marginally Acceptable (Disagree) (2.0), Unacceptable (Strongly disagree) (1.0)

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 6. Weighting Factors for Core Physics SLOs, AY 21

	Enrollment			
Course	AY 21-1	AY 21-2	AY 21 Total	Weighting Factor
PH205	447	436	883	3.2
PH255	144	135	279	1.0
PH206	156	168	324	2.1
PH256	71	82	153	1.0

5. The AY 21 assessment includes assessment of indirect indicators for all SLOs based on CEF surveys. The CEF survey results tend to be high, which has the potential to skew the assessment results. Table 7 shows the Core Physics SLO Assessment, AY 21, considering direct indicators only for all SLOs.

Appendix B – Assessment Methodology for Core Physics Program Student Learning Outcomes

Table 7. Core Physics SLOs Assessment Summary, AY 21, Direct Indicators Only

SLOs	Assessment					
Cadets who successfully complete the Core Physics program can:	PH205	PH255	PH206	PH256	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.0	4.0	1.0	2.0	1.99	1.5
b. Apply physical principles and mathematical competencies with logical processes.	3.0	4.0	4.0	4.5	3.63	3.5
c. Analyze and solve problems of increasing complexity.	2.0	4.5	4.0	4.5	3.26	3.0
d. Develop the ability to learn independently.	N/A	N/A	N/A	N/A	N/A	N/A
e. Examine physical phenomena, analyze data, and communicate results.	4.5	4.5	4.5	4.0	4.43	4.0

Direct:

Excellent
5.0

Good
4.0

Acceptable
3.0

Marginally Acceptable
2.0

Unacceptable
1.0

Indirect:

Excellent (Strongly agree)
5.0

Good (Agree)
4.0

Acceptable (Neutral)
3.0

Marginally Acceptable (Disagree)
2.0

Unacceptable (Strongly disagree)
1.0

6. Table 8 shows a comparison of the SLO assessment based on direct and indirect indicators vs. direct indicators only.

Table 8. Core Physics SLOs Assessment Summary, AY 21, Comparison of Indicators

SLOs	Direct and Indirect Indicators		Direct Indicators Only	
Cadets who successfully complete the Core Physics program can:	Weighted Average	Program Director	Weighted Average	Program Director
a. Comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	2.77	2.5	1.99	1.5
b. Apply physical principles and mathematical competencies with logical processes.	3.85	3.5	3.63	3.5
c. Analyze and solve problems of increasing complexity.	3.48	3.0	3.26	3.0
d. Develop the ability to learn independently.	4.00	4.0	N/A	N/A
e. Examine physical phenomena, analyze data, and communicate results.	4.00	4.0	4.43	4.0

Direct:

Excellent
5.0

Good
4.0

Acceptable
3.0

Marginally Acceptable
2.0

Unacceptable
1.0

Indirect:

Excellent (Strongly agree)
5.0

Good (Agree)
4.0

Acceptable (Neutral)
3.0

Marginally Acceptable (Disagree)
2.0

Unacceptable (Strongly disagree)
1.0

7. Examination of SLO a in Table 8 shows a weighted average of 2.77 for when considering both direct and indirect indicators compared to a weighted average of 1.99 when only considering direct indicators. This comparison shows that inclusion of indirect indicators slightly skews the assessment higher. However, the weighted average of 1.99 is very close to 2.00, which would have resulted in an overall assessment of Marginally Acceptable. Therefore, the inclusion of indirect indicators on the assessment is minimal, and the assessment in Table 5 will be used.

Appendix C – Core Physics Program APG-WGCD to SLO Crosswalk and Assessment

1. The Core Physics program APG-WGCD to SLO crosswalk and assessment for AY 21 is shown in Table 9. For WGCD statements that were based on the assessment of multiples SLO, the lower SLO assessment was used for the overall assessment of the WGCD statement. For example, the assessment of WGCD statement 2.2 is based on the assessment of SLO a (Marginally Acceptable) and SLO c (Acceptable). The overall assessment of WGCD statement 2.2 is therefore Marginally Acceptable.

Table 9. Core Physics Program APG-WGCD to SLO Crosswalk and Assessment, AY 20

APG	WGCD Statement and Course Responsibility	SLO Crosswalk and SLO Assessment	WGCD Statement Assessment
1. Communication	1.2 Speak and write using Standard American English. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
	1.5 Use sound logic and relevant evidence to make convincing arguments. PH206 (I)	e. examine physical phenomena, analyze data, and communicate results.	Good
2. Critical and Creative Thinking	2.2 Integrate knowledge and skills from a variety of disciplines PH205 (I)	a. comprehend physical concepts relevant to follow-on disciplines and future service as an officer.	Marginally Acceptable
		c. analyze and solve problems of increasing complexity.	
	2.4 Reason both quantitatively and qualitatively PH205 (I)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
3. Lifelong Learning	3.1 Demonstrate the willingness and ability to learn independently. PH205 (I), PH206 (R)	d. develop the ability to learn independently.	Good
	3.3 Pursue self-awareness and embrace the responsibility for personal intellectual development PH206 (R)	d. develop the ability to learn independently.	Good
5. STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes or behaviors PH205 (R), PH206 (R)	b. apply physical principles and mathematical competencies with logical processes.	Acceptable
		c. analyze and solve problems of increasing complexity.	
	5.2 Apply the scientific method PH205 (R), PH206 (R)	e. examine physical phenomena, analyze data, and communicate results.	Good
	5.3 Collect and analyze data in support of decision making R—PH205 and PH206	e. examine physical phenomena, analyze data, and communicate results.	Good

Appendix D – Planned Curricular Changes

1. There is one planned curricular change for AY 22. The change is to realign the course content for PH205 and PH206 to the traditional topical coverage of mechanics in PH205 and electricity and magnetism in PH206 and to increase the time spent in laboratory. This will allow for more in-depth and cohesive treatment of mechanics in PH205 and electricity and magnetism in PH206.
2. The curricular change is currently underway. In AY 22-1, the Core Physics program is piloting 8 sections of PH201X. PH201X is being taught on a 40 lesson by 55-minute schedule, with 120-minute labs. In comparison, PH205 is taught on a 30 lesson by 75-minute schedule, with 75-minute labs.
3. In AY 22-2, the Core Physics program will pilot 6 sections of PH202X. PH202X will be taught on a 40 lesson by 55-minute schedule, with 120-minute labs. In comparison, PH206 is taught on a 30 lesson by 75-minute schedule, with 75-minute labs



SECTION II

EXAMINATIONS



TAB E WPR#1



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #1 HANDOUT

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Written Partial Review I

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- b. Check your exam for 7 problems. Check for the Physics Reference Card provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to 75 minutes to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **06 March 2026 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	35	
2	35	
3	45	
4	35	
5	55	
6	55	
7	40	

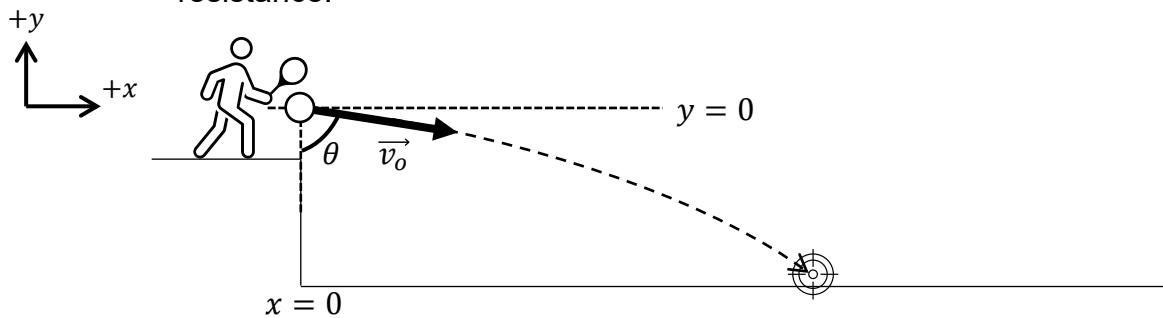
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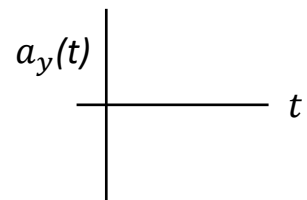
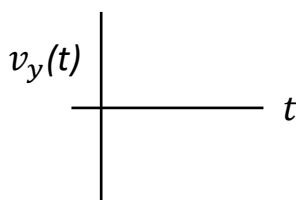
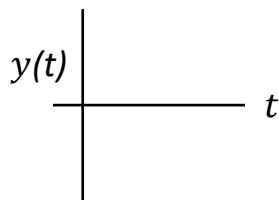
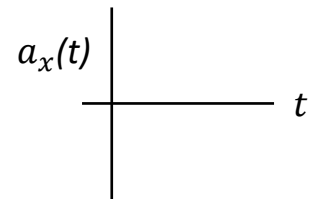
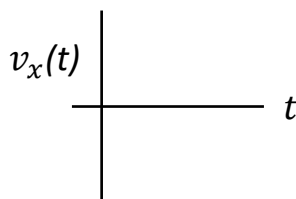
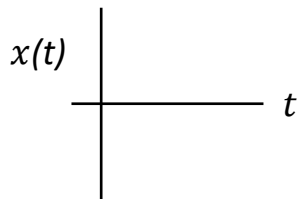
Wt. **No.**

Cadet: _____

- 35 1. A tennis player on a hilltop hits a tennis ball at a target that is at an elevation lower than the player. The trajectory of the ball is shown below. Neglect air resistance.



- (25) 1a. In the following diagrams, sketch the expected functions of position, velocity, and acceleration as a function of time for the tennis ball. Assume that the origin is the point at which the tennis ball is no longer in contact with the tennis racket.



- (10) 1b. The velocity of the tennis ball just before hitting the target is given by:

$$\vec{v} = (10.1\hat{i} - 37.5\hat{j}) \text{ m/s}$$

Calculate the speed of the projectile, $|\vec{v}|$, at that time.

- ☐ 47.6 m/s
- ☐ 0.00 m/s
- ☐ 38.8 m/s
- ☐ 27.4 m/s

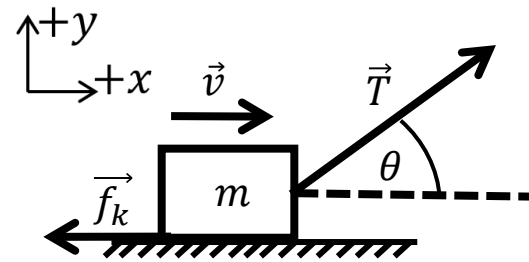
Wt. **No.**

Cadet: _____

35

2.

A block of mass m on a rough horizontal surface is pulled by a rope at a constant speed. The rope makes an angle θ above the horizontal and applies a constant force. The rough surface imparts a frictional force $\vec{f}_k$ on the block.



- (15) 2a. Derive an expression for the magnitude of the tension force T while the block moves at a constant speed.

☐ $T = f_k \sin(\theta)$

☐ $T = f_k$

☐ $T = \frac{f_k}{\cos(\theta)}$

☐ $T = ma$

- (10) 2b. Select the most correct statement for the block's acceleration in the y-direction while on the horizontal surface.

☐ $a_y = g$

☐ a_y depends on the mass m .

☐ a_y depends on the magnitude of the tension force.

☐ $a_y = 0$

- (10) 2c. Select which of the following is a Newton's Third Law Force Pair.

☐ The normal force from the ground on the box and the block's weight force

☐ The tension force on the block and the friction force on the block

☐ The normal force on the block from the surface and the normal force on the surface from the block

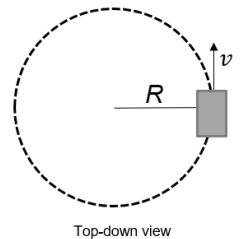
☐ The block's weight force and the tension force on the block

Wt. **No.**

Cadet: _____

- 45 3. A 1250-kg car rounds an unbanked curve of radius $R = 176$ m. The coefficient of static friction between the road and the tires is $\mu_s = 0.693$.

- (30) 3a. Calculate the maximum speed at which the car can turn without slipping.



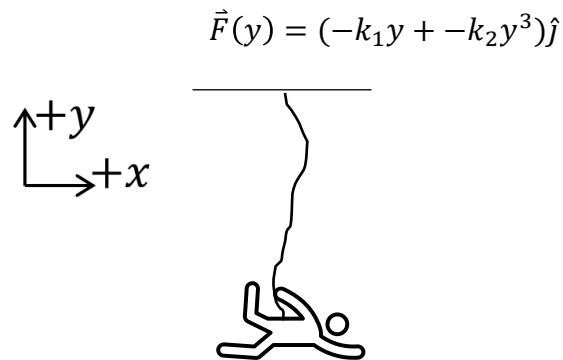
Report your answer in this box.	Report units in this box.
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- (15) 3b. You are tasked with redesigning this road to permit drivers to travel at the same speed during icy conditions. Briefly describe how you will change the road using the physics learned in this class. You may use words and/or equations; you **must** use physics. You only need to describe one modification. Solutions must be legible.

Wt. **No.**

Cadet: _____

- 35 4. A 78.4-kg Cadet goes bungee jumping on spring break. The Cadet falls from rest straight down. The force exerted by the bungee cord is given by the function below, assuming the bungee cord is unstretched when the Cadet jumps.



- (15) 4a. Calculate the work done by gravity on the Cadet after she has fallen 125 meters.

- ☐ 96,100 J
- ☐ - 96,100 J
- ☐ 60,100 J
- ☐ - 61,100 J

- (10) 4b. Select the expression that best describes the work done on the cadet, by the bungee cord, while the cadet falls.

- ☐ $W = -k_1 - 3k_2y^2$
- ☐ $W = \frac{k_1}{2}y^2 + \frac{k_2}{4}y^4$
- ☐ $W = k_1y + k_2y^3$
- ☐ $W = -\frac{k_1}{2}y^2 - \frac{k_2}{4}y^4$
- ☐ $W = 0$

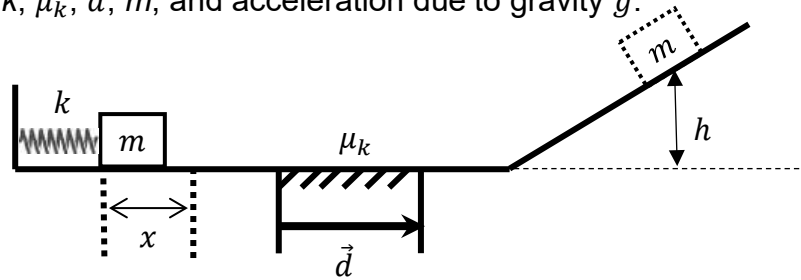
- (10) 4c. Select the option that best describes the forces acting on the Cadet while falling.

- ☐ The force of gravity is conservative, but the force of the bungee is non-conservative.
- ☐ The force of gravity is non-conservative, but the force of the bungee cord is conservative.
- ☐ Both gravity and the bungee cord's force are conservative forces.
- ☐ Both gravity and the bungee cord's force are non-conservative forces.

Wt. **No.**

Cadet: _____

- 55 5. An object of mass m is initially at rest and held against a spring of spring constant k . The spring is compressed a distance x from the relaxed length. Then, the block is released and travels on a track that is frictionless except for a rough region of length d , as shown. The coefficient of kinetic friction between the object and the rough portion of the track is μ_k . The object slides through the rough region and then up a frictionless ramp to a final height h , where the object momentarily comes to rest.
- (50) 5a. Derive an algebraic expression for the maximum height the mass reaches, h , in terms of x , k , μ_k , d , m , and acceleration due to gravity g .



Report your answer in this box.

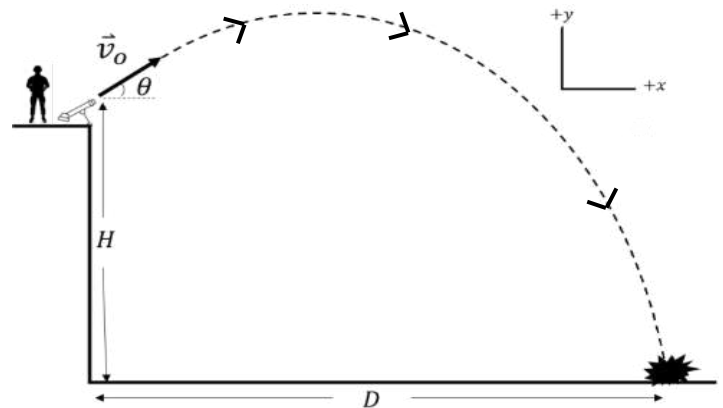
- (5) 5b. In a new scenario, the coefficient of kinetic friction between the object and the rough region is decreased. Determine how the maximum height of the object on the hill changes, if at all.
- ☐ The maximum height decreases.
 - ☐ The maximum height does not change.
 - ☐ The maximum height increases.

Wt. **No.**

Cadet: _____

- 55 6. A Soldier launches a mortar round from a hilltop that is a height H above an enemy position. The mortar round is launched with an initial velocity $\vec{v}_0 = 158$ m/s at an angle $\theta = 25.5^\circ$, travels along a parabolic path as shown, and lands after traveling a distance $D = 2250$ m. Air resistance is negligible.

- (50) 6a. Calculate H , the height of the hill.



Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

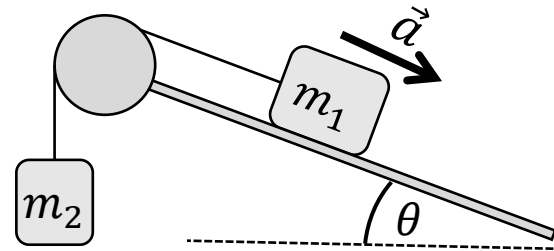
- (5) 6b. Select the correct option.
- ☐ The round reaches terminal velocity at its maximum height.
 - ☐ The speed of the round as it strikes the ground is 0 m/s.
 - ☐ The force of gravity only acts on the round after it reaches its highest point.
 - ☐ If the hilltop were taller, the round would travel further.

Wt. **No.**

Cadet: _____

- 40 7. Two blocks are connected by an ideal rope that passes over a massless, frictionless pulley. Block 1 has a mass m_1 and slides down a ramp. Neglect frictional forces. The ramp forms an angle θ with the horizontal. Block 2 has mass m_2 and is pulled up by the rope. Both blocks accelerate with a magnitude of acceleration a as shown.

- (10) 7a. Draw a free body diagram of Block 1.



- (30) 7b. Derive an algebraic expression for the magnitude of the tension force between the blocks.

Report your answer in this box.
--



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #1 SOLUTION

Time of completion _____ Cadet _____
 25 Feb 2026 (First) (Last)

Section _____
 Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Written Partial Review I

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Check your exam for 7 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **06 March 2026 at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	35	
2	35	
3	45	
4	35	
5	55	
6	55	
7	40	

Total:	300
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A: 33 (11%)

B: 91 (30%)

D: 176 (59%)

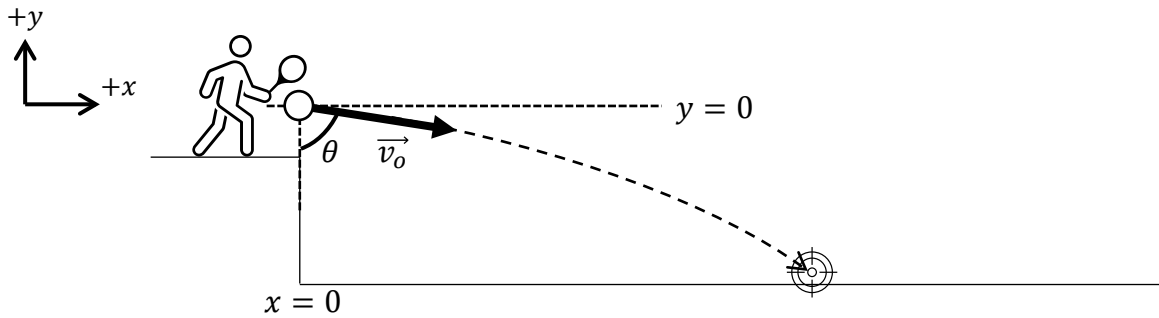
Total: 300

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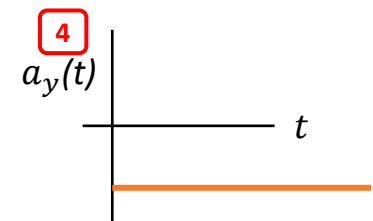
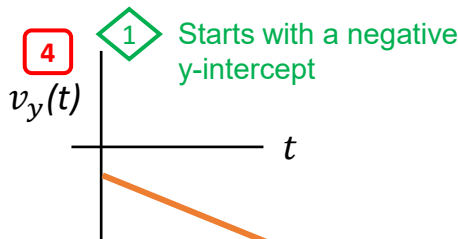
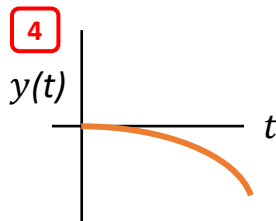
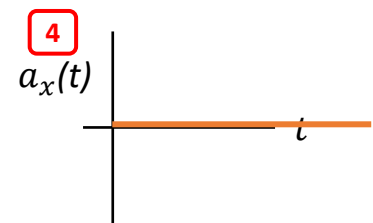
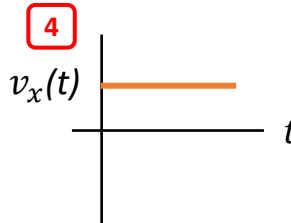
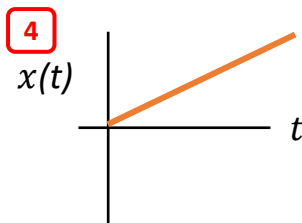
Wt. No.

Cadet: _____

- 35 1. A tennis player on a hilltop hits a tennis ball at a target that is at an elevation lower than the player. The trajectory of the ball is shown below. Neglect air resistance.



- (25) 1a. In the following diagrams, sketch the expected functions of position, velocity, and acceleration as a function of time for the tennis ball. Assume that the origin is the point at which the tennis ball is no longer in contact with the tennis racket.



- (10) 1b. The velocity of the tennis ball just before hitting the target is given by:

$$\vec{v} = (10.1\hat{i} - 37.5\hat{j}) \text{ m/s}$$

Calculate the speed of the projectile, $|\vec{v}|$, at that time.

☐ 47.6 m/s

☐ 0.00 m/s

6 3 1 ☒ 38.8 m/s

☐ 27.4 m/s

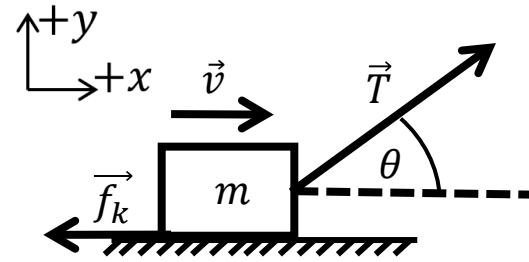
Wt. No.

Cadet: _____

35

2.

A block of mass m on a rough horizontal surface is pulled by a rope at a constant speed. The rope makes an angle θ above the horizontal and applies a constant force. The rough surface imparts a frictional force $\vec{f}_k$ on the block.



- (15) 2a. Derive an expression for the magnitude of the tension force T while the block moves at a constant speed.

4

☐ $T = f_k \sin(\theta)$

8

☐ $T = f_k$

10

3

2

☒ $T = \frac{f_k}{\cos(\theta)}$

☐ $T = ma$

- (10) 2b. Select the most correct statement for the block's acceleration in the y-direction while on the horizontal surface.

☐ $a_y = g$

☐ a_y depends on the mass m .

2

☐ a_y depends on the magnitude of the tension force.

6

3

1

☒ $a_y = 0$

- (10) 2c. Select which of the following is a Newton's Third Law Force Pair.

☐ The normal force from the ground on the box and the block's weight force

☐ The tension force on the block and the friction force on the block

6

3

1

☒ The normal force on the block from the surface and the normal force on the surface from the block

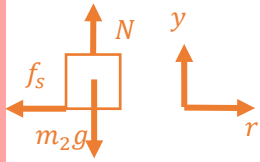
☐ The block's weight force and the tension force on the block

Wt. No.

Cadet: _____

- 45 3. A 1250-kg car rounds an unbanked curve of radius $R = 176$ m. The coefficient of static friction between the road and the tires is $\mu_s = 0.693$.

- (30) 3a. Calculate the maximum speed at which the car can turn without slipping.



4 Eq. 12, a_c

4 Eq. 17, static friction

6 Eq. 15, $F = ma$

2 Attempts to Solve in r and y

3 Subs a_c

$$\sum F_r = ma_r$$

$$f_s = m * \frac{v^2}{r}$$

$$\mu_s N = m * \frac{v_{max}^2}{R}$$

$$\mu_s mg = m * \frac{v_{max}^2}{R}$$

$$v = \sqrt{\mu_s g R}$$

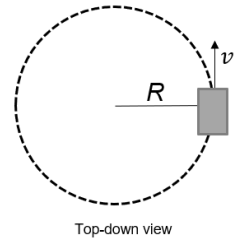
$$v = 34.6 \text{ m/s}$$

$$\sum F_y = ma_y$$

$$N - W = 0$$

$$N = mg$$

4 Attempts to solve for v



3 Correct ANS 1 SF 1 Units

$v = 34.6$	m/s
Report your answer in this box.	Report units in this box.

- (15) 3b. You are tasked with redesigning this road to permit drivers to travel at the same speed during icy conditions. Briefly describe how you will change the road using the physics learned in this class. You may use words and/or equations; you **must** use physics. You only need to describe one modification. Solutions must be legible.

Possible Correct Answers:

- Bank the road
- Increase the radius of the road
- Increase μ_s in spite of the ice
- Add salt (must mention increased μ_s)
- Plow Road (must mention increased μ_s)

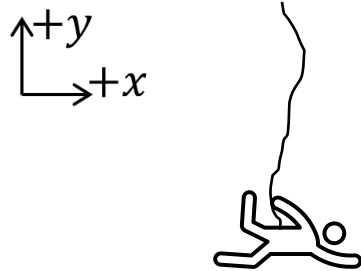
- 3 Correct modification with correct physics
- 2 Correct modification, poorly justified with physics
- 2 Correct modification, no justification
- 8 Incorrect Modification, but attempts to justify with physics
- 0 Blank or Illegible

Wt. No.

Cadet: _____

- 35 4. A 78.4-kg Cadet goes bungee jumping on spring break. The Cadet falls from rest straight down. The force exerted by the bungee cord is given by the function below, assuming the bungee cord is unstretched when the Cadet jumps.

$$\vec{F}(y) = (-k_1y + -k_2y^3)\hat{j}$$



- (15) 4a. Calculate the work done by gravity on the Cadet after she has fallen 125 meters.

- ☒ 10 ☐ 4 ☒ 1 ☐ 96,100 J
☐ ☒ 8 ☐ - 96,100 J
☐ ☒ 4 ☐ 60,100 J
☐ - 61,100 J

- (10) 4b. Select the expression that best describes the work done on the cadet, by the bungee cord, while the cadet falls.

- ☐ $W = -k_1 - 3k_2y^2$
☒ 5 ☐ $W = \frac{k_1}{2}y^2 + \frac{k_2}{4}y^4$
☐ $W = k_1y + k_2y^3$
☒ 7 ☒ 3 ☐ $W = -\frac{k_1}{2}y^2 - \frac{k_2}{4}y^4$
☐ $W = 0$

- (10) 4c. Select the option that best describes the forces acting on the Cadet while falling.

- ☒ 4 ☐ The force of gravity is conservative, but the force of the bungee is non-conservative.
☒ 3 ☐ The force of gravity is non-conservative, but the force of the bungee cord is conservative.
☒ 4 ☐ 3 ☒ 3 ☐ Both gravity and the bungee cord's force are conservative forces.
☐ Both gravity and the bungee cord's force are non-conservative forces.

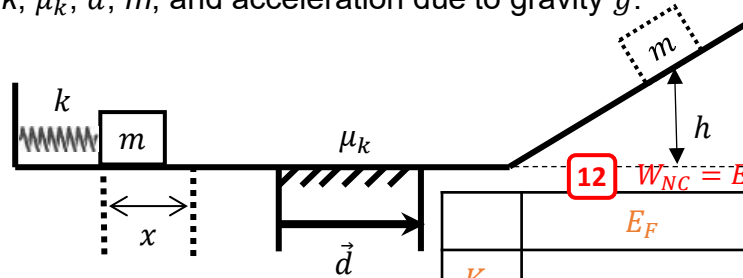
A: 7 (20%)
 B: 14 (40%)
 D: 14 (40%)
 Total: 35

Wt. No.

Cadet: _____

- 55 5. An object of mass m is initially at rest and held against a spring of spring constant k . The spring is compressed a distance x from the relaxed length. Then, the block is released and travels on a track that is frictionless except for a rough region of length d , as shown. The coefficient of kinetic friction between the object and the rough portion of the track is μ_k . The object slides through the rough region and then up a frictionless ramp to a final height h , where the object momentarily comes to rest.

- (50) 5a. Derive an algebraic expression for the maximum height the mass reaches, h , in terms of x , k , μ_k , d , m , and acceleration due to gravity g .



12 $W_{NC} = E_f - E_0$ or COE attempted

	E_f	E_0
K_T	--	--
K_R	—	—
U_s	0	$\frac{1}{2}kx^2$
U_g	mgh	0

Non-cons. Work

$$\Sigma F_y = ma_y \rightarrow N = mg$$

$$\Sigma F_x = ma_x \rightarrow f_k = \mu_k mg$$

$$W_{NC} = \mu_k mg d \cos(180^\circ)$$

$$W_{NC} = -\mu_k mg d$$

$$W_{NC} = E_f - E_0$$

$$W_{NC} = mgh - \frac{1}{2}kx^2$$

$$-\mu_k mg d = mgh - \frac{1}{2}kx^2$$

$$h = \frac{-\mu_k mg d + \frac{1}{2}kx^2}{mg}$$

4 $U_g = mgh$

4 $U_s = \frac{1}{2}kx^2$

4 $W = \vec{F} \cdot \vec{d}$

2 $\Sigma F = ma$

3 $f_k = \mu_k N$

3 Attempts any W_{nc}

3 Any N2L for W_{nc}

4 Correct $W_{NC} = -\mu_k mgL$

2 Attempts to solve for h

4 Energy terms present (1 pt) and correct (1 pt)

5 Correct ANS

$$h = \frac{-\mu_k mg d + \frac{1}{2}kx^2}{mg}$$

Report your answer in this box.

- (5) 5b. In a new scenario, the coefficient of kinetic friction between the object and the rough region is decreased. Determine how the maximum height of the object on the hill changes, if at all.

☐ The maximum height decreases.

☐ The maximum height does not change.

3 1 1 ☒ The maximum height increases.

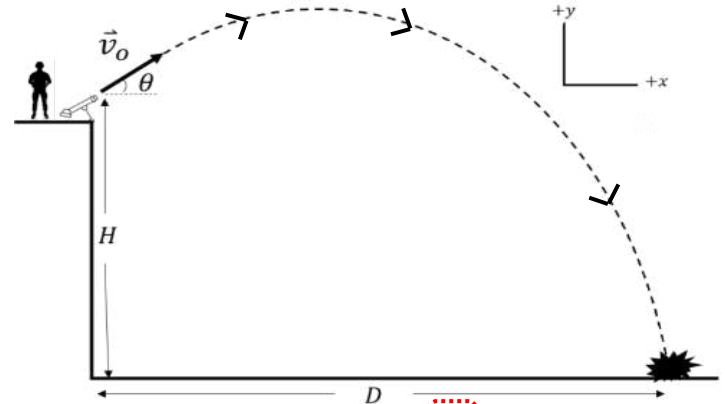
Wt. No.

Cadet: _____

- 55 6. A Soldier launches a mortar round from a hilltop that is a height H above an enemy position. The mortar round is launched with an initial velocity $\vec{v}_0 = 158 \text{ m/s}$ at an angle $\theta = 25.5^\circ$, travels along a parabolic path as shown, and lands after traveling a distance $D = 2250 \text{ m}$. Air resistance is negligible.

- (50) 6a. Calculate H , the height of the hill.

$x_0 = 0$	$y_0 = H$
$x = D$	$y = 0$
$v_{0x} = v_0 \cos(\theta)$	$v_{0y} = v_0 \sin \theta$
$v_{0x} = v_0 \cos(\theta)$	$v_y = ?$
$a_x = 0$	$a_y = -g$



Get time from x

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$t = \frac{D}{v_{0x}} \quad \text{where } v_{0x} = v_0 \cos(\theta)$$

$$t = 15.78 \text{ s}$$

Use t from x in y

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

to solve for H

$$H = -v_0 \sin \theta t + \frac{g}{2}(t)^2$$

$$h = 147 \text{ m}$$

14 Kinematics one dim.

7 Kinematics second dim.

6 SOHCAHTOA - Any trig

4 Correct components of initial velocity (x and y) (2 each)

2 $x = D$

2 Δy term is correctly applied

2 $a_y = -g$

2 Attempts to solve for time.

2 Uses/attempts to sub t

2 Solves for h

7 First dim PRC 2

7 First dim PRC 3

4 Second dim PRC 2

3 First dim PRC 3

5 Correct

1 SF

1 Units

$h = 147$	m
Report your answer in this box.	Report units in this box.

- (5) 6b. Select the correct option.

- ☐ The round reaches terminal velocity at its maximum height.
☐ The speed of the round as it strikes the ground is 0 m/s.
☐ The force of gravity only acts on the round after it reaches its highest point.

3

1

1

1

If the hilltop were taller, the round would travel further.

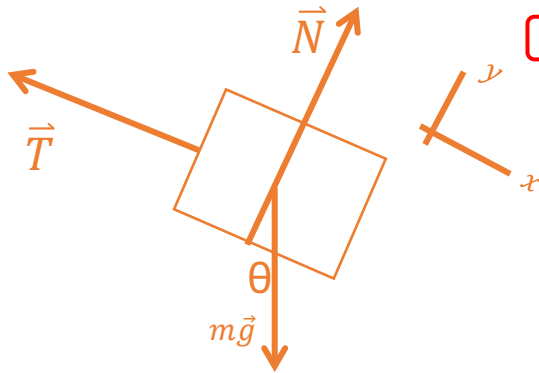
A: 6 (11%)
 B: 17 (31%)
 D: 32 (58%)
 Total: 55

Wt. No.

Cadet: _____

- 40 7. Two blocks are connected by an ideal rope that passes over a massless, frictionless pulley. Block 1 has a mass m_1 and slides down a ramp. Neglect frictional forces. The ramp forms an angle θ with the horizontal. Block 2 has mass m_2 and is pulled up by the rope. Both blocks accelerate with a magnitude of acceleration a as shown.

- (10) 7a. Draw a free body diagram of Block 1.



2 Isolate Body

2 Any Forces present

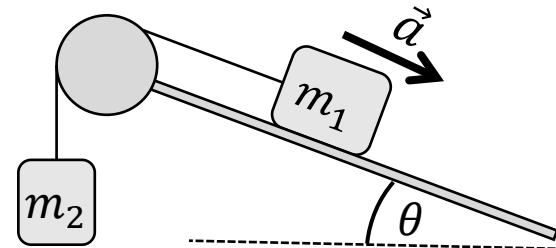
1 Normal force correct

1 Weight force correct

1 Tension force correct

2 Coordinates present

1 Critical angle correct



- (30) 7b. Derive an algebraic expression for the magnitude of the tension force between the blocks.

3 Attempts N2L on Mass 1

8 N2L Equation or attempt present

3 Attempts N2L on Mass 2

Mass 1

Mass 2

x	y
$\sum F_x = m_1 a_x$	$\sum F_y = m a_y$
$m_1 g \sin(\theta) - T = m_1 a_x$	$N - m_1 g \cos(\theta) = 0$

x	y
$\sum F_x = m_2 a_x$	$\sum F_y = m a_y$
$T - m_2 g = m_2 a_x$	$0 = 0$

2 $m_1 g \sin(\theta)$: +1 present, +1 correct

2 T : +1 present, +1 correct

2 $m_2 g$: +1 present, +1 correct

2 T : +1 present, +1 correct

$$m_1 g \sin(\theta) - T = m_1 \frac{T - m_2 g}{m_2}$$

$$m_2 m_1 g \sin(\theta) - m_2 T = m_1 T - m_1 m_2 g$$

$$m_2 m_1 g \sin(\theta) + m_1 m_2 g = m_1 T + m_2 T$$

$$T = \frac{m_1 m_2 g (\sin(\theta) + 1)}{m_1 + m_2}$$

3 Subs from 1 to 2 or 2 to 1

3 Attempts to solve for T

2 Correct ANS

$$T = \frac{m_1 m_2 g (\sin(\theta) + 1)}{m_1 + m_2}$$

Report your answer in this box.

A: 2 (5%)
B: 18 (45%)
D: 20 (50%)
Total: 40



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #2 HANDOUT

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X1, Physics I
Written Partial Review II

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices (including glasses and watches) are allowed in the examination room.**
- Check your exam for 6 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. Report answers in the answer boxes provided for each question.
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **24 April 2026** at **1700**.
- Cadet Signature: _____
- Cadet Name (print): _____

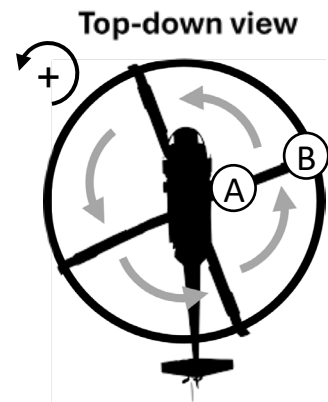
PROBLEM	WEIGHT
1	50
2	50
3	35
4	55
5	55
6	55
Total:	300

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Wt. **No.**

Cadet: _____

- 50 1. A UH-60 helicopter is depicted in the picture below. Points A and B on one of the four rotor blades are shown.
- (30) 1a. During startup, the rotor blades experience a constant angular acceleration. The blades are brought from rest to an angular speed of 18.5 rad/s in 24.6 s . **Calculate** the total angular displacement during the 24.6 s .



Report your numeric answer in this box.	Report units in this box.

- (10) 1b. Select the correct direction of the angular acceleration of the rotor blades on the picture above.

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙	None

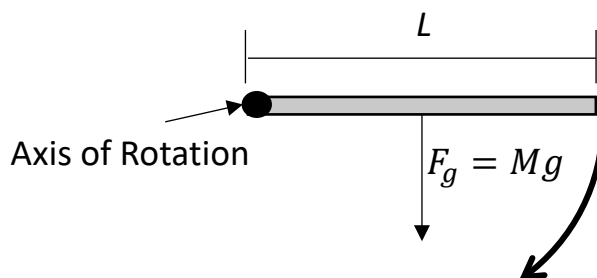
- (5) 1c. Correctly compare the angular speed at Point A to that of Point B.
- ☐ Point A has a smaller angular speed than Point B.
 - ☐ Point A has the same angular speed as Point B.
 - ☐ Point A has a larger angular speed than Point B.
- (5) 1d. Correctly compare the tangential speed at Point A to that of Point B.
- ☐ Point A has a smaller tangential speed than Point B.
 - ☐ Point A has the same tangential speed as Point B.
 - ☐ Point A has a larger tangential speed than Point B.

Wt. **No.**

Cadet: _____

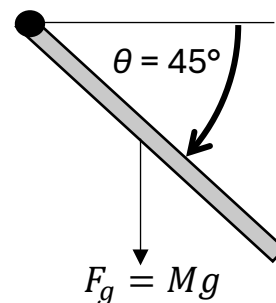
- 50 2. A thin uniform rod of mass M and length L is free to swing about a hinge at one end.
- (15) 2a. In Scenario 1, the rod is parallel to the Earth's surface, as shown. **Derive** an expression for the magnitude of the torque applied by the force due to gravity in this scenario.

- ☐ $\tau_g = MgL$
- ☐ $\tau_g = \frac{1}{2}MgL$
- ☐ $\tau_g = \frac{1}{2}MgL^2$
- ☐ $\tau_g = \frac{1}{12}MgL^2$

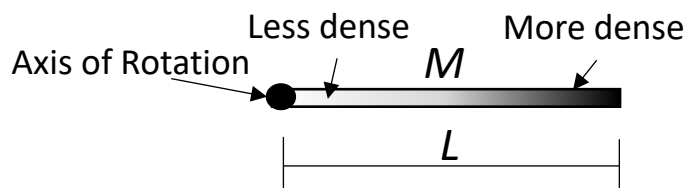


- (5) 2b. In Scenario 2, the same rod has swung halfway between being parallel to the Earth's surface and perpendicular to the Earth's surface. Which of the following best describes the torque applied due to the force of gravity?

- ☐ $\tau_{g, \text{Scenario 1}} < \tau_{g, \text{Scenario 2}}$
- ☐ $\tau_{g, \text{Scenario 1}} > \tau_{g, \text{Scenario 2}}$
- ☐ $\tau_{g, \text{Scenario 1}} = \tau_{g, \text{Scenario 2}}$
- ☐ $\tau_{g, \text{Scenario 2}} = 0$

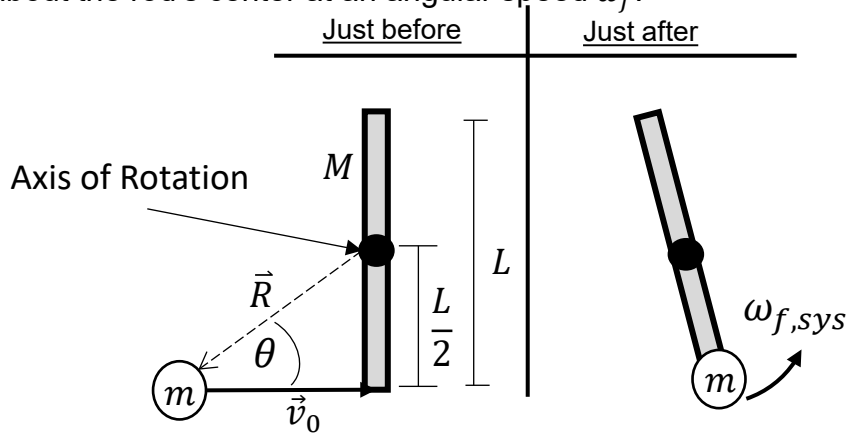


- (30) 2c. A different rod of mass M and length L is free to swing about a hinge at one end. The mass density of this new rod is not uniform. The mass density increases as a function of distance from the hinge. **Describe** how this non-uniform mass distribution affects the **moment of inertia** of this new rod compared to that of the original rod with uniform mass distribution. Justify your answer using physics concepts you have learned in this course. Limit your answer to a few sentences and/or equations.



35

3. A uniform thin rod with mass M and length L is initially at rest and is free to rotate about an axis through the center of the rod. A point particle with mass m travels with a constant initial velocity $\vec{v}_0$. As the particle travels, let $\vec{R}$ be the displacement between the particle and the rod's center, and let θ be the angle between $\vec{R}$ and $\vec{v}_0$. The particle impacts the rod, and the particle-rod system rotates about the rod's center at an angular speed ω_f .



- (10) 3a. **Derive** an equation for the magnitude of the angular momentum of the particle relative to the rod's axis of rotation prior to the collision.

- ☐ $l_o = L(m + M)v_o \sin(180 - \theta)$
☐ $l_o = Rmv_o \sin(180 - \theta)$
☐ $l_o = 0$
☐ $l_o = Rmv_o \cos(180 - \theta)$

- (10) 3b. Select the direction that most closely matches the direction of the particle's angular momentum relative to the rod's axis of rotation just before the collision.

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙	None

- (15) 3c. **Derive** an equation for the total moment of inertia of the particle-rod system just after the collision.

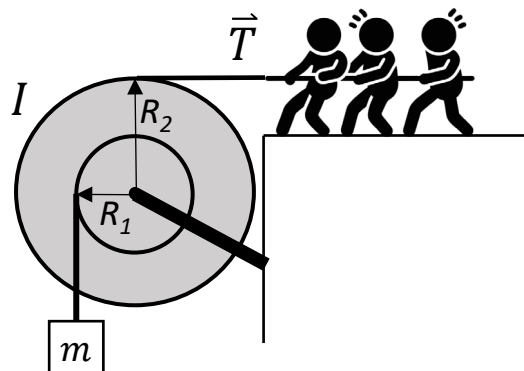
- ☐ $I = \frac{1}{12}(M + m)L^2$
☐ $I = \frac{1}{12}ML^2 + \frac{1}{4}mL^2$
☐ $I = \frac{1}{3}ML^2$
☐ $I = \frac{1}{12}ML^2 + \frac{1}{2}mL$

Wt. **No.**

Cadet: _____

- 55 4. The pulley system shown is used to lift heavy stones. The pulley can be modelled as two concentric solid cylinders. The compound pulley is supported by an axle through the pulley's center connected to the scaffolding where the workers are pulling.

- (10) 4a. Draw a free body diagram of the pulley.



- (40) 4b. The compound pulley has an inner radius R_1 of 0.235 m, an outer radius R_2 of 1.75 m, and a moment of inertia $I = 327 \text{ kg} \cdot \text{m}^2$. The workers lift a rock with a mass 578 kg by pulling with a constant tension force of 852 Newtons. **Calculate** the magnitude of the angular acceleration of the pulley system.

Report your numeric answer in this box.	Report units in this box.

- (5) 4c. Select the correct direction of the net torque on the pulley.

○	○	○	○	○	○	○	○	○	○	○
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙	None

Wt. **No.**

Cadet: _____

55

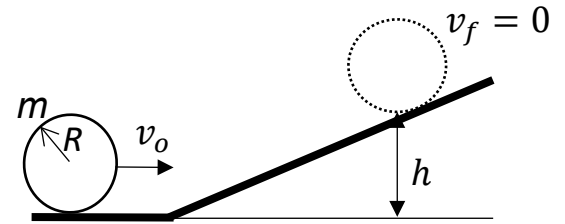
5.

A uniform solid sphere with radius R and mass m rolls without slipping with an initial translational speed of v_o . The sphere then rolls up an incline to a maximum height of h .

(50)

5a.

Derive an algebraic expression for the maximum height of the sphere, h , in terms of v_o and the acceleration due to gravity, g .



Report your algebraic answer in this box.

(5)

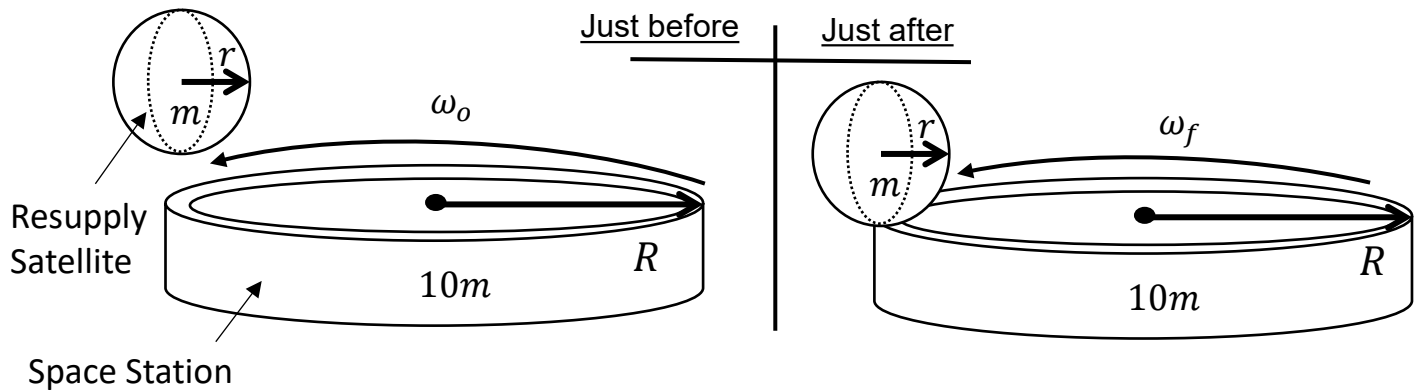
5b.

Select the correct statement regarding friction in rolling motion.

- ☐ Both static and kinetic friction are acting on an object that is rolling without slipping.
- ☐ Only static friction is acting on an object that is rolling without slipping.
- ☐ Only kinetic friction is acting on an object that is rolling without slipping.
- ☐ No friction acts on an object that is rolling without slipping.

- 55 6. A ring-shaped space station, approximated as a hoop about a central axis of mass $10m$ and radius R , rotates about its central axis at an angular speed of ω_0 . A resupply satellite, approximated as a solid sphere of mass m and radius r , docks with the space station and sticks to it. The satellite's speed while docking with the station is negligible. The satellite sticks such that the center of the satellite is a distance R from the space station's center.

Derive an algebraic expression for the final angular speed of the system, ω_f , after the space station and asteroid stick together in terms of R , r , and ω_0 .



Report your algebraic answer in this box.



UNITED STATES MILITARY ACADEMY
WEST POINT.

WPR #2 SOLUTION

Time of completion _____ Cadet _____
 17 April 2026 (First) (Last)

Section _____
 Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Written Partial Review II

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smart devices (including glasses and watches) are allowed in the examination room.**
- Check your exam for 6 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. Report answers in the answer boxes provided for each question.
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **24 April 2026** at **1700**.
- Cadet Signature: _____
- Cadet Name (print): _____

A: 29 (10%)
 B: 91 (30%)
 D: 180 (60%)

PROBLEM	WEIGHT
1	50
2	50
3	35
4	55
5	55
6	55
Total:	300

A: 0 (0%)
 B: 0 (0%)
 D: 0 (0%)

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PR2, AY26-2, Version 1	
A cuts: 0	
B cuts: 0	
D cuts: 0	
Total: 0	

Wt. No.

Cadet: _____

- 50 1. A UH-60 helicopter is depicted in the picture below. Points A and B on one of the four rotor blades are shown.
- (30) 1a. During startup, the rotor blades experience a constant angular acceleration. The blades are brought from rest to an angular speed of 18.5 rad/s in 24.6 s. **Calculate** the total angular displacement during the 24.6 s.

10 1st Rotational Kinematics Equation

6 2nd Rotational Kinematics Equation

② Attempts to apply 2 kinematic equations

② Solves for α

② Substitutes α into other equation

③ Solves for θ

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$\alpha = \frac{\omega - 0}{t}$$

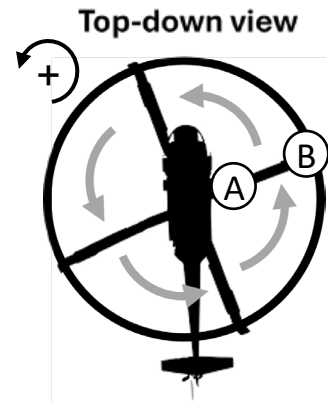
$$\theta = \frac{1}{2} * \frac{\omega}{t} * t^2$$

$$\Delta\theta = \frac{\omega^2 - \omega_0^2}{2 * \alpha}$$

$$\omega = \omega_0 + \alpha t$$

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$\Delta\theta = \frac{\omega^2}{2 * \frac{\omega}{t}}$$



③ Correct answer

228
1.30 · 10⁴

① Sig fig

radians

degrees

① Units

36.2	Revolutions/Rotations
Report your numeric answer in this box.	Report units in this box.

- (10) 1b. Select the correct direction of the angular acceleration of the rotor blades on the picture above.

☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐
										None

- (5) 1c. Correctly compare the angular speed at Point A to that of Point B.

☐ Point A has a smaller angular speed than Point B.

③ ② ☒ Point A has the same angular speed as Point B.

☐ Point A has a larger angular speed than Point B.

- (5) 1d. Correctly compare the tangential speed at Point A to that of Point B.

③ ① ① ☒ Point A has a smaller tangential speed than Point B.

☐ Point A has the same tangential speed as Point B.

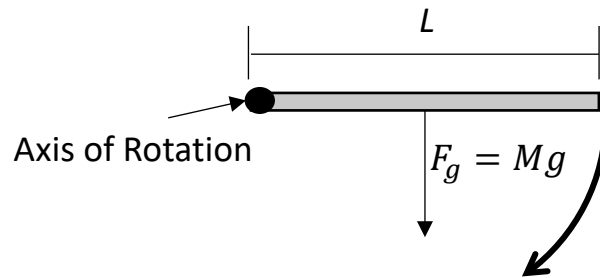
① ☐ Point A has a larger tangential speed than Point B.

Wt. No.

Cadet: _____

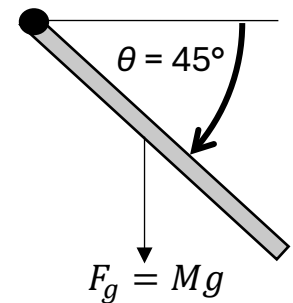
- 50 2. A thin uniform rod of mass M and length L is free to swing about a hinge at one end.
- (15) 2a. In Scenario 1, the rod is parallel to the Earth's surface, as shown. **Derive** an expression for the magnitude of the torque applied by the force due to gravity in this scenario.

- 8 ☐ $\tau_g = MgL$
- 8 5 2 ☒ $\tau_g = \frac{1}{2}MgL$
- 5 ☐ $\tau_g = \frac{1}{2}MgL^2$
- ☐ $\tau_g = \frac{1}{12}MgL^2$

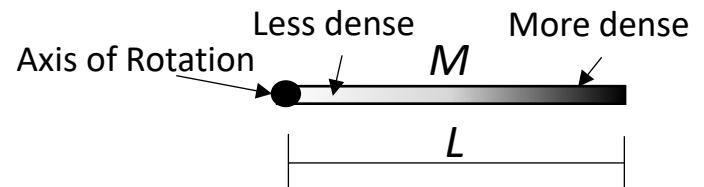


- (5) 2b. In Scenario 2, the same rod has swung halfway between being parallel to the Earth's surface and perpendicular to the Earth's surface. Which of the following best describes the torque applied due to the force of gravity?

- ☐ $\tau_{g, \text{Scenario 1}} < \tau_{g, \text{Scenario 2}}$
- 4 1 ☒ $\tau_{g, \text{Scenario 1}} > \tau_{g, \text{Scenario 2}}$
- ☐ $\tau_{g, \text{Scenario 1}} = \tau_{g, \text{Scenario 2}}$
- ☐ $\tau_{g, \text{Scenario 2}} = 0$



- (30) 2c. A different rod of mass M and length L is free to swing about a hinge at one end. The mass density of this new rod is not uniform. The mass density increases as a function of distance from the hinge. **Describe** how this non-uniform mass distribution affects the **moment of inertia** of this new rod compared to that of the original rod with uniform mass distribution. Justify your answer using physics concepts you have learned in this course. Limit your answer to a few sentences and/or equations.



Is there a legible answer?

Is the answer correct? (Mol increases)

No

Yes

Does cadet attempt to justify with relevant physics?

Does cadet justify with relevant physics?

Yes, physics is relevant

No 2

No, physics is irrelevant or completely wrong 10

No justification attempted

3 3 No, physics is irrelevant or completely wrong

3 9 Does the justification contain errors in understanding?

Yes: correct physics, wrong conclusion 13

Yes, there are some misunderstandings shown

No, CUT

A: 5 (10%)
B: 15 (30%)
D: 30 (60%)
Total: 50

Wt. No.

Cadet: _____

- 55 4. The pulley system shown is used to lift heavy stones. The pulley can be modelled as two concentric solid cylinders. The compound pulley is supported by an axle through the pulley's center connected to the scaffolding where the workers are pulling.

- (10) 4a. Draw a free body diagram of the pulley.

- 2 Isolate Body

2 Tension Forces Present (1 each)

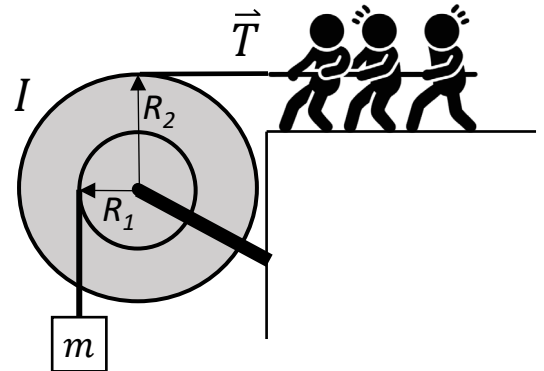
2 Positive Direction Indicated

1 Pulley's Weight Force Present

1 Pulley's Normal Force Present

1 Pulley Radii Present

1 Angles between R and T Present



- (40) 4b. The compound pulley has an inner radius R_1 of 0.235 m, an outer radius R_2 of 1.75 m, and a moment of inertia $I = 327 \text{ kg} \cdot \text{m}^2$. The workers lift a rock with a mass 578 kg by pulling with a constant tension force of 852 Newtons. **Calculate** the magnitude of the angular acceleration of the pulley system.

N2L Tension Force

$$\Sigma F_{ext} = m a$$

$$T = T$$

Pulley

$$\Sigma \tau = I \alpha$$

- 9 N2L-R 2 $\tau = R_n T_n \sin(90^\circ)$ (one point per term)

- 5 N2L

4 $\tau = r \times F$

- 4 $a_t = \alpha r$ 2 Applies link equation $a_t = \alpha r$

$$R_2 T - R_1 T_m = I \alpha$$

$$R_2 T - R_1 (m a + m g) = I \alpha$$

$$R_2 T - R_1 (m \alpha R_1 + m g) = I \alpha$$

$$R_2 T - m \alpha R_1^2 - R_1 m g = I \alpha$$

$$R_2 T - R_1 m g = I \alpha + m \alpha R_1^2$$

3 Attempts to solve for α
$$\alpha = \frac{R_2 T - R_1 m g}{I + m R_1^2}$$

- 3 Applies N2L to hanging mass
N2L Mass

$$\Sigma F_{ext} = m a$$

$$T_m = m g + m a$$

- 2 Subs from N2L to N2L-R (hanging mass)

- 2 Uses I for pulley

- 2 Correct ANS 1 Sig Figs 1 Units

$\alpha = 0.443$	rad/s^2
Report your numeric answer in this box.	Report units in this box.

- (5) 4c. Select the correct direction of the net torque on the pulley.

☐	☐	☐	☐	☐	☐	☐	☐	5 ☒	2 ☐	☐
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙	None

A: 4 (7%)
B: 16 (29%)
D: 35 (64%)
Total: 55

Wt. No.

55

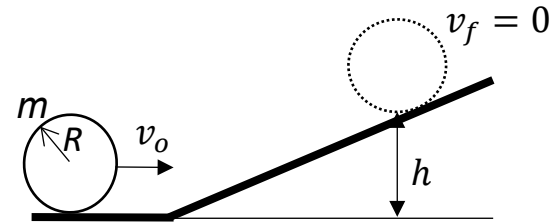
5.

A uniform solid sphere with radius R and mass m rolls without slipping with an initial translational speed of v_o . The sphere then rolls up an incline to a maximum height of h .

(50)

5a.

Derive an algebraic expression for the maximum height of the sphere, h , in terms of v_o and the acceleration due to gravity, g .



1 Any moment of Inertia equation

2 $I = \frac{2}{5}mR^2$

10 COE

5 $K_R = \frac{1}{2}I\omega^2$

2 $v_t = \omega r$

5 $U_g = mgh$

5 $K_T = \frac{1}{2}mv^2$

$W_{NC} = E_f - E_o$

$W_{NC} = E_f - E_o$

$0 = mgh - \frac{1}{2}mv^2 - \frac{1}{2}I\omega^2$

$0 = mgh - \frac{1}{2}mv_f^2 - \frac{1}{2}mR^2 \frac{v_f^2}{R^2}$

$h = \frac{7v^2}{10g}$

3 $W_{NC} = 0$

3 Energy term points present (1 per term)

3 Energy term points correct (1 per term)

1 Applies $v_t = \omega r$

1 Applies $I = \frac{2}{5}mR^2$

6 Attempts alg soln for h

3 Correct ANS

$h = \frac{7v^2}{10g}$

Report your algebraic answer in this box.

(5)

5b.

Select the correct statement regarding friction in rolling motion.

☐ Both static and kinetic friction are acting on an object that is rolling without slipping.

5 ☒ Only static friction is acting on an object that is rolling without slipping.

☐ Only kinetic friction is acting on an object that is rolling without slipping.

☐ No friction acts on an object that is rolling without slipping.

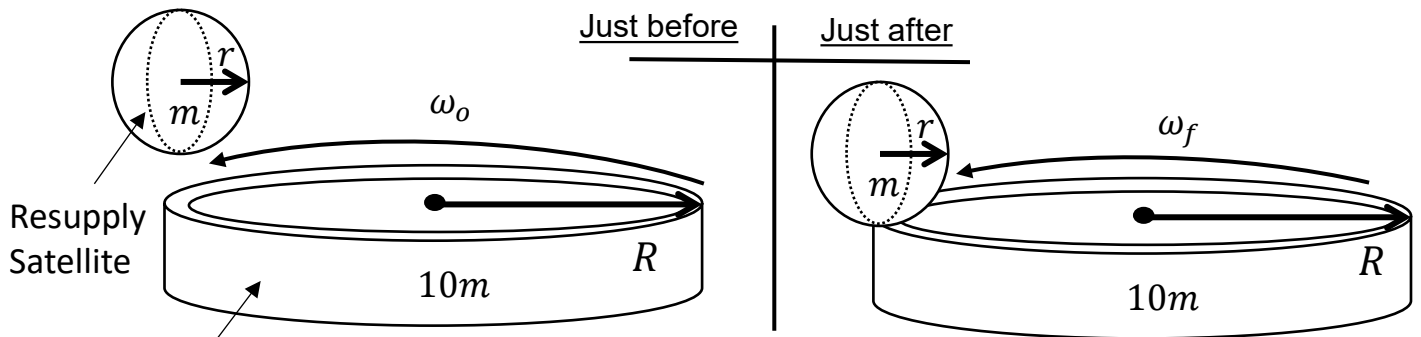
A: 5 (9%)
B: 20 (36%)
D: 30 (55%)
Total: 55

Wt. No.

Cadet: _____

- 55 6. A ring-shaped space station, approximated as a hoop about a central axis of mass $10m$ and radius R , rotates about its central axis at an angular speed of ω_0 . A resupply satellite, approximated as a solid sphere of mass m and radius r , docks with the space station and sticks to it. The satellite's speed while docking with the station is negligible. The satellite sticks such that the center of the satellite is a distance R from the space station's center.

Derive an algebraic expression for the final angular speed of the system, ω_f , after the space station and asteroid stick together in terms of R , r , and ω_0 .



Space Station

$$\sum L_0 = \sum L_f$$

$$L = I\omega$$

$$I_{PA} = I_{COM} + md^2$$

3 Applies I_{hoop} for L_0

1 Correct I_{hoop} for L_0

$$I_{hoop} = 10mR^2$$

$$\sum L_f = (I_{hoop} + I_{sphere, IPA})\omega_f$$

$$I_{hoop}\omega_0 = (I_{hoop} + I_{sphere, IPA})\omega_f$$

$$10mR^2\omega_0 = \left(10mR^2 + \frac{2}{5}mr^2 + mR^2\right)\omega_f$$

$$\omega_f = \frac{10mR^2\omega_0}{\left(11mR^2 + \frac{2}{5}mr^2\right)}$$

14 COAM

$$3 I = \frac{2}{5}mR^2$$

$$3 I = mR^2$$

$$10 L = I\omega$$

$$3 I_{PA} = I_{COM} + md^2$$

2 Attempts I_{PA}

$$1 I_{satellite} = I_{COM} + md^2 \text{ (correct)}$$

5 Attempts to combine /

$$2 I_{tot} = I_{Hoop} + I_{satellite} \text{ (correct)}$$

$$I_{PA} = I_{COM} + md^2$$

$$I_{PA} = \frac{2}{5}mr^2 + mR^2$$

6 Solves for ω_f

2 Correct ANS

$$\omega_f = \frac{10R^2\omega_0}{\left(11R^2 + \frac{2}{5}r^2\right)}$$

Report your algebraic answer in this box.

A: 5 (9%)
B: 17 (31%)
D: 33 (60%)
Total: 55



TAB G

TEE



UNITED STATES MILITARY ACADEMY
WEST POINT.

TEE HANDOUT

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Term End Exam

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Do not write on the Physics Reference Card.
- Check your exam for 30 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions choose the **one** selection that best answers each question and **completely fill in your selection.**
- For free response questions show all work; partial credit may be given for correct work shown. Report answers in the answer boxes provided for each question.
- Take up to 3.5 hours to complete the examination.
- An instructor is in the hall or in the classroom for any questions.

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **30 May 2026** at **2359**.
- Cadet Signature: _____
- Cadet Name (print): _____

Problem	Wt.	Score
1	10	
2	10	
3	5	
4	45	
5	10	
6	10	
7	50	
8	10	
9	5	
10	10	
11	45	
12	30	

Subtotal: 240

Problem	Wt.	Score
13	30	
14	45	
15	10	
16	5	
17	10	
18	10	
19	10	
20	5	
21	45	
22	45	
23	45	
24	5	

Subtotal: 265

Problem	Wt.	Score
25	5	
26	5	
27	10	
28	10	
29	35	
30	30	

Subtotal: 95

Total:	600	
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Wt. **No.**

Cadet: _____

- 10 1. A train speeds up along a one-dimensional path with an initial speed of 1.2 m/s at $t = 0$ s and a constant positive acceleration of 0.85 m/s^2 . Calculate the distance travelled by the train from $t = 0$ s to $t = 5.4$ s.

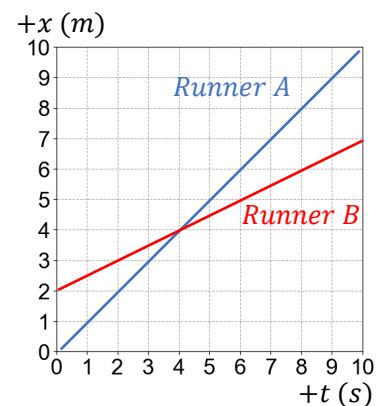
- ☐ 31 m
- ☐ 19 m
- ☐ 14 m
- ☐ 12 m

- 10 2. A train travelling at 13.0 m/s in the positive $\hat{i}$ direction slows down at a constant acceleration for 145 m and then comes to rest. Calculate the train's acceleration during this time.

- ☐ $0.583 \hat{i} \text{ m/s}^2$
- ☐ $-1.75 \hat{i} \text{ m/s}^2$
- ☐ $-0.0448 \hat{i} \text{ m/s}^2$
- ☐ $-0.583 \hat{i} \text{ m/s}^2$

- 5 3. The x-dimension position functions over time for two runners are shown. Select the statement correctly comparing their speeds.

- ☐ $v_{Ax} > v_{Bx}$
- ☐ $v_{Ax} < v_{Bx}$
- ☐ $v_{Ax} = v_{Bx}$ at $t = 4 \text{ s}$

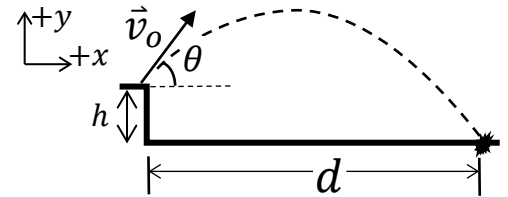


Wt. **No.**

Cadet: _____

- 45 4. A cannon launches a projectile from a cliff of a height $h = 635$ m above a valley with an initial velocity $\vec{v}_0$ at an angle $\theta = 32.7^\circ$ above the horizontal as depicted. The projectile travels a horizontal distance $d = 4270$ m before landing on the ground. Assume air resistance is negligible.

Calculate the magnitude of v_0 .



Report your numeric answer in this box.	Report units in this box.
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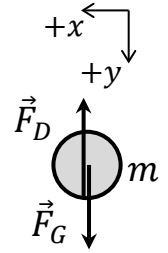
Wt. **No.**

Cadet: _____

A 16.5 kg-ball falls straight down from a cliff and air resistance acts on it. The coefficient of drag is 0.500 and the effective cross-sectional area is 0.0154 m².

10. 5. Select the most correct Newton's Second Law equation for the y-dimension of the ball while it is falling before it reaches terminal velocity using the provided coordinate axes

- ☐ $F_D + mg = ma_y$
- ☐ $-F_D + mg = ma_y$
- ☐ $F_D = mg$
- ☐ $-F_D + mg = \frac{mv^2}{r}$

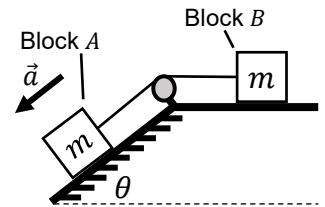


- 10 6. Calculate the magnitude of the drag force acting on the object when it is falling at a speed of 95.6 m/s.
- ☐ 42.6 N
- ☐ 95.6 N
- ☐ 162 N
- ☐ 31.3 N

Wt. **No.**

Cadet: _____

- 50 7. Two blocks are connected by an ideal rope that passes over a massless, frictionless pulley. Block *A* has mass m and slides down a ramp with a coefficient of kinetic friction between the block and the ramp μ_k . The ramp forms an angle θ with the horizontal. Block *B* has mass m and slides along a horizontal and frictionless surface. Both blocks accelerate with a magnitude of acceleration a as shown.

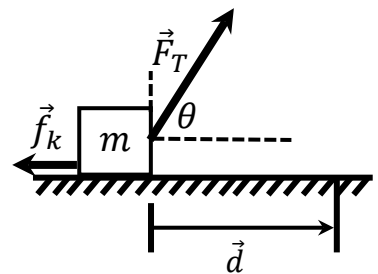


- (5) a. Draw a free-body diagram for Block *A*.
- (45) b. Derive an algebraic expression for the magnitude of the acceleration, a , of the two-block system. Express your answer in terms of μ_k , θ , and the acceleration due to gravity, g .

Report your algebraic answer in this box.

Answer the questions on this page using the provided scenario.

A block of mass $m = 11.5 \text{ kg}$ is pulled by a rope with tension force $F_T = 105 \text{ N}$ at an angle $\theta = 60.0^\circ$ for a distance $d = 1.50 \text{ m}$ along a rough horizontal surface. The friction force $\vec{f}_k$ does -45.1 J of work on the block across $\vec{d}$. See the figure below.



- 10 8. Calculate the work by the tension force on the block across the displacement, $\vec{d}$.
- ☐ 78.8 J
 - ☐ 56.4 J
 - ☐ 136 J
 - ☐ 45.1 J
- 5 9. Select which of the following is true during the block's motion over displacement, $\vec{d}$.
- ☐ The work on the block by the tension force equals its change in kinetic energy.
 - ☐ The work on the block by the friction force equals its change in kinetic energy.
 - ☐ The net work on the block equals its change in kinetic energy.
 - ☐ The net work on the block equals zero because the work by friction cancels the work by the tension force.
- 10 10. Select which of the following is true if the mass of the block is increased while $\vec{F}_T$, θ , and $\vec{d}$ remain constant.
- ☐ The acceleration of the block decreases.
 - ☐ The acceleration of the block increases.
 - ☐ The acceleration of the block goes to zero.

Wt. **No.**

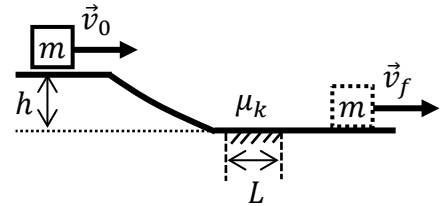
Cadet: _____

45

11.

A block of mass m slides along a track with an initial speed v_0 and continues down a hill of height h as shown. The track is frictionless everywhere except for a rough horizontal segment of length L and coefficient of friction μ_k between the block and the surface. The block exits the rough segment with a final speed v_f .

Derive an algebraic expression for the final speed, v_f , of the block after it exits the rough segment in terms of h , μ_k , L , v_0 , and acceleration due to gravity, g .

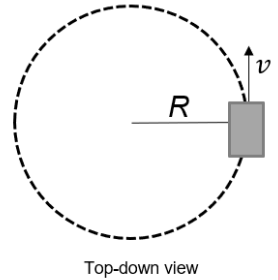


Report your algebraic answer in this box.

Wt. **No.**

Cadet: _____

- 30 12. A car rounds an unbanked curve of radius $R = 105$ m. The car rounds the curve without slipping at a maximum speed $v = 28.0$ m/s. The coefficient of static friction between the road and the car is μ_s . Calculate the coefficient of static friction, μ_s , between the road and the car.



Report your numeric answer in this box.

- 30 13. A moose is standing on a road in front of a moving motorcycle. The motorcycle has mass $m = 245$ kg and is traveling at a speed $v = 12.6$ m/s just before it collides with the moose. The stationary moose exerts an average force of magnitude $F_{ave} = 991$ N, on the motorcycle to bring it to a complete stop. Calculate the time, t , it takes for the motorcycle to come to a complete stop.

Report your numeric answer in this box.

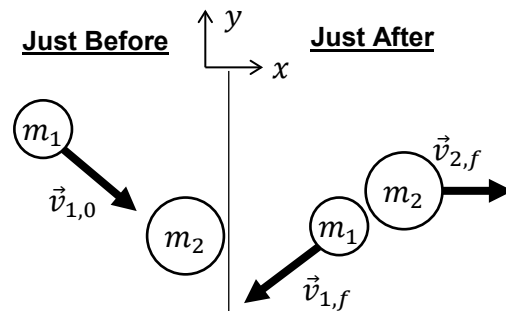
Report units in this box.

Wt. **No.**

Cadet: _____

- 45 14. Two disks collide on a frictionless horizontal surface. Disk 1 has mass $m_1 = 2.87$ kg and disk 2 has mass $m_2 = 4.23$ kg. Just before the collision, disk 1 has a velocity $\vec{v}_{1,0} = (8.72 \hat{i} - 3.15 \hat{j})$ m/s and disk 2 is initially at rest. Just after the collision disk 2 has a velocity $\vec{v}_{2,f} = 9.51 \hat{i}$ m/s.

Calculate the velocity of disk 1, $\vec{v}_{1,f}$, just after the collision.
Report your answer in unit-vector notation.



Report your numeric answer in this box.

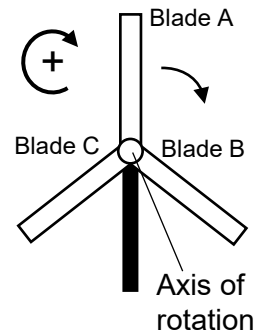
Report units in this box.

Wt. **No.**

Cadet: _____

Answer the questions on this page using the provided scenario.

A windmill with three blades, approximated as thin rods, is rotating clockwise (defined as positive) as shown in the figure to the right.



- 10 15. Blade A has an initial angular velocity $\omega_0 = 1.93 \text{ rad/s}$. It speeds up with an angular acceleration $\alpha = 1.21 \text{ rad/s}^2$, to a final angular velocity $\omega_f = 5.14 \text{ rad/s}$. Calculate the angular displacement during the acceleration.

- ☐ 18.8 rad
- ☐ 3.21 rad
- ☐ 2.65 rad
- ☐ 9.38 rad

- 5 16. Select the direction of the angular acceleration while the windmill speeds up.

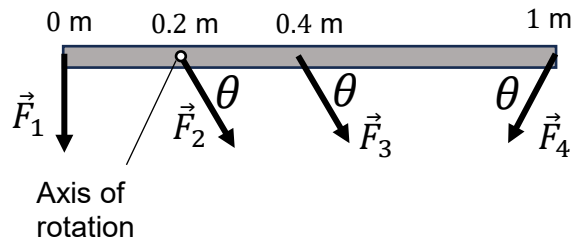
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙

- 10 17. At an arbitrary point in time the angular speed of the windmill is 2.67 rad/s and the windmill blade is 50.5 m long from the axis of rotation to the tip. Calculate the tangential speed at the tip of the blade at that time.

- ☐ 135 m/s
- ☐ 0.141 m/s
- ☐ 0.0529 m/s
- ☐ 119 m/s

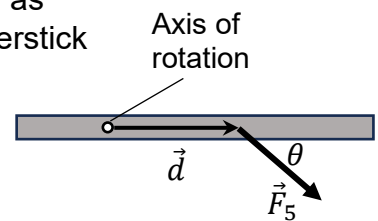
- 10 18. Below is an overhead view of a meterstick that can pivot about a point 0.20 m from one end. All forces on the stick have the same magnitude, $F_1 = F_2 = F_3 = F_4$. Compare the magnitudes of the torques created by the respective forces about the pivot point when $\theta = 60^\circ$.

- ☐ $\tau_4 > \tau_3 > \tau_2 > \tau_1$
☐ $\tau_4 > \tau_3 > \tau_1 > \tau_2$
☐ $\tau_4 > \tau_1 > \tau_3 > \tau_2$
☐ $\tau_4 > \tau_1 > \tau_2 > \tau_3$
☐ $\tau_4 > \tau_3 = \tau_1 > \tau_2$



- 10 19. A fifth force $F_5 = 15 \text{ N}$ is applied to the meterstick a distance, $d = 0.30 \text{ m}$, from the axis of rotation at an angle, $\theta = 25^\circ$, as shown. Calculate the magnitude of the torque on the meterstick due to $\vec{F}_5$.

- ☐ 4.5 N m
☐ 1.9 N m
☐ 4.1 N m
☐ 110 N m



- 5 20. A spherical shell, a solid sphere, and a solid cylinder are rotating at the same angular speed ω about their respective central axes. All objects have the same mass M and radius R . Compare the angular momentum between the spherical shell L_{shell} , solid sphere $L_{\text{solid sphere}}$, and solid cylinder L_{cyl} .

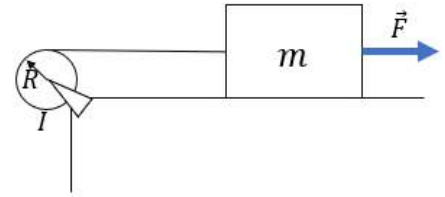
- ☐ $L_{\text{shell}} > L_{\text{solid sphere}} > L_{\text{cyl}}$
☐ $L_{\text{shell}} > L_{\text{cyl}} > L_{\text{solid sphere}}$
☐ $L_{\text{solid sphere}} > L_{\text{cyl}} > L_{\text{shell}}$
☐ $L_{\text{solid sphere}} > L_{\text{shell}} > L_{\text{cyl}}$
☐ $L_{\text{cyl}} = L_{\text{solid sphere}} = L_{\text{shell}}$

Wt. **No.**

Cadet: _____

- 45 21. A solid disk pulley with a radius $R = 0.234$ m and mass $M_p = 1.25$ kg is mounted on a frictionless horizontal axle. A massless cord wrapped around the pulley is attached to a block of mass, $m = 2.56$ kg. A horizontal force of magnitude, $F = 8.00$ N, is applied to the block which causes the block to accelerate to the right along a horizontal and frictionless surface.

Calculate the magnitude of the angular acceleration, α , of the pulley.



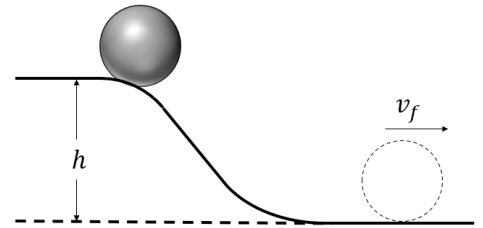
Report your numeric answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 45 22. A uniform solid sphere is at the top of a ramp of height, $h = 9.90$ m. The sphere is released from rest and rolls without slipping down the ramp. At the bottom of the ramp, the sphere has a final translational speed, v_f .

Calculate the final translational speed, v_f , of the sphere at the bottom of the ramp.



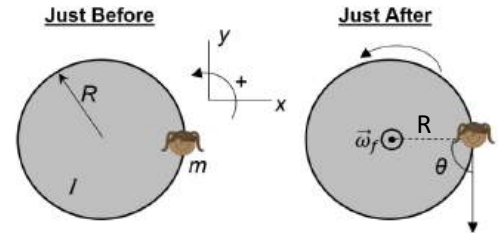
Report your numeric answer in this box.	Report units in this box.
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Wt. **No.**

Cadet: _____

- 45 23. A child of mass m is at the edge of a merry-go-round. The merry-go-round has a radius R and mass M and can be approximated as a disk. The merry-go-round and child are initially at rest. The child then jumps off the merry-go-round with velocity v at an angle $\theta = 90.0^\circ$ in the direction indicated. Immediately after, the merry-go-round rotates counter-clockwise (when viewed from above) with an angular speed ω_f . The child can be approximated as a point mass.

Derive an algebraic expression for the angular speed, ω_f , of the merry-go-round immediately after the child jumps off in terms of m , v , R and M .



Report your algebraic answer in this box.

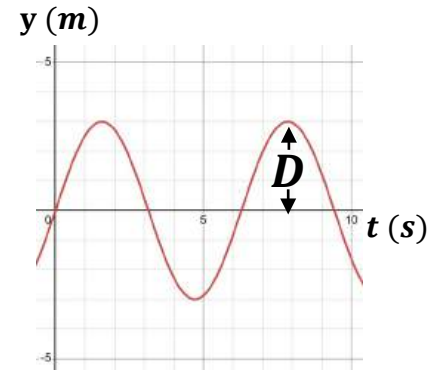
Wt. **No.**

Cadet: _____

- 5 24. The time to complete one full oscillation of a wave is defined as the _____.
○ Frequency
○ Wavelength
○ Period
○ Angular frequency

- 5 25. In the graph to the right the variable “D” represents the _____.

- Frequency
○ Wavelength
○ Period
○ Amplitude



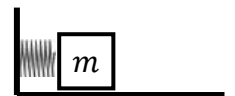
- 5 26. The variable k in the equation $k = \frac{2\pi}{\lambda}$ represents the _____.

- Frequency
○ Angular wave number
○ Spring constant
○ Angular frequency

- 10 27. A string under a tension of 121 N has a linear mass density of 5.82×10^{-4} kg/m. Calculate the speed of a transverse mechanical wave on the string.

- 0.0704 m/s
○ 208000 m/s
○ 456 m/s
○ 4.81×10^{-6} m/s

- 10 28. A spring-mass system rests on a frictionless surface as shown to the right and is in simple harmonic motion. The spring constant is 95.7 N/m and the angular frequency of the system is 3.51 rad/s. Calculate the mass of the block in the spring-mass system.

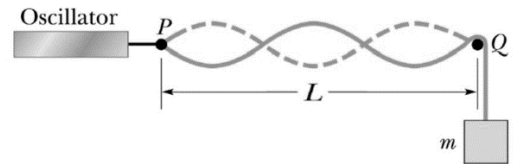


- 27.3 kg
○ 7.77 kg
○ 336 kg
○ 0.129 kg

Wt. **No.**

Cadet: _____

- 35 29. A string with a linear density of μ is tied to a string oscillator at P and passes over a frictionless support at Q . A block of mass m is suspended from the end of the string. The string's length between points P and Q is L . The amplitude of the motion at P is small enough for that point to be considered a node. A node also exists at Q .
- (30) a. Derive an algebraic expression for the frequency, f , of the string oscillator in terms of L , m , g , μ , and n where n is the number of anti-nodes.



Report your algebraic answer in this box.

- (5) b. Determine the number of antinodes in the diagram above. Write your answer in the provided box below.

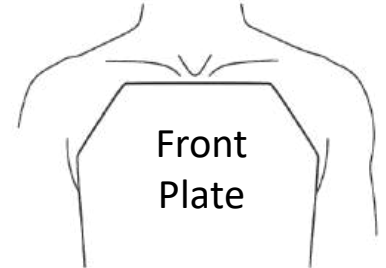
Report your answer here

Wt. **No.**

Cadet: _____

- 30 30. Your commander wants you to give a brief describing how the front plate of individual body armor stops bullets. You must use physics to **describe** how a plate carrier brings a bullet with a mass of m from an initial speed of v_o to rest after impact. **State** the overarching PH2X1 course concept(s) you will use in your description, **justify** your answer using **at least one equation** used in this course, and limit your answer to this page (additional work on scratch paper will not be evaluated for this question).

Physics Concept(s):



Equation(s):

Justification:



UNITED STATES MILITARY ACADEMY
WEST POINT.

TEE SOLUTION

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Term End Exam

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Do not write on the Physics Reference Card.
- Check your exam for 30 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions choose the **one** selection that best answers each question and **completely fill in your selection.**
- For free response questions show all work; partial credit may be given for correct work shown. Report answers in the answer boxes provided for each question.
- Take up to 3.5 hours to complete the examination.
- An instructor is in the hall or in the classroom for any questions.

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **30 May 2026** at **2359**.
- Cadet Signature: _____
- Cadet Name (print): _____

Problem	Wt.	Score
1	10	
2	10	
3	5	
4	45	
5	10	
6	10	
7	50	
8	10	
9	5	
10	10	
11	45	
12	30	

Subtotal: 240

Problem	Wt.	Score
13	30	
14	45	
15	10	
16	5	
17	10	
18	10	
19	10	
20	5	
21	45	
22	45	
23	45	
24	5	

Subtotal: 265

Problem	Wt.	Score
25	5	
26	5	
27	10	
28	10	
29	35	
30	30	

Subtotal: 95

Total:

A: 69 (12%)

B: 166 (28%)

D: 365 (61%)

Total: 600

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Wt. No.

Cadet: _____

- 10 1. A train speeds up along a one-dimensional path with an initial speed of 1.2 m/s at $t = 0$ s and a constant positive acceleration of 0.85 m/s^2 . Calculate the distance travelled by the train from $t = 0$ s to $t = 5.4$ s.

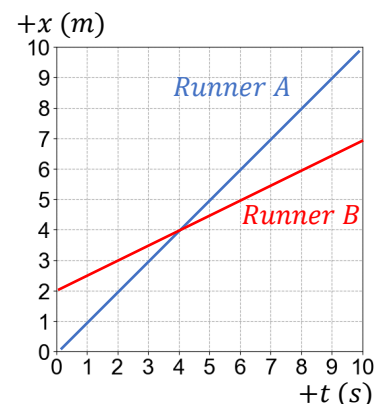
- ☐ 31 m $at^2 + v_0t$ [6] Dropped 1/2
☒ 19 m $x = v_0t + \frac{1}{2}a_xt^2$ [6] [3] [1]
☐ 14 m
☐ 12 m $0.5at^2$ [4] Dropped first term

- 10 2. A train travelling at 13.0 m/s in the positive $\hat{i}$ direction slows down at a constant acceleration for 145 m and then comes to rest. Calculate the train's acceleration during this time.

- ☐ $0.583 \hat{i} \text{ m/s}^2$ $a = \frac{v_0^2}{2x}$ [4] flipped initial and final
☐ $-1.75 \hat{i} \text{ m/s}^2$
☐ $-0.0448 \hat{i} \text{ m/s}^2$ $a = \frac{-v_0^1}{2x}$ [6] Dropped exponent
☒ $-0.583 \hat{i} \text{ m/s}^2$ $a = \frac{-v_0^2}{2x}$ [6] [3] [1]

- 5 3. The x-dimension position functions over time for two runners are shown. Select the statement correctly comparing their speeds.

- ☒ $v_{Ax} > v_{Bx}$ [3] [1] [1]
☐ $v_{Ax} < v_{Bx}$
☐ $v_{Ax} = v_{Bx}$ at $t = 4$ s

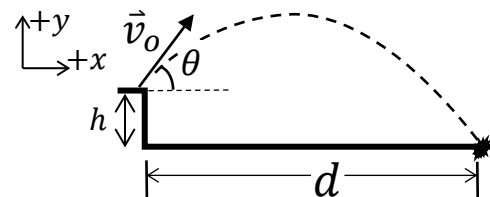


Wt. No.

Cadet: _____

- 45 4. A cannon launches a projectile from a cliff of a height $h = 635$ m above a valley with an initial velocity $\vec{v}_0$ at an angle $\theta = 32.7^\circ$ above the horizontal as depicted. The projectile travels a horizontal distance $d = 4270$ m before landing on the ground. Assume air resistance is negligible.

Calculate the magnitude of v_0 .



- 14 Kinematics one dim. 7 First dim PRC 2
7 Kinematics second dim. 7 First dim PRC 3
6 SOHCAHTOA - Any trig 4 Second dim PRC 2
3 First dim PRC 3

1 $\Delta x = d$

- 4 Identifies correct components of initial velocity and $v_{yo} = v_o \sin(\theta)$ (2 each)

$x_0 = 0$	$y_0 = h$
$x = d$	$y = 0$
$v_{0x} = v_0 \cos(\theta)$	$v_{0y} = v_0 \sin \theta$
$v_{0x} = v_0 \cos(\theta)$	$v_y = ?$
$a_x = 0$	$a_y = -g$

1 $\Delta y = h$

1 $a_y = -g$

- 2 Attempts to solve for time.

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$t = \frac{d}{v_{0x}} \quad \text{where } v_{0x} = v_0 \cos(\theta)$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$y - y_0 = v_0 \sin \theta t - \frac{g}{2}(t)^2$$

$$\Delta y = v_0 \sin \theta \frac{d}{v_0 \cos \theta} - \frac{g}{2} \left(\frac{d}{v_0 \cos \theta} \right)^2$$

$$v_0 = \sqrt{\frac{g * d^2}{2 * \cos^2(\theta) * (d * \tan(\theta) - \Delta y)}}$$

- 2 Attempts to sub t from one direction into the other

- 2 Correct substitution of $t = \frac{d}{v_{0x}}$ from x into y terms

- 2 Solves for v_0

1 Correct

1 SF

1 Units

$v_0 = 245$	m/s
Report your numeric answer in this box.	Report units in this box.

A: 5 (11%)
B: 11 (24%)
D: 29 (64%)
Total: 45

Wt. No.

Cadet: _____

A 16.5 kg-ball falls straight down from a cliff and air resistance acts on it. The coefficient of drag is 0.500 and the effective cross-sectional area is 0.0154 m².

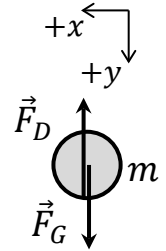
10. 5. Select the most correct Newton's Second Law equation for the y-dimension of the ball while it is falling before it reaches terminal velocity using the provided coordinate axes

☐ $F_D + mg = ma_y$ 6 Sign off on Drag

☒ $-F_D + mg = ma_y$ 6 3 1

☐ $F_D = mg$ 4 Sets a=0

☐ $-F_D + mg = \frac{mv^2}{r}$



- 10 6. Calculate the magnitude of the drag force acting on the object when it is falling at a speed of 95.6 m/s.

☒ 42.6 N correct 6 3 1

☐ 95.6 N

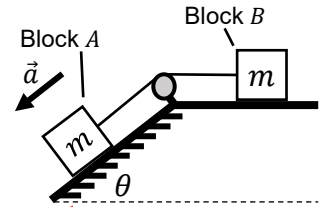
☐ 162 N mg 4 At terminal velocity

☐ 31.3 N no ρ 4 No ρ

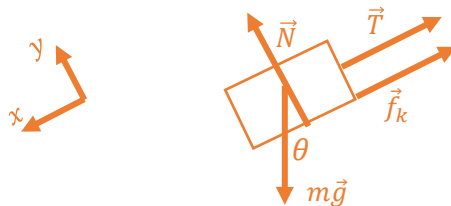
Wt. **No.**

Cadet: _____

- 50 7. Two blocks are connected by an ideal rope that passes over a massless, frictionless pulley. Block A has mass m and slides down a ramp with a coefficient of kinetic friction between the block and the ramp μ_k . The ramp forms an angle θ with the horizontal. Block B has mass m and slides along a horizontal and frictionless surface. Both blocks accelerate with a magnitude of acceleration a as shown.



- (5) a. Draw a free-body diagram for Block A.



- 1 Isolate body
- 1 Coordinates present
- 1 Coordinate follows ramp
- 1 Critical angle present
- 1 All forces completely correct

- (45) b. Derive an algebraic expression for the magnitude of the acceleration, a , of the two-block system. Express your answer in terms of μ_k , θ , and the acceleration due to gravity, g .

N2L Mass A

$$\Sigma F_{ext} = m a_x$$

$$mg \sin \theta - f_k - T = m a_x$$

$$f_k = \mu_k mg \cos \theta$$

N2L Mass B

$$\Sigma F_{ext} = m a$$

$$T = m_B a_x$$

Sub T=mbax from Mass B

$$mg \sin \theta - \mu_k mg \cos \theta - m a_x = m a_x$$

$$mg \sin \theta - \mu_k mg \cos \theta = m a_x + m a_x$$

$$a_x = \frac{mg \sin \theta - \mu_k mg \cos \theta}{2m}$$

- 16 N2L Present
- 3 N2L on Mass A
- 3 N2L on Mass B
- 6 $f_k = \mu_k N$
- 3 SOHCAHTOA, any trig

- 2 Attempts to get N for friction

- 2 Correct normal force
 $N = mg \cos(\theta)$

- 4 Combines Tension from both bodies

- 4 Attempts to solve for a_x

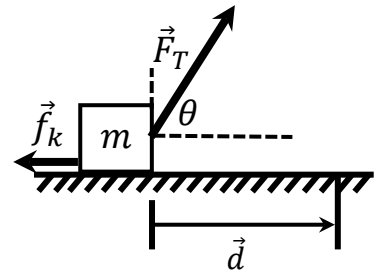
- 2 Correct Answer

$$a_x = \frac{g \sin \theta - \mu_k g \cos \theta}{2}$$

Report your algebraic answer in this box.

Answer the questions on this page using the provided scenario.

A block of mass $m = 11.5 \text{ kg}$ is pulled by a rope with tension force $F_T = 105 \text{ N}$ at an angle $\theta = 60.0^\circ$ for a distance $d = 1.50 \text{ m}$ along a rough horizontal surface. The friction force $\vec{f}_k$ does -45.1 J of work on the block across $\vec{d}$. See the figure below.



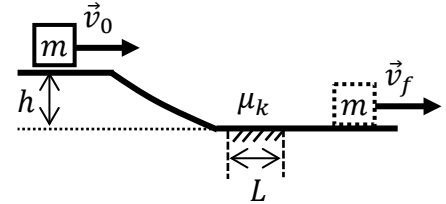
- 10 8. Calculate the work by the tension force on the block across the displacement, $\vec{d}$.
- ☒ 78.8 J $Td\cos(60)$ 6 3 1
- ☐ 56.4 J $Mg\cos(60)$ 4 Incorrect force
- ☐ 136 J $Td\cos(30)$ or $Td\sin(60)$ 6 Incorrect angle
- ☐ 45.1 J $-W_{f_k}$
- 5 9. Select which of the following is true during the block's motion over displacement, $\vec{d}$.
- ☐ The work on the block by the tension force equals its change in kinetic energy.
- ☐ The work on the block by the friction force equals its change in kinetic energy.
- ☒ The net work on the block equals its change in kinetic energy. 1 2 2
- ☐ The net work on the block equals zero because the work by friction cancels the work by the tension force.
- 10 10. Select which of the following is true if the mass of the block is increased while $\vec{F}_T$, θ , and $\vec{d}$ remain constant.
- ☒ The acceleration of the block decreases. 5 3 2
- ☐ The acceleration of the block increases.
- ☐ The acceleration of the block goes to zero.

Wt. No.

Cadet: _____

- 45 11. A block of mass m slides along a track with an initial speed v_0 and continues down a hill of height h as shown. The track is frictionless everywhere except for a rough horizontal segment of length L and coefficient of friction μ_k between the block and the surface. The block exits the rough segment with a final speed v_f .

Derive an algebraic expression for the final speed, v_f , of the block after it exits the rough segment in terms of h , μ_k , L , v_0 , and acceleration due to gravity, g .



$$E_0 = \frac{1}{2}mv_0^2 + mgh$$

$$E_F = \frac{1}{2}mv_f^2$$

$$W_{NC} = f_k L \cos 180^\circ$$

$$\boxed{10} \quad W_{NC} = E_f - E_0$$

$$\boxed{6} \quad K_T = \frac{1}{2}mv^2$$

$$\boxed{6} \quad U_g = mgh$$

$$\boxed{5} \quad W = \vec{F} \cdot \vec{d}$$

$$\boxed{3} \quad \text{Attempts any } W_{nc}$$

$$W_{NC} = E_F - E_0$$

$$-mg\mu_k L = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_0^2 - mgh$$

$$v_f = \sqrt{2gh - 2g\mu_k L + v_0^2}$$

$$\boxed{2} \quad \text{Correct } W_{nc} = -mg\mu_k L$$

$$\boxed{6} \quad \text{Energy terms present (1 pt) and correct (1 pt)}$$

$$\boxed{4} \quad \text{Attempts to solve for } v_f$$

$$\boxed{3} \quad \text{Correct ANS}$$

$$v_f = \sqrt{2gh - 2g\mu_k L + v_0^2}$$

Report your algebraic answer in this box.

Wt. **No.**

Cadet: _____

- 30 12. A car rounds an unbanked curve of radius $R = 105$ m. The car rounds the curve without slipping at a maximum speed $v = 28.0$ m/s. The coefficient of static friction between the road and the car is μ_s . Calculate the coefficient of static friction, μ_s , between the road and the car.

$$\Sigma F_{ext} = m a$$

$$mg\mu_s = \frac{mv^2}{r}$$

$$\mu_s = \frac{v^2}{gr}$$

8 $\Sigma F_{ext} = m a$

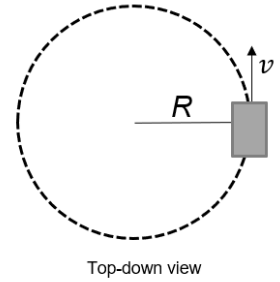
5 $a_c = v^2/r$

4 $f_s = \mu_s N$

4 Sub $\frac{v^2}{r}$ for a

2 $N = mg$

4 Attempts to solve for μ_s



$$\mu_s = 0.761$$

Report your numeric answer in this box.

2 Correct ANS

1 SF

- 30 13. A moose is standing on a road in front of a moving motorcycle. The motorcycle has mass $m = 245$ kg and is traveling at a speed $v = 12.6$ m/s just before it collides with the moose. The stationary moose exerts an average force of magnitude $F_{ave} = 991$ N, on the motorcycle to bring it to a complete stop. Calculate the time, t , it takes for the motorcycle to come to a complete stop.

$$\vec{J} = \Delta \vec{p} \quad \& \quad \vec{J} = \vec{F}_{ave} \Delta t$$

$$\Delta \vec{p} = \vec{F}_{ave} \Delta t$$

6 $\vec{p} = m\vec{v}$

6 $\vec{J} = \Delta \vec{p}$

$$v_f = 0$$

$$m(v_f - v_i) = F_{ave} \Delta t \quad F_{A,B} = -F_{B,A}$$

6 $\vec{J} = \vec{F}_{ave} \Delta t$

$$\Delta t = \frac{m(-v_i)}{-F_{ave}}$$

2 Sets $\Delta \vec{p} = \vec{F}_{ave} \Delta t$

2 Subs $\vec{p} = m\vec{v}$

$$\Delta t = \frac{-mv_i}{-F_{ave}}$$

2 Correct Algebraic Solution

2 $\Delta p = mv_f - mv_i$ or $(0 - mv)$

$$\Delta t = 3.12s$$

1 Solves for Δt

$$\Delta t = 3.12$$

Report your numeric answer in this box.

$$s$$

Report units in this box.

1 SF

1 uni

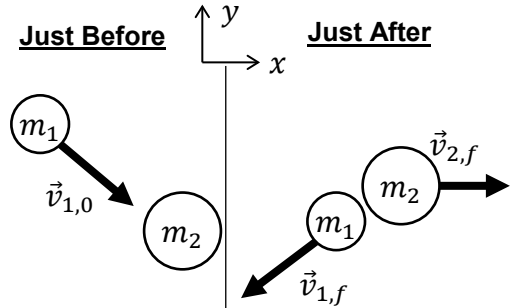
A: 7 (12%)
B: 15 (25%)
D: 38 (63%)
Total: 60

Wt. No.

Cadet: _____

- 45 14. Two disks collide on a frictionless horizontal surface. Disk 1 has mass $m_1 = 2.87$ kg and disk 2 has mass $m_2 = 4.23$ kg. Just before the collision, disk 1 has a velocity $\vec{v}_{1,0} = (8.72 \hat{i} - 3.15 \hat{j})$ m/s and disk 2 is initially at rest. Just after the collision disk 2 has a velocity $\vec{v}_{2,f} = 9.51 \hat{i}$ m/s.

Calculate the velocity of disk 1, $\vec{v}_{1,f}$, just after the collision.
Report your answer in unit-vector notation.



$$\Sigma P_{0x} = \Sigma P_{fx}$$

$$p_{10x} + p_{20x} = p_{1fx} + p_{2fx}$$

$$m_1 v_{10x} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$v_{1fx} = \frac{m_1 v_{10x} - m_2 v_{2fx}}{m_1} \quad \text{X correct algebraic solution}$$

$$v_{1fx} = -5.30 \text{ m/s}$$

$$\Sigma P_{0y} = \Sigma P_{fy}$$

$$p_{10y} + p_{20y} = p_{1fy} + p_{2fy}$$

$$m_1 v_{10y} = m_1 v_{1fy}$$

$$v_{10y} = v_{1fy} \quad \text{Y correct algebraic solution}$$

$$v_{1fy} = -3.15 \text{ m/s}$$

16 $\Sigma P_0 = \Sigma P_f$

10 $\vec{p} = m\vec{v}$

6 Applies COLM to both dimensions (3 pt each)

3 $m_1 v_{10x} = m_1 v_{1fx} + m_2 v_{2fx}$ - terms present

2 $m_1 v_{10y} = m_1 v_{1fy}$ - terms present

2 X correct (1), Y correct (1)

2 Correct unit vector notation

$v_{1f} = -5.30\hat{i} - 3.15\hat{j}$ Report your numeric answer in this box.	m/s Report units in this box.
--	---

1 SF
1 units

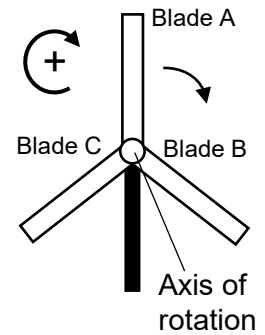
A: 4 (9%)
B: 13 (29%)
D: 28 (62%)
Total: 45

Wt. No.

Cadet: _____

Answer the questions on this page using the provided scenario.

A windmill with three blades, approximated as thin rods, is rotating clockwise (defined as positive) as shown in the figure to the right.



- 10 15. Blade A has an initial angular velocity $\omega_0 = 1.93 \text{ rad/s}$. It speeds up with an angular acceleration $\alpha = 1.21 \text{ rad/s}^2$, to a final angular velocity $\omega_f = 5.14 \text{ rad/s}$. Calculate the angular displacement during the acceleration.

- ☐ 18.8 rad $(\omega_f^2 - \omega_0^2)/\alpha$ 6 Dropped 2
☐ 3.21 rad $\omega_f - \omega_0$
☐ 2.65 rad *time* 4
☒ 9.38 rad $(\omega_f^2 - \omega_0^2)/2\alpha$ 6 3 1

- 5 16. Select the direction of the angular acceleration while the windmill speeds up.

☐	☐	☐	☐	☐	☐	☐	☐	☒	☐
↑	↗	→	↘	↓	↙	←	↖	⊗	⊙

2 LHR

3 1 1

- 10 17. At an arbitrary point in time the angular speed of the windmill is 2.67 rad/s and the windmill blade is 50.5 m long from the axis of rotation to the tip. Calculate the tangential speed at the tip of the blade at that time.

- ☒ 135 m/s ωr 6 3 1
☐ 0.141 m/s ω^2/r
☐ 0.0529 m/s ω/r 6 bad algebra
☐ 119 m/s $2\pi r/\omega$

Wt. No.

Cadet: _____

- 10 18. Below is an overhead view of a meterstick that can pivot about a point 0.20 m from one end. All forces on the stick have the same magnitude, $F_1 = F_2 = F_3 = F_4$. Compare the magnitudes of the torques created by the respective forces about the pivot point when $\theta = 60^\circ$.

☐ $\tau_4 > \tau_3 > \tau_2 > \tau_1$

☐ $\tau_4 > \tau_3 > \tau_1 > \tau_2$ 6

☒ $\tau_4 > \tau_1 > \tau_3 > \tau_2$ 4 3 3

☐ $\tau_4 > \tau_1 > \tau_2 > \tau_3$

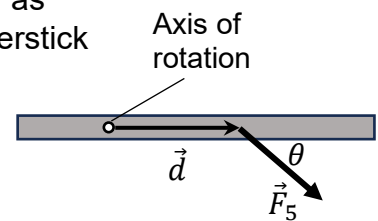
☐ $\tau_4 > \tau_3 = \tau_1 > \tau_2$ 4

0 m 0.2 m 0.4 m 1 m

Axis of rotation

- 10 19. A fifth force $F_5 = 15 \text{ N}$ is applied to the meterstick a distance, $d = 0.30 \text{ m}$, from the axis of rotation at an angle, $\theta = 25^\circ$, as shown. Calculate the magnitude of the torque on the meterstick due to $\vec{F}_5$.

- ☐ 4.5 N m $F_5 d$ 4
- ☒ 1.9 N m $F_5 d \sin \theta$ 5 3 2
- ☐ 4.1 N m $F_5 d \cos \theta$ 6
- ☐ 110 N m $F_5 d \theta$



- 5 20. A spherical shell, a solid sphere, and a solid cylinder are rotating at the same angular speed ω about their respective central axes. All objects have the same mass M and radius R . Compare the angular momentum between the spherical shell L_{shell} , solid sphere $L_{\text{solid sphere}}$, and solid cylinder L_{cyl} .

- ☐ $L_{\text{shell}} > L_{\text{solid sphere}} > L_{\text{cyl}}$
- ☒ $L_{\text{shell}} > L_{\text{cyl}} > L_{\text{solid sphere}}$ 1 2 2
- ☐ $L_{\text{solid sphere}} > L_{\text{cyl}} > L_{\text{shell}}$
- ☐ $L_{\text{solid sphere}} > L_{\text{shell}} > L_{\text{cyl}}$
- ☐ $L_{\text{cyl}} = L_{\text{solid sphere}} = L_{\text{shell}}$

$\text{shell}(2/3) > \text{cyl}\left(\frac{1}{2}\right)$
 $> \text{Solid sphere } (2/5)$

Wt. **No.**

Cadet: _____

- 45 21. A solid disk pulley with a radius $R = 0.234$ m and mass $M_p = 1.25$ kg is mounted on a frictionless horizontal axle. A massless cord wrapped around the pulley is attached to a block of mass, $m = 2.56$ kg. A horizontal force of magnitude, $F = 8.00$ N, is applied to the block which causes the block to accelerate to the right along a horizontal and frictionless surface.

Calculate the magnitude of the angular acceleration, α , of the pulley.

$$\Sigma \vec{F}_{ext} = m\vec{a}$$

$$F - T = ma$$

$$T = F - m\alpha R$$

$$\Sigma \vec{\tau}_{ext} = I\alpha$$

$$\tau_T = I\alpha$$

$$R \times T = \frac{1}{2} M_p R^2 \alpha$$

$$RT \sin 90 = \frac{1}{2} M_p R^2 \alpha$$

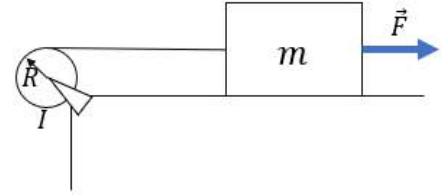
$$T = \frac{1}{2} M_p R \alpha$$

$$F - m\alpha R = \frac{1}{2} M_p R \alpha$$

$$\alpha = \frac{F}{mR + \frac{1}{2} M_p R}$$

$$\alpha = 10.7 \text{ rad/s}^2$$

$$\boxed{8} \quad \Sigma \vec{F}_{ext} = ma$$



$\boxed{2}$ Attempts to solve for T with N2L

$$\boxed{12} \quad \Sigma \vec{\tau}_{ext} = I\alpha$$

$$\boxed{7} \quad \vec{\tau} = \vec{r} \times \vec{F}$$

$\boxed{3}$ Applies cross product $\tau = RT$

$\boxed{2}$ Applies $I = \frac{1}{2} MR^2$

$\boxed{2}$ Applies link equation

$\boxed{4}$ Solves for α

$\boxed{2}$ Correct Algebraic Solution

If they just use F and mass to get a (bad):

$$a = \frac{F}{m} = 3.125$$

$$\alpha = \frac{a}{R} = 13.35$$

$\boxed{1}$ Correct

$\boxed{1}$ SF

$\boxed{1}$ units

$\alpha = 10.7$	rad/s^2
Report your numeric answer in this box.	Report units in this box.

Wt. **No.**

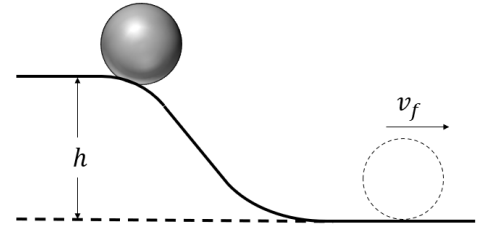
Cadet: _____

45

22.

A uniform solid sphere is at the top of a ramp of height, $h = 9.90$ m. The sphere is released from rest and rolls without slipping down the ramp. At the bottom of the ramp, the sphere has a final translational speed, v_f .

Calculate the final translational speed, v_f , of the sphere at the bottom of the ramp.



10 $W_{NC} = E_f - E_0$

5 $K_T = \frac{1}{2}mv^2$

5 $K_R = \frac{1}{2}I\omega^2$

5 $U_g = mgh$

3 Terms – Present (1 pt ea)

2 $W_{NC} = 0$

2 Correct moment of inertia applied $I = \frac{2}{5}MR^2$

4 Applies link equation

4 Solves for v

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$2mgh = mv^2 + \frac{2}{5}\frac{mr^2v^2}{r^2}$$

$$2gh = \frac{7}{5}v^2$$

$$v = \sqrt{\frac{10}{7}gh}$$

2 Correct Algebraic Solution

$$v = 11.8 \text{ m/s}$$

1 Correct

1 SF

1 units

$v = 11.8$ Report your numeric answer in this box.	m/s Report units in this box.
--	--

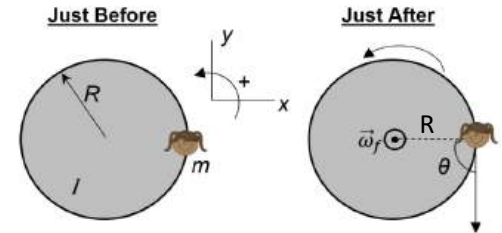
A: 5 (11%)
B: 13 (29%)
D: 27 (60%)
Total: 45

Wt. No.

Cadet: _____

- 45 23. A child of mass m is at the edge of a merry-go-round. The merry-go-round has a radius R and mass M and can be approximated as a disk. The merry-go-round and child are initially at rest. The child then jumps off the merry-go-round with velocity v at an angle $\theta = 90.0^\circ$ in the direction indicated. Immediately after, the merry-go-round rotates counter-clockwise (when viewed from above) with an angular speed ω_f . The child can be approximated as a point mass.

Derive an algebraic expression for the angular speed, ω_f , of the merry-go-round immediately after the child jumps off in terms of m , v , R and M .



14 $\Sigma L_0 = \Sigma L_f$

5 $L = I\omega$

5 $\vec{l} = \vec{r} \times \vec{p}$

4 $\vec{p} = m\vec{v}$

$\Sigma L_0 = \Sigma L_F \quad \text{CCW} +$

$0 = -\vec{r} \times \vec{p} + I_M \omega_f$

$0 = -Rmv \sin 90 + \frac{1}{2}MR^2\omega_f$

$\omega_f = \frac{2Rmv}{MR^2}$

- 6 Terms –
Child Present and correct (1 pt ea)
MGR Present and correct (1 pt ea)
 $L_0 = 0$ (2 pt)

- 3 Attempts cross-product

- 4 Solves for ω_f

3 Correct

1 units

A: 3 (7%)
B: 13 (29%)
D: 29 (64%)
Total: 45

$$\omega_f = \frac{2mv}{MR}$$

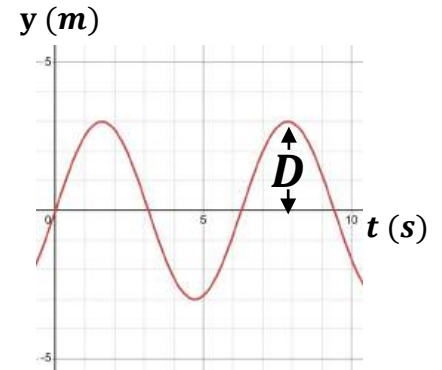
Report your algebraic answer in this box.

Wt. No.

Cadet: _____

- 5 24. The time to complete one full oscillation of a wave is defined as the _____.
☐ Frequency
☐ Wavelength
☒ Period 4 1
☐ Angular frequency

- 5 25. In the graph to the right the variable "D" represents the _____.
☐ Frequency
☐ Wavelength
☐ Period
☒ Amplitude 3 1 1

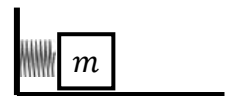


- 5 26. The variable k in the equation $k = \frac{2\pi}{\lambda}$ represents the _____.
☐ Frequency
☒ Angular wave number 1 2 2
☐ Spring constant
☐ Angular frequency

- 10 27. A string under a tension of 121 N has a linear mass density of 5.82×10^{-4} kg/m. Calculate the speed of a transverse mechanical wave on the string.

- ☐ 0.0704 m/s μT
☐ 208000 m/s T/μ 6 Dropped square root
☒ 456 m/s $\sqrt{T/\mu}$ 6 3 1
☐ 4.81×10^{-6} m/s μ/T

- 10 28. A spring-mass system rests on a frictionless surface as shown to the right and is in simple harmonic motion. The spring constant is 95.7 N/m and the angular frequency of the system is 3.51 rad/s. Calculate the mass of the block in the spring-mass system.

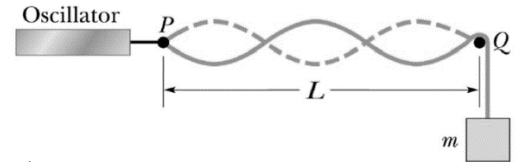


- ☐ 27.3 kg k/ω 6 Dropped exponent
☒ 7.77 kg k/ω^2 6 3 1
☐ 336 kg $k\omega$
☐ 0.129 kg ω^2/k 3

Wt. No.

Cadet: _____

- 35 29. A string with a linear density of μ is tied to a string oscillator at P and passes over a frictionless support at Q . A block of mass m is suspended from the end of the string. The string's length between points P and Q is L . The amplitude of the motion at P is small enough for that point to be considered a node. A node also exists at Q .
- (30) a. Derive an algebraic expression for the frequency, f , of the string oscillator in terms of L , m , g , μ , and n where n is the number of anti-nodes.



4 $\sum \vec{F}_{ext} = m\vec{a}$

9 $f_n = \frac{nv}{2L}$

9 $v = \lambda f$

9 $v = \sqrt{\frac{F_T}{\mu}}$

9 $\lambda_n = \frac{2L}{n}$

$F = m a_y$

$T - mg = m a_y$

$T = mg$

$v = \sqrt{F_T/\mu}$

$v = \sqrt{mg/\mu}$

$f_n = \frac{nv}{2L}$

3 Subs mg for F_T

2 Subs v into f_n

2 Solves for f

If they do NO N2L:
-4 PRC
-3 Subs
-2 ANS

26/35
74%

That is reasonable

$f_n = \frac{n}{2L} \sqrt{mg/\mu}$

Report your algebraic answer in this box.

1

$f_n = \sqrt{\frac{n^2 mg}{4L^2 \mu}}$

- (5) b. Determine the number of antinodes in the diagram above. Write your answer in the provided box below.

3

Report your answer here

1 2 2

A: 3 (9%)
B: 9 (26%)
D: 23 (66%)
Total: 35

Wt. No.

Cadet: _____

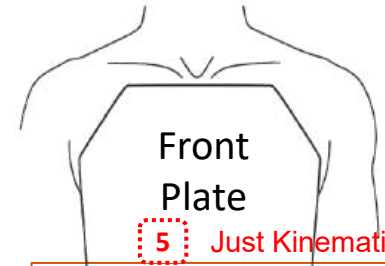
- 30 30. Your commander wants you to give a brief describing how the front plate of individual body armor stops bullets. You must use physics to **describe** how a plate carrier brings a bullet with a mass of m from an initial speed of v_o to rest after impact. **State** the overarching PH2X1 course concept(s) you will use in your description, **justify** your answer using **at least one equation** used in this course, and limit your answer to this page (additional work on scratch paper will not be evaluated for this question).

Physics Concept(s):

8

Work-Kinetic Energy

Impulse-Momentum



Newton's Laws/Kinematics

Equation(s):

10

$$\sum W = \Delta K$$

$$W = \vec{F} \cdot \vec{d}$$

$$K_T = \frac{1}{2}mv^2$$

$$\vec{J} = \vec{F}_{avg}\Delta t$$

$$\vec{J} = \Delta \vec{p}$$

$$F = ma$$

$$v^2 = v_o^2 + 2a(x - x_o)$$

5 Just Kinematics

Justification:

Does the justification match the chosen concept and equations?

No justification attempted

No

Yes

3

Does cadet's justification use a different, potentially correct concept than the stated chosen concept?

Does cadet's justification demonstrate an understanding of the relevant concept?

No

No

Yes

6

No, the cadet's justification does not demonstrate understanding of the concept

Does the justification contain errors in understanding?

Yes, but justification is poorly explained, vague, or contains more than 2 noticeable errors in understanding.

Yes, there are some misunderstandings shown

No, 100% correct

3

Yes: correct physics for a concept not the same as the one stated in the first section



TAB H FINAL COURSE GRADES

AY 26-1 PH2X1 Final OML

Person of Interest
Physics Major
Nuclear Engineering Major
Space Major

A+ on event/course (>96.00%)
F on event/course (<60.00%)

A's:	186
F's:	1
5 or more Absences	41

PANE
WANTF

CID	Name	Year	Hour Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C70729151	Young, Lucas	2029	E1F1 1	CSC1	0	A+	100.47%		1407	1400	90.81%	99.23%	99.67%	100%	100.00%	99.00%		
C23877298	Liera-Cohn, Aidan	2029	A1B1 3	MEN3	0	A+	99.41%		1988	2000	93.15%	100.00%	97.00%	97%	99.00%	98.67%		
C41326026	Zhang, Lillian	2029	A1B1 8	BIO0	2	A+	99.02%		1980	2000	92.28%	97.89%	96.33%	100%	100.00%	99.33%	TENNIS (WOMEN)	
C84101034	Pace, Hudson	2029	E1F1 8	MEN3	3	A+	98.65%		1973	2000	91.11%	99.11%	97.33%	94%	99.33%	99.17%		
C83881921	Smith, Keagan	2029	A2B2 2	ESC3	2	A+	98.56%		1971	2000	97.90%	100.00%	99.00%	100%	94.67%	96.33%		ML100 N/C
C67969256	Pak, Joshua	2029	E2F2 10	PAP3T	6	A+	98.55%		1956	1985	97.52%	96.44%	98.67%	100%	90.00%	99.83%		
C25344941	Kim, Avery	2029	E2F2 2	HMH3T	0	A+	98.38%		1977	2010	92.50%	95.64%	97.33%	98%	98.33%	99.17%		
C39141777	Bauer, Braden	2029	C2D2 2	MEN3	0	A+	98.01%		1970	2010	83.33%	93.33%	98.00%	98%	94.00%	98.67%		
C19500807	Moore, Ty	2029	E1F1 5	MSC1	0	A+	97.79%		1956	2000	99.31%	105.33%	95.00%	98%	94.67%	98.33%		
C87117979	Chuang, Lu Wen Eden	2029	C1D1 4	CHM2A	3	A+	97.73%		1955	2000	89.69%	100.87%	97.67%	98%	94.00%	99.00%	SWIMMING (WOMEN)	
C17987586	Ryan, Devin	2029	A1B1 8	ENM1	2	A+	97.72%		1954	2000	86.74%	100.00%	96.67%	100%	100.00%	99.00%	WRESTLING	
C42698555	Konopka, Benjamin	2029	C2D2 2	BIO0	1	A+	97.70%		1964	2010	88.33%	100.00%	97.67%	97%	91.00%	98.50%	FOOTBALL	ML100 N/C
C77362881	Lim, Elijah	2029	A1B1 3	MEN3	0	A+	97.47%		1949	2000	94.16%	92.30%	97.67%	100%	94.33%	98.50%		
C43372800	Zanella, Clark	2029	E1F1 5	CVN3	0	A+	97.36%		1947	2000	97.52%	96.56%	94.00%	100%	99.00%	98.00%		
C39046695	Connors, Aidan	2029	A2B2 2	ECN2	1	A+	97.29%		1922	1975	97.90%	100.00%	90.00%	98%	95.67%	97.67%		
C04168263	Watson, Maddix	2029	E1F1 1	MEN3	0	A+	97.24%		1945	2000	90.52%	98.85%	98.00%	98%	94.33%	96.50%		
C14073664	McCloskey, Lukas	2029	A1B1 5	CSC1	9	A+	97.23%		1925	1980	90.36%	100.00%	99.67%	98%	93.00%	97.83%	HOCKEY	
C96659744	Yu, Jacob	2029	E2F2 3	MEN3	0	A+	97.16%		1938	1995	88.00%	98.67%	100.00%	82%	98.33%	99.00%		ML100 N/C
C87615385	Baek, Blake	2029	E2F2 10	MEN3	3	A+	97.02%		1911	1970	96.84%	100.00%	98.00%	99%	98.00%	95.00%		
C04564538	Suresh, Mia	2029	C2D2 6	MEN3	1	A+	96.93%		1919	1980	94.43%	92.42%	96.00%	96%	92.33%	98.17%		
C23729913	Carbaugh, Nathan	2029	A2B2 10	HUS3T	2	A+	96.39%		1913	1985	94.36%	100.00%	87.67%	100%	98.00%	96.00%		
C16765980	Moctezuma, Giselle	2028	A2B2 2	PIF0 FLS2	1	A+	96.21%		1934	2010	91.67%	97.26%	91.33%	100%	89.33%	99.00%		
C09318099	Gillis, Cooper	2029	A2B2 2	SEN2	2	A+	96.19%		1934	2010	90.00%	85.33%	98.33%	100%	92.33%	96.83%		
C92782256	Kaplan, Ari	2029	E2F2 3	MEN3	0	A+	96.15%		1918	1995	93.28%	90.97%	94.67%	100%	92.00%	96.17%		
C62631388	Bin Haswadi, Luthfil Haika	2029	C1D1 1	SEN2	3	A+	96.10%		1912	1990	91.31%	97.26%	92.33%	100%	91.33%	96.50%		
C69791224	Nies, Keirsten	2029	A1B1 3	BIO0	2	A+	96.09%		1912	1990	94.33%	83.04%	96.00%	85%	94.00%	98.00%		
C15765238	Giannotti, Andrew	2029	A1B1 8	MEN3	1	A+	96.07%		1921	2000	92.28%	94.08%	97.67%	98%	91.33%	96.83%		
C22678875	Chiu, Austin	2029	E1F1 5	MEN3	5	A+	96.03%		1921	2000	94.64%	97.33%	99.00%	100%	87.67%	95.83%		
C00045647	Hisle, Reed	2029	C2D2 2	ORE2	0	A+	96.02%		1930	2010	86.67%	93.19%	88.00%	100%	94.33%	98.83%	FOOTBALL (SPRINT)	
C39092947	Benjaminson, Ethan	2029	E1F1 8	MEN3	1	A	95.97%		1919	2000	85.31%	92.56%	94.33%	100%	94.33%	98.17%		
C44242307	Choi, Sungjoon	2029	E1F1 1	CVN3	1	A	95.90%		1918	2000	86.89%	70.58%	97.67%	95%	94.67%	98.17%		
C97042383	Estevez, Lucas	2029	A2B2 2	ENM1	0	A	95.83%		1917	2000	94.97%	86.78%	97.00%	98%	88.33%	96.83%	FOOTBALL (SPRINT)	
C09044504	Salas, Isabel	2028	A1B1 3	ENL2	1	A	95.51%		1882	1970	95.28%	77.33%	95.67%	79%	97.00%	98.33%	RUGBY (WOMEN)	
C87479863	Fahey, Neilan	2029	C1D1 8	BIO0	1	A	95.48%		1910	2000	84.23%	98.60%	92.00%	100%	98.33%	98.83%	RIFLE	
C14678957	Maslov, Maxim	2029	A1B1 3	MSC1	0	A	95.44%		1909	2000	93.28%	94.44%	96.33%	93%	91.33%	95.67%		
C92174744	Frank, Noah	2029	E1F1 5	CYS1	1	A	95.41%		1908	2000	97.52%	102.00%	95.33%	100%	77.00%	98.17%		
C48968425	Penna, Damon	2029	E1F1 1	NEN1	1	A	95.37%		1907	2000	85.73%	100.00%	91.00%	99%	95.33%	96.83%		
C25898832	Harbeson, Sebastian	2029	C1D1 1	NEN1	2	A	95.36%		1907	2000	90.30%	91.67%	95.33%	98%	92.33%	97.33%	Rabble Rousers (Varsity I)	
C33647928	Ballard, Kenyon	2029	C2D2 3	PII0	1	A	95.32%		1878	1970	92.74%	92.89%	87.33%	100%	94.33%	96.33%		
C55136829	Popelka, Kyle	2029	C1D1 5	EEN1	1	A	95.24%		1886	1980	94.80%	88.94%	87.33%	100%	92.33%	98.00%		
C04889739	Lee, Aaron	2029	E1F1 1	MGN0	0	A	95.05%		1901	2000	89.99%	105.58%	92.00%	99%	92.00%	94.67%		
C29652003	Mitchell, William	2029	C2D2 2	EEN1	1	A	94.93%		1908	2010	81.67%	97.33%	81.33%	100%	93.67%	99.83%		
C06581873	Belveal, John	2029	A2B2 10	ENM1	0	A	94.87%		1897	2000	93.31%	93.45%	92.00%	100%	89.00%	97.00%	FOOTBALL (SPRINT)	
C12983866	Coomer, Arthur	2029	A2B2 10	MEN3	0	A	94.81%		1896	2000	91.97%	75.67%	94.67%	97%	97.67%	98.17%		
C32459460	Kehler, Bradley	2029	E2F2 2	CEN2	0	A	94.81%		1906	2010	81.67%	93.11%	89.67%	98%	96.67%	96.33%		
C13691856	Grote, Koleson	2029	A2B2 2	CVN3	2	A	94.68%		1903	2010	94.17%	98.67%	90.67%	98%	85.67%	97.00%		
C01550262	Means, Ace	2029	C1D1 1	MEN3	0	A	94.67%		1893	2000	92.62%	97.26%	84.67%	100%	95.33%	95.50%		
C43055810	Abraham, Joel	2029	C1D1 1	MEN3	1	A	94.65%		1893	2000	90.46%	90.64%	99.00%	99%	81.67%	95.83%		
C22649252	Srivastav, Kabir	2029	E1F1 8	MEN3	0	A	94.50%		1890	2000	87.51%	82.67%	94.67%	98%	92.67%	93.67%		
C64532000	Lim, Kyle	2029	A2B2 2	SEN2	3	A	94.47%		1795	1900	97.90%	99.11%	87.67%	91%	95.33%	91.17%		
C12694604	Niang, Ousseynou	2029	A2B2 2	WST0	6	A	94.32%		1896	2010	95.00%	61.33%	93.67%	88%	95.00%	97.67%		
C12372154	Ure, Isaac	2029	E2F2 2	MEN3	0	A	94.31%		1896	2010	81.67%	74.00%	95.00%	100%	87.00%	97.50%		
C24817494	Soldato, Antonio	2029	E2F2 2	NEN1	0	A	94.23%		1894	2010	85.00%	98.60%	88.33%	94%	92.33%	93.17%		
C68239941	El-Bashir, Kareem	2029	C2D2 6	ENM1	1	A	94.16%		1864	1980	94.52%	100.00%	84.00%	100%	84.33%	97.50%	HOCKEY	
C75775128	Driscoll, Andrew	2029	C1D1 9	AEN0	0	A	94.11%		1882	2000	85.57%	98.67%	95.67%	94%	95.33%	93.50%		
C10751472	Lessard, Aidan	2029	C2D2 2	PHY2	0	A	94.11%		1892	2010	81.67%	98.67%	83.33%	100%	88.67%	97.33%	FOOTBALL (SPRINT)	
C12302770	Nguyen, Hai	2029	C2D2 4	CYS1	1	A	94.04%		1881	2000	90.42%	119.33%	85.67%	98%	85.33%	95.67%		
C21432654	Kim, Andrew	2029	C2D2 10	EEN1	0	A	93.99%		1880	2000	85.96%	93.71%	97.00%	97%	95.00%	95.67%		
C82780108	Liquard, Dylan	2029	C1D1 1	SEN2	0	A	93.94%		1879	2000	83.70%	97.89%	89.67%	97%	91.00%	97.00%		
C70920406	Nutwell, Natalie	2029	A1B1 8	FSU2	3	A	93.90%		1850	1970	91.59%	100.00%	91.00%	100%	91.00%	93.67%	Rabble Rousers (Varsity I)	
C33152851	Batten, William	2029	A1B1 5	ENM1	9	A	93.83%		1858	1980	90.36%	100.00%	94.67%	98%	91.33%	90.33%	HOCKEY	
C14818443	Rian, Bjorn	2027	A1B1 3	CY00	5	A	93.79%		1876	2000	92.82%	93.71%	85.67%	94%	91.33%	93.67%	RIFLE	
C20246142	Kim, Nathaniel	2029	C1D1 1	MEN3	2	A	93.66%		1864	1990	90.49%	96.00%	88.67%	100%	87.33%	97.17%	Rabble Rousers (Varsity I)	
C51575696	Malich, Dominic	2029	A2B2 2	BIO0	4	A	93.64%		1882	2010	91.67%	98.67%	97.00%	97%	89.67%	95.33%		
C61703523	Chaffin, Jake	2029	E1F1 1	ESC3	1	A	93.61%		1872	2000	80.26%	85.89%	91.67%	100%	97.67%	96.50%		
C18515332	McGowan, Petra	2029	C2D2 2	KIN1	3	A	93.52%		1880	2010	86.67%	100.00%	84.00%	88%	90.00%	97.00%	TRACK/INDOOR TRACK	
C69088621	Greco, Caden	2028	A2B2 3	CYS1	1	A	93.48%		1841	1970	91.59%	93.68%	87.67%	100%	90.00%	95.83%		
C11997339	Barone, Andrew	2029	C1D1 1	NEN1	0	A	93.44%		1869	2000	90.56%	79.49%	96.67%	98%	92.00%	93.67%		
C31057853	Johnson, Nolan	2029	A1B1 8	ENM1	0	A	93.41%		1868	2000	91.59%	97.33%	84.67%	98%	92.67%	92.67%	FOOTBALL	
C56760243	Anderson, Coby	2029	E2F2 6	FFR2	0	A	93.40%		1868	2000	80.05%	73.89%	97.67%	95%	95.67%	97.50%		

CID	Name	Year	Hour Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C91146101	Sincavage, Massimo	2029	E1F1 1	EEN1	0	A	93.40%		1868	2000	73.31%	96.44%	96.33%	100%	95.67%	97.17%		
C70863099	Chen, Kuan-Po	2029	E2F2 10	SEN2	1	A	93.35%		1867	2000	89.34%	87.56%	96.33%	94%	95.00%	92.83%		ML100 N/C
C22373366	Self, Aden	2028	C1D1 1	MGN0	2	A	93.26%		1856	1990	88.43%	73.46%	89.33%	98%	93.33%	96.50%	FOOTBALL	ML100
C40896417	Bahar, Shafi	2029	C2D2 2	SEN2	0	A	93.24%		1874	2010	81.67%	97.33%	83.00%	95%	90.00%	95.83%		
C82909858	Aguilar, Andrew	2029	A2B2 7	NEN1	2	A	93.18%		1864	2000	85.41%	79.11%	94.33%	97%	96.33%	98.17%		HI158W
C69140405	Ewers, Luke	2029	A2B2 2	LLS3	2	A	93.15%		1840	1975	94.97%	92.56%	85.33%	100%	79.33%	98.17%		
C42981398	Corcoran, Ryan	2029	E1F1 1	ASD0	2	A	92.83%		1857	2000	80.26%	91.11%	91.67%	100%	93.33%	97.33%		
C83270599	Dafoe, Logan	2029	A1B1 5	AEN0	1	A	92.77%		1855	2000	87.98%	80.94%	89.67%	97%	94.00%	95.67%	FOOTBALL (SPRINT)	
C46147471	Wilkerson, Steven	2029	A2B2 2	EEN1	2	A	92.62%		1862	2010	85.00%	93.33%	93.33%	91%	78.33%	95.67%	TRACK/INDOOR TRACK	
C31023220	Francis, Andrew	2028	C1D1 1	MGN0	4	A	92.50%		1850	2000	90.95%	93.12%	86.33%	98%	91.33%	92.67%	RUGBY (MEN)	
C05227774	Hatcher, Lindsey	2029	A1B1 5	AEN0	3	A	92.48%		1850	2000	86.26%	99.20%	90.33%	98%	90.33%	92.67%		ML100 N/C
C68392256	O'Koniewski, Tait	2029	C1D1 1	BEA0	1	A	92.48%		1850	2000	91.75%	67.60%	91.00%	96%	88.67%	92.83%		
C97750973	Kirk, Robert	2029	C2D2 10	NEN1	1	A	92.41%		1848	2000	91.09%	60.49%	96.67%	94%	90.00%	94.67%		
C60273416	Haley, Timothy	2029	C1D1 1	HMH3T	2	A	92.40%		1848	2000	89.70%	60.37%	87.67%	94%	93.67%	95.33%		
C87308805	Gancarz, Jonathan	2029	A2B2 10	CHM2A	0	A	92.38%		1848	2000	94.88%	100.00%	80.00%	98%	82.33%	95.67%		ML100 N/C
C18171358	Harzbecker, Theo	2029	E1F1 1	FAS2	2	A	92.38%		1848	2000	81.21%	97.11%	94.00%	97%	86.33%	94.33%		ML100 N/C
C78588789	Gunn, Matthew	2029	A2B2 2	LLS3	0	A	92.36%		1847	2000	90.87%	87.42%	88.67%	95%	88.00%	93.00%		
C92730340	Rivera Rodriguez, Eduard	2029	E1F1 1	CVN3	2	A	92.32%		1846	2000	93.80%	88.49%	80.67%	93%	92.00%	96.50%		
C96266781	Holloway, Kendall	2029	A2B2 6	LLS3	4	A	92.26%		1813	1965	87.44%	97.33%	91.00%	93%	90.67%	93.67%	SOFTBALL (WOMEN)	
C59139352	Jennings, Masud	2029	E1F1 1	CVN3	0	A	92.20%		1844	2000	83.31%	91.78%	92.00%	93%	88.00%	96.67%		
C53517031	King, Derek	2029	E2F2 3	KIN1	0	A	92.11%		1814	1970	88.00%	95.67%	92.00%	98%	81.00%	94.33%		
C08829640	Tran, Nam	2029	A2B2 3	HNT3T	1	A	92.07%		1763	1915	91.45%	86.67%	90.00%	98%	86.67%	94.33%		
C15306817	Cadden, William	2029	E1F1 1	CY00	3	A	92.03%		1841	2000	91.87%	96.89%	93.67%	99%	86.67%	91.17%		
C29157914	Girardi, Nicola	2029	E1F1 1	CY00	0	A	92.03%		1841	2000	90.92%	95.11%	95.67%	98%	74.67%	94.50%		
C21536297	Halford, Adam	2029	C2D2 2	MEN3	0	A	92.03%		1850	2010	81.67%	70.00%	97.33%	99%	90.00%	91.17%		
C53454336	Ford, Robert	2029	E2F2 10	SEN2	1	A-	91.94%		1825	1985	90.30%	92.00%	89.33%	95%	86.00%	95.50%		
C18411586	Fechtmann, Karinna	2029	A2B2 2	PIF0	0	A	91.93%		1848	2010	94.17%	96.44%	83.00%	90%	84.67%	95.33%		
C49742057	Bode, Daniel	2028	C1D1 1	CSC1	4	A-	91.82%		1836	2000	89.38%	97.19%	87.00%	97%	86.67%	92.00%		
C83131442	Shin, Joel	2029	E1F1 1	CSC1	1	A	91.71%		1834	2000	91.84%	92.67%	88.33%	98%	73.33%	98.67%		
C02350587	Kim, Daeyong	2029	A2B2 2	MGN0	4	A-	91.70%		1843	2010	95.00%	97.33%	71.67%	96%	90.00%	97.83%		
C85045476	Sollie, Hunter	2029	A2B2 2	MGN0	0	A-	91.69%		1834	2000	92.02%	92.44%	89.00%	96%	85.00%	90.83%		
C33851618	Gelineau, Campbell	2029	C1D1 5	ENM1	1	A-	91.66%		1833	2000	94.80%	98.67%	90.00%	95%	85.33%	92.83%		
C42473161	Grube, Matthew	2029	A2B2 10	NEN1	0	A-	91.65%		1805	1970	95.41%	88.87%	93.67%	80%	79.67%	95.67%		
C13734873	Mujica, Ricardo	2029	C2D2 11	ORE2	0	A-	91.64%		1833	2000	79.95%	95.78%	96.00%	95%	89.00%	97.50%		
C92959821	Buss, Griffin	2029	C1D1 5	EPS1	4	A-	91.61%		1814	1980	81.37%	93.33%	96.00%	100%	83.33%	94.50%	FOOTBALL (SPRINT)	
C08634222	Steffen, Nathan	2029	E2F2 10	ESC3	3	A-	91.58%		1818	1985	90.96%	92.22%	89.33%	88%	83.67%	94.83%		
C27274603	Matin, Zain	2029	A2B2 2	EPS1	0	A-	91.53%		1840	2010	85.00%	96.00%	85.67%	91%	86.33%	92.83%		
C31946788	Bowcutt, Asher	2029	C1D1 1	EEN1	3	A-	91.46%		1829	2000	91.34%	91.33%	86.67%	98%	93.00%	89.67%	TRACK/INDOOR TRACK	
C08072693	Osborn, Lauren	2028	A1B1 3	HNT3T RSE1N	0	A-	91.46%		1829	2000	93.02%	84.30%	85.00%	91%	84.00%	94.17%		
C43462890	Lai, Johnston	2029	E2F2 6	CVN3	1	A-	91.45%		1811	1980	89.34%	92.42%	85.00%	95%	91.33%	92.00%		
C42908929	Curtis, Jordan	2029	C2D2 2	ECN2	0	A-	91.38%		1837	2010	88.33%	87.04%	78.67%	87%	84.67%	95.00%		
C06392564	Hokanson, Ivar	2029	C2D2 2	NEN1	1	A-	91.36%		1836	2010	88.33%	93.04%	73.00%	96%	89.00%	94.83%	CROSS COUNTRY (MEN)	
C97128389	Quirk, Sydney	2028	C2D2 2	FFS2	1	A-	91.26%		1834	2010	88.33%	92.42%	79.33%	100%	75.67%	95.67%		
C30399127	Choi, Young	2029	E2F2 6	PYL2	0	A-	90.96%		1819	2000	89.44%	89.82%	94.33%	86%	89.67%	94.17%		
C03865145	Davis, Collin	2029	C1D1 5	MEN3	0	A-	90.89%		1818	2000	94.72%	97.01%	76.67%	98%	87.00%	91.17%		
C26727774	Farmer, Cameron	2029	A1B1 3	MEN3	0	A-	90.84%		1817	2000	88.23%	86.20%	90.00%	99%	85.33%	91.67%		
C07122083	Singh, Suyash	2028	E2F2 6	CEN2	0	A-	90.79%		1816	2000	81.11%	93.33%	94.33%	87%	84.33%	96.00%		EN101
C20935481	Biancheri, Jackson	2028	E1F1 8	KIN1	3	A-	90.74%		1815	2000	89.66%	94.53%	83.67%	98%	85.33%	90.83%		
C07115612	Lundell, James	2029	E2F2 2	EEN1	0	A-	90.72%		1823	2010	85.00%	82.97%	80.00%	98%	79.67%	96.17%		
C96037877	Geck, Leo	2029	C1D1 9	EEN1	0	A-	90.67%		1813	2000	84.26%	70.94%	91.67%	83%	86.33%	97.83%		
C38821354	Kang, Eric	2029	C1D1 4	SEN2	0	A-	90.67%		1813	2000	79.79%	98.73%	94.00%	97%	83.67%	92.17%		
C37445715	DiCicco, Matteo	2029	A2B2 6	CVN3	0	A-	90.61%		1812	2000	86.80%	97.33%	83.67%	99%	89.00%	93.00%		
C62583994	Oldroyd, Pierce	2029	E2F2 11	NEN1	2	A-	90.47%		1809	2000	76.38%	88.34%	95.33%	66%	96.33%	95.00%		
C78110246	Stacy-Brandenburgh, Alej	2028	A2B2 2	CSC1	1	A-	90.47%		1818	2010	85.00%	99.11%	85.00%	98%	82.00%	88.00%		
C44360843	Rice, Samuel	2029	A1B1 3	EEN1	0	A-	90.43%		1809	2000	88.10%	67.09%	95.00%	96%	89.67%	95.50%		
C45307694	Stone, Rebekah	2029	E1F1 1	CHM2A	1	A-	90.42%		1808	2000	78.97%	96.92%	93.33%	96%	73.00%	94.67%		ML100 N/C
C37000238	Wong, Christopher	2029	C2D2 3	MGN0	0	A-	90.42%		1804	1995	91.59%	98.60%	83.67%	100%	74.67%	94.50%		ML100 N/C
C82537613	Woolfolk, Kai	2029	C1D1 1	ENM1	0	A-	90.42%		1808	2000	87.77%	97.33%	83.67%	95%	89.33%	92.67%	FOOTBALL	ML100 N/C
C30570320	Clifford, Jackson	2029	E1F1 5	SEN2	2	A-	90.36%		1807	2000	97.52%	90.67%	69.00%	97%	94.00%	91.00%		
C19737926	Shalom, Prince	2029	C1D1 9	SEN2	0	A-	90.34%		1807	2000	85.90%	91.42%	84.67%	98%	86.67%	92.17%		
C68967409	Wegner, Dylan	2028	C1D1 4	LLS3	9	A-	90.33%		1807	2000	82.67%	106.33%	97.67%	99%	82.00%	84.50%	HOCKEY	
C09320065	Bao, Yuwei	2028	E2F2 2	WST0 RSA1N	1	A-	90.32%		1815	2010	85.00%	82.34%	72.00%	98%	96.00%	93.50%		
C76734629	Cho, Jonathan	2029	C2D2 6	AEN0	1	A-	90.29%		1806	2000	79.52%	88.30%	86.33%	97%	88.67%	93.67%		
C82060479	Persteins, Marshall	2029	E2F2 11	CVN3	0	A-	90.27%		1805	2000	75.65%	75.67%	93.00%	95%	89.33%	95.67%	FOOTBALL (SPRINT)	
C68195385	Rodriguez, Roberto	2029	C2D2 4	MEN3	2	A-	90.25%		1805	2000	75.52%	95.60%	84.67%	98%	88.33%	95.83%		
C20672756	Astone, Andrew	2029	C1D1 9	MEN3	0	A-	90.23%		1805	2000	84.59%	90.34%	85.33%	98%	94.33%	95.33%	FOOTBALL	ML100 N/C
C90674436	Hotta, Akimasa	2029	A2B2 10	GSC0	2	A-	90.23%		1791	1985	92.00%	97.33%	84.67%	88%	78.33%	93.83%		
C35594403	Beaver, Nathaniel	2029	C2D2 2	AEN0	3	A-	90.21%		1813	2010	81.67%	80.42%	81.67%	98%	84.00%	91.00%		
C67829283	Knapp, Brian	2029	C1D1 1	KIN1	4	A-	90.09%		1784	1980	88.16%	83.75%	90.00%	97%	79.00%	93.33%	LACROSSE (MEN)	
C72040943	Murillo, Samuel	2029	C2D2 2	PAP3	0	A-	90.08%		1779	1975	91.75%	80.56%	69.00%	100%	82.33%	95.67%		ML100 N/C
C09585972	Putra, Haidar	2029	E2F2 10	WST0	1	A-	90.04%		1787	1985	90.30%	92.00%	83.67%	91%	90.00%	88.33%		
C46015805	Linzey, Darren	2029	E1F1 5	ASD0	0	A-	89.98%		1800	2000	99.31%	91.79%	80.00%	98%	91.67%	88.17%		
C54678079	Luter, Bryson	2029	A1B1 8	EPS1	0	A-	89.92%		1799	2000	86.74%	93.26%	88.33%	98%	84.67%	88.67%	FOOTBALL	ML100 N/C
C61057038	Edwards, Henry	2029	C2D2 4	CVN3	0	A-	89.85%		1797	2000	90.71%	102.46%	83.00%	94%	77.67%	92.33%	TRACK/INDOOR TRACK	
C42685632	Molina, Kiana	2029	A1B1 8	EEN1	3	A-	89.85%		1784	1985	79.92%	73.04%	95.					

CID	Name	Year	Hour Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C46453041	Park, Sangwoo	2028	A2B2 2	CYS1	5	A-	89.75%		1804	2010	90.00%	93.78%	63.67%	90%	91.67%	95.33%		
C58957634	Ferraro, Emma	2029	C2D2 10	CVN3	2	A-	89.74%		1795	2000	89.17%	93.82%	78.67%	87%	81.67%	96.00%		
C21942862	Jiang, Daniel	2029	E2F2 6	AEN0	0	A-	89.71%		1794	2000	83.05%	86.67%	81.33%	98%	83.33%	97.00%		
C05203503	Gacad, Brinon	2029	C2D2 2	LLS3	0	A-	89.64%		1793	2000	91.66%	93.33%	75.00%	91%	85.00%	91.50%		
C30645389	Smith, Gennaro	2028	E1F1 1	CYS1	0	A-	89.61%		1792	2000	78.40%	75.38%	86.33%	99%	85.67%	92.00%		
C45724374	Leahy, William	2029	C2D2 4	CEN2	6	A-	89.57%		1791	2000	78.84%	95.45%	82.00%	97%	92.00%	89.33%	BASEBALL	
C16642121	Savala, Colton	2029	C2D2 10	CEN2	0	A-	89.56%		1791	2000	89.15%	84.68%	80.67%	91%	84.33%	96.00%		
C03977868	Carreras Merced, Juan	2029	E2F2 3	MEN3	3	A-	89.55%		1760	1965	88.82%	100.00%	79.00%	94%	85.33%	94.50%		
C95200718	Twomey, Aidan	2029	C2D2 3	MGN0	0	A-	89.50%		1763	1970	92.74%	95.04%	83.00%	98%	79.67%	89.00%		
C97240595	Pearce, Austin	2029	A2B2 6	AEN0	0	A-	89.47%		1789	2000	87.49%	89.71%	86.33%	94%	84.67%	92.00%		
C02481855	Jones, Isabel	2028	C2D2 10	ESC3	2	A-	89.45%		1789	2000	91.11%	98.22%	77.67%	95%	86.00%	92.17%		
C43348436	Evans, Annabelle	2029	A2B2 3	CYO0	1	A-	89.42%		1762	1970	94.72%	76.73%	85.00%	100%	73.33%	94.17%		
C91868722	Castell, Leila	2029	E1F1 5	ENM1	1	A-	89.32%		1786	2000	82.90%	95.93%	79.33%	98%	86.33%	93.00%	SOCCER (WOMEN)	ML100 N/C
C87048970	Freedman, Crawford	2028	A1B1 3	HUS3	1	A-	89.32%		1786	2000	90.33%	92.49%	82.33%	95%	84.67%	89.33%		
C37942819	Coppel, Grace	2029	C1D1 8	EPS1	4	A-	89.16%		1783	2000	72.38%	92.67%	79.00%	98%	91.00%	95.17%	CROSS COUNTRY (WOI	
C03511905	Kyle, Kambria	2029	C2D2 3	HUS3T	2	A-	89.12%		1756	1970	94.69%	96.60%	77.00%	83%	80.67%	91.83%		
C39759965	Park, Minjun	2029	E1F1 8	ENM1	0	A-	89.10%		1782	2000	84.08%	95.56%	87.00%	94%	83.67%	89.67%		
C45461483	Nelson, Joseph	2029	C2D2 2	AEN0	1	A-	89.09%		1760	1975	91.75%	90.56%	79.33%	96%	84.00%	88.83%		
C55542501	Chivers, Lane	2029	C1D1 1	ENM1	1	A-	88.98%		1780	2000	90.10%	89.64%	79.67%	98%	87.67%	87.83%	TRACK/INDOOR TRACK	
C70339276	Mladenovic, Nikolai	2029	C1D1 8	CEN2	1	A-	88.97%		1779	2000	76.23%	75.89%	91.33%	98%	87.00%	94.67%		
C31284550	Behan, Tyler	2028	A1B1 3	EPS1	0	A-	88.81%		1776	2000	90.20%	75.53%	86.00%	98%	81.00%	89.50%		
C86932315	Rodriguez, Jack	2029	E2F2 2	CEN2	0	A-	88.76%		1784	2010	85.00%	96.00%	77.67%	88%	75.33%	93.17%		
C70060564	Tran, Andrew	2027	E2F2 2	ASDO FLC2	0	A-	88.75%		1784	2010	85.00%	80.82%	85.67%	98%	76.33%	90.33%		
C64113593	Gray, John Paul	2029	C2D2 6	FLC2	0	A-	88.67%		1773	2000	79.82%	76.05%	81.67%	95%	88.67%	96.17%		
C87279163	McNee, Cooper	2029	C1D1 8	EEN1	1	A-	88.64%		1773	2000	83.30%	79.01%	89.33%	98%	76.33%	93.83%		
C53100355	Wigginton, Brantley	2029	C1D1 8	CVN3	1	A-	88.64%		1773	2000	70.89%	98.00%	87.33%	100%	83.00%	92.50%	SOCCER (MEN)	ML100 N/C
C28699949	Williams, Sydney	2029	E1F1 5	EPS1	4	A-	88.62%		1772	2000	82.87%	100.00%	84.33%	87%	82.67%	90.50%		
C77215135	Swanson, Hatcher	2029	C2D2 3	PIF0	0	A-	88.61%		1768	1995	86.62%	82.20%	79.00%	95%	85.67%	93.83%		
C00709873	Nava, Jasmine	2029	E1F1 8	BEA0	0	A-	88.55%		1771	2000	85.31%	99.11%	84.00%	98%	72.33%	92.17%		
C44790015	Moore, Logan	2029	C1D1 8	MEN3	3	A-	88.53%		1771	2000	81.61%	95.93%	76.67%	95%	84.00%	92.17%		
C38644906	Smith, Kellen	2029	C2D2 4	ENM1	1	A-	88.52%		1770	2000	89.02%	95.19%	78.67%	93%	78.00%	91.17%	WRESTLING	
C95667010	Vojta, Christopher	2028	E1F1 1	MSC1	1	A-	88.52%		1770	2000	76.74%	85.38%	85.33%	100%	86.00%	89.33%		
C07271551	Stoddard, Lucas	2027	E1F1 5	LLS3	9	A-	88.51%		1753	1980	96.77%	98.00%	84.67%	98%	72.33%	89.50%	WRESTLING	EN101
C81630358	Javate, Ryu	2029	E2F2 2	HNT3H	2	A-	88.47%		1778	2010	81.67%	80.00%	74.67%	95%	86.67%	90.17%		
C00304391	Georgiovany, Jason	2028	E2F2 2	CYS1	0	A-	88.38%		1776	2010	85.00%	73.33%	78.67%	95%	85.00%	96.17%		MD201
C39198108	Verzwylvet, Cade	2029	A2B2 10	EEN1	0	A-	88.32%		1766	2000	92.66%	97.89%	78.67%	100%	80.67%	91.33%		
C79500418	Alvarez, Noah	2028	C1D1 1	MGN0	2	A-	88.30%		1766	2000	68.36%	98.00%	98.33%	99%	75.33%	91.00%	HOCKEY	
C22074382	Crawford, Maximilian	2028	A1B1 3	FLC2 PIF0	3	A-	88.25%		1756	1990	93.02%	93.89%	87.00%	97%	78.67%	85.33%		
C54509577	Ferrell, Maxwell	2029	C2D2 2	ECN2	1	A-	88.10%		1771	2010	88.33%	73.26%	74.33%	99%	79.33%	88.83%		
C86087887	Chen, Lance	2029	E2F2 2	CVN3	0	A-	88.01%		1769	2010	92.50%	94.00%	81.00%	91%	93.33%	87.83%		ML100 N/C
C93460842	Letcher, Tyler	2029	C2D2 3	CYO0	0	A-	88.01%		1734	1970	83.05%	96.67%	87.67%	98%	80.67%	89.50%		
C17496507	Marquardt, Lukas	2029	C1D1 8	ECN2	0	B+	87.98%		1760	2000	73.49%	86.20%	84.67%	90%	92.67%	91.17%		
C39730653	Bolainez, Angel	2029	C1D1 8	CVN3	1	B+	87.88%		1744	1985	82.23%	88.35%	70.67%	100%	85.33%	94.67%		
C26144588	Peterson, Hudson	2029	E1F1 1	ORE2	0	B+	87.88%		1758	2000	89.74%	84.05%	79.33%	92%	75.00%	93.67%		
C02123974	Schmelzer, Ethan	2029	A1B1 3	NEN1	1	B+	87.87%		1749	1990	93.15%	88.68%	82.67%	56%	84.00%	90.50%	TRACK/INDOOR TRACK	
C86346133	McLachlan, Alexander	2029	A2B2 6	ORE2	0	B+	87.77%		1755	2000	85.21%	65.26%	89.33%	98%	88.33%	92.00%		
C20278760	Castro Enamorado, Diego	2029	C2D2 2	CVN3	3	B+	87.69%		1710	1950	87.08%	97.33%	83.00%	92%	85.67%	86.33%		
C56462757	Kim, Minchan	2029	E2F2 11	CHM2A	1	B+	87.63%		1753	2000	77.61%	97.33%	82.67%	91%	92.33%	88.67%		
C68784200	Ivey, Benjamin	2028	C1D1 5	CYS1	8	B+	87.61%		1735	1980	85.17%	100.00%	94.33%	100%	89.00%	77.33%	HOCKEY	
C04630914	Pezan, Dennis	2029	A2B2 7	ORE2	10	B+	87.58%		1752	2000	96.97%	94.60%	78.00%	92%	80.67%	86.00%		
C41811319	Owen, John	2029	C1D1 5	MGN0	3	B+	87.53%		1733	1980	94.82%	95.11%	80.67%	100%	72.33%	89.33%	SWIMMING (MEN)	
C73798801	Magoto, Louis	2029	E2F2 3	NEN1	1	B+	87.51%		1702	1945	88.82%	69.17%	82.00%	99%	88.33%	90.83%		
C64236148	Cano, Justin	2029	C1D1 4	CYS1	1	B+	87.43%		1749	2000	74.55%	98.86%	88.00%	95%	86.67%	88.50%		
C64746103	Cooney, Edward	2029	C1D1 1	BEO0	1	B+	87.41%		1748	2000	85.48%	48.73%	82.33%	98%	87.33%	89.00%		
C84779793	Brisco, Nathaniel	2029	C1D1 1	MGN0	0	B+	87.39%		1748	2000	86.98%	100.00%	81.67%	78%	86.33%	85.83%		
C45471197	Rathnam, Kareena	2029	E2F2 6	FLA2	4	B+	87.35%		1730	1980	75.74%	85.64%	80.67%	99%	83.67%	94.00%		
C57571199	Franz, Lukas	2029	C1D1 8	LLS3	2	B+	87.26%		1732	1985	83.30%	92.60%	79.00%	91%	77.33%	92.00%		
C40657085	Cameron, Kieran	2028	E1F1 8	FLC2	0	B+	87.25%		1745	2000	84.08%	83.11%	73.33%	96%	96.33%	86.50%		
C91282829	Hendricks, William	2029	A2B2 2	ENM1	0	B+	87.17%		1752	2010	85.00%	87.98%	79.33%	95%	80.00%	84.33%		
C84941910	Pontzer, Adam	2029	A2B2 7	MEN3	2	B+	87.05%		1741	2000	88.20%	66.67%	90.00%	97%	91.00%	92.50%		
C32880933	Davidyock, Justin	2028	E1F1 1	KIN1	0	B+	87.03%		1741	2000	77.23%	89.80%	83.33%	91%	76.00%	91.50%	SOCCER (MEN)	ML100
C65785357	Byas, Caden	2028	A1B1 3	CYS1	1	B+	86.93%		1739	2000	92.98%	93.33%	84.00%	88%	77.00%	83.33%		
C76144868	Suh, Dean	2029	E2F2 10	MGN0	1	B+	86.83%		1724	1985	90.30%	96.37%	72.67%	94%	86.00%	87.83%		
C15111372	Juenger, Joseph	2029	E2F2 6	MSC1	1	B+	86.81%		1736	2000	86.89%	59.93%	87.33%	91%	91.67%	88.50%		
C62658327	Sanders, Jack	2028	E1F1 1	CYS1	0	B+	86.61%		1732	2000	86.09%	97.69%	80.00%	96%	82.00%	80.83%		
C39930058	Bowman, Addison	2029	C2D2 2	BEO0	5	B+	86.58%		1740	2010	83.33%	87.11%	65.67%	98%	78.00%	86.33%	RUGBY (WOMEN)	
C12483346	Conti, Annabella	2028	C2D2 3	BEO0	1	B+	86.54%		1727	1995	91.26%	95.11%	72.00%	98%	72.33%	89.67%	SOCCER (WOMEN)	ML100
C98097821	Jacobs, Charles	2029	C2D2 10	SEN2	1	B+	86.52%		1730	2000	89.15%	66.15%	69.00%	93%	87.67%	88.83%		
C59865190	Wilson, Silas	2029	C1D1 1	HNT3T	0	B+	86.41%		1728	2000	59.48%	98.00%	82.33%	88%	88.00%	95.67%	FOOTBALL (SPRINT)	
C37236327	Mellynychuk, Olivia	2029	A2B2 10	HNT3T	3	B+	86.24%		1712	1985	93.34%	95.11%	62.00%	79%	80.00%	91.50%	TENNIS (WOMEN)	
C49326814	Hodges, Edward	2029	A1B1 8	NEN1	1	B+	86.18%		1724	2000	78.61%	78.71%	83.00%	98%	84.33%	87.00%		
C06603237	Wynn, Tade	2029	C1D1 4	SDS1	5	B+	86.18%		1724	2000	69.75%	98.15%	83.67%	98%	81.00%	92.67%	LACROSSE (MEN)	
C60612176	Johnson, Hayden	2029	A2B2 7	SEN2	0	B+	86.17%		1723	2000	90.33%	82.60%	73.00%	96%	90.67%	83.17%		
C52954050	Son, Ethan	2029	C1D1 5	CYO0	0	B+	86.15%		1723	2000	94.80%	95.93%	79.33%	97%	78.67%	78.67%		
C12904306	Ryu, Timothy	2029	E1F1 5	EEN1	1	B+	86.10%		1722	2000	83.11%	97.78%	77.00%	100%	87.00%	81.83%		

CID	Name	Year	Hour Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C10610265	Pass, Luca	2028	E1F1 1	WST0 HMH3H	5	B+	86.09%		1722	2000	84.70%	77.43%	78.67%	98%	88.33%	81.33%		
C91479258	Cullison, Dain	2029	C2D2 4	CVN3	0	B+	86.05%		1721	2000	76.10%	97.19%	85.00%	94%	70.33%	90.83%		
C58807068	Shenoy, Lavanya	2029	A2B2 2	AEN0	0	B+	86.03%		1729	2010	91.67%	95.93%	74.00%	93%	67.00%	88.50%		ML100 N/C
C34936224	Speers, Wyatt	2029	A2B2 3	ENM1	1	B+	85.89%		1713	1995	84.54%	96.44%	67.00%	100%	86.00%	87.33%		
C15611523	Brown, Elijah	2029	A2B2 3	HMH3T	1	B+	85.88%		1670	1945	86.56%	89.26%	78.67%	97%	83.67%	84.33%		
C53372823	Range, Garrett	2029	C1D1 9	SEN2	1	B+	85.82%		1716	2000	87.21%	94.15%	84.67%	88%	80.00%	86.83%		
C24172351	Esterly, Lewis	2029	E1F1 8	PYL2	2	B+	85.78%		1716	2000	76.97%	95.56%	82.00%	94%	81.33%	87.83%	SWIMMING (MEN)	
C86813114	Rowe, Ailyn	2029	C2D2 2	SEN2	0	B+	85.77%		1715	2000	87.08%	91.53%	65.00%	96%	81.00%	88.00%	SOCCER (WOMEN)	ML100 N/C
C38085750	Weber, Brayden	2029	C2D2 10	CVN3	1	B+	85.75%		1702	1985	85.96%	89.24%	60.00%	92%	92.67%	89.67%		
C85600013	Baker, Christopher	2029	E1F1 8	MEN3	1	B+	85.68%		1714	2000	76.97%	59.09%	91.67%	98%	93.33%	91.67%		
C88892398	Choi, Ryan	2029	C2D2 3	ORE2	1	B+	85.65%		1709	1995	86.15%	94.97%	76.00%	94%	86.00%	84.33%		
C69661675	Oleszko, Tadeusz	2028	E1F1 8	CYS1 MSM0N	1	B+	85.46%		1709	2000	89.66%	74.34%	82.67%	98%	87.33%	77.83%		PE117
C73051872	Alpaugh, Jack	2029	C2D2 4	SDS1	2	B+	85.45%		1709	2000	89.02%	91.78%	78.67%	97%	66.00%	87.33%		
C72757458	Guest, Destin	2029	C2D2 3	LLS3	1	B+	85.44%		1683	1970	86.62%	85.04%	77.00%	98%	62.67%	92.83%	RIFLE	
C57698589	McCall, Draven	2029	C2D2 4	CYO0	1	B+	85.36%		1707	2000	76.81%	96.67%	73.00%	91%	79.00%	87.67%		
C66719554	Andersen, Mallorie	2029	C1D1 5	CYO0	1	B+	85.25%		1705	2000	81.37%	92.97%	58.67%	100%	81.67%	92.50%		
C18642254	Smith, Mason	2029	C1D1 1	MGN0	0	B+	85.15%		1703	2000	68.16%	96.89%	83.00%	93%	78.33%	91.83%	FOOTBALL	ML100 N/C
C69334884	Baggish, Owen	2029	C2D2 11	SEN2	2	B+	85.10%		1685	1980	74.90%	84.24%	85.67%	92%	73.67%	92.50%		
C66682449	Letso, Damian	2028	E1F1 5	BEA0	1	B+	85.08%		1702	2000	94.64%	93.33%	68.00%	94%	84.00%	82.67%		
C92318245	Hochstrasser, Hayden	2029	A1B1 8	ECN2	1	B+	84.97%		1699	2000	61.11%	98.67%	79.00%	98%	92.00%	87.00%	WRESTLING	
C03309230	Jirak, Layton	2029	A2B2 2	EEN1	1	B+	84.93%		1669	1965	91.43%	69.07%	67.00%	99%	84.33%	86.17%		
C43397804	Son, Kyle	2029	C2D2 2	SEN2	0	B+	84.93%		1699	2000	87.16%	97.19%	70.00%	97%	82.33%	81.33%		
C76764387	Vann, Nehemiah	2028	C1D1 1	LLS3	0	B+	84.83%		1697	2000	89.51%	95.33%	69.33%	98%	73.00%	84.83%	FOOTBALL	
C91897564	Pineiro, Joanna	2028	C2D2 11	LLS3 RSM1N	0	B+	84.79%		1696	2000	85.46%	94.34%	80.00%	86%	77.00%	85.67%		
C34834589	Stewart, George	2029	E2F2 3	GSC0	0	B+	84.79%		1670	1970	93.44%	89.78%	65.00%	100%	83.67%	86.17%		
C71333960	Lundeen, Carter	2029	C1D1 1	LLS3	0	B+	84.66%		1693	2000	60.95%	97.78%	82.67%	97%	76.67%	88.00%		
C61250670	Taura, Masayoshi	2029	A1B1 8	HNT3T	0	B+	84.66%		1693	2000	78.61%	78.95%	62.33%	98%	86.00%	91.17%		
C60031526	Thompson, Samuel	2029	E1F1 8	GSC0	2	B+	84.66%		1693	2000	75.07%	87.26%	65.33%	94%	84.33%	94.33%		
C52716275	Hahn-Vadstein, Alexander	2029	A2B2 3	GSC0	1	B+	84.62%		1625	1920	83.08%	62.18%	80.33%	94%	81.00%	89.33%	CROSS COUNTRY (MEN)	EN101 N/C
C71924590	Sombeck, Adam	2029	E2F2 10	MGN0	2	B+	84.62%		1680	1985	92.54%	69.78%	62.33%	98%	79.00%	91.83%		
C23546905	Hagberg, Alex	2029	E1F1 5	AEN0	1	B+	84.60%		1692	2000	83.11%	83.79%	75.67%	96%	76.00%	89.00%		
C05744697	Fuentespina, Lauren Taylor	2029	A2B2 10	GEO1	4	B+	84.57%		1691	2000	94.84%	89.75%	72.67%	88%	77.33%	84.17%	TRACK/INDOOR TRACK	
C93934041	Davis, Addison	2029	C2D2 4	PIF0	6	B+	84.52%		1690	2000	77.82%	94.19%	80.00%	97%	80.00%	86.83%		
C98200954	Harris, Jackson	2029	A2B2 7	ECN2	0	B+	84.52%		1690	2000	86.39%	82.67%	80.00%	94%	74.67%	90.50%		
C31114913	Roll, Philip	2029	A2B2 6	CVN3	1	B+	84.50%		1690	2000	72.59%	61.64%	81.67%	100%	81.00%	94.00%	RUGBY (MEN)	
C93759077	Vinton, Sawyer	2029	E1F1 1	PAP3	2	B+	84.50%		1690	2000	72.79%	82.56%	88.33%	95%	81.00%	84.50%		
C24488477	Tepyuha, Kunakorn	2029	A2B2 6	CVN3	1	B+	84.48%		1690	2000	80.43%	87.93%	82.67%	83%	70.67%	96.33%		ML100 N/C
C70360802	Madison, Connor	2028	C2D2 2	ESC3	1	B+	84.46%		1668	1975	87.16%	70.82%	83.33%	80%	74.67%	85.17%		
C95577429	McLaughlin, Delaney	2028	E2F2 6	ESC3	1	B+	84.39%		1671	1980	80.82%	91.93%	77.00%	85%	85.00%	84.50%	SWIMMING (WOMEN)	
C96406244	Brooks, Mason	2029	E2F2 3	LLS3	3	B+	84.29%		1618	1920	90.79%	96.75%	67.00%	94%	80.67%	87.17%		
C03934564	Stewart, Keilani	2028	C1D1 9	PII0	5	B+	84.26%		1685	2000	87.46%	83.67%	65.00%	90%	78.33%	92.50%		
C74677335	Garcia, Micaela	2029	C1D1 9	CVN3	9	B+	84.11%		1682	2000	82.62%	67.13%	81.67%	88%	77.33%	92.67%	SOFTBALL (WOMEN)	
C82833287	Oh, Jeremiah	2028	E1F1 1	LLS3	0	B+	84.10%		1682	2000	69.61%	93.64%	76.67%	92%	80.67%	86.17%		
C22839013	Poe, Daniel	2029	C1D1 1	MGN0	1	B+	84.10%		1682	2000	68.39%	93.78%	69.33%	90%	86.67%	88.17%		
C69507451	Edmondson, Eli	2029	C1D1 4	SEN2	1	B+	84.08%		1682	2000	89.69%	98.73%	66.33%	97%	76.33%	80.83%	SWIMMING (MEN)	
C88465822	Jesse, Benjamin	2029	E2F2 6	BE00	3	B+	84.06%		1648	1960	75.49%	91.16%	79.33%	93%	78.33%	91.17%		
C42212042	Forsellus, Nils	2028	C1D1 1	MGN0	10	B	83.99%		1671	1990	87.34%	72.00%	95.67%	92%	69.67%	78.33%	HOCKEY	
C52552031	Gerak, Moses	2029	E1F1 5	ORE2	0	B	83.99%		1680	2000	97.52%	88.53%	66.67%	98%	77.33%	78.50%		
C41968573	Wasserman, Max	2029	E2F2 3	SEN2	1	B	83.98%		1654	1970	88.89%	82.08%	78.67%	97%	73.33%	85.83%		
C33816300	Arne, Liam	2029	A1B1 5	MEN3	0	B	83.97%		1679	2000	90.00%	68.37%	75.00%	97%	88.00%	83.67%		
C24987518	Caldwell, Caleb	2029	A2B2 7	SDS1	2	B	83.94%		1679	2000	87.05%	92.49%	72.33%	93%	89.67%	82.17%	LACROSSE (MEN)	
C59157155	Jones, Brock	2029	C2D2 6	SDS1	5	B	83.88%		1640	1955	93.23%	98.67%	66.33%	84%	82.67%	86.33%	HOCKEY	
C03350380	Brooke, Coleman	2029	C1D1 9	EPS1	0	B	83.69%		1674	2000	80.00%	82.15%	74.00%	97%	76.33%	89.67%	RUGBY (MEN)	
C64444024	Showan, Grant	2029	E2F2 2	NEN1	1	B	83.68%		1682	2010	85.00%	89.04%	85.67%	85%	72.00%	77.83%		
C75060745	Hartley, Jason	2029	C2D2 10	WST0T	4	B	83.49%		1657	1985	88.29%	59.11%	78.00%	91%	65.00%	93.00%		
C08939995	Donovan, Sean	2028	C1D1 4	MGN0	0	B	83.38%		1668	2000	66.52%	96.33%	82.33%	93%	74.33%	86.50%	FOOTBALL	ML100
C46796155	Porter, Grace	2029	A2B2 10	BEA0	2	B	83.38%		1655	1985	95.02%	93.26%	71.00%	88%	71.00%	82.67%	Rabble Rousers (Varsity I)	
C93079515	Mathews, Tyler	2029	A2B2 6	GEO1	2	B	83.32%		1666	2000	84.13%	94.00%	73.33%	92%	72.33%	92.67%	LACROSSE (MEN)	
C35163668	Niemoth, Jackson	2029	C1D1 5	WST0T	0	B	83.29%		1649	1980	84.93%	86.13%	81.00%	94%	82.67%	82.50%		
C26815242	Hutcheson, Gregory	2029	A2B2 10	MGN0	0	B	83.23%		1665	2000	94.88%	96.37%	57.00%	93%	78.67%	86.17%		
C37158506	Lee, Andrew	2029	C1D1 9	SEN2	3	B	83.16%		1663	2000	84.59%	96.00%	78.67%	92%	76.00%	88.67%	CROSS COUNTRY (MEN)	
C90459880	Wilson, Elias	2029	C2D2 4	MEN3	2	B	83.15%		1663	2000	71.80%	90.30%	91.00%	82%	78.67%	79.00%		
C46220213	Mathis, Skyler	2028	A1B1 5	FLF2	2	B	83.13%		1663	2000	87.98%	81.23%	77.33%	83%	73.33%	81.67%		
C28611228	Park, Noah	2029	E1F1 8	EEN1	1	B	83.10%		1662	2000	76.97%	82.00%	83.00%	98%	85.00%	78.00%		
C99634937	Dembowski, Jake	2029	E1F1 5	PIF0	0	B	83.05%		1644	1980	83.11%	83.65%	72.00%	93%	78.00%	83.83%		
C44170579	Chun, April	2029	E1F1 5	EVE2	6	B	83.03%		1644	1980	82.90%	98.67%	64.67%	98%	66.67%	87.00%	SWIMMING (WOMEN)	GE203
C80533478	Roessler, Jordan	2028	A1B1 5	BEA0	2	B	82.95%		1659	2000	80.41%	97.33%	77.33%	96%	71.67%	82.17%	LACROSSE (WOMEN)	
C34837279	Cho, Issac	2028	C1D1 9	BEA0	2	B	82.94%		1659	2000	83.61%	83.56%	72.67%	93%	67.67%	88.83%		
C13657835	Regennitter, Benjamin	2029	E1F1 1	PAP3	0	B	82.83%		1657	2000	83.97%	94.67%	57.67%	90%	65.00%	91.00%		
C73998346	Ogunlola, Zainab	2029	C2D2 3	GEO1	2	B	82.82%		1656	2000	93.80%	83.68%	63.00%	90%	79.00%	82.83%	RUGBY (WOMEN)	
C95519426	Mensah, Deborah	2029	C2D2 2	CVN3	0	B	82.78%		1656	2000	92.31%	58.13%	68.00%	84%	74.67%	84.00%		ML100 N/C
C83315510	Blackwell, Emma	2029	A2B2 2	AEN0	4	B	82.73%		1663	2010	94.17%	93.33%	60.33%	95%	65.33%	84.50%		
C24043648	Lucas, Nicholas	2029	E2F2 3	NEN1	1	B	82.68%		1629	1970	93.44%	85.71%	77.00%	80%	63.33%	89.33%		
C29024378	Dwornicki, Riley	2028	C1D1 1	LLS3	3	B	82.67%		1653	2000	84.59%	89.78%	76.00%	93%	72.67%	81.17%		
C07799850	Dossey, Beckett	2029	C2D2 4	ECN2	1	B	82.61%		1652	2000	76.10%							

CID	Name	Year	Hour	Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C49956300	Jones, Dawson	2028	A1B1	3	SDS1	1	B	82.58%		1652	2000	95.28%	83.45%	66.67%	84%	65.33%	89.50%	FOOTBALL	ML100
C11233755	Rothstein, James	2028	A1B1	5	MGN0	0	B	82.57%		1651	2000	84.97%	86.13%	75.33%	94%	68.00%	84.17%	FOOTBALL	ML100
C58711315	Hill, Daniel	2029	C2D2	3	EEN1	0	B	82.56%		1627	1970	94.69%	96.00%	70.67%	80%	76.67%	80.00%		
C21414860	Dees, Abel	2029	E2F2	11	SEN2	0	B	82.51%		1650	2000	69.66%	76.49%	82.67%	100%	76.00%	87.50%		
C26225772	Ophardt, Elizabeth	2029	C1D1	4	BEO0	2	B	82.45%		1649	2000	88.32%	98.28%	70.33%	88%	64.33%	84.17%		
C75368561	Lanham, Logan	2028	E1F1	1	HMH3T	2	B	82.39%		1648	2000	85.32%	88.98%	72.67%	90%	71.00%	85.17%		
C52174272	Reisinger, Avery	2029	E2F2	11	MEN3	0	B	82.38%		1648	2000	71.64%	93.93%	60.00%	82%	83.00%	92.17%		ML100 N/C
C68686554	Hinton, Jacob	2029	A2B2	3	ENM1	0	B	82.34%		1643	1995	86.56%	84.00%	68.67%	97%	69.33%	86.67%	FOOTBALL	ML100 N/C
C69605433	Swierkosz, Marissa	2029	A1B1	5	EPS1	1	B	82.33%		1647	2000	80.41%	96.75%	72.00%	100%	66.00%	85.00%		
C88237758	Murray, Timothy	2029	A2B2	2	CVN3	0	B	82.30%		1646	2000	93.98%	68.89%	52.67%	98%	80.33%	84.17%		
C63501849	Smith, Javon	2028	C2D2	11	BEA0	0	B	82.30%		1646	2000	77.11%	94.67%	66.33%	90%	80.67%	86.17%	FOOTBALL	ML100
C71291760	Renslow, Mason	2028	C2D2	10	FRS2	0	B	82.17%		1643	2000	89.15%	95.11%	76.00%	91%	78.33%	74.67%		
C30227098	Suarez, Jayden	2029	C2D2	4	CY00	3	B	82.14%		1643	2000	75.52%	69.07%	88.33%	91%	71.00%	84.67%		
C92726491	Lund, Gabriel	2029	A2B2	3	ECN2	3	B	82.07%		1637	1995	92.26%	100.00%	80.33%	95%	74.33%	74.33%	BASKETBALL (WOMEN)	
C50027281	Flowers, Carson	2029	C2D2	10	BEA0	0	B	81.98%		1640	2000	89.15%	97.33%	63.33%	93%	76.67%	85.00%		
C541555251	Robinson, James	2029	C1D1	9	ENM1	0	B	81.96%		1639	2000	85.90%	73.04%	75.67%	81%	81.33%	85.67%	FOOTBALL (SPRINT)	
C43135660	McKenzie, Melinda	2028	C2D2	10	LLS3	2	B	81.94%		1627	1985	91.11%	88.60%	71.67%	79%	82.67%	78.00%		
C70644655	Little, Joshua	2029	A2B2	2	LLS3	1	B	81.93%		1639	2000	94.97%	87.67%	56.33%	91%	73.67%	81.00%		
C08434117	Vonderheide, Henry	2028	E1F1	1	FLG2 HMH3	0	B	81.92%		1638	2000	87.45%	84.03%	72.00%	94%	67.67%	87.17%		
C66096995	McCurdy, Evelyn	2029	E1F1	8	ENL2	1	B	81.88%		1638	2000	89.66%	92.67%	55.00%	99%	82.67%	79.00%		
C51604754	Murray, Christopher	2029	E1F1	8	ASD0	2	B	81.88%		1638	2000	76.97%	84.67%	85.33%	98%	55.00%	88.67%		
C44997734	Lenteja, Jaedon Clyde	2029	E2F2	3	AEN0	2	B	81.83%		1592	1945	88.82%	91.33%	64.00%	93%	68.33%	87.83%		
C94591377	Hicks, Arianna	2029	C2D2	6	MGN0	1	B	81.81%		1620	1980	90.79%	85.33%	64.00%	95%	78.67%	81.33%	TRACK/INDOOR TRACK	
C23302918	Newcomb, Christopher	2029	C1D1	5	SEN2	3	B	81.79%		1636	2000	88.98%	33.26%	77.33%	100%	79.33%	83.00%	RUGBY (MEN)	
C23062439	Verma, Shaan	2029	C2D2	2	EEN1	3	B	81.77%		1615	1975	87.16%	76.67%	76.33%	95%	72.33%	78.83%	SOCCER (MEN)	
C01363258	Crosstef, William	2029	C1D1	4	CYS1	3	B	81.76%		1635	2000	63.40%	97.29%	79.00%	95%	82.00%	81.17%		
C77443684	Snipes, Ryan	2028	C2D2	2	SDS1	0	B	81.64%		1641	2010	81.67%	71.60%	68.00%	88%	69.33%	87.00%	FOOTBALL	ML100
C04028885	Jones, Aiden	2028	C1D1	1	MGN0	0	B	81.63%		1633	2000	66.92%	90.60%	69.00%	96%	77.00%	88.50%		
C19159446	Schaff, Noah	2028	A1B1	3	HNT3H	0	B	81.63%		1633	2000	90.52%	79.65%	64.67%	91%	70.00%	83.33%		PE117
C31924452	Wade, Jacob	2029	C2D2	3	ECN2	0	B	81.62%		1628	1995	94.36%	63.23%	73.00%	88%	81.00%	80.83%		
C24023203	Grayson, Wilson	2029	C2D2	3	WST0	0	B	81.53%		1626	1995	81.59%	93.26%	74.67%	96%	71.33%	84.67%		
C13522482	Zuernemann, Andrew	2029	A2B2	10	ORE2	0	B	81.42%		1628	2000	84.89%	80.00%	72.67%	93%	64.67%	88.83%		
C14505920	Kim, Dylan	2028	E1F1	1	PYL2	1	B	81.37%		1627	2000	89.02%	73.97%	74.67%	98%	62.33%	86.00%		
C23688564	Nielsen, Scarlet	2029	E2F2	3	SDS1	1	B	81.25%		1580	1945	88.03%	93.78%	65.67%	97%	72.67%	80.17%		
C39094293	Kuruvilla, Joseph	2029	E2F2	6	CY00	4	B	81.23%		1596	1965	81.79%	71.86%	68.00%	83%	65.00%	93.67%		
C06361352	Denson, Daniel	2029	E2F2	6	CYS1	0	B	81.18%		1624	2000	82.69%	92.00%	70.67%	75%	83.67%	84.50%		
C28741315	Del Busto, Joseph	2028	A1B1	8	CHM2	4	B	81.17%		1623	2000	78.61%	66.15%	71.33%	94%	81.67%	79.17%		
C24567541	Roberts, John	2029	C2D2	11	MGN0	3	B	81.03%		1621	2000	75.33%	87.56%	63.67%	98%	77.00%	85.67%		
C52016078	Belcher, Christina	2028	A1B1	5	MGN0	1	B	80.98%		1611	1990	83.66%	99.74%	73.00%	93%	71.67%	76.33%	TRACK/INDOOR TRACK	
C69191377	Griggs, Jonah	2029	C1D1	8	FSG1	1	B	80.97%		1619	2000	84.23%	65.68%	82.00%	93%	70.00%	82.33%		
C98346680	Nolan, Corinne	2029	A2B2	2	LLS3	2	B	80.91%		1598	1975	90.87%	94.41%	47.67%	97%	87.33%	76.33%	LACROSSE (WOMEN)	
C43419930	Stauffer, Jameson	2028	C2D2	2	CSC1	0	B	80.81%		1624	2010	86.67%	81.31%	74.33%	95%	69.33%	69.83%		
C60491079	Hicks, Owen	2029	C1D1	4	ENM1	1	B	80.78%		1616	2000	87.12%	90.84%	65.67%	97%	74.67%	75.50%	WRESTLING	
C22635703	Riordan, Kellye	2029	A1B1	8	HNT3	6	B	80.65%		1601	1985	91.10%	78.44%	67.67%	100%	87.67%	73.17%	LACROSSE (MEN)	
C08068142	Kleinjan, Jacob	2028	A2B2	2	CYS1	1	B	80.62%		1621	2010	90.00%	98.60%	70.67%	94%	71.67%	69.50%		
C79223946	Rossi, Brayden	2029	C2D2	10	AEN0	0	B	80.58%		1612	2000	91.11%	72.89%	62.33%	90%	77.33%	81.50%		
C46969357	Connor, Cole	2029	C1D1	4	SDS1	1	B	80.56%		1611	2000	89.42%	91.67%	71.67%	79%	71.33%	77.67%	BASEBALL	MD101
C84796679	Gupta, Aarav	2028	E1F1	1	CYS1	3	B	80.47%		1609	2000	91.67%	91.15%	77.00%	95%	69.67%	69.50%		
C84852353	Fitts, Faith	2028	A2B2	2	LLS3 TST1N	0	B	80.40%		1616	2010	85.00%	84.23%	62.00%	98%	70.00%	74.00%		
C32130516	Delucia, Joseph	2029	A2B2	6	ECN2	0	B	80.38%		1608	2000	70.57%	46.46%	73.67%	91%	80.33%	91.83%	FOOTBALL (SPRINT)	EN101 N/C
C08956689	Carmody, Nolan	2029	A1B1	8	EEN1	4	B	80.37%		1607	2000	58.85%	83.09%	78.33%	98%	74.67%	83.17%		ML100 N/C
C77852492	Johnson, Devion	2029	A2B2	3	BEA0	4	B	80.36%		1583	1970	94.72%	74.83%	72.00%	100%	65.67%	77.50%	FOOTBALL (SPRINT)	
C65661162	Negus, Troy	2029	E2F2	6	BEO0	1	B	80.35%		1595	1985	80.43%	96.00%	70.00%	104%	75.67%	86.67%		
C61752236	Johnson, Charles	2028	A2B2	2	SDS1	1	B	80.34%		1615	2010	91.67%	80.56%	63.67%	89%	67.00%	79.33%	FOOTBALL	ML100
C82580441	Merideth, Judson	2028	C1D1	5	LLS3	0	B	80.32%		1606	2000	84.93%	91.60%	81.00%	98%	70.33%	78.50%		PE117
C54913622	Chung, Daniel	2029	E1F1	5	ENM1	2	B	80.22%		1604	2000	83.11%	93.82%	80.00%	93%	71.33%	78.50%		
C07163107	Reilly, Shane	2029	A2B2	7	ENM1	0	B	80.20%		1604	2000	81.64%	89.26%	60.67%	91%	77.67%	83.33%	WRESTLING	
C20265856	Song, Ian	2029	C2D2	11	ENM1	1	B	80.20%		1604	2000	67.23%	95.78%	70.67%	73%	88.67%	85.00%		
C62214412	Resendez, Damian	2029	C2D2	4	ENM1	0	B	80.15%		1603	2000	76.15%	100.67%	73.33%	85%	81.67%	72.83%	WRESTLING	
C10973744	Arbogast, Marek	2029	C1D1	9	SDS1	1	B	80.14%		1603	2000	83.28%	93.33%	59.67%	90%	84.67%	80.50%	FOOTBALL (SPRINT)	
C80813082	Thomazini, Giovanni	2029	A2B2	10	EPS1	0	B	80.12%		1602	2000	91.97%	73.26%	61.67%	86%	79.00%	77.17%		
C28912658	Le, Alexander	2029	C2D2	11	CYS1	1	B	80.05%		1593	1990	76.03%	79.16%	84.00%	87%	68.67%	85.83%		
C70793726	Dudas, Paloma	2029	C2D2	11	SEN2	3	B	80.02%		1600	2000	76.75%	90.60%	70.00%	89%	77.00%	81.00%		
C25066102	Schroeder, Xander	2029	E2F2	3	PAP3	2	B-	79.96%		1575	1970	88.89%	97.78%	71.67%	96%	69.67%	75.50%		
C17812653	Myers, Thomas	2029	E2F2	6	ENM1	3	B-	79.87%		1566	1960	79.34%	88.00%	73.33%	99%	73.67%	84.83%		
C24486272	Ryan, Andrew	2029	E2F2	6	LLS3	0	B-	79.82%		1596	2000	77.07%	87.09%	67.67%	81%	73.67%	86.17%		
C65579486	McDonnell, Michael	2029	C2D2	2	LLS3	1	B-	79.76%		1575	1975	94.28%	94.67%	66.67%	93%	68.67%	72.50%		
C19656010	Salisbury, Clark	2029	C2D2	2	PAP3	1	B-	79.74%		1575	1975	91.75%	58.53%	66.00%	84%	63.67%	82.33%		
C60504316	Beckstrand, Kohler	2029	A2B2	6	CEN2	3	B-	79.73%		1575	1975	73.02%	64.87%	74.67%	97%	73.67%	91.50%		
C48233553	Tsuneyoshi, Cal	2029	A2B2	7	CEN2	5	B-	79.70%		1594	2000	89.18%	86.90%	56.67%	95%	58.67%	87.67%	BASKETBALL (WOMEN)	
C47534836	Joyce, Jake	2029	E2F2	3	WST0	1	B-	79.52%		1547	1945	88.00%	90.39%	54.33%	95%	85.33%	77.00%		
C86615068	Grady, Samuel	2028	A2B2	2	FLG2	1	B-	79.41%		1596	2010	94.17%	88.00%	60.67%	94%	74.33			

CID	Name	Year	Hour	Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C42635192	Aquino, Noah	2029	C1D1	9	CYS1	1	B-	79.01%		1580	2000	83.28%	83.54%	82.33%	86%	68.00%	79.67%		
C34718165	Ciocon, Emma	2029	C2D2	6	PIF0	2	B-	78.95%		1547	1960	83.16%	100.00%	51.00%	95%	74.67%	83.67%	LACROSSE (WOMEN)	
C14185552	Player, Samuel	2028	E1F1	1	CSC1	2	B-	78.90%		1578	2000	79.67%	66.83%	75.67%	100%	71.33%	76.17%		
C66486895	Mazzola, Rodolfo	2029	A2B2	2	PAP3	4	B-	78.82%		1576	2000	95.93%	98.60%	63.67%	83%	65.00%	70.83%	TENNIS (MEN)	
C89015783	Gear, Brandon	2029	C2D2	6	SDS1	0	B-	78.80%		1576	2000	92.48%	96.67%	59.33%	89%	86.00%	69.00%	SWIMMING (MEN)	
C88754261	Foley, Evan	2029	A1B1	8	CVN3	0	B-	78.70%		1574	2000	91.59%	28.42%	87.33%	98%	83.67%	82.17%		
C73644119	Von Eschenbach, Thomas	2029	C1D1	1	ORE2	1	B-	78.67%		1573	2000	89.02%	91.60%	64.67%	96%	83.33%	67.67%	WRESTLING	
C28420480	Rios, Diego	2029	C1D1	5	NEN1	1	B-	78.66%		1573	2000	83.27%	102.80%	59.00%	87%	53.00%	85.83%		
C32258765	Lee, Lawrence	2029	A2B2	6	CSC1	5	B-	78.62%		1517	1930	80.31%	97.26%	59.33%	89%	74.00%	83.83%		
C11325612	Fountain, Brayden	2028	C1D1	5	ASD0	4	B-	78.59%		1556	1980	83.27%	98.67%	77.67%	100%	69.67%	72.00%	LACROSSE (MEN)	
C60686555	Rolands, Quinn	2028	E1F1	5	CSC1 RSE1N	0	B-	78.33%		1567	2000	82.90%	79.23%	74.33%	98%	70.33%	75.50%		
C27982665	Simkins, Owen	2029	C2D2	6	WST0	3	B-	78.24%		1522	1945	73.39%	36.42%	70.00%	80%	70.67%	90.17%		
C98423786	Levitt, Sophia	2027	A1B1	5	HNT3H RSM1N	1	B-	78.21%		1533	1960	80.41%	94.60%	72.33%	96%	73.67%	67.33%		PE117
C97222865	Sheets, Ella	2029	E2F2	10	SEN2	0	B-	78.21%		1564	2000	88.28%	93.78%	65.33%	92%	64.67%	76.67%		
C86159419	Paolucci, Grant	2029	A1B1	5	LLS3	5	B-	78.15%		1532	1960	87.26%	81.64%	60.00%	98%	64.00%	82.33%	GOLF	
C49647382	Porter, Julian	2029	A1B1	5	SEN2	0	B-	78.14%		1563	2000	87.98%	60.82%	74.67%	97%	62.33%	77.83%	FOOTBALL	ML100 N/C
C75546693	Odell, Daniel	2029	C2D2	10	SDS1	3	B-	77.97%		1555	1995	83.84%	97.00%	46.67%	81%	75.67%	82.67%	LACROSSE (MEN)	
C58123458	Laylo, Cohen	2029	A2B2	10	EVE2	4	B-	77.86%		1546	1985	85.16%	32.89%	66.33%	95%	81.00%	91.50%	DIVING (MEN)	ML100 N/C
C14614880	Tsoukatos, Joaquin	2028	A2B2	2	WST0	0	B-	77.85%		1565	2010	95.00%	97.78%	60.33%	98%	52.33%	73.00%		
C83217253	Renner, Lawrence	2029	E2F2	11	MGN0	0	B-	77.84%		1557	2000	61.34%	39.09%	62.33%	98%	77.67%	88.33%		
C59423375	Acinapura, Vincent	2029	C1D1	4	PIF0	2	B-	77.70%		1554	2000	81.82%	86.46%	46.67%	89%	87.33%	73.50%	TRACK/INDOOR TRACK	
C06868693	Blakely, William	2029	E2F2	3	NEN1	1	B-	77.70%		1511	1945	88.03%	83.09%	77.00%	99%	58.00%	74.33%		
C10990457	Schultz, Lillian	2029	C2D2	11	BEA0	1	B-	77.58%		1552	2000	71.03%	94.00%	57.33%	91%	75.67%	82.50%		
C08117481	Norton, Justin	2029	A2B2	10	LLS3	2	B-	77.55%		1551	2000	85.55%	84.28%	69.33%	91%	64.33%	75.67%	FOOTBALL (SPRINT)	
C83094378	Monk, Christopher	2028	E2F2	11	GEO1 RSL1N	0	B-	77.52%		1550	2000	75.07%	97.78%	70.33%	95%	66.00%	74.00%		
C50446709	Rios, Christopher	2029	C1D1	8	CHM2A	3	B-	77.48%		1538	1985	73.49%	70.53%	74.00%	95%	51.33%	82.33%		
C71622527	Cole, Eliot	2029	E1F1	1	MSC1	1	B-	77.41%		1548	2000	76.36%	69.75%	71.67%	80%	80.67%	78.50%		ML100 N/C
C62648163	Quinn, Kathryn	2029	E1F1	8	GSC0	2	B-	77.41%		1537	1985	85.31%	85.71%	65.67%	100%	88.33%	60.67%		
C81878176	Michael, Campbell	2028	C1D1	9	PIF0	3	B-	77.37%		1547	2000	85.90%	83.71%	68.33%	98%	62.33%	78.00%		
C24252520	Larkin, Jessica	2028	A1B1	8	FLS2	2	B-	77.35%		1547	2000	92.28%	94.64%	68.67%	93%	62.33%	68.33%	LACROSSE (WOMEN)	CH101
C23187985	Gibson, Jonathan	2029	A1B1	8	CEN2	0	B-	77.28%		1546	2000	86.74%	92.00%	66.67%	92%	72.00%	69.00%		ML100 N/C
C96210718	Newell, Nathan	2029	A1B1	5	ENM1	0	B-	77.20%		1544	2000	86.26%	82.90%	63.33%	88%	63.33%	74.67%		
C02712228	Tillman, Simone	2029	A1B1	8	LLS3	1	B-	77.16%		1543	2000	91.59%	93.38%	78.00%	94%	61.33%	62.67%	TRACK/INDOOR TRACK	
C44138283	Lazo Cazares, Alma	2028	C1D1	1	PIF0	5	B-	76.98%		1540	2000	70.85%	88.89%	73.67%	90%	58.33%	81.33%	CROSS COUNTRY (WOMEN)	PH201
C41377053	Brown, Miles	2029	E2F2	3	ESC3	0	B-	76.96%		1497	1945	88.00%	88.89%	68.00%	87%	78.33%	71.00%		
C67103970	Povarchuk, Julia	2029	C2D2	3	ECN2	3	B-	76.90%		1515	1970	94.36%	83.23%	66.33%	97%	52.33%	80.17%		ML100 N/C
C78488988	Anderson, Brady	2028	C1D1	5	MGN0	2	B-	76.86%		1537	2000	84.93%	95.60%	65.33%	85%	68.33%	72.17%	FOOTBALL	
C09826608	Johnson, Jeremiah	2029	C1D1	9	SEN2	0	B-	76.86%		1537	2000	87.54%	75.93%	70.00%	80%	74.67%	72.50%		ML100 N/C
C57015456	Plavetzky, Katherine	2029	C2D2	2	CYS1	0	B-	76.85%		1518	1975	92.31%	68.75%	53.00%	95%	70.33%	71.67%		ML100 N/C
C27200115	Worth, Todd	2029	A2B2	7	GSC0	4	B-	76.85%		1537	2000	86.61%	64.56%	57.33%	97%	74.67%	75.00%	CROSS COUNTRY (MEN)	
C07442561	Lukowsky, Jesse	2028	E2F2	3	PIF0 TST1N	0	B-	76.69%		1530	1995	88.89%	96.00%	47.67%	94%	62.00%	79.67%		
C42235667	Zito, Ryan	2029	A1B1	5	WST0	1	B-	76.68%		1518	1980	86.26%	98.67%	69.67%	96%	24.33%	88.67%		
C36012958	Jorjani, Nickolas	2029	E2F2	10	PIF0	2	B-	76.65%		1522	1985	88.17%	79.00%	63.33%	86%	63.67%	73.83%		ML100 N/C
C33672577	Golotyuk Morales, Phillip	2029	A2B2	3	EPS1	1	B-	76.62%		1536	2005	81.64%	97.33%	70.00%	85%	65.33%	77.17%		
C78364984	Perez Montoya, Helena	2029	C1D1	1	LLS3	0	B-	76.62%		1532	2000	56.33%	85.72%	64.67%	93%	73.33%	83.83%		
C20688279	Boyle, Turner	2029	E2F2	3	PIF0	0	B-	76.35%		1523	1995	86.92%	96.15%	57.67%	88%	65.33%	79.83%		ML100 N/C
C97537978	Lawrence, Karina	2028	C2D2	2	FGR2	3	B-	76.29%		1533	2010	88.33%	37.61%	58.33%	93%	60.67%	75.50%		
C44929314	Mundell, Zachary	2026	A2B2	2	LLS3	1	B-	76.20%		1505	1975	91.43%	88.53%	52.67%	88%	70.67%	71.83%	FOOTBALL	MA104
C16693510	Rigley, Maura	2029	A1B1	5	MGN0	5	B-	76.12%		1522	2000	77.00%	97.33%	79.67%	94%	66.33%	62.00%	LACROSSE (WOMEN)	
C56974256	Torre, Antonio	2029	E1F1	1	MGN0	2	B-	76.08%		1522	2000	88.79%	73.60%	58.00%	83%	56.67%	81.50%		
C63922597	Morris, Hunter	2028	C2D2	6	WST0	2	B-	76.01%		1505	1980	93.00%	93.19%	73.33%	76%	56.67%	75.83%	TRACK/INDOOR TRACK	
C65391066	Martini, Jake	2029	C1D1	4	SDS1	4	C+	75.94%		1519	2000	65.55%	100.80%	63.67%	85%	75.00%	77.83%	LACROSSE (MEN)	
C22328061	Cochran, Liam	2029	C2D2	6	ORE2	0	C+	75.73%		1515	2000	70.44%	91.60%	61.33%	88%	66.33%	83.83%		
C11558166	Heideman, Tyler	2029	E2F2	10	WST0	1	C+	75.60%		1512	2000	81.59%	93.19%	56.00%	100%	44.33%	82.50%		
C25337768	Ocampo, Will	2029	E1F1	1	SDS1	2	C+	75.60%		1501	1985	89.17%	74.53%	55.33%	74%	67.33%	74.83%		
C08742240	Edwards, Caelan	2029	C2D2	11	LLS3	1	C+	75.58%		1512	2000	75.82%	96.67%	51.00%	90%	63.33%	79.33%	WRESTLING	
C73288909	Corso, Margaret	2029	C2D2	11	BEA0	1	C+	75.56%		1511	2000	82.62%	100.00%	54.00%	76%	58.67%	79.17%	DIVING (WOMEN)	
C31977629	Bross, Aiden	2029	A2B2	3	LLS3	2	C+	75.50%		1468	1945	84.34%	97.33%	61.67%	94%	73.67%	71.00%	LACROSSE (MEN)	
C06730312	Fang, Vincent	2029	E1F1	8	PYL2	4	C+	75.47%		1509	2000	85.31%	48.75%	78.33%	90%	61.67%	70.17%		
C74784068	Hansing, Evan	2029	C2D2	6	ORE2	3	C+	75.47%		1483	1965	78.90%	92.00%	79.67%	85%	65.67%	71.17%		
C30158472	Marr, Aedan	2029	A2B2	6	AEN0	4	C+	75.42%		1459	1935	81.95%	34.67%	70.00%	85%	72.33%	80.17%		
C57644778	Perez Montoya, Sophia	2029	A2B2	3	SDS1	2	C+	75.38%		1504	1995	91.59%	96.44%	59.67%	94%	68.33%	71.00%		
C97877536	Mavity, Brenner	2028	E1F1	1	CHM2	2	C+	75.23%		1505	2000	90.79%	71.11%	64.00%	89%	68.33%	68.33%		
C19411165	Demarco, Benjamin	2029	E2F2	11	ECN2	1	C+	75.19%		1504	2000	66.28%	60.56%	66.00%	98%	59.00%	92.00%		
C88047807	Dial, Nadia	2028	A2B2	3	FLP2	3	C+	75.13%		1469	1955	91.59%	89.12%	58.67%	78%	58.00%	78.00%		MD101
C73603718	Madera, David	2029	E1F1	5	PIF0	5	C+	75.06%		1456	1940	95.62%	69.04%	49.67%	98%	70.67%	71.33%		
C75633963	Flores, Alexis	2028	E2F2	6	MGN0	4	C+	75.04%		1486	1980	85.05%	84.87%	74.33%	80%	55.67%	73.50%		
C59766174	Adejokun, Nathan	2029	C2D2	2	EPS1	0	C+	75.03%		1482	1975	87.08%	80.00%	70.67%	83%	59.33%	67.83%	FOOTBALL	ML100 N/C
C61964357	Drucks, Gabrielle	2029	C2D2	6	LLS3	1	C+	74.98%		1485	1980	87.03%	93.33%	73.00%	88%	72.33%	64.50%		
C98989626	Hampton, Braden	2028	A2B2	10	FLF2	0	C+	74.88%		1498	2000	84.89%	87.67%	66.33%	87%	50.33%	76.50%		PH201
C47134256	Kasanganay, Victoria	2028	C1D1	5	LLS3	3	C+	74.84%		1467	1960	94.72%	98.60%	54.33%	89%	46.67%	74.33%		
C44152643	Laszcz, Olivia	2029	C2D2	6	ECN2	0	Olivia	74.74%		1495	2000	88.08%	97.89%	56.33%	82%	62.33%	78.33%		
C41074687	Seo, Aiden	2029	E2F2	11	ORE2	3	C+	74.7											

CID	Name	Year	Hour Section	Major	bsence	Grade	Percent	PTG	Curr Pts	Max Pts	Labs - Grp	Labs - Ind	WPR 1	COLM PLW	WPR 2	TEE	Athletes	Graybook
C91379281	Hennen, Aidan	2029	E2F2 11	LLS3	1	C+	74.61%		1474	1975	63.52%	65.12%	61.67%	80%	70.67%	81.83%	RUGBY (MEN)	
C31853280	Farley, Lawson	2029	C1D1 5	ECN2	1	C+	74.60%		1492	2000	81.37%	95.93%	58.00%	100%	58.67%	69.83%		
C99063682	Hooper, Robert	2029	C1D1 4	CEN2	0	C+	74.50%		1490	2000	71.31%	93.48%	58.33%	81%	79.67%	71.67%		
C09967108	Cramer, Constantine	2029	C1D1 1	WST0	2	C+	74.39%		1488	2000	83.74%	84.14%	46.67%	86%	46.33%	88.50%		
C35730095	Wooton, Matthew	2029	E2F2 2		0	C+	74.36%		1495	2010	92.50%	37.09%	61.67%	85%	74.67%	74.00%		
C04873860	Poynton, Jacob	2029	C1D1 8	ENM1	3	C+	74.29%		1486	2000	85.67%	87.79%	47.00%	98%	72.67%	71.17%	FOOTBALL	ML100 N/C
C51144569	Finke, Nicholas	2028	E2F2 2	GSC0 TST1N	5	C+	74.27%		1493	2010	85.00%	91.04%	73.33%	78%	73.67%	52.83%		
C05229101	Berardi, Nicholas	2029	C1D1 1	MGN0	1	C+	74.25%		1485	2000	68.16%	91.71%	54.67%	65%	67.33%	77.00%	SOCCER (MEN)	ML100 N/C



SECTION III

ASSIGNMENTS



TAB I LAB QUIZ



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB QUIZ HANDOUT

Time of completion _____

Cadet _____

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X1, Physics I
Laboratory Phase Line Writ 1

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- b. Check your exam for 23 problems. Check for the Laboratory Physics Reference Card provided with the examination.
- c. Show all work; partial credit may be given for correct work shown.
- d. Take up to 35 minutes to complete the examination.

2. Exam Security:

- a. I will not discuss this Phase Line Writ with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

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- 2 1. **Match** the step of the scientific method with the description which most accurately describes it.

Observe _____

A. An experimenter reviews current literature and publications to understand what is known about the phenomenon in question.

Research _____

B. An experimenter conducts multiple trials to collect data to test their theory.

Hypothesize _____

C. An experimenter interprets results in this step to determine the outcome of the experiment.

Experiment _____

D. An experimenter predicts what is expected to occur during the experiment.

Analyze _____

E. An experimenter communicates their findings to advance the understanding of the phenomenon.

Report _____

F. An experimenter identifies a phenomenon and creates a research question.

- 4 2. A researcher is attempting to determine the acceleration due to gravity near the surface of the earth (g). To accomplish this they intend to drop a ball from varying heights (h) and measure the time it takes for the ball to hit the ground (t).

The researcher completes a linearized equation as part of their Appendix A in $y = mx + b$ form:

$$t^2 = \frac{2}{g}h$$

Match the quantities below to their definitions.

Dependent Variable _____

A. height (h)

Independent Variable/Quantity _____

B. time (t)

Objective Quantity _____

C. time-squared (t^2)

Dependent Quantity _____

D. acceleration due to gravity (g)

Wt. No.

Cadet _____

Prompt: A Cadet performs a linear regression using the Excel analysis toolpak and the following linear regression data is generated. Use this figure to answer questions 3 and 4. The units of the slope term are in meters [m].

Regression Statistics					
Multiple R	0.998146601				
R Square	0.996296637				
Adjusted R Square	0.992593274				
Standard Error	0.094241205				
Observations	3				
ANOVA					
	df	SS	MS	F	Significance F
Regression	1	2.389318595	2.389318595	269.0248547	0.038765615
Residual	1	0.008881405	0.008881405		
Total	2	2.3982			
	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	-0.083671525	0.195721119	-0.427503814	0.742812305	-2.570544137
X Variable 1	0.437814225	0.026692771	16.40197716	0.038765615	0.098650408

2 3. **Select** which of the following represents the slope of the linear regression based on the data provided (all responses are rounded to three significant figures).

- ☐ 0.998 m
- ☐ 2.39 m
- ☐ -0.0837 m
- ☐ 0.196 m
- ☐ 0.438 m
- ☐ 0.0267 m

2 4. **Select** which of the following represents the absolute uncertainty of the slope for the linear regression based on the data provided (all responses are rounded to one significant figure).

- ☐ 0.09 m
- ☐ 0.009 m
- ☐ 0.08 m
- ☐ 0.2 m
- ☐ 0.4 m
- ☐ 0.03 m

Wt. No.

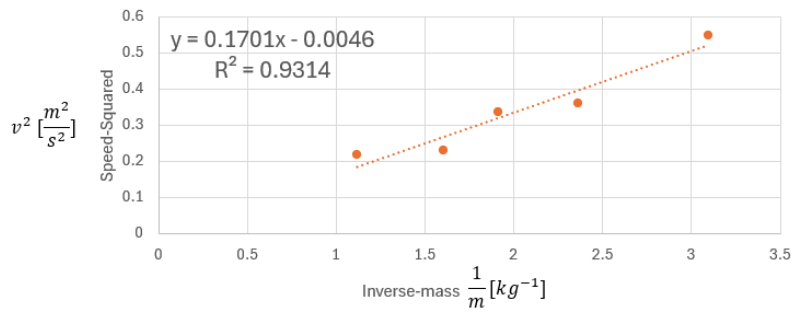
Cadet _____

Prompt: A group of Cadets run an experiment to determine the spring constant of a spring. They do so by varying the mass of the cart and recording the launch speed. Their linearized equation from Appendix A is:

$$v^2 = k x^2 \left(\frac{1}{m} \right)$$

Where v is the launch speed of a pasco cart, k is the spring constant, x is the spring displacement, and m is the mass of the cart. Their data forms the following graph and linear regression. Elements of the graph have been removed for clarity. Use this graph to answer the following questions.

Plot of speed-squared vs inverse mass to determine a spring constant



4 5. **Select** the independent quantity for this experiment.

- ☐ v^2
- ☐ x^2
- ☐ $\frac{1}{m}$
- ☐ kx^2

2 6. **Select** the correct units of the slope of this graph.

- ☐ $\frac{m^2}{s^2}$
- ☐ $\frac{m^2 kg}{s^2}$
- ☐ $\frac{m^2}{s^2 kg}$
- ☐ $\frac{s^2}{m^2 kg}$
- ☐ $\frac{s^2 kg}{m^2}$

Wt. No. Cadet _____

1 7. While working alone to complete a lab notebook, your group member told you to check their work for the propagation of uncertainty calculation for the absolute uncertainty of the objective quantity. **Determine** which appendix of the lab notebook you will review based on this request:

- ☐ Appendix A
- ☐ Appendix B
- ☐ Appendix C
- ☐ Appendix D
- ☐ Appendix E
- ☐ Appendix F

1 8. While working with your group your group member asked you to double check their greatest factor of uncertainty for the height measurement. **Determine** which appendix of the lab notebook you will review based on this request:

- ☐ Appendix A
- ☐ Appendix B
- ☐ Appendix C
- ☐ Appendix E
- ☐ Appendix F

3 9. Which of the following is **NOT** one of the four factors of uncertainty discussed in the PH201 lab program:

- ☐ Nature of the quantity
- ☐ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☐ Conditions under which the measurement was performed
- ☐ The limitations of humans

Wt. No.

Cadet _____

Prompt: A Cadet is measuring the length of an M4 rifle using a tape measure. The rifle is well defined, has a clear beginning at the butt of the rifle and ends at the muzzle. The tape measure is easy to hold along the length of the rifle. The Cadet performs this measurement in the middle of the night while pulling security in the patrol base, it is difficult to read the tape measure under their red lens light. The Cadet makes a best guess that the rifle is 0.84 meters long, however they admit that it could be as short as 0.83 m or as long as 0.85 m. Using this prompt answer questions 9 and 10.

3 10. Determine which of the following factors of uncertainty dominates in this measurement, based on the description provided.

- ☐ Nature of the quantity
- ☐ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☐ Conditions under which the measurement was performed
- ☐ The limitations of humans (it is not this one)

3 11. **Determine** which of the following confidence intervals is most correct based on the description. Consider the numbers reported, the significant figures, the formatting, and the units.

- ☐ $0.84\text{ cm} \pm 0.1$
- ☐ $84\text{ cm} \pm 0.01\text{ m}$
- ☐ $(0.84 \pm 0.01)\text{ m}$
- ☐ $(0.84 \pm 0.010)\text{ m}$

2 12. A Cadet is told to measure the length of rope that is to be used to tie their rappelling harness. They are given a tape measure which can be easily placed along the rope. The ends of the rope are very frayed and it is difficult to determine where they start and stop. Also the rope is very elastic, and if they pull too hard on the rope it can increase by a few centimeters.

Determine which factor of uncertainty dominates in this measurement, based on the description provided.

- ☐ Nature of the quantity
- ☐ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☐ Conditions under which the measurement was performed.
- ☐ The limitations of humans (it is not this one)

Prompt: Use the example Appendix D to answer questions 13-15.

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (App A)	$g = \frac{2 (v_{0x})^2}{slope}$		
	(Obj qty)	= (expression in terms of 'slope')	
Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$	4.1	0.2	m
v_{0x}	4.4	0.1	$\frac{m}{s}$

4 13. **Calculate** the objective quantity, g (acceleration due to gravity), using the data and equation provided in the example appendix D. Report your answer in units of $\frac{m}{s^2}$ to 2 significant figures.

3 14. **Determine** which of the following is a correct equation to calculate the absolute uncertainty of the objective quantity using propagation of uncertainty based on the Appendix D provided.

- ☐ $\delta g = \sqrt{\left(\frac{\partial g}{\partial slope} \delta slope\right)^2}$
- ☐ $\delta g = \sqrt{\left(\frac{\partial g}{\partial v_{0x}} \delta v_{0x}\right)^2}$
- ☐ $\delta g = \sqrt{\left(\frac{\partial g}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial g}{\partial v_{0x}} \delta v_{0x}\right)^2}$
- ☐ $\delta g = \sqrt{\left(\frac{\partial v_{0x}}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial v_{0x}}{\partial v_{0x}} \delta v_{0x}\right)^2}$

Prompt: Use the example Appendix D to answer questions 12-14.

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (App A)	$g = \frac{2 (v_{0x})^2}{slope}$		
	(Obj qty)	= (expression in terms of 'slope')	
Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$	4.1	0.2	m
v_{0x}	4.4	0.1	$\frac{m}{s}$

4 15. **Calculate** the absolute uncertainty of the objective quantity, g (acceleration due to gravity), using propagation of uncertainty and the data provided in the example appendix D. Report your answer in units of and to the appropriate number of significant figures.

Prompt: Use the example Appendix E to answer questions 16 and 17.

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity

170
(experimental objective quantity)

±

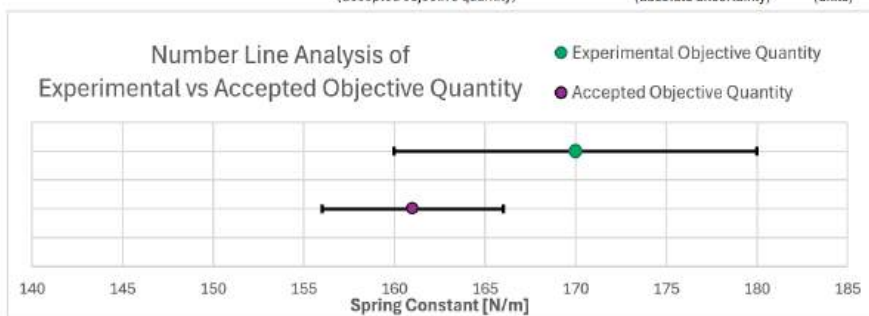
10	$\frac{N}{m}$
(absolute uncertainty)	(units)

Accepted Objective Quantity

161
(accepted objective quantity)

±

5	$\frac{N}{m}$
(absolute uncertainty)	(units)



3 16. **Determine** the dominant form of experimental error in the number line provided:

- ☐ Systematic
- ☐ Random
- ☐ Limiting
- ☐ Relative

2 17. **Calculate** the relative uncertainty for the experimentally determined objective quantity based on the values provided. Report your answer as a percent to 2 significant figures.

2 18. Five groups of Cadets conducted an experiment to determine the acceleration due to gravity. Their instructor compiled their percent differences into the table below. **Select** which group had the **most accurate** experimental result based on the provided data.

- ☐ Group A
- ☐ Group B
- ☐ Group C
- ☐ Group D
- ☐ Group E

Group Name	Reported Percent Difference
Group A	5.6 %
Group B	6.7 %
Group C	2.3 %
Group D	4.6 %
Group E	16 %

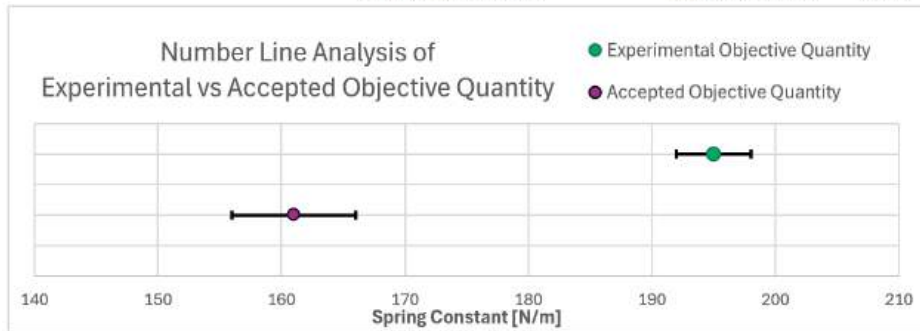
Wt. No.

Cadet _____

Prompt: Use the example Appendix E to answer questions 19 and 20.

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	195 (experimental objective quantity)	$\pm$	3 (absolute uncertainty)	$\frac{N}{m}$ (units)
Accepted Objective Quantity	161 (accepted objective quantity)	$\pm$	5 (absolute uncertainty)	$\frac{N}{m}$ (units)



3 19. **Determine** the dominant form of experimental error in the number line provided:

- ☐ Systematic
- ☐ Random
- ☐ Limiting
- ☐ Relative

2 20. **Calculate** the percent difference for the experimentally determined objective quantity based on the values provided. Report your answer as a percent to 2 significant figures.

3 21. A certain tape measure has tick marks to the nearest 0.001 m. Determine the absolute uncertainty for this tape measure due to the limitations of the measuring device based on the information provided. Report your absolute uncertainty (in meters) to the appropriate number of significant digits:

3 22. A Cadet is attempting to experimentally determine the spring constant (k) of a spring on a PASCO Smart Cart. To accomplish this, they design an experiment to vary the mass of the cart (m) [kg] and measure the launch speed of the cart (v) [m/s] for a constant spring displacement (x) [m] (control variable). They derive the following linearized equation:

$$v^2 = kx^2 \frac{1}{m}$$

Determine which of the following represents the objective quantity equation for this linearized equation.

☐ $k = v^2 (\text{slope})$

☐ $k = \frac{\text{slope}}{x^2}$

☐ $k = \frac{x^2}{\text{slope}}$

☐ $k = \frac{\text{slope}}{m}$

2 23. A Cadet conducts an experiment and determines the following linear regression equation fits their data:

$$y = \left(0.209 \left[\frac{s^2}{m} \right] \right) x + 0$$

Calculate the objective quantity (g) given their appendix A, objective quantity equation is given as:

$$g = \frac{2}{\text{slope}}$$

Report your answer in units of acceleration ($\frac{m}{s^2}$) to 3 significant figures.



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB QUIZ SOLUTION

Time of completion _____

Cadet _____

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
PH2X1, Physics I
Laboratory Phase Line Writ 1

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. No phones, cameras, or smartwatches are allowed in the examination room.
- b. Check your exam for 21 problems. Check for the Laboratory Physics Reference Card provided with the examination.
- c. Show all work; partial credit may be given for correct work shown.
- d. Take up to 35 minutes to complete the examination.

2. Exam Security:

- a. I will not discuss this Phase Line Writ with anyone until it is released from academic security on _____ (date) at _____ (time).
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE	PROBLEM	WEIGHT	SCORE
1	3		12	4	
2	4		13	3	
3	2		14	4	
4	2		15	3	
5	4		16	2	
6	2		17	3	
7	3		18	3	
8	3		19	2	
9	3		20	3	
10	3		21	2	
11	2				
				Total	/60

_____ %

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Wt. No.

Cadet _____

- 2 1. **Match** the step of the scientific method with the description which most accurately describes it.

+0.5
per
correct

Observe __F__

A. An experimenter reviews current literature and publications to understand what is known about the phenomenon in question.

Research __A__

B. An experimenter conducts multiple trials to collect data to test their theory.

Hypothesize __D__

C. An experimenter interprets results in this step to determine the outcome of the experiment.

Experiment __B__

D. An experimenter predicts what is expected to occur during the experiment.

Analyze __C__

E. An experimenter communicates their findings to advance the understanding of the phenomenon.

Report __E__

F. An experimenter identifies a phenomenon and creates a research question.

- 4 2. A researcher is attempting to determine the acceleration due to gravity near the surface of the earth (g). To accomplish this they intend to drop a ball from varying heights (h) and measure the time it takes for the ball to hit the ground (t).

The researcher completes a linearized equation as part of their Appendix A in $y = mx + b$ form:

$$t^2 = \frac{2}{g}h$$

+1 per
correct

Match the quantities below to their definitions.

Dependent Variable __B__

A. height (h)

Independent Variable/Quantity __A__

B. time (t)

Objective Quantity __D__

C. time-squared (t^2)

Dependent Quantity __C__

D. acceleration due to gravity (g)

Wt. No.

Cadet _____

Prompt: A Cadet performs a linear regression using the Excel analysis toolpak and the following linear regression data is generated. Use this figure to answer the associated questions. The units of the slope term are in meters [m].

Regression Statistics					
Multiple R	0.998146601				
R Square	0.996296637				
Adjusted R Square	0.992593274				
Standard Error	0.094241205				
Observations	3				
ANOVA					
	df	SS	MS	F	Significance F
Regression	1	2.389318595	2.389318595	269.0248547	0.038765615
Residual	1	0.008881405	0.008881405		
Total	2	2.3982			
	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	-0.083671525	0.195721119	-0.427503814	0.742812305	-2.570544137
X Variable 1	0.437814225	0.026692771	16.40197716	0.038765615	0.098650408

2 3. **Select** which of the following represents the slope of the linear regression based on the data provided (all responses are rounded to three significant figures).

- ☐ 0.998 m
☐ 2.39 m
☐ -0.0837 m
☐ 0.196 m
☒ 0.438 m
☐ 0.0267 m

2 4. **Select** which of the following represents the absolute uncertainty of the slope for the linear regression based on the data provided (all responses are rounded to one significant figure).

- ☐ 0.09 m
☐ 0.009 m
☐ 0.08 m
☐ 0.2 m
☐ 0.4 m
☒ 0.03 m

Wt. No.

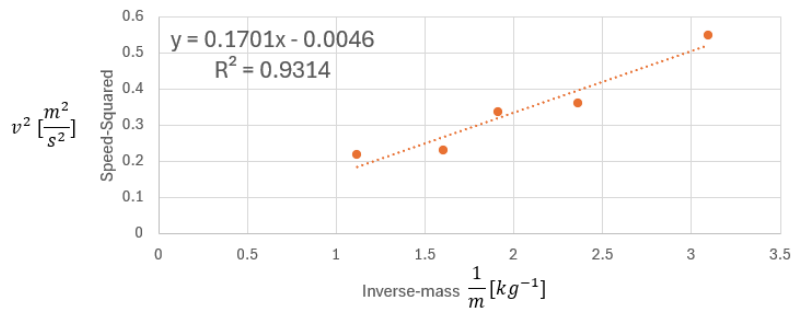
Cadet _____

Prompt: A group of Cadets run an experiment to determine the spring constant of a spring. They do so by varying the mass of the cart and recording the launch speed. Their linearized equation from Appendix A is:

$$v^2 = k x^2 \left(\frac{1}{m} \right)$$

Where v is the launch speed of a pasco cart, k is the spring constant, x is the spring displacement, and m is the mass of the cart. Their data forms the following graph and linear regression. Elements of the graph have been removed for clarity. Use this graph to answer the following questions.

Plot of speed-squared vs inverse mass to determine a spring constant



4 5. **Select** the independent quantity for this experiment.

- ☐ v^2 +1
- ☐ x^2
- ☒ $\frac{1}{m}$ +4
- ☐ kx^2

2 6. **Select** the correct units of the slope of this graph.

- ☐ $\frac{m^2}{s^2}$
- ☒ $\frac{m^2 kg}{s^2}$ +2
- ☐ $\frac{m^2}{s^2 kg}$
- ☐ $\frac{s^2}{m^2 kg}$ +1
- ☐ $\frac{s^2 kg}{m^2}$

Wt. No. Cadet _____

1 7. While working alone to complete a lab notebook, your group member told you to check their work for the propagation of uncertainty calculation for the absolute uncertainty of the objective quantity. **Determine** which appendix of the lab notebook you will review based on this request:

- ☐ Appendix A
- ☐ Appendix B
- ☐ Appendix C
- ☒ Appendix D
- ☐ Appendix E
- ☐ Appendix F

1 8. While working with your group your group member asked you to double check their greatest factor of uncertainty for the height measurement. **Determine** which appendix of the lab notebook you will review based on this request:

- ☐ Appendix A
- ☒ Appendix B
- ☐ Appendix C
- ☐ Appendix E
- ☐ Appendix F

3 9. Which of the following is **NOT** one of the four factors of uncertainty discussed in the PH201 lab program:

- ☐ Nature of the quantity
- ☒ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☐ Conditions under which the measurement was performed
- ☐ The limitations of humans

Wt. No.

Cadet _____

Prompt: A Cadet is measuring the length of an M4 rifle using a tape measure. The rifle is well defined, has a clear beginning at the butt of the rifle and ends at the muzzle. The tape measure is easy to hold along the length of the rifle. The Cadet performs this measurement in the middle of the night while pulling security in the patrol base, it is difficult to read the tape measure under their red lens light. The Cadet makes a best guess that the rifle is 0.84 meters long, however they admit that it could be as short as 0.83 m or as long as 0.85 m. Using this prompt answer questions 9 and 10.

3 10. Determine which of the following factors of uncertainty dominates in this measurement, based on the description provided.

- ☐ Nature of the quantity
- ☐ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☒ Conditions under which the measurement was performed +3
- ☐ The limitations of humans (it is not this one) +1

3 11. **Determine** which of the following confidence intervals is most correct based on the description. Consider the numbers reported, the significant figures, the formatting, and the units.

- ☐ $0.84\text{ cm} \pm 0.1$
- ☐ $84\text{ cm} \pm 0.01\text{ m}$
- ☒ $(0.84 \pm 0.01)\text{ m}$
- ☐ $(0.84 \pm 0.010)\text{ m}$

2 12. A Cadet is told to measure the length of rope that is to be used to tie their rappelling harness. They are given a tape measure which can be easily placed along the rope. The ends of the rope are very frayed and it is difficult to determine where they start and stop. Also the rope is very elastic, and if they pull too hard on the rope it can increase by a few centimeters.

Determine which factor of uncertainty dominates in this measurement, based on the description provided.

- ☒ Nature of the quantity
- ☐ Uncertainty in the precision of the measurement
- ☐ Quality of the measuring device
- ☐ Conditions under which the measurement was performed.
- ☐ The limitations of humans (it is not this one)

Prompt: Use the example Appendix D to answer questions 12-14.

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (App A)	$g = \frac{2 (v_{0x})^2}{slope}$		
	(Obj qty)	= (expression in terms of 'slope')	
Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope	4.1	0.2	m
v_{0x}	4.4	0.1	$\frac{m}{s}$

4 13. **Calculate** the objective quantity, g (acceleration due to gravity), using the data and equation provided in the example appendix D. Report your answer in units of $\frac{m}{s^2}$ to 2 significant figures.

$g = 9.4$ (within
2%), SF and units
not graded

3 14. **Determine** which of the following is a correct equation to calculate the absolute uncertainty of the objective quantity using propagation of uncertainty based on the Appendix D provided.

- ☐ $\delta g = \sqrt{\left(\frac{\partial g}{\partial slope} \delta slope\right)^2}$
- ☐ $\delta g = \sqrt{\left(\frac{\partial g}{\partial v_{0x}} \delta v_{0x}\right)^2}$
- ☒ $\delta g = \sqrt{\left(\frac{\partial g}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial g}{\partial v_{0x}} \delta v_{0x}\right)^2} +3$
- ☐ $\delta g = \sqrt{\left(\frac{\partial v_{0x}}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial v_{0x}}{\partial v_{0x}} \delta v_{0x}\right)^2} +1$

Wt. No.

Cadet _____

Prompt: Use the example Appendix D to answer questions 12-14.

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (App A)	$g = \frac{2 (v_{0x})^2}{slope}$		
	(Obj qty)	= (expression in terms of 'slope')	
Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$	4.1	0.2	m
v_{0x}	4.4	0.1	$\frac{m}{s}$

4. 15. **Calculate** the absolute uncertainty of the objective quantity, g (acceleration due to gravity), using propagation of uncertainty and the data provided in the example appendix D. Report your answer in units of and to the appropriate number of significant figures.

$dg = 0.6$ (within
2%), SF and units
not graded

Wt. No.

Cadet _____

Prompt: Use the example Appendix E to answer questions 15-16.

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity

170
(experimental objective quantity)

±

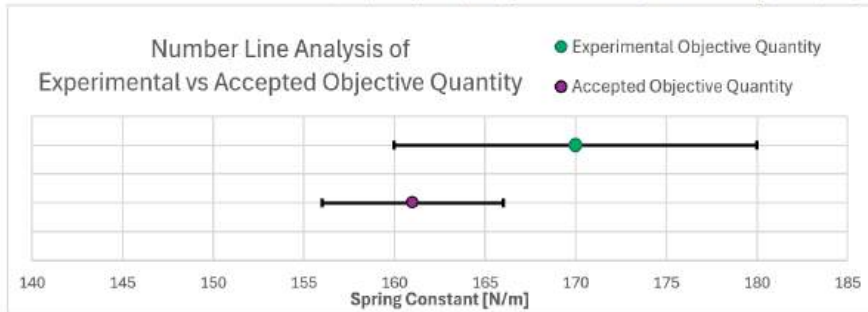
10	$\frac{N}{m}$
(absolute uncertainty)	(units)

Accepted Objective Quantity

161
(accepted objective quantity)

±

5	$\frac{N}{m}$
(absolute uncertainty)	(units)



3 16. **Determine** the dominant form of experimental error in the number line provided:

- ☐ Systematic
- ☒ Random
- ☐ Limiting
- ☐ Relative

2 17. **Calculate** the relative uncertainty for the experimentally determined objective quantity based on the values provided. Report your answer as a percent to 2 significant figures.

Ru= 5.9 (within 2%), SF and units not graded

3 18. Five groups of Cadets conducted an experiment to determine the acceleration due to gravity. Their instructor compiled their percent differences into the table below. **Select** which group had the **most accurate** experimental result based on the provided data.

- ☐ Group A
- ☐ Group B
- ☒ Group C
- ☐ Group D
- ☐ Group E

Group Name	Reported Percent Difference
Group A	5.6 %
Group B	6.7 %
Group C	2.3 %
Group D	4.6 %
Group E	16 %

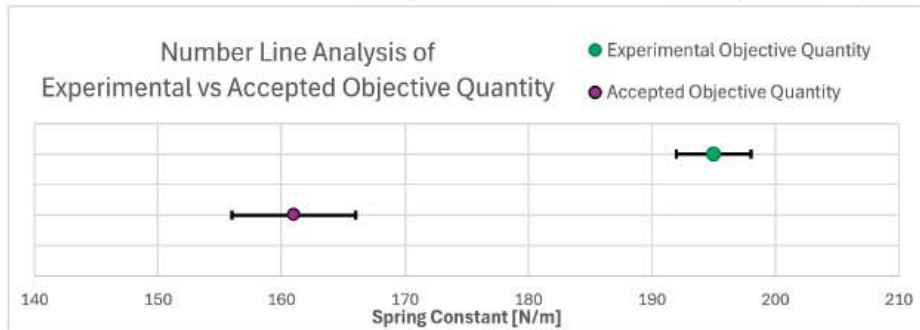
Wt. No.

Cadet _____

Prompt: Use the example Appendix E to answer questions 18-19.

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	195 (experimental objective quantity)	$\pm$	3 (absolute uncertainty)	$\frac{N}{m}$ (units)
Accepted Objective Quantity	161 (accepted objective quantity)	$\pm$	5 (absolute uncertainty)	$\frac{N}{m}$ (units)



3 19. **Determine** the dominant form of experimental error in the number line provided:

- ☒ Systematic
☐ Random
☐ Limiting
☐ Relative

2 20. **Calculate** the percent difference for the experimentally determined objective quantity based on the values provided. Report your answer as a percent to 2 significant figures.

PD = 21% (within 2%), SF and units not graded

Wt. No.

Cadet _____

3 21. A certain tape measure has tick marks to the nearest 0.001 m. Determine the absolute uncertainty for this tape measure due to the limitations of the measuring device based on the information provided. Report your absolute uncertainty (in meters) to the appropriate number of significant digits:

0.0005 m

3 22. A Cadet is attempting to experimentally determine the spring constant (k) of a spring on a PASCO Smart Cart. To accomplish this, they design an experiment to vary the mass of the cart (m) [kg] and measure the launch speed of the cart (v) [m/s] for a constant spring displacement (x) [m] (control variable). They derive the following linearized equation:

$$v^2 = kx^2 \frac{1}{m}$$

Determine which of the following represents the objective quantity equation for this linearized equation.

☐ $k = v^2 (\text{slope})$

☒ $k = \frac{\text{slope}}{x^2} +3$

☐ $k = \frac{x^2}{\text{slope}} +1$

☐ $k = \frac{\text{slope}}{m}$

2 23. A Cadet conducts an experiment and determines the following linear regression equation fits their data:

$$y = \left(0.209 \left[\frac{s^2}{m} \right] \right) x + 0$$

Calculate the objective quantity (g) given their appendix A, objective quantity equation is given as:

$$g = \frac{2}{\text{slope}}$$

G=9.57 (within 2%), SF and units not graded

Report your answer in units of acceleration ($\frac{m}{s^2}$) to 3 significant figures.



TAB J

COLM QUIZ



UNITED STATES MILITARY ACADEMY
WEST POINT.

COLM HANDOUT

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Block IV Phase Line Writ

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Check your exam for 5 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- Take up to 35 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **20 March at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	5	
2	20	
3	25	
4	45	
5	5	

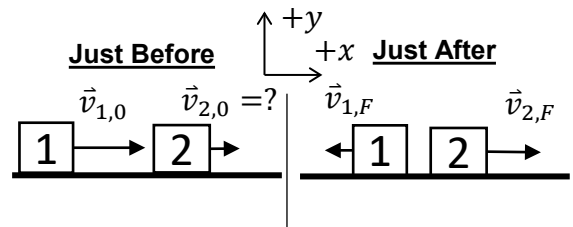
Total:	100	
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Wt. **No.**

Cadet: _____

- 5 1. A tank has a mass of 5.4×10^4 kg. The magnitude of the tank's linear momentum is 8.7×10^5 kg-m/s. Calculate the tank's speed.
- ☐ 16 m/s
- ☐ 0.062 m/s
- ☐ 5.7 m/s
- ☐ 25 m/s
-
- 20 2. Two carts move along a straight horizontal track where friction is negligible. Cart 1 has a mass of 2.65 kg and moves at a speed of 3.32 m/s to the right. Cart 2 has a mass of 4.89 kg moves to the right at an unknown speed. Cart 1 collides with Cart 2. After the collision, Cart 1 is moving to the left at 0.354 m/s, and Cart 2 moves to the right at 2.26 m/s. Calculate the initial velocity of Cart 2. Report your answer in unit vector notation.



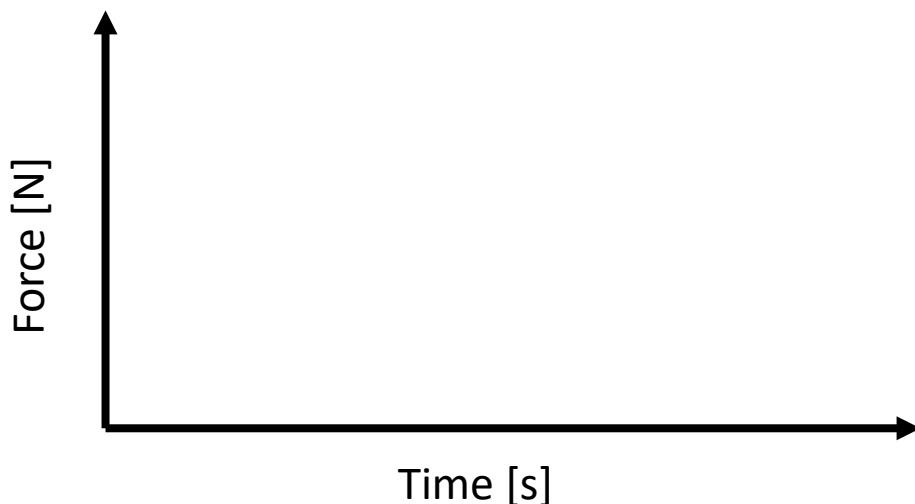
Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

Wt. **No.**

Cadet: _____

- 20 3a. A passenger in a car experiences a head-on collision. The passenger is moving with a speed of 22 m/s and eventually comes rest.
Explain how an airbag will make this collision safer for the passenger.
You may use words, equations, and diagrams in your explanation.
Your explanation **must** include relevant physics concepts covered in this course.
Solutions must be legible or no credit will be awarded.

- 5 3b. On the axes below, sketch the force as a function of time that acts on the passenger for a collision with an airbag and without an airbag (two different curves). Label which curves represent the collision with and without an airbag.



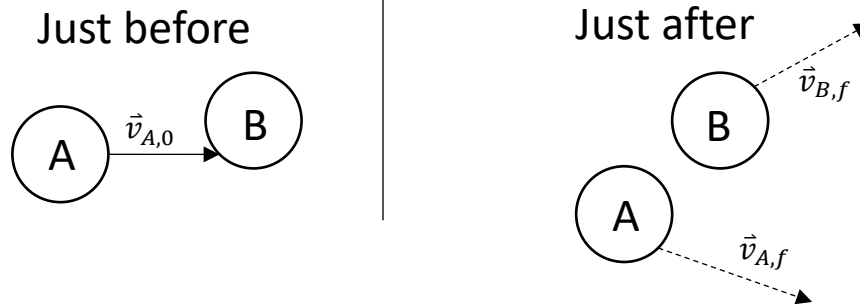
Wt. **No.**

Cadet: _____

45

4.

A cadet is playing pool. The cadet strikes Ball A, with a mass of 0.396 kg, at stationary Ball B, which has a mass of 0.485 kg. Ball A has an initial velocity of $(12.4 \hat{i} + 0.00 \hat{j})$ m/s. Ball A then collides with Ball B, and after the collision Ball A has a velocity of $(2.16 \hat{i} - 5.91 \hat{j})$ m/s. Calculate the final velocity of ball B just after the collision. Report your answer in unit-vector notation.



Report your answer in this box.	Report units in this box.
---------------------------------	---------------------------

5

5.

Select which statement is true for a completely inelastic collision in a closed and isolated system.

- ☐ Both linear momentum and kinetic energy are conserved
- ☐ Linear momentum is conserved, but kinetic energy is not conserved.
- ☐ Kinetic energy is conserved, but linear momentum is not conserved.
- ☐ None of the above.



UNITED STATES MILITARY ACADEMY
WEST POINT.

COLM SOLUTION

Time of completion _____ Cadet _____
 11 March 2026 (First) (Last)

Section _____
 Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
 PH2X1, Physics I
 Block IV Phase Line Writ

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Check your exam for 5 problems. Check for the Physics Reference Card provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- Take up to 35 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **20 March at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

PROBLEM	WEIGHT	SCORE
1	5	
2	20	
3	25	
4	45	
5	5	

Total:	100	
---------------	-----	--

A: 10 (10%)

B: 27 (27%)

D: 63 (63%)

Total: 100

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Wt. No.

Cadet: _____

- 5 1. A tank has a mass of 5.4×10^4 kg. The magnitude of the tank's linear momentum is 8.7×10^5 kg-m/s. Calculate the tank's speed.

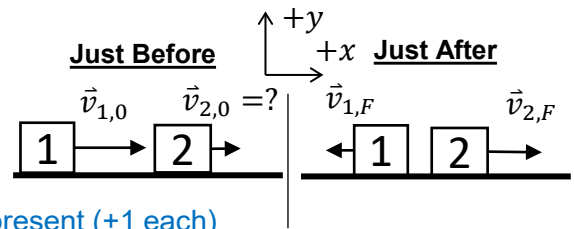
- 5** ● 16 m/s *pv*
☐ 0.062 m/s
☐ 5.7 m/s
☐ 25 m/s

- 20 2. Two carts move along a straight horizontal track where friction is negligible. Cart 1 has a mass of 2.65 kg and moves at a speed of 3.32 m/s to the right. Cart 2 has a mass of 4.89 kg moves to the right at an unknown speed. Cart 1 collides with Cart 2. After the collision, Cart 1 is moving to the left at 0.354 m/s, and Cart 2 moves to the right at 2.26 m/s. Calculate the initial velocity of Cart 2. Report your answer in unit vector notation.

6 COLM $\Sigma \vec{P}_0 = \Sigma \vec{P}_F$ **4** $\vec{p} = m\vec{v}$
 $m_1 v_{10x} + m_2 v_{20x} = m_1 v_{1fx} + m_2 v_{2fx}$

$v_{20x} = \frac{-m_1 v_{10x} + m_2 v_{20x} + m_1 v_{1fx}}{m_2}$ **4** Terms present (+1 each)

3 Attempts to solve for v_{20x}



2 ANS w/ 3 SF

1 Units

$v_{20x} = 0.269 \hat{i}$ Report your answer in this box.	m/s Report units in this box.
---	---

A: 2 (8%)
 B: 7 (28%)
 D: 16 (64%)
 Total: 25

Wt. No.

Cadet: _____

- 20 3a. A passenger in a car experiences a head-on collision. The passenger is moving with a speed of 22 m/s and eventually comes rest. Explain how an airbag will make this collision safer for the passenger. You may use words, equations, and diagrams in your explanation. Your explanation **must** include relevant physics concepts covered in this course. Solutions must be legible or no credit will be awarded.

Overarching Concepts

- Impulse/Momentum Theorem
- Work/KE also allowed

Equations

- $39 + 40/41$
- $32 + 27 + 33/34$

Terms indicating good understanding

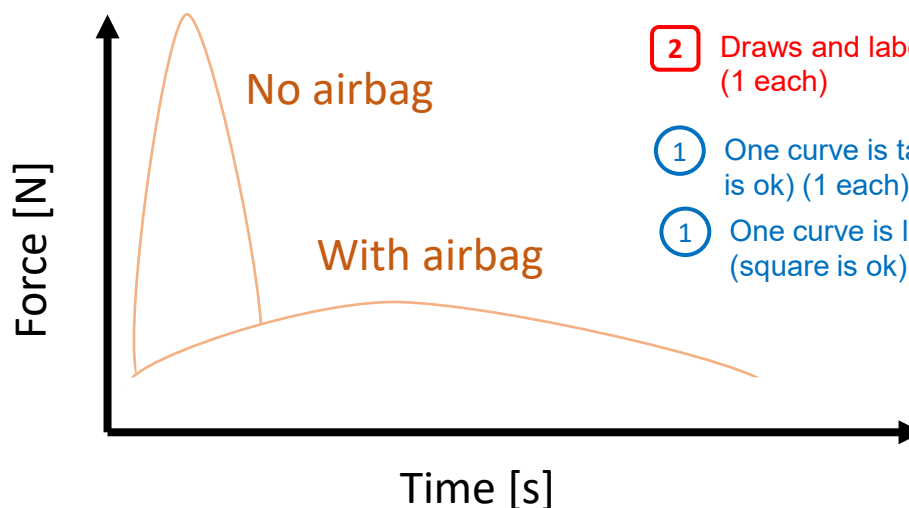
- Less average force
- Longer time interval/displacement
- Same impulse / Δp

Terms indicating limited understanding

- Harder surface

- ② Answer is correct and well-justified with proper physics
- ③ Answer is correct and attempts to justify with relevant physics, but still indicates a small misunderstanding
- ⑤ Answer is generally correct, but does not include relevant physics or shows limited understanding
- ⑥ Wrong answer, but uses PRC equations or physics to attempt to justify
- ④ Wrong answer, attempts no justification
- ⑦ Blank or Illegible

- 5 3b. On the axes below, sketch the force as a function of time that acts on the passenger for a collision with an airbag and without an airbag (two different curves). Label which curves represent the collision with and without an airbag.



- ② Draws and labels two separate graphs (1 each)

- ① One curve is taller than the other (square is ok) (1 each)

- ① One curve is longer than the other (square is ok) (1 each)

- ① Completely correct

A: 3 (12%)
B: 5 (20%)
D: 17 (68%)
Total: 25

Wt. No.

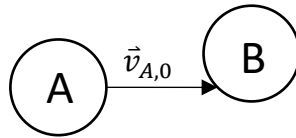
Cadet: _____

45

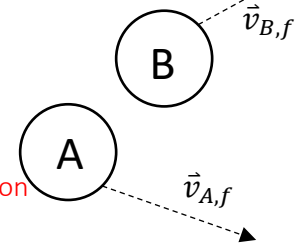
4.

A cadet is playing pool. The cadet strikes Ball A, with a mass of 0.396 kg, at stationary Ball B, which has a mass of 0.485 kg. Ball A has an initial velocity of $(12.4 \hat{i} + 0.00 \hat{j})$ m/s. Ball A then collides with Ball B, and after the collision Ball A has a velocity of $(2.16 \hat{i} - 5.91 \hat{j})$ m/s. Calculate the final velocity of ball B just after the collision. Report your answer in unit-vector notation.

Just before



Just after



12 COLM

6 $\vec{p} = m\vec{v}$

4 Applies COLM to X directions

4 Applies COLM to Y direction

$$\Sigma \vec{P}_0 = \Sigma \vec{P}_f$$

X-dim

$$m_a v_{a0x} = (m_a v_{A,Fx}) + m_b v_{b,Fx}$$

$$v_{bfx} = \frac{(m_a) v_{A,Fx} - m_a v_{a0x}}{m_b}$$

Y-dim

$$0 = (-m_a v_{A,Fy}) + m_b v_{b,Fy}$$

$$v_{bfy} = \frac{-(m_a) v_{AB,Fy}}{m_b}$$

2 ANS X

2 ANS Y

Present terms – x dim

1 Initial A

1 Final A

1 Final B

Present terms – y dim

2 Initial 0

1 Final A

2 Final A

4 Attempts to solve for final velocity B

1 Unit-vector notation applied

1 SF

1 Units

$$\vec{v}_{Bf} = 8.36 \hat{i} + 4.83 \hat{j}$$

Report your answer in this box.

m/s

Report units in this box.

5

5. Select which statement is true for a completely inelastic collision in a closed and isolated system.

1 ☐ Both linear momentum and kinetic energy are conserved

1 **3** **1** ☒ Linear momentum is conserved, but kinetic energy is not conserved.

☐ Kinetic energy is conserved, but linear momentum is not conserved.

☐ None of the above.

A: 5 (10%)
B: 15 (30%)
D: 30 (60%)
Total: 50



TAB K

LAB



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB#1 HANDOUT

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

Note: Items in orange are placeholders that you must update.

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

Type equation here.

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV) [X on graph]

independent variable

IV

units

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV) [Y on graph]

dependent variable

DV

units

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)(Control Variable)(units)

control variable 2

CV2

units

(Control Variable Name)(Control Variable)(units)

control variable 3

CV3

units

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

type equation here

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

(y)

(Dependent Qty)

=

m

(m)

(slope)

IQ

(x)

(Independent Qty)

+

(b)

4b. Rewrite the slope term below (which includes your objective quantity):

slope =

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty) = (expression in terms of 'slope')

5. Write a hypothesis statement.

As we (increase/decrease) the (independent quantity), we expect to observe a (n) (linear/exponential/logarithmic/quadratic) (increase/decrease) in the (dependent quantity) based on (Physics Principle).

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.

6d. Calculate objective quantity and uncertainty using appendix D.

6e. Plot experimental results on a number line to analyze precision and accuracy of findings.

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment designed based on one dimensional kinematics. This experiment involves dropping a ball from varying heights (independent variable) and measuring the time for the ball to strike the ground (dependent variable). Then you will determine the acceleration due to gravity using a linearized model.

Objective Quantity:

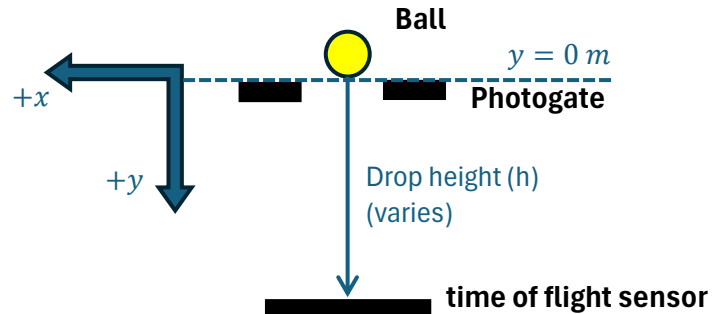
acceleration due to gravity, g

Varying (independent variable):

Drop height (h)

Measuring (dependent variable):

Time for ball to hit the ground (t)

Diagram:

Note: Items in orange are placeholders that you must update.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)

(OQ variable)

(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

Type equation here.

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV) [X on graph]

independent variable

IV

units

(Independent Variable Name)

(Indendent Variable)

(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV) [Y on graph]

dependent variable

DV

units

(Dependent Variable Name)

(Dependent Variable)

(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)

(Control Variable)

(units)

control variable 2

CV2

units

(Control Variable Name)

(Control Variable)

(units)

control variable 3

CV3

units

(Control Variable Name)

(Control Variable)

(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

type equation here

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope =

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expression in terms of slope

(Obj qty)

=

(expression in terms of 'slope')

5. Write a hypothesis statement.

As we

(increase/decrease)

 the

(independent quantity)

, we expect to observe a

(n)

(linear/exponential/logarithmic/quadratic)

(increase/decrease)

 in the

(dependent quantity)

 based on

(Physics Principle)

.

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6b. Complete appendix B detailing measurement procedures and uncertainties.

6d. Calculate objective quantity and uncertainty using appendix D.

Note: Items in orange are placeholders that you must update.

Note: Items in orange are placeholders that you must update.

Note: Items in orange are placeholders that you must update.

Appendix B: Measured quantity

B1. Physical quantity measured:

Height of ball (at photogate sensor)	h		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	$units$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	$units$
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		$units$
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Time of flight	t		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	$units$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	$units$
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		$units$
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
------------------------------------	-----------------------------	---	-----------------

(Objective Quantity Name)

(OQ variable)

(OQ units)

Objective quantity equation (From App A)	<i>OQ Eqn from App A</i>
---	--------------------------

(Obj qty)

=

(expression in terms of 'slope')

Variable from equation above	Measured Value	Absolute Uncertainty	Units
<i>slope</i>			<i>units</i>

Calculate the objective quantity using the variables and equation above:

List OQ equation with variables here

List OQ equation with values here

OQ

=

use excel to calculate here

units

(Obj qty)

(value)

(units)

Calculate the confidence interval for the objective quantity:

Note: Items in orange are placeholders that you must update.

Variable from equation above	Associated Partial Derivative	Quadrature
<i>slope</i>	$\frac{\partial OQ}{\partial slope} = \text{FILL}$	
0	Type equation here	

NOT COMPLETED THIS LAB

δOQ

=

use excel to calculate here

units

(Obj qty, links from above)

=

(value)

(units)

Report the objective quantity as a confidence interval
--

	$\pm$	
(objective quantity)		(absolute uncertainty) (units)

Discussion Questions

1. Write a short paragraph which describes a.) the goal of the experiment and b.) a brief explanation of how your group conducted the experiment. Use the space below:

2. Describe the difference between these four terms:

a.) Independent variable

b.) Dependent variable

c.) Independent quantity

3. The accepted value for the acceleration due to gravity is 9.807 m/s^2 .

a.) Describe how close your result came to the accepted value.

b.) If you had to do this experiment different what would you change?



UNITED STATES MILITARY ACADEMY
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LAB#1 SOLUTION

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

The goal of this experiment is to determine the acceleration due to gravity near the surface of the Earth.

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Acceleration due to gravity

g

m/s²

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2$

2b. Apply equations from 2a into a model equation:

$y = y_0 + v_{0y}t + \frac{1}{2}gt^2$ where: $v_{0y} = 0 \text{ m/s}$
 $y = y_0 + \frac{1}{2}gt^2$
 $y - y_0 = \frac{1}{2}gt^2$ where: $y - y_0 = h$
 $h = \frac{1}{2}gt^2$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

height of drop

h

m

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable, which we will measure: (Dependent variable, DV)

time to floor

t

s

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

N/A

None

(Control Variable Name)(Control Variable)(units)

N/A

None

(Control Variable Name)(Control Variable)(units)

N/A

None

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$h = \frac{1}{2}gt^2$
 $\frac{1}{2}gt^2 = h$
 $t^2 = \frac{2}{g}h$
 $t^2 = \frac{2}{g}h$
 $(y) = (m)(x) + (b)$
(Dependent Qty)(slope)(Independent Qty)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

4b. Rewrite the slope term below (which includes your objective quantity):

slope = $\frac{2}{g}$

4c. Rearrange and solve for your objective quantity in terms of slope:

$g = \frac{2}{slope}$
(Obj qty) = (expression in terms of 'slope')

5. Experiment planning and execution

5a. Review appendix A and design your experiment around your chosen IV and DV

5b. Complete appendix B detailing measurement procedures and uncertainties.

5c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.

5d. Calculate objective quantity and uncertainty using appendix D.

5e. Plot experimental results on a number line to analyze precision and accuracy of findings.

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment designed based on one dimensional kinematics. This experiment involves dropping a ball from varying heights (independent variable) and measured the time for the ball to strike the ground (dependent variable). Then by using a linearized model the acceleration due to gravity can be calculated.

The purpose of this lab is to introduce you to lab procedures and the lab notebook. This experiment can cause high variability in the final calculation for the acceleration due to gravity. As we progress through the first three labs we will improve our methods for calculating the acceleration due to gravity.

Objective Quantity:

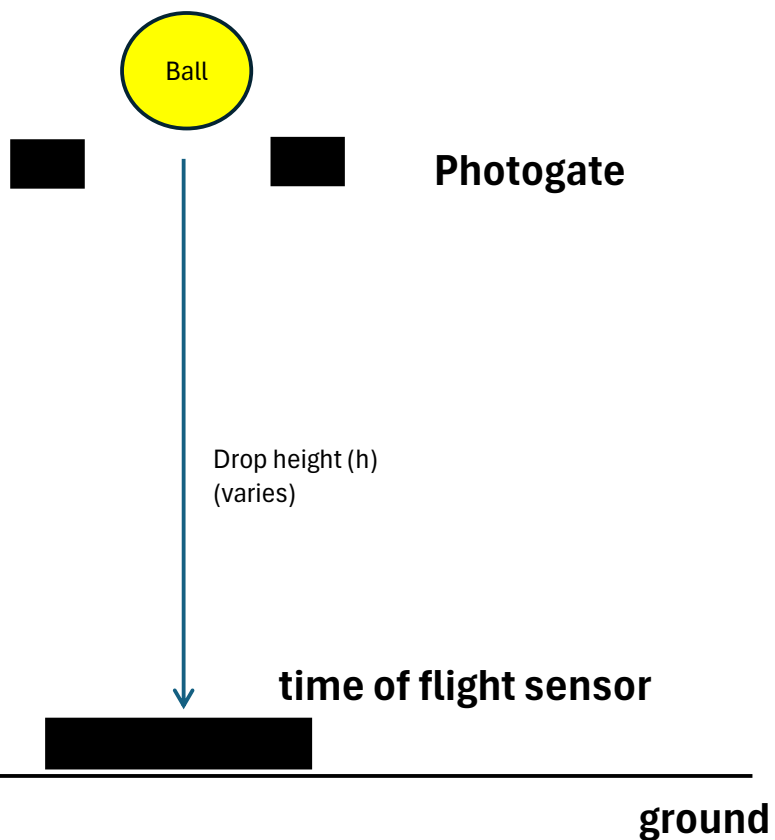
acceleration due to gravity, g

Varying (independent variable):

Drop height (h)

Measuring (dependent variable):

Time for ball to hit the ground (t)

Diagram:

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

The goal of this experiment is to determine the acceleration due to gravity near the surface of the Earth.

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Acceleration due to gravity

g

2s

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2$

2b. Apply equations from 2a into a model equation:

$y = y_0 + v_{0y}t + \frac{1}{2}gt^2$ where: $v_{0y} = 0\text{ m/s}$
 $y = y_0 + \frac{1}{2}gt^2$
 $y - y_0 = \frac{1}{2}gt^2$ where: $y - y_0 = h$
 $h = \frac{1}{2}gt^2$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

height of drop

h

m

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable, which we will measure: (Dependent variable, DV)

time to floor

t

s

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

N/A

None

(Control Variable Name)(Control Variable)(units)

N/A

None

(Control Variable Name)(Control Variable)(units)

N/A

None

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$h = \frac{1}{2}gt^2$
 $\frac{1}{2}gt^2 = h$
 $t^2 = \frac{2}{g}h$
 $t^2 = \frac{2}{g}h$
 $(y) = (m)(x) + (b)$
(Dependent Qty)(slope)(Independent Qty)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

4b. Rewrite the slope term below (which includes your objective quantity):

$slope = \frac{2}{g}$

4c. Rearrange and solve for your objective quantity in terms of slope:

$g = \frac{2}{slope}$
(Obj qty) = (expression in terms of 'slope')

5. Experiment planning and execution

5a. Review appendix A and design your experiment around your chosen IV and DV

5b. Complete appendix B detailing measurement procedures and uncertainties.

5c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.

5d. Calculate objective quantity and uncertainty using appendix D.

5e. Plot experimental results on a number line to analyze precision and accuracy of findings.

Appendix B: Measured quantity

B1. Physical quantity measured:

Height of ball (at photogate sensor)	<i>h</i>	Measuring Tape	☒ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

1.101	<i>m</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Clearly describes height measurement. Includes detailed instructions that are repeatable.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	The size of the photogate is about 2 mm, so we can define this measuring point to within +/- 1 mm.	0.001 m
Quality of the measuring device	The tape measure has a resolution to millimeters. So the uncertainty is 1mm/2 = +/-0.0005 m.	0.0005 m
Conditions under which the measurement was performed	The tape measure is difficult to hold steady and can vary within about +/- 2 mm.	0.002 m
Limitations of humans	The tape measure is easy to read and does not vary based on a human's measurement abilities. The ball may drop above the photogate but the ball time always starts from the photogate height.	0 m

B5. Select the largest of the four factors:

0.002	<i>m</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

0.18%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

1.101	$\pm$	0.002	<i>m</i>
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☒ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Time of flight	t	PASCO Time of Flight Sensor	☒ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.350	s	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Clearly describes time of flight. Includes detailed instructions that are repeatable. Should mention using PASCO and how they used it.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty
Nature of the quantity	Time of flight is clearly defined each time.	0s
Quality of the measuring device	PASCO reports to many significant figures, but we will assume the default 2 SF gives a reasonably accurate value.	0.005 s
Conditions under which the measurement was performed	There are no gusts, winds, or other condition factors that would add variance to the time of flight between trials.	0 s
Limitations of humans	No human limitaions impact this measurement.	0 s

B5. Select the largest of the four factors:

0.005	s
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

1.4%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

0.350	$\pm$	0.005	s
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

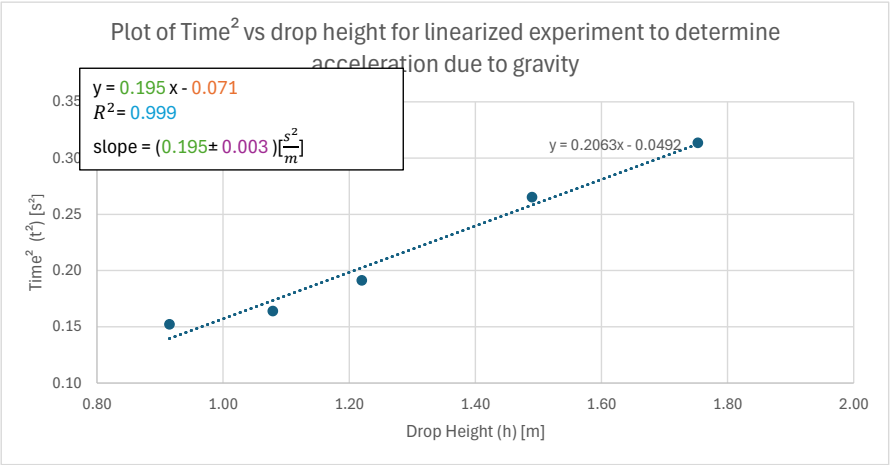
☒ Yes
☐ No

Appendix C: Experimental Results and Linear Regression

Control Variable	Value	Units

	Independent Variable	IV Absolute Uncertainty	Independent Quantity	IQ Absolute Uncertainty	Dependent Variable	DV Absolute Uncertainty	Dependent Quantity	DQ Absolute Uncertainty
Trial number	Height (h) [meter]	δIV	Height (h) [meter]	δQ	Time (t) [seconds]	δDV	Time^2 (t^2) [s^2]	δDV
1	0.92		0.92		0.39		0.15	
2	1.08		1.08		0.405		0.16	
3	1.22		1.22		0.44		0.19	
4	1.49		1.49		0.52		0.27	
5	1.75		1.75		0.56		0.31	
6								
7								
8								
9								
10								

Plot



Linear Regression results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.989016
R Square	0.978152
Adjusted R Square	0.97087
Standard Error	0.011868
Observations	5

	df	SS	MS	F	Significance F
Regression	1	0.018917	0.018917	134.3133	0.00138
Residual	3	0.000423	0.000141		
Total	4	0.019339			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-0.0492	0.023594	-2.08528	0.128354	-0.12429	0.025887	-0.12429	0.025887
X Variable 1	0.206317	0.017802	11.58936	0.00138	0.149662	0.262971	0.149662	0.262971

0.20631662

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity
(Links from App A)

Acceleration due to gravity

g

$\frac{m}{s^2}$

(Objective Quantity Name)

(OQ variable)

(OQ units)

Objective quantity equation
(App A)

$g = \frac{2}{slope}$

(Obj qty)

=

(expression in terms of 'slope')

Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$	0.206	0.003	$\frac{s^2}{m}$

Calculate the objective quantity using the variables and equation above: (You can use excel to calculate OQ)

$g = \frac{2}{0.206}$

g

=

9.7

$\frac{m}{s^2}$

(Obj qty)

(value)

(units)

Calculate the combined absolute uncertainty of the objective quantity: (You can use excel to final abs. unc)

Type δOQ equation with variables here

Variable from equation above	AssociatedPartial Derivative	Quadrature
$slope$	$\frac{\partial OQ}{\partial slope} = FILL$	
0	Type equation here	

NOT COMPLETED THIS LAB

δOQ

=

use excel to calculate here

$units$

(Obj qty, links from above)

=

(value)

(units)

Report the objective quantity as a confidence interval

$\pm$

(objective quantity)

(absolute uncertainty)

(units)

Discussion Questions

1. Write a short paragraph which describes a.) the goal of the experiment and b.) a brief explanation of how your group conducted the experiment. Use the space below:

Answer should include:

- States the goal (determine acceleration due to gravity)
- Describes how conducted (IV, DV, linearized model, what was changed / measured)

0	Empty
0.6	Minimal
0.7	Marginal
1.0	Maximal

2. Describe the difference between these four terms:

- a.) Independent variable and dependent variable
b.) Independent quantity and dependent quantity.

Answer should indicate some of the following:

- IV: variable that is changed by the experimenter
- DV: variable that depends on changes in the IV
- IQ: Manipulation of independent variable
- DQ: Manipulation of dependent variable
- Compares which axis they are plotted on (when in quantity form)
- May use examples from this lab.

0	Empty
0.6	Minimal
0.7	Marginal
1.0	Maximal

3. The accepted value for the acceleration due to gravity is 9.807 m/s^2 .

a.) Describe how close your result came to the accepted value.

b.) If you used a stop watch instead of a PASCO Time of Flight Sensor would your results have been more accurate, less accurate, or the same? Justify your response.

Answer should include:

- Qualitative description (close, far, way-off etc.) or quantitative (percent difference)
- Stop watch is less accurate due to human error (The reading does not include info on this so answers should vary) This is designed to get them thinking about error (covered in Lab 2)

Rubric

Apply the points that best fit each Group's Answer relative to the expected responses and your expertise as an instructor.

0.0-0.6

Minimal:

- Incomplete answer
- Poor explanation
- Does not demonstrate understanding

0.61-

Marginal:

- Generally complete answer

0	Empty
0.6	Minimal
0.8	Marginal
1.0	Maximal

0.8	- Attempts reasoning with an explanation - Marginal understanding with some inaccuracies
0.81-1.0	Maximal: - Complete repsonse - Well reasoned explanation - Demonstrates understanding

1. Complete the appendices for your experiment
2. For multiple Appendix B's right click the sheet and click 'move or copy', select create a copy and put it in this world
3. To compile your report click:
 - File
 - Save as adobe pdf
 - select all appropriate sheets
 - Review the pdf to ensure all required information is present.
4. If you need to adjust pdf spacing use the 'page layout' tab in Excel

kbook.



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LAB#1 ONLINE POST LAB

1



Ordering 1 point Scientific Method Ordering

Order the correct sequence of the steps of the scientific method. Order from first to last in chronological order.

1

 Observe

2

 Research

3

 Hypothesize

4

 Experiment

5

 Analyze

6

 Report

2



Matching 1 point Linearization Terms

Match the key term with the definition that most closely matches.

Objective Quantity

The physical constant that tl 

Independent Variable

The variable within the expe 

Dependent Variable

The variable within the expe 

Independent Quantity

Typically plotted on the x-ax 

Dependent Quantity

Typically plotted on the y-ax 

Control Variable

Variables that do not change 

Possible answers

⋮ Typically plotted on the y-axis of the Appendix C graph; a mathematical manipulation of the corresponding variable

⋮ The physical constant that the experiment is attempting to determine.

⋮ The variable within the experiment that is adjusted directly by the experimenter.

⋮ Variables that do not change from trial to trial over the course of an experiment.

⋮ Typically plotted on the x-axis of the Appendix C graph; a mathematical manipulation of the corresponding variable

⋮ A method of experiment design which simplifies complex relationships between variables.

⋮ The variable within the experiment that is measured as a result of changes made by the experimenter.

3



Multiple Choice 1 point The Lab Notebook

Select the lab notebook appendix which outlines the experiment and defines independent, dependent, and control variables based on a model equation:

- ☒ Appendix A
- ☐ Appendix B
- ☐ Appendix C
- ☐ Appendix D
- ☐ Appendix E
- ☐ Appendix F

4



Matching 1 point Applying Linearization terms

An experiment is designed to determine the acceleration due to gravity based on varying the height a ball is launched from and measuring the distance the ball travels for a fixed launch velocity. The following model equation is derived:

$$d^2 = \frac{-2(v_0)^2}{g} h$$

where d is the distance a projectile travels, v_0 is the launch velocity, g is the acceleration due to gravity, and h is the height the ball is launched from.

Match the variables to the correct terms for this linearized model:

v_0	Control Variable
d	dependent variable
h	independent variable / independent quantity
d^2	Dependent quantity

Possible answers

⋮ Dependent quantity

⋮ dependent variable

⋮ independent variable / independent quantity

⋮ Control Variable

5



Multiple Choice 1 point Determining the objective quantity equation

An experiment is designed to determine the acceleration due to gravity based on varying the height a ball is launched from and measuring the distance the ball travels for a fixed launch velocity. The following model equation is derived:

$$d^2 = \frac{-2(v_0)^2}{g} h$$

where d is the distance a projectile travels, v_0 is the launch velocity, g is the acceleration due to gravity, and h is the height the ball is launched from.

Select which of the following is the correct equation for the objective quantity in terms of **slope**.

☐

$$g = \frac{-2(v_0)^2}{d^2} h$$

☐

$$g = \frac{-2(v_0)^2}{slope} h$$

☒

$$g = \frac{-2(v_0)^2}{slope}$$

☐

$$g = \frac{2}{slope}$$

6



Multiple Choice 0 points Documentation and Acknowledgement of Assistance

Take a moment to REFLECT on your efforts when completing this assignment.
Did you receive any assistance in completing this assignment?



No: I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.



Yes: I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

7



Essay 0 points Document Assistance and Sources

Take a moment to REFLECT on your efforts when completing this assignment.
Did you receive any assistance in completing this assignment?

Copy and paste which response you choose. If you chose 'Yes' then use the space below to acknowledge assistance and document sources. Reference the question number in your documentation.

NO: I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

YES: I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

Edit View Insert Format Tools Table

12pt Paragraph | **B** *I* U **A** ▼ ▼ T^2 ▼ |

▼ ▼ ▼ ▼ | ▼ | ⋮





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LAB#2 HANDOUT

Overview:

Calculate the acceleration due to gravity (objective quantity) and analyze sources of uncertainty (error) in measured and calculated quantities. In this experiment you will fire a projectile from varying heights (independent variable) and measure the distance the projectile travels (dependent variable) for a fixed launch velocity (control variable).

Objective Quantity:

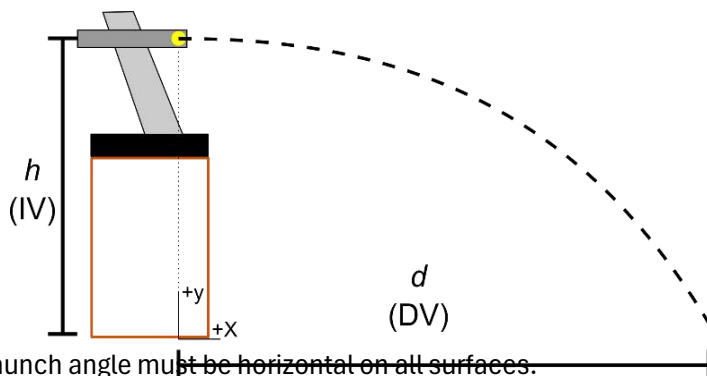
Acceleration due to gravity, g [m/s^2]

Varying (independent variable):

Launch Height, h

Measuring (dependent variable):

Launch distance to floor, d

Diagram:**Grading Rubric (15 Points)**

Deliverable	Description	Points
Cover Sheet, Notebook Formatting, and Writing Style	Formatting, well written, and presentable	2
Appendix A	Linearization Table	1
Appendix B	Height/Distance/Initial Velocity	3
Appendix C	Data, Plot, and Regression	3
Appendix D	Objective Quantity Calculation	1
Appendix D	Propagation of Uncertainty	2
Discussion Questions	Question 1	1
Discussion Questions	Question 2	1
Discussion Questions	Question 3	1

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

The goal for this experiment is to determine the acceleration due to gravity near the surface of the Earth using a projectile launcher.

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Acceleration due to gravity

g

m/s^2

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$
 $x_0 = 0$
 $x = d$
 $v_{0x} = v_{0x}$
 $v_f = v_{0x}$
 $a_x = 0$

$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$
 $y_0 = h$
 $y = 0$
 $v_{0y} = 0$
 $v_f = ?$
 $a_y = -g$

2b. Apply equations from 2a into a model equation:

$x = v_{0x}t$
 $t = \frac{x}{v_{0x}}$

$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$
 $y - y_0 = -\frac{1}{2}g t^2$
 $y - y_0 = -\frac{1}{2}g \left(\frac{x}{v_{0x}}\right)^2$
where:
 $y - y_0 = -h$
 $x = d$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

varunits

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

varunits

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:

varunits

(Control Variable Name)(Dependent Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$d^2 = \frac{2(v_{0x})^2}{g} h$

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQmIQ

(y) = (m)(x) + (b)

(Dependent Qty)(slope)(Independent Qty)

Note: Items in orange are placeholders that you must update.

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

g = (expr in terms of slope)

(Obj qty, links from above)(expression in terms of 'slope')

5. Write a hypothesis statement.

As we (increase/decrease) the (independent quantity), we expect to observe a(n) (linear/exponential/logarithmic/quadratic) (increase/decrease) in the (dependent quantity) based on (Physics Principle).

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

5b. Complete appendix B detailing measurement procedures and uncertainties.

6d. Calculate objective quantity and uncertainty using appendix D.

Note: Items in orange are placeholders that you must update.

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (Links from App A)	acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)

Objective quantity equation (App A)	$OQ\ Eqn$
	(Obj qty) = (expression in terms of 'slope')

Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$			$units$
var			$units$

Calculate the objective quantity using the variables and equation above: (You can use excel to calculate OQ)

$OQ\ Eqn$			
g	=	#DIV/0!	$units$
(Obj qty, links from above)		(value)	(units)

Note: Items i
that you mus

Calculate the combined absolute uncertainty of the objective quantity: (You can use excel to final abs. unc)

$$\delta g = \sqrt{\left(\frac{\partial g}{\partial slope} \delta(slope)\right)^2 + \left(\frac{\partial g}{\partial var} \delta(var)\right)^2}$$

Variable from equation above	AssociatedPartial Derivative	Quadrature (partial derivate * abs unc)^2
$slope$	$\frac{\partial g}{\partial slope} = ?$	
var	$\frac{\partial OQ}{\partial (var)} = ?$	

δg	=	0.0	$units$
(Obj qty, links from above)	=	(value)	(units)

Report the objective quantity as a confidence interval

#DIV/0!	$\pm$	0.0	$units$
(objective quantity)		(absolute uncertainty)	(units)

1. To complete this lab we made a few assumptions about our setup that were not explicitly stated in the theory section. List at least two assumptions and describe how they would impact the experiment if they were not valid assumptions.

2. Further describe the factor of uncertainty that dominated in your distance measurement. Explain how you could reduce that uncertainty in a future iteration of this experiment.

3. The accepted value for the acceleration due to gravity is $9.807 \frac{m}{s^2}$.

Is this accepted value within the confidence interval of your results?

In Appendix D which variable contributed most to your objective quantity's absolute uncertainty?

Recommend one change to your experiment that could reduce the uncertainty in that measurement.



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LAB#2 SOLUTION

1. To complete this lab we made a few assumptions about our setup that were not explicitly stated in this section. List at least two assumptions and describe how they would impact the experiment if they were not made.

- Some possibilities: drag/air resistance, constant launch velocity, level launch angle,
- List an assumption and provides a reasonable explanation about impacts to the experiment.
- For example: Drag would cause the ball to travel a shorter distance than our kinematics model predicts.

0	Empty
0.6	Minimal
0.8	Marginal
1.0	Maximal

2. Further describe the factor of uncertainty that dominated in your distance measurement. Explain how you could reduce that uncertainty in a future iteration of this experiment.

- They should talk about the difficulty in knowing where the ball landed, the distance the tape measure was extended, etc.
- In that case they should describe equipment they could use to record the distance better (graph paper, a laser, a camera, etc.)

0	Empty
0.6	Minimal
0.8	Marginal
1.0	Maximal

3. The accepted value for the acceleration due to gravity is $9.807 \frac{m}{s^2}$.

Is this accepted value within the confidence interval of your results?

- The provided solution does cover the acceleration due to gravity.
- Cadets may or may not, assess if they do or not.

In Appendix D which variable contributed most to your objective quantity's absolute uncertainty?

- In the final uncertainty calculation the greatest quadrature component should be the initial velocity.

Recommend one change to your experiment that could reduce the uncertainty in that measurement.

- If they choose this they should describe an experimental adjustment to consider the initial velocity as an independent variable (described on page 4 of the lab worksheet)

0	Empty
0.6	Minimal
0.8	Marginal
1.0	Maximal

Rubric

Apply the points that best fit each Group's Answer relative to the expected responses and your expertise as an instructor.

0-0.6	Minimal: - Incomplete answer - Poor explanation - Does not demonstrate understanding
0.6-0.8	Marginal: - Generally complete answer - Attempts reasoning with an explanation - Marginal understanding with some inaccuracies
0.8-1.0	Maximal: - Complete response - Well reasoned explanation - Demonstrates understanding



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LAB# ONLINE POST LAB

1



Matching 1 point Factors of Uncertainty

Match the factor of uncertainty to the description or example given.

Quality of the measuring device

Using calipers would improv ☐

Nature of the quantity

Minor variations in a measur ☐

Conditions under which the measurement was performed

Taking a measurement in a lab ☐

Limitations of humans

Human reaction time is an ex ☐

Possible answers

⋮ Using calipers would improve this factor of uncertainty compared to using a meter stick.

⋮ Taking a measurement in a lab improves this factor of uncertainty compared to driving on a bumpy road

⋮ Minor variations in a measurement lead to inherent difficulties in being able to sample uniformly.

⋮ Human reaction time is an example of this factor of uncertainty.

2



Multiple Choice 0.5 points Absolute Uncertainty

Determine which of the following symbols is used to identify '**absolute uncertainty**' in this course.

☐ ∂ ☒ δ ☐ Δ

3



Multiple Choice 1 point Identifying the dominant factor of uncertainty

A group of Cadets measure the height of the highest point on the library using a tape measure. They estimate that the height does not naturally fluctuate and so the uncertainty due to the nature of the quantity is minimal. Their tape measure has a resolution to 1 cm and estimate the absolute uncertainty due to the quality of their measuring device to be 0.5 cm. It is a windy day outside and their tape measure is blowing a bit in the wind. Based on the amount of deflection they estimate the absolute uncertainty due to the conditions under which the measurement was performed to be 2 cm. The group does not identify any uncertainty associated with the limitations of humans and assume that is minimal.

Determine which factor of uncertainty contributes the most to the uncertainty (or error) in their measurement.

☐ Nature of the quantity☐ Quality of the measuring device☒ Conditions under which the measurement was performed☐ The limitations of humans

4

Multiple Choice0.5 pointsConfidence Intervals

A group of cadets measure the height of a soda can to be 0.1230 m with an absolute uncertainty due to the quality of the measuring device of 0.5 mm. Determine which of the following is a correctly formatted confidence interval for the height of the soda can:

- ☐ .1230m ± .5mm
- ☒ 0.1230 ± 0.0005m
- ☐ 0.123 ± 0.005m
- ☐ 0.1230 + 0.0005m

Stimulus

Propagation of Uncertainty

In Lab 1 we used the following objective quantity equation in terms of slope:

$$g = \frac{2}{slope}$$

Assume your Appendix D slope value: (Shown in the figure below:

$$slope = (0.195 \pm 0.003) \frac{s^2}{m}$$

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty			
Objective quantity (Links from App A)	Acceleration due to gravity	g	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (App A)	$g = \frac{2}{slope}$		
	(Obiety)	(incresation in terms of 'slope')	
Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope	0.195	0.003	$\frac{s^2}{m}$

5

Num...1 p...Calculate Objective Q...

Calculate the acceleration due to gravity using the provided values.

10.3

6

Num...1 po...Propagation of Uncer...

Calculate the absolute uncertainty in the objective quantity calculation above using the values provided in the sample appendix D and propagation of uncertainty techniques.

0.2

7



Essay 0 points Document Assistance and Sources

Take a moment to REFLECT on your efforts when completing this assignment.

Did you receive any assistance in completing this assignment?

Copy and paste which response you choose. If you chose 'Yes' then use the space below to acknowledge assistance and document sources. Reference the question number in your documentation.

NO: I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

YES: I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

Edit View Insert Format Tools Table

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UNITED STATES MILITARY ACADEMY
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LAB#3 HANDOUT

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment based on Newton's Laws. For this experiment you will setup a two mass system with a pulley. Mass one (m_1 , PASCO Smart cart) is on a frictionless track which is connected to a hanging mass (m_2) over an ideal (massless) pulley. You will vary the mass of the PASCO smart cart (independent variable) and measure the acceleration of the system (dependent variable) to determine the acceleration due to gravity (objective quantity).

Objective Quantity:

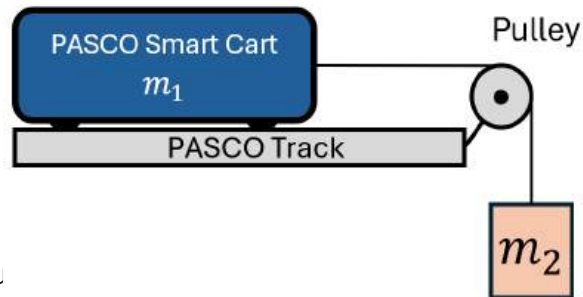
Acceleration due to gravity, g [m/s^2]

Varying (independent variable):

Mass of PASCO Smart Cart

Measuring (dependent variable):

Acceleration of the system

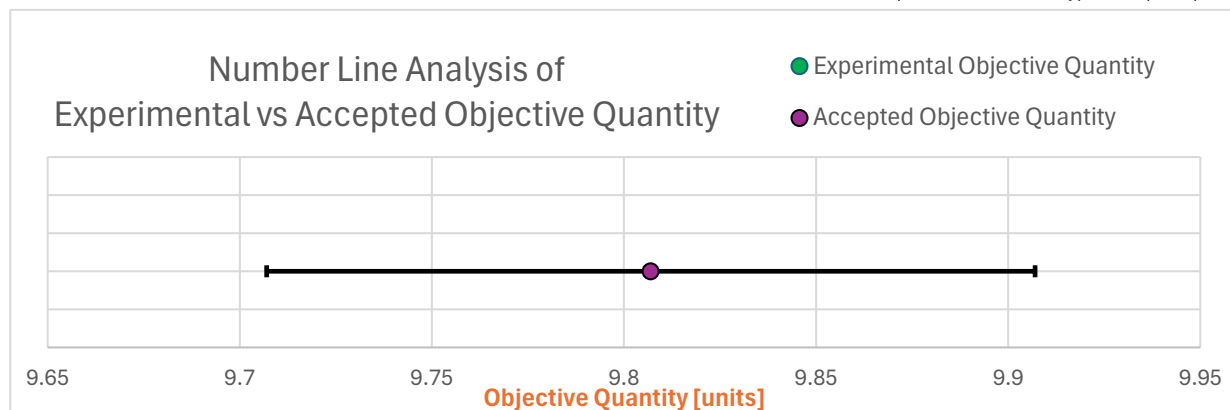
Diagram:**Pre-Lab**

1. Read the sections included in the lab manual.
2. In your own words, describe the difference between systematic and random error.
3. Using your results from **Lab 2** (if you haven't completed Lab 2 then use your Lab 1 results and assume your objective quantity uncertainty was 0.5 m/s^2).
 - 3a. Complete an Appendix E using your results for the objective quantity calculation.
 - 3b. Complete an Appendix F to analyze the error.
4. Complete an Appendix A in preparation for this lab following the experiment overview below and the lab worksheet.

In your own words, describe the difference between systematic and random error.

Appendix E: Precision and Accuracy Table FROM LAB 2 RESULTS

Experimental Objective Quantity		$\pm$		units
	(experimental objective quantity)		(absolute uncertainty)	(units)
Accepted Objective Quantity	9.8	$\pm$	0.1	$\frac{m}{s^2}$
	(accepted objective quantity)		(absolute uncertainty)	(units)



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level
of Precision: 1.0 %

Acceptable level of
Accuracy: 5.0 %

Calculate Relative Uncertainty

$RU =$

RU = 99%

Calculate Percent Difference

$PD =$

PD = 99%

Appendix F: Error Analysis Table FROM LAB 2 RESULTS

Move the **check** to the dominant form of error and follow that flow chart: 

Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

var

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$\Sigma F_x = m a_x$

2b. Apply equations from 2a into a model equation:

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

independent variable

IV

units

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

dependent variable

DV

units

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)(Control Variable)(units)

control variable 2

CV2

units

(Control Variable Name)(Control Variable)(units)

control variable 3

CV3

units

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty) = (expression in terms of 'slope')

5. Write a hypothesis statement.

As we (increase/decrease) the (independent quantity), we expect to observe a(n) (linear/exponential/logarithmic/quadratic) (increase/decrease) in the (dependent quantity) based on (Physics Principle).

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6b. Complete appendix B detailing measurement procedures and uncertainties.

6c. Calculate objective quantity and uncertainty using appendix D.

6d. Plot experimental results on a number line to analyze precision and accuracy of findings.

6e. Evaluate and report error found in step 6d.

Note: Items in orange are placeholders that you must update.

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment designed based on Newton's Laws. In this experiment you will setup a two mass system with a pulley. Mass one (m_1 , PASCO Smart cart) is on a frictionless track which is connected to a hanging mass (m_2) over an ideal pulley. You will vary the mass of the PASCO smart cart (independent variable) and measure the acceleration of the system (dependent variable) to determine the acceleration due to gravity (objective quantity).

Objective Quantity:

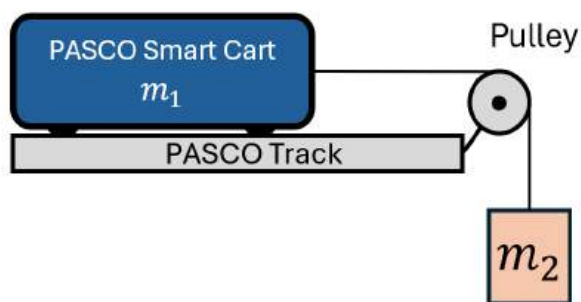
Acceleration due to gravity, g [m/s^2]

Varying (independent variable):

Mass of PASCO Smart Cart

Measuring (dependent variable):

Acceleration of the system

Diagram:**Grading Rubric (35 Points)**

Deliverable	Description	Points
Cover Sheet, Notebook Formatting, and Writing Style	Formatting, well written, and presentable	2
Appendix A	Linearization Table	2
Appendix B	mass 1, mass 2, acceleration	6
Appendix C	Data, Plot, and Regression	6
Appendix D	Objective Quantity Calculation	3
Appendix D	Propagation of Uncertainty	4
Appendix E	Number Line	4
Appendix F	Error Analysis	2
Discussion Questions	Question 1	2
Discussion Questions	Question 2	2
Discussion Questions	Question 3	2

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$\Sigma F_x = m a_x$

2b. Apply equations from 2a into a model equation:

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

independent variable

IV

units

(Independent Variable Name)(Indendent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

dependent variable

DV

units

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)(Control Variable)(units)

control variable 2

CV2

units

(Control Variable Name)(Control Variable)(units)

control variable 3

CV3

units

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty) = (expression in terms of 'slope')

5. Write a hypothesis statement.

As we

(increase/decrease)

 the

(independent quantity)

, we expect to observe a

(n)

(linear/exponential/logarithmic/quadratic)

(increase/decrease)

 in the

(dependent quantity)

 based on

(Physics Principle)

.

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6b. Complete appendix B detailing measurement procedures and uncertainties.

6c. Calculate objective quantity and uncertainty using appendix D.

6d. Plot experimental results on a number line to analyze precision and accuracy of findings.

6e. Evaluate and report error found in step 6d.

Note: Items in orange are placeholders that you must update.

Note: Items in orange are placeholders that you must update.

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass 2 - Hanging Mass	h		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	$units$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	$units$
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		$units$
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass 1 (Smart Cart)	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	units	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	units	
(absolute uncertainty, 1 significant figure)	(units)	

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	±		units	
(measured value)		(absolute uncertainty, 1 SF)	(units)	

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Acceleration of System	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent:

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(measured value)		(absolute uncertainty, 1 SF)	(units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	acceleration due to gravity	g	$\frac{m}{s^2}$
------------------------------------	-----------------------------	-----	-----------------

(Objective Quantity Name)

(OQ variable)

(OQ units)

Objective quantity equation (From App A)	$OQ = \text{expr in terms of slope (From App A)}$
---	---

(Obj qty)

=

(expression in terms of 'slope')

Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope			units
var			

Calculate the objective quantity using the variables and equation above:

List OQ equation with variables here

List OQ equation with values here

OQ

=

use excel to calculate here

units

(Obj qty, links from above)

(value)

(units)

Calculate the combined absolute uncertainty of the objective quantity:

$$\delta f(x, \text{slope}) = \sqrt{\left(\frac{\partial f}{\partial \text{slope}} \delta \text{slope}\right)^2 + \left(\frac{\partial f}{\partial x} \delta x\right)^2} \text{ [EXAMPLE]}$$

Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2
slope	$\frac{\partial OQ}{\partial \text{slope}} = \text{FILL}$	
var	Type equation here.	
var	Type equation here.	
var	Type equation here.	

δOQ

=

use excel to calculate here

units

(Obj qty, links from above)

=

(value)

(units)

Report the objective quantity as a confidence interval

OQ =

use excel to calculate here

±

use excel to calculate here

units

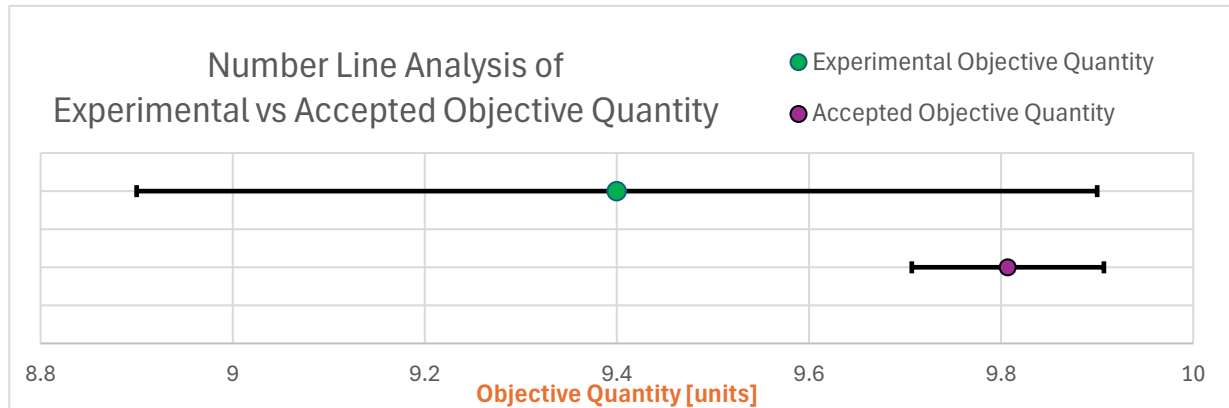
(objective quantity)

(absolute uncertainty)

(units)

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	9.4	$\pm$	0.5	units
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	9.8	$\pm$	0.1	$\frac{m}{s^2}$
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level
of Precision: 1.0 %

Acceptable level of
Accuracy: 5.0 %

Calculate Relative Uncertainty

$RU =$

RU = 99%

Calculate Percent Difference

$PD =$

PD = 99%

Appendix F: Error Analysis Table	
Move the check to the dominant form of error and follow that flow chart: 	
	Random Error Dominates
R1. Identify term in Appendix D with highest quadrature term:	<i>var</i>
R2. Discuss ways to reduce the uncertainty in that variable in the space below:	
<i>Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.</i>	
<i>Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B</i>	
	Systematic Error Dominates
S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.	
S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.	
S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.	
S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.	

1. Discuss the precision of your measurements in Lab 3 compared to the measurements made in Lab 2. Comment on which experiment had 'better' procedures to determine the acceleration due to gravity.

2. Is your experimental value higher or lower than the accepted value? Discuss elements of your Appendix F and how those may have contributed to the results being higher or lower.

3. If you were to conduct this experiment again, describe one way you could reduce the dominant form of error from your experiment.



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LAB#3 SOLUTION

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment designed based on Newton's Laws. In this experiment you will setup a two mass system with a pulley. Mass one (m_1 , PASCO Smart cart) is on a frictionless track which is connected to a hanging mass (m_2) over an ideal pulley. You will vary the mass of the PASCO smart cart (independent variable) and measure the acceleration of the system (dependent variable) to determine the acceleration due to gravity (objective quantity).

Objective Quantity:

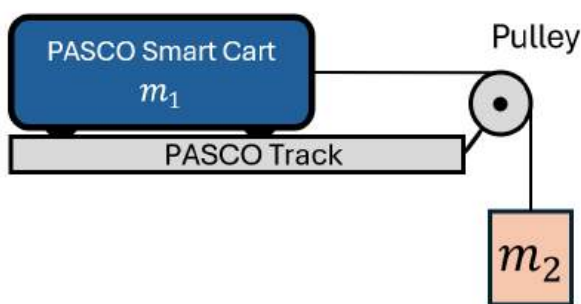
Acceleration due to gravity, g [m/s^2]

Varying (independent variable):

Mass of PASCO Smart Cart

Measuring (dependent variable):

Acceleration of the system

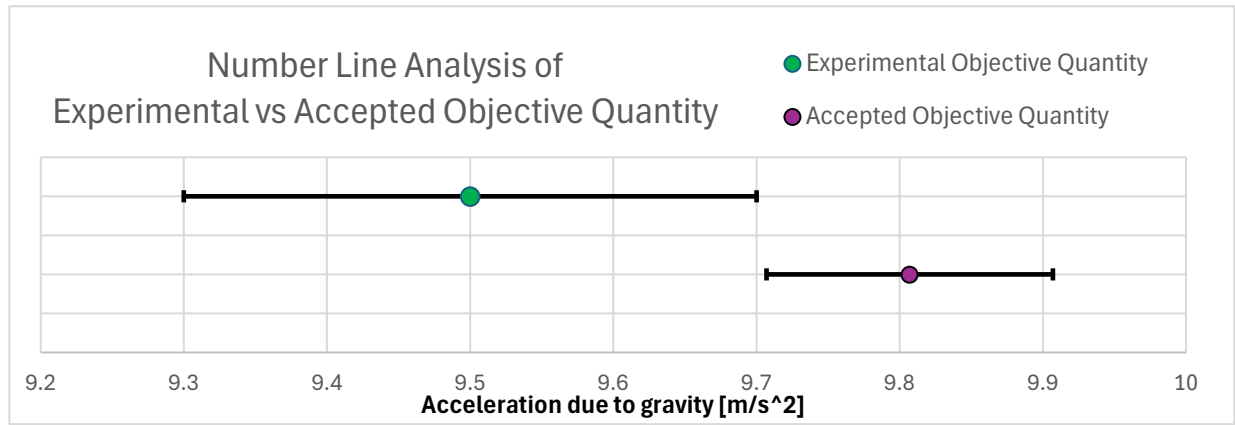
Diagram:**Pre-Lab**

1. Read the sections inc
2. In your own words, de
3. Using your results from Lab 2 (if you haven't completed Lab 2 then use your Lab 1 results and assume your objective quantity uncertainty was 0.5 m/s^2).
 - 3a. Complete an Appendix E using your results for the objective quantity calculation.
 - 3b. Complete an Appendix F to analyze the error.
4. Complete an Appendix A in preparation for this lab following the experiment overview below and the lab worksheet.

In your own words, describe the difference between systematic and random error.

Appendix E: Precision and Accuracy Table FROM LAB 2 RESULTS

Experimental Objective Quantity	9.5	±	0.2	$\frac{m}{s^2}$
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	9.8	±	0.1	$\frac{m}{s^2}$
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

- ☒ Systematic Confidence intervals do not overlap
☐ Random Confidence intervals do overlap

Acceptable level of Precision: 1.0 %

Acceptable level of Accuracy: 5.0 %

Calculate Relative Uncertainty

$$RU = \frac{\delta g}{g}$$

RU = 2.1%

Calculate Percent Difference

$$PD = \frac{|g_{exp} - g|}{g}$$

PD = 3.1%

Appendix F: Error Analysis Table FROM LAB 2 RESULTS

Move the check to the dominant form of error and follow that flow chart:

Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

var

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B



Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

Drag and constant launch velocity are examples that could be used.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

The launcher probably wasn't level on all surfaces. We took an average of velocities separately from our distance measurements.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

It was difficult to measure the launch distance with the equipment provided. It was hard to determine both where the ball was launch relative to the ground (horizontally) and where the ball landed exactly, and then control for the straight line between those points and measure with the tape measure.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

Carbon paper would improve distance estimation. A digital level would help to ensure launcher was level before each launch. Log launch velocity for each trial and incorporate into independent quantity. Some sort of laser levels wi determine launch location and a high speed camera setup to determine landing location would be beneficial.

Overview:

In this laboratory you will determine the acceleration due to gravity near the surface of the Earth (objective quantity) using an experiment designed based on Newton's Laws. In this experiment you will setup a two mass system with a pulley. Mass one (m_1 , PASCO Smart cart) is on a frictionless track which is connected to a hanging mass (m_2) over an ideal pulley. You will vary the mass of the PASCO smart cart (independent variable) and measure the acceleration of the system (dependent variable) to determine the acceleration due to gravity (objective quantity).

Objective Quantity:

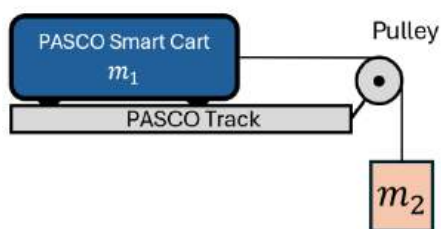
Acceleration due to gravity, g [m/s^2]

Varying (independent variable):

Mass of PASCO Smart Cart

Measuring (dependent variable):

Acceleration of the system

Diagram:**Grading Rubric (35 Points)**

Deliverable	Description	Points	A	B	D	
Cover Sheet, Notebook Formatting, and Writing Style	Formatting, well written, and presentable	3		0.2	0.8	2
Appendix A	Linearization Table	2		0.2	0.8	1
Appendix B	mass 1, mass 2, acceleration	6		0.6	1.8	3.6
Appendix C	Data, Plot, and Regression	6		1	1.5	3.5
Appendix D	Objective Quantity Calculation	3		1.5	1.5	3
Appendix D	Propagation of Uncertainty	3				
Appendix E	Number Line	3		0.5	1	1.5
Appendix F	Error Analysis	3		0.5	1	1.5
Discussion Questions	Question 1	2		0.2	0.6	1.2
Discussion Questions	Question 2	2		0.2	0.6	1.2
Discussion Questions	Question 3	2		0.2	0.6	1.2

Overall Submission Formatting, Presentation, and Organization

These points are for an instructor to critique the professionalism and effort of a submission outside of the normal cut sheet requirements

0-2.0	Minimal: - Many typos and/or grammatical errors, - Many formatting issues present -
2.0-2.8	Marginal: - Generally presentable though clearly lacking any revisions. - Some typos or grammatical errors are present and noticeable. - Formatting issues do not distract from the content
2.8-3.0	Maximal: - Well formatted and professionally presented. - Free of most typos and grammatical errors.

5.1 10.2 19.7
15% 29% 56%

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective			
1a. Describe the goal for this experiment:			
In this experiment we will determine the acceleration due to gravity by varying the mass of a PASCO Cart in a two mass pulley system.			
1b. Objective quantity for this experiment: (objective quantity, OQ)	acceleration due to gravity	<i>g</i>	$\frac{m}{s^2}$
	(Objective Quantity Name)	(OQ variable)	(OQ units)
2. Physics Model Equations - Create equation which includes OQ			
2a. List the physics equations associated with the OQ		2b. Apply equations from 2a into a model equation:	
$\Sigma F_x = m a_x$		<i>Mass 1:</i> $T = m_1 a_x$ <i>Mass 2:</i> $m_2 g - T = m_2 a_x$ <i>From Mass 2:</i> $m_2 g - m_2 a_x = m_1 a_x$ $m_2 g = a_x (m_1 + m_2)$	
3. Determining Independent and dependent variables for your experiment			
3a. The variable changing between trials: (Independent variable, IV)	mass 1	<i>m₁</i>	<i>kg</i>
	(Independent Variable Name)	(Independent Variable)	(units)
3b. Changing the IV will change the following variable: (Dependent variable, DV)	Acceleration of system	<i>a_x</i>	$\frac{m}{s^2}$
	(Dependent Variable Name)	(Dependent Variable)	(units)
3c. The variables in the model equation which will not change between trials: Fill in or blank out as appropriate.	mass 2	<i>m₂</i>	<i>kg</i>
	(Control Variable Name)	(Control Variable)	(units)
	(Control Variable Name)	(Control Variable)	(units)
	(Control Variable Name)	(Control Variable)	(units)
4. Linearized Equation and Objective Quantity Calculation			
4a. Rearrange 2b. into a Linearized Equation (y=mx+b)			
$m_2 g = a_x (m_1 + m_2)$ $a_x = m_2 g \frac{1}{m_1 + m_2}$ $a_x = m_2 g \frac{1}{m_1 + m_2}$ The 'm' term must include your objective quantity variable from 1b. 'x' term must include your independent variable from 3b. 'y' term includes DV from 3a. a_x (y) (Dependent Qty) = $m_2 g$ (m) (slope) $\frac{1}{m_1 + m_2}$ (x) (Independent Qty) + (b) (Intercept)			
4b. Rewrite the slope term below (which includes your objective quantity):			
<i>slope = m₂g</i>			
4c. Rearrange and solve for your objective quantity in terms of slope:			
$g = \frac{slope}{m_2}$ (Obj qty) = (expression in terms of 'slope')			
5. Write a hypothesis statement.			
As we _increase_ the _inverse mass of Cart 1_, we expect to observe a _linear_ _increase_ in the _acceleration of Cart 1_ based on _Newton's 2nd Law_.			
6. Experiment planning and execution		0.1	
6a. Review appendix A and design your experiment around your chosen IV and DV		Correct hypothesis statement that you must update.	
6b. Complete appendix B detailing measurement procedures and uncertainties.			
6c. Calculate objective quantity and uncertainty using appendix D.			
6d. Plot experimental results on a number line to analyze precision and accuracy of findings.			

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass 2 - Hanging Mass	m_2	Digital Scale	☐ Independent Variable ☐ Dependent Variable ☒ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.050080	kg	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Zeroed the mass balance first, then after the scale read 0.00 we placed the hanger mass onto the surface of the mass balance and recorded the results.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	well defined	0
Quality of the measuring device	limited to rounding of output	0.00003 kg
Conditions under which the measurement was performed	no impacts	0
Limitations of humans	no impacts	0

B5. Select the largest of the four factors:

0.00003	kg	
(absolute uncertainty, 1 significant figure)	(units)	

B6. Calculate your relative uncertainty, report as a percent.

0.060%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

0.050080	$\pm$	0.00003	kg
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass 1 (Smart Cart)	m_1		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.33088	kg	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Zeroed the mass balance first, then after the scale read 0.00 we placed the cart onto the surface of the mass balance. The cart was laid on its side to prevent rolling. The cart had the mass holder attached to it.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	well defined	0
Quality of the measuring device	limited to rounding of output	0.00003 kg
Conditions under which the measurement was performed	no impacts	0
Limitations of humans	no impacts	0

B5. Select the largest of the four factors:

0.00003	kg	
(absolute uncertainty, 1 significant figure)	(units)	

B6. Calculate your relative uncertainty, report as a percent.

0.0091%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

0.33088	$\pm$	0.00003	kg
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Acceleration of System	a	PASCO Smart Cart	☐ Independent Variable ☒ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

1.206	$\frac{m}{s^2}$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

The PASCO Cart was bluetooth connected to a computer running PASCO Capstone. The acceleration of the cart was selected to be 'zero' at the start. A graph was created in Capstone with acceleration on the y axis and position on the x axis. The data record button was clicked and then the cart was allowed to roll, following the experiment setup in the lab 3 overview and procedures. The cart was stopped and the recording was stopped. The acceleration of the first 15 centimeters of travel was highlighted and averaged. This measurement was repeated for each mass variation in the appendix C.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	the acceleration varies based on the minor interactions with the track, elasticity of the string, friction on the axles etc.	Std dev (0.01 m/s ²)
Quality of the measuring device	accurate down to 2 decimals	0.005 m/s ²
Conditions under which the measurement was performed	no impacts	0
Limitations of humans	no impacts	0

B5. Select the largest of the four factors:

0.01	m/s^2
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

0.83%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

1.21	±	0.01	m/s^2
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

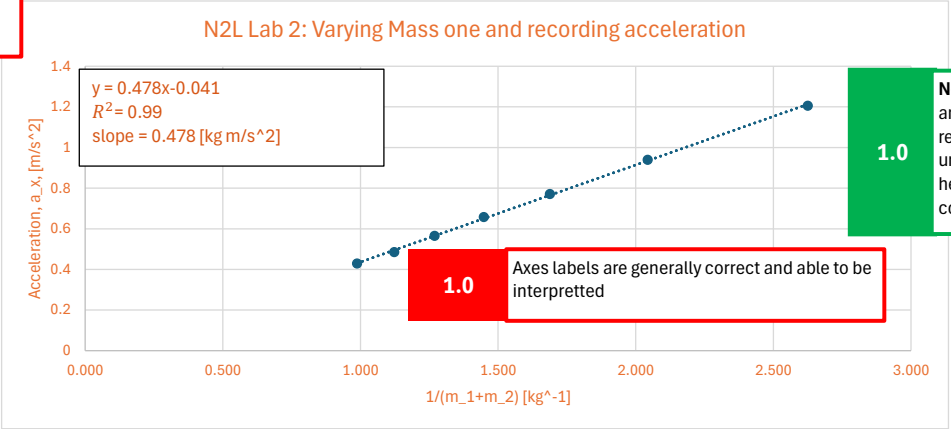
- ☐ Yes
☐ No

Appendix C: Experimental Results and Linear Regression

Control Variable	Value	Abs Unc	Units
Hanging Mass	0.05005	0.00003	kg
NA			

0.5 Hanging mass data is present and reasonable

Plot



	Independent Variable	IV Absolute Uncertainty	Independent Quantity		Dependent Variable	DV Absolute Uncertainty	Dependent Quantity	
Trial number	mass 1 [kg]	δIV	$1/(m_1+m_2)$ [kg ⁻¹]		a_x	δa_x	a_x	
1	0.33088	0.00003	2.625		1.206	0.001	1.206	
2	0.43932	0.00003	2.043		0.9399	0.0009	0.9399	
3	0.54214	0.00003	1.689		0.771	0.001	0.771	
4	0.64057	0.00003	1.448		0.658	0.002	0.658	
5	0.73785	0.00003	1.269		0.565	0.002	0.565	
6	0.84070	0.00003	1.123		0.485	0.001	0.485	
7	0.96224	0.00003	0.988		0.4289	0.0007	0.4289	
8								
9								
10								

1.0 All table data is present

0.75 Table labels are generally completed and identify the data they represent

0.75 Absolute uncertainties of IV and DV and CV are correct

Linear Regression results

1.0 Regression data table is present and slope value is consistent

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.999711
R Square	0.999423
Adjusted R Square	0.999307
Standard Error	0.007248
Observations	7

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.454696	0.45469632	8654.616757	2.72049E-09

Note: Items in orange are placeholders that you must update.

Note: Adjust this page as needed to effectively record your data.

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	acceleration due to gravity	g	$\frac{m}{s^2}$
------------------------------------	-----------------------------	-----	-----------------

(Objective Quantity Name) (OQ variable) (OQ units)

Objective quantity equation (From App A)	$g = \frac{slope}{m_2}$
---	-------------------------

(Obj qty) = (expression in terms of 'slope')

Variable from equation above	Measured Value	Absolute Uncertainty	Units
$slope$	0.478	0.005	$kg\ m/s^2$
m_2	0.05005	0.00003	kg

Calculate the objective quantity using the variables and equation above:

$g = \frac{slope}{m_2}$

$g = 9.6\ m/s^2$

(Obj qty, links from above) (value) (units)

Calculate the combined absolute uncertainty of the objective quantity:
--

$\delta g(m_2, slope) = \sqrt{\left(\frac{\partial g}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial g}{\partial m_2} \delta m_2\right)^2}$

Variable from equation above	Associated Partial Derivative	Quadrature (partial derivate * abs unc)^2
$slope$	$\frac{\partial g}{\partial slope} = \frac{1}{m_2}$	0.00998003
m_2	$\frac{\partial g}{\partial m_2} = \frac{-slope}{m_2^2}$	3.27704E-05

$\delta g = 0.1\ m/s^2$

(Obj qty, links from above) (value) (units)

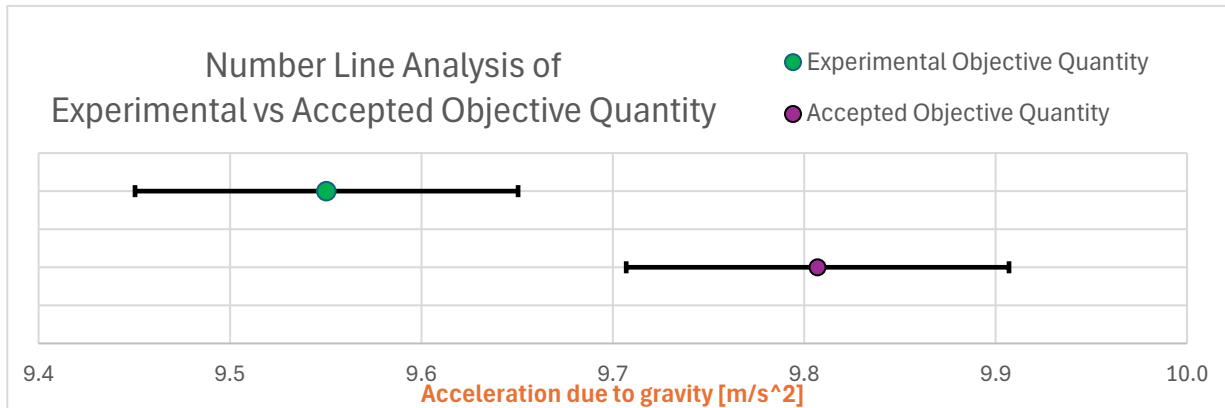
Report the objective quantity as a confidence interval
--

$g =$	9.6	$\pm$	0.1	m/s^2
-------	-----	-------	-----	---------

(objective quantity) (absolute uncertainty) (units)

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	9.6	$\pm$	0.1	m/s^2
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	9.8	$\pm$	0.1	$\frac{m}{s^2}$
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

- ☒ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level
of Precision: 1.0 %

Acceptable level of
Accuracy: 5.0 %

Calculate Relative Uncertainty

Calculate Percent Difference

$RU =$

$PD =$

0.5

Attempts both Calculations

$RU =$ 1.0%

$PD =$ 2.6%

Appendix F: Error Analysis Table	
Move the check to the dominant form of error and follow that flow chart:	
	Random Error Dominates
R1. Identify term in Appendix D with highest quadrature term:	<i>var</i>
R2. Discuss ways to reduce the uncertainty in that variable in the space below:	
IF RANDOM: Identifies the greatest measurement uncertainty and some ways to improve it, should be SLOPE.	
<i>Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.</i>	
<i>Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B</i>	
✓	Systematic Error Dominates
S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.	
IF SYSTEMATIC: Discusses friction or drag or other stated assumptions.	
S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.	
Identifies track levelling	
S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.	
Identifies accelerometer data is noisy and jittery	
S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.	
Make sure track is level or discusses another point from one of their three items above.	

Rubric	
Should complete at least one of the errors, based on their appendix E. Error should match to Appendix E.	
0-1.8	Minimal: - Incomplete answer - Poor explanation / no critical thinking applied - Does not demonstrate understanding
1.9-2.5	Marginal: - Generally complete thought - Attempts reasoning with an explanation - Marginal understanding with some
2.5-3.0	Maximal: - Complete response with critical thinking - Well reasoned explanation - Demonstrates understanding

Answer the questions from the discussion section of the lab notebook here or in a separate word document.

1. Discuss the precision of your measurements in Lab 3 compared to the measurements made in Lab 2. Comment on which experiment had 'better' procedures to determine the acceleration due to gravity.

Lab 3 had much more precision / less error prone measurements. The experiment should have been 'better' in more precise and possibly accurate measurements, though even some slope or friction could cause systematic errors.

2. Is your experimental value higher or lower than the accepted value? Discuss elements of your Appendix F and how those may have contributed to the results being higher or lower.

Should be a 'technical writing' exercise that they can extract the elements from Appendix E and F and form a coherent thought that is supported by evidence. Again apply grading rubric along their critical thinking abilities.

3. If you were to conduct this experiment again, describe one way you could reduce the dominant form of error?

Another 'technical writing' exercise that they can extract the elements from their experiment and draw conclusions on how to improve the experiment. The answer should be something tangible, based on their data, and ideally include an expectation of what would change.

0	Empty
1.2	Minimal
1.6	Marginal
2.0	Maximal

0	Empty
1.2	Minimal
1.6	Marginal
2.0	Maximal

0	Empty
1.2	Minimal
1.6	Marginal
2.0	Maximal

Rubric

Apply the points that best fit each Group's Answer relative to the expected responses and your expertise as an instructor.

0-1.2	Minimal: - Incomplete answer - Poor explanation / no critical thinking applied - Does not demonstrate understanding
1.2-1.6	Marginal: - Generally complete thought - Attempts reasoning with an explanation - Marginal understanding with some inaccuracies
1.6-2.0	Maximal: - Complete response with critical thinking - Well reasoned explanation - Demonstrates understanding



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1



Fill in the Blank 1 point Types of Error

Of the two types of experimental error, errors are inherent in every measurement. These errors are primarily due to the uncertainties related to the experiment and primarily impact the precision of an experiment's results.

2



Numeric 1 point Relative Uncertainty

The length of an M4 rifle is measured as **0.838 m**. The relative uncertainty of the measurement is determined to be 1.2 %. **Calculate** the absolute uncertainty of the measurement of the rifle and **report** your answer in meters and the appropriate number of significant figures.

3



Numeric 1 point Percent Difference

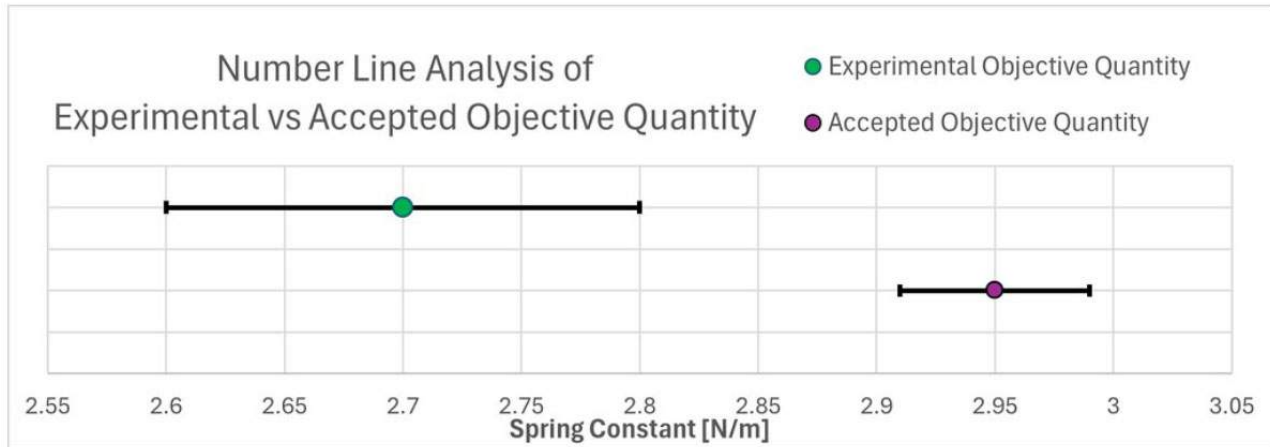
The acceleration due to gravity is determined as: **$10.32 \frac{m}{s^2}$** . **Calculate** the percent difference from the accepted value of the acceleration due to gravity: **$9.807 \frac{m}{s^2}$** . Report your answer as a percent with the correct number of significant figures.

4



Multiple Choice 1 point Type of Error

An experiment is performed to determine the spring constant of a spring and an Appendix E is prepared (shown below).



Determine which form of error dominates in this experiment based on the Appendix E.

- ☐ Random Error
- ☐ Absolute Error
- ☐ Relative Error
- ☒ Systematic Error

5



Multiple Choice 1 point Error Analysis

During lab 3 we assumed that the cart (mass 1) was on a frictionless track. The objective quantity was calculated using:

$$g = \frac{\text{slope}}{m_2}$$

where mass 2 is the hanging mass. The slope was determined from graphing the independent quantity, $\frac{1}{m_1 + m_2}$ on the x axis and the dependent quantity, a_x , the acceleration of the system, on the y axis.

Select which of the following statements is most correct if the assumption of a frictionless surface **was not** valid (i.e. friction impacts the results of this experiment):

- ☐ The acceleration of our system would be larger relative to a frictionless assumption
- ☒ The acceleration of our system would be smaller relative to a frictionless assumption
- ☐ The acceleration of our system would be the same relative to a frictionless assumption

6



Essay 0 points Document Assistance and Sources

Take a moment to REFLECT on your efforts when completing this assignment.

Did you receive any assistance in completing this assignment?

Copy and paste which response you choose. If you chose 'Yes' then use the space below to acknowledge assistance and document sources. Reference the question number in your documentation.

NO: I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

YES: I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

Edit View Insert Format Tools Table

12pt Paragraph **B** *I* U **A** T^2





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LAB#4 HANDOUT-SEE LAB UI



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LAB#4 SOLUTION-SEE LAB UI SOLUTION



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LAB#5 HANDOUT

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

Determine the spring constant of a pasco smart cart by varying the mass and measuring the velocity

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Spring Constant pasco cart

k

N/m

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

COE

$W_{nc} = E_f - E_i \text{ where } W_{nc} = 0$

$E_f = \frac{1}{2}mv^2 \text{ and } E_i = \frac{1}{2}kx^2$

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

mass

m

kg

(Independent Variable Name)(Independent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

velocity (speed)

v

m/s

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

Spring displacement

x

m

(Control Variable Name)(Control Variable)(units)

(Control Variable Name)(Control Variable)(units)

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$ $mv^2 = kx^2$

v^2 $=$ kx^2 $\frac{1}{m}$

(y)(m)(x)(b)

(Dependent Qty)(slope)(Independent Qty)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

4b. Rewrite the slope term below (which includes your objective quantity):

$slope = kx^2$

4c. Rearrange and solve for your objective quantity in terms of slope:

$k = \frac{slope}{x^2}$

(Obj qty) = (expression in terms of 'slope')

5. Write a hypothesis statement.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)

(OQ variable)

(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

Type equation here.

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

independent variable

IV

units

(Independent Variable Name)

(Indendent Variable)

(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

dependent variable

DV

units

(Dependent Variable Name)

(Dependent Variable)

(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)

(Control Variable)

(units)

control variable 2

CV2

units

(Control Variable Name)

(Control Variable)

(units)

control variable 3

CV3

units

(Control Variable Name)

(Control Variable)

(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty) = (expression in terms of 'slope')

5. Write a hypothesis statement.

As we (increase/decrease) the (independent quantity), we expect to observe a(n) (linear/exponential/logarithmic/quadratic) (increase/decrease) in the (dependent quantity) based on (Physics Principle).

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6b. Complete appendix B detailing measurement procedures and uncertainties.

6c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.

6d. Calculate objective quantity and uncertainty using appendix D.

6e. Plot experimental results on a number line to analyze precision and accuracy of findings.

Note: Items in orange are placeholders that you must update.

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

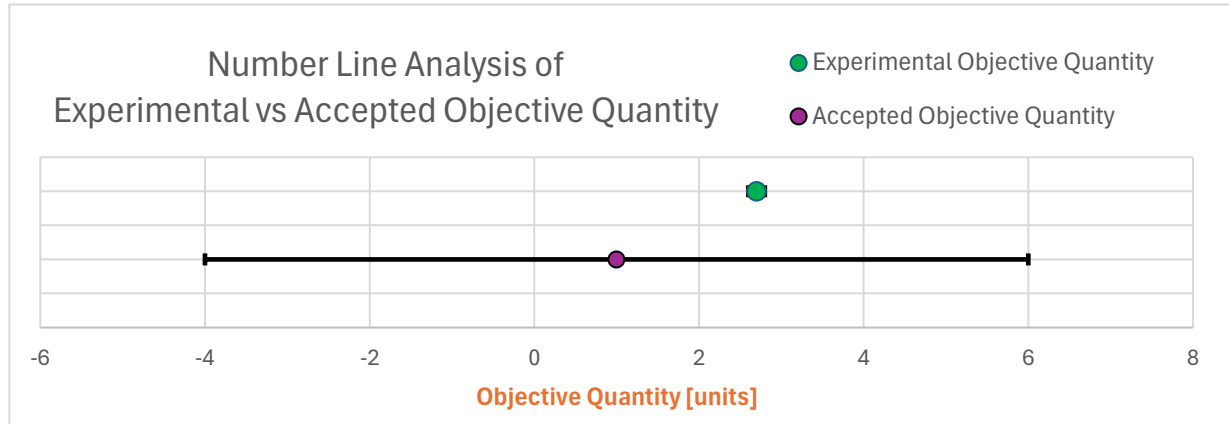
B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty				
Objective quantity (From App A)	Objective quantity		OQ	units
		(Objective Quantity Name)	(OQ variable)	(OQ units)
Objective quantity equation (From App A)	OQ = expr in terms of slope (From App A)			
		(Obj qty) = (expression in terms of 'slope')		
Variable from equation above	Measured Value	Absolute Uncertainty	Units	
slope			units	
var				
Calculate the objective quantity using the variables and equation above:				
List OQ equation with variables here				
List OQ equation with values here				
OQ = use excel to calculate here units				
(Obj qty)		(value)	(units)	
Calculate the combined absolute uncertainty of the objective quantity:				
$\delta f(x, slope) = \sqrt{\left(\frac{\partial f}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial x}{\partial x} \delta x\right)^2}$ [EXAMPLE]				
Variable from equation above	Associated Partial Derivative		Quadrature (partial derivative * abs unc)^2	
slope	$\frac{\partial OQ}{\partial slope} = FILL$			
var	Type equation here.			
var	Type equation here.			
var	Type equation here.			
δOQ = use excel to calculate here units				
(Obj qty)		(value)	(units)	
Report the objective quantity as a confidence interval				
	OQ =	use excel to calculate here	±	use excel to calculate here units
	(objective quantity)			(absolute uncertainty, 1SF) (units)

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	2.7 (experimental objective quantity)	$\pm$	0.1 (absolute uncertainty)	units (units)
Accepted Objective Quantity	(Follow link in lab worksheet to find this) (accepted objective quantity)	$\pm$	5 (absolute uncertainty)	$\frac{N}{m}$ (units)



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level of Precision: 5.0 %
(default value for PH2X1 experiments)

Acceptable level of Accuracy: 5.0 %
(default value for PH2X1 experiments)

Calculate Relative Uncertainty (2SF)

$RU =$

RU = 99%

Calculate Percent Difference (2SF)

$PD =$

PD = 99%

Appendix F: Error Analysis Table

Move the **check** to the dominant form of error and follow that flow chart:



Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

var

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

1. Complete the appendices for your experiment
2. For multiple Appendix B's right click the sheet and click 'move or copy', select create a copy and put it in this wc
3. To compile your report click:
 - File
 - Save as adobe pdf OR print to pdf
 - select all appropriate sheets (except this one)
 - Review the pdf to ensure all required information is present.
4. If you need to adjust pdf spacing use the 'page layout' tab on the sheet you need to adjust

orkbook.



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LAB#5 SOLUTION

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

Determine the spring constant of a pasco smart cart by varying the mass and measuring the velocity

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Spring Constant pasco cart

k

N/m

(Objective Quantity Name)(OQ variable)(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

COE

$W_{nc} = E_f - E_i \text{ where } W_{nc} = 0$

$E_f = \frac{1}{2}mv^2 \text{ and } E_i = \frac{1}{2}kx^2$

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

mass

m

kg

(Independent Variable Name)(Independent Variable)(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

velocity (speed)

v

m/s

(Dependent Variable Name)(Dependent Variable)(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

Spring displacement

x

m

(Control Variable Name)(Control Variable)(units)

(Control Variable Name)(Control Variable)(units)

(Control Variable Name)(Control Variable)(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$mv^2 = kx^2$

v^2

=

kx^2

$\frac{1}{m}$

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

4b. Rewrite the slope term below (which includes your objective quantity):

$slope = kx^2$

4c. Rearrange and solve for your objective quantity in terms of slope:

$k = \frac{slope}{x^2}$

(Obj qty)

=

(expression in terms of 'slope')

5. Write a hypothesis statement.

Lab 5 score distribution and Appendix Cuts

Deliverable	Description	Points
Entire Report	Formatting and adherence to style guide	2
Theory	Content follows guidelines in reading and formatted appropriately	10
Procedures	Content follows guidelines in reading and formatted appropriately	10
Results	Content follows guidelines in reading and formatted appropriately	10
Appendix A	Linearization Table	1
Appendix B	Displacement, mass, speed	3
Appendix C	Data, Plot, and Regression	3
Appendix D	Objective Quantity Calculation	2
Appendix D	Propagation of Uncertainty	2
Appendix E	Number Line	1
Appendix F	Error Analysis	1

Appendix A

0.5

Attempts to complete Appendix.

0.5

Correct OQ eqn

You may also see them sqrt the eqn, still correct just make sure they plot the correct IQ.

Appendix B (3x)

0.5

App B is generally completed.

0.5

Reasonable justification of uncertainty

Appendix C

1.0

Appendix C is generally completed correctly

0.5

Control variable is reported and reasonable

0.5

Table is completely filled including variables and units in headers and absolute uncertainty of velocity changes from trial to trial or is reasonable.

1.0

Near perfect graph: Equation, R-squared, and slope are accurately and correctly reported to include SF and rounding (Slope uncertainty does not need to be reported here) Axes have units and are labelled correctly

Appendix D

1.0

Fills in all App D Data.

1.0

Attempts Objective quantity calculation.

0.5

Attempts propagation of uncertainty

1.0

Correct propagation of uncertainty

0.5

Completely correct confidence interval (rounding, SF, units)

Appendix E

0.5

Appendix E generally complete

0.5

Correct RU and PD with 2SF

Appendix F

0.5

Error type follows from App E.

0.5

Well reasoned / complete responses.

Lab 5 Grading Rubric distribution

Max Points		Poor		Marginal		Good		Exceptional	
		(Pts)	Description	(Pts)	Description	(Pts)	Description	(Pts)	Description
Formatting	2	(0)	No attempt made to conform to style guide.			(1)	Formatting is generally consistent, though does not adhere to guide.	(2)	Formatting completely follows the style guide.
Theory	10	(0-5)	Minimal or unclear theory that lacks substance or is generally incorrect.	(6-7)	Theory present but vague, lacks sufficient explanation / not effectively communicated.	(8-9)	Theory explained well, with minor omissions of key information from theory checklist.	(10)	Thorough, clear, and relevant theory explained with assumptions. Uses correct physics to support theory. Nearly all elements of checklist discussed
Procedures	10	(0-5)	<i>Nearly impossible to repeat experiment.</i> Incomplete or unclear procedures.	(6-7)	<i>Difficult to repeat experiment.</i> Steps taken are included but missing important details (e.g. entire variables not described).	(8-9)	<i>Very possible to repeat experiment.</i> Descriptive steps but limited discussion of uncertainty in measurements/confidence intervals/ other essential elements from the checklist.	(10)	<i>Extremely easy to repeat experiment.</i> Complete, detailed, and well-organized procedures. Nearly all checklist elements discussed.
Results	10	(0-6)	<i>Does not report objective quantity results and/or lacks any effort.</i> Unclear results reported. Lacks tables, graphs, figures, or any data.	(6-7)	<i>Does not report objective quantity results though some effort to report results is evident.</i> Results are unclear/not supported by writing and/or provides sparse data.	(8-9)	<i>Reports the objective quantity results and writing supports results.</i> Results are clear and introduced in the written report, though missing some details e.g. one table is missing, a plot isn't well described or incorporated, fails to report the slope, etc.	(10)	<i>Reports the objective quantity full confidence interval completely correctly with all key supporting data.</i> Clear results, well-formatted tables/graphs are error-free and introduced in the report.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

Determine the spring constant of a pasco smart cart by varying the mass and measuring the velocity

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Spring Constant pasco cart

k

N/m

(Objective Quantity Name)

(OQ variable)

(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

COE

$W_{nc} = E_f - E_i \text{ where } W_{nc} = 0$

$E_f = \frac{1}{2}mv^2 \text{ and } E_i = \frac{1}{2}kx^2$

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

mass

m

kg

(Independent Variable Name)

(Indendent Variable)

(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

velocity (speed)

v

m/s

(Dependent Variable Name)

(Dependent Variable)

(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

Spring displacement

x

m

(Control Variable Name)

(Control Variable)

(units)

(Control Variable Name)

(Control Variable)

(units)

(Control Variable Name)

(Control Variable)

(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$
 $mv^2 = kx^2$

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

v^2

=

kx^2

$\frac{1}{m}$

(y)

=

(m)

(x)

+

(b)

(Depende Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

$slope = kx^2$

0.5

Attempts to complete Appendix.

4c. Rearrange and solve for your objective quantity in terms of slope:

$k = \frac{slope}{x^2}$

0.4

Correct OQ eqn
You may also see them sqrt the eqn
still correct just make sure they plc
the correct IQ.

5. Write a hypothesis statement.

Appendix B: Measured quantity

B1. Physical quantity measured:

Displacement of spring	x	Calipers, Tape measure, meter stick	☐ Independent Variable ☐ Dependent Variable ☒ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.0330	m	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Detailed description that is repeatable (including the detent they measured to). The number reported above includes adding 0.5 cm as instructed by the lab worksheet.

NOTE: The lab worksheet tells them to add 0.5 cm to their measurement. They should indicate whether or not they did that.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	There is some bounce and play in how the plungers set, maybe at a tenth of a mm	0.0001 m
Quality of the measuring device	Varies based on device used, if using calipers	0.00003 m
Conditions under which the measurement was performed	The smart cart doesn't let us measure the actual spring displacement so we use the plunger but this does not impact the measurement	0
Limitations of humans	none	0

B5. Select the largest of the four factors:

0.0001	mm
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

0.30%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

0.0330	$\pm$	0.0001	mm
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☒ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass of cart	<i>m</i>	digital scale	☒ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.32350	<i>kg</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Describes thoroughly including taring scale. May also include details about how they added masses.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	well defined. They may try to describe how the mass says one thing on it but the scale conflicts with the stamped mass value.	0
Quality of the measuring device	Scale uncertainty	0.00003 kg
Conditions under which the measurement was performed	Nothing about conditions impacts the measurement	0
Limitations of humans	none	0

B5. Select the largest of the four factors:

0.00003	<i>kg</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

0.0093%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

0.03300	$\pm$	0.00003	<i>kg</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☒ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Speed of Cart	v	PASCO	☐ Independent Variable ☒ Dependent Variable ☐ Control Variable Variable type (Check one)
(name)	(variable)	(tool used)	

B2. Measured Value:

0.740	m/s	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Describes thoroughly how they used PASCO to obtain the measurement.

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity	Variance here, as the cart and spring have different interactions each time. Sampled based on average of multiple iterations.	0.002
Quality of the measuring device	They could interpret the jitter in velocity as the quality of the measuring device, which could also be a fair assumption as we don't really know the absolute uncertainty of the PASCO speed measurements as a derivative of the wheel position msmt.	0.00003 kg
Conditions under which the measurement was performed	The conditions do not impact the measurement itself.	0
Limitations of humans	none	0

B5. Select the largest of the four factors:

0.002	m/s
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

0.27%	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Write your measurement with absolute uncertainty as a confidence interval:

0.740	$\pm$	0.002	$\frac{m}{s}$
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☒ Yes
☐ No

0.5

Control variable is reported and reasonable

Control Variable	Value	Abs Unc	Units
x	0.033	0.005	m
m_cart	0.32350	0.00003	kg

Note: I chose to use this fixed m and measured the mass I added, which makes the IV Abs Unc a little worse (not much)

1.0

Appendix C is generally completed correctly

Plot

Determining PASCO Spring Constant from Speed and Mass Measurements

1.0

Near perfect graph: Equation, R-squared, and slope are accurately and correctly reported to include SF and rounding (Slope uncertainty does not need to be reported here) Axes have units and are labelled correctly

Trial number	m [kg]	Δm [kg]	1/m[kg^-1]	v [m/s]	Δv [m/s]	v^2 [m^2/s^2]
1	0	4E-05	3.091190108	0.740	0.002	0.5476
2	0.10005	4E-05	2.36099634	0.600	0.003	0.3600
3	0.2	4E-05	1.910219675	0.580	0.003	0.3364
4	0.3	4E-05	1.603849238	0.480	0.003	0.2304
5	0.572	4E-05	1.116694584	0.467	0.003	0.2181
6						
7						
8						
9						
10						

0.5

Table is completely filled including variables and units in headers and absolute uncertainty of velocity changes from trial to trial or is reasonable.

Note: Items in orange are placeholders that you must update.

Linear Regression results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.965104
R Square	0.931426
Adjusted R Square	0.908568
Standard Error	0.040108
Observations	5

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.065548	0.065548	40.74814	0.007784
Residual	3	0.004826	0.001609		
Total	4	0.070374			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-0.00457	0.056657	-0.08061	0.940826	-0.18488	0.175741	-0.18488	0.175741
X Variable 1	0.170121	0.02665	6.383427	0.007784	0.085308	0.254935	0.085308	0.254935

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	Spring constant	k	N/m
(Objective Quantity Name)		(OQ variable)	(OQ units)
Objective quantity equation (From App A)	$k = \frac{slope}{x^2}$		
(Obj qty) = (expression in terms of 'slope')			
Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope	0.170121403	0.026650482	$\frac{m^2}{s^2} kg$
x	0.033	0.001	m
1.0	Fills in all App D Data.		
1.0	Attempts Objective quantity calculation.		
0.5	Attempts propagation of uncertainty		
Objective quantity using the variables and equation above:			
$k = 156.218001 \text{ N/m}$ (Obj qty) (value) (units)			
Calculate the combined absolute uncertainty of the objective quantity:			
$\delta k(x, slope) = \sqrt{\left(\frac{\partial k}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial k}{\partial x} \delta x\right)^2}$			
Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2	
slope	$\frac{\partial k}{\partial slope} = \frac{1}{x^2}$	598.9001	
1.0	$\frac{\partial k}{\partial x} = -2slope/x^3$	89.63843469	
var	Type equation here.	0.5	Completely correct confidence interval (rounding, SF, units)
var	Type equation here.		
$\delta k = 26.24001781 \text{ N/m}$ (Obj qty) (value) (units)			
Report the objective quantity as a confidence interval			
OQ =	160	±	30 N/m
(objective quantity)		(absolute uncertainty, 1SF) (units)	

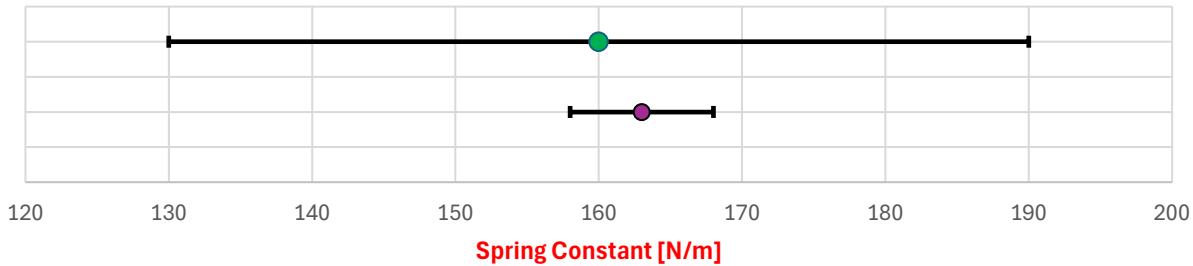
Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	160	$\pm$	30	$\frac{N}{m}$
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	163	$\pm$	5	$\frac{N}{m}$
(accepted objective quantity)			(absolute uncertainty)	(units)

Number Line Analysis of Experimental vs Accepted Objective Quantity

0.5

Appendix E generally complete



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level of Precision: 2.0 %
(default value for PH2X1 experiments)

Acceptable level of Accuracy: 5.0 %
(default value for PH2X1 experiments)

Calculate Relative Uncertainty (2SF)

Calculate Percent Difference (2SF)

0.5

Correct RU and PD with 2SF

RU = 19%

PD = 1.8%

Appendix F: Error Analysis Table	
Move the check to the dominant form of error and follow that flow chart:	
✓	Random Error Dominates
R1. Identify term in Appendix D with highest quadrature term:	
	<i>slope</i>
R2. Discuss ways to reduce the uncertainty in that variable in the space below:	
More measurements, improved speed measuring procedures (sampling closer to launch, using a different sensor)	
Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.	
Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B	
	Systematic Error Dominates
S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.	
Frictionless surface and level surface	
S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.	
Spring may not have contacted surface perfectly, release mechanism on cart is prone to friction if not careful	
S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.	
The track doesn't hold the cart perfectly so it may not contact ideally.	
S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.	
Something from the above is listed.	

0.5

Error type f

0.5

Well reason

follows from App E.

ied / complete responses.



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LAB# ONLINE POST LAB

1



Ordering 2 points Lab Report Structure

Arrange the lab report sections in the order in which they appear in a physics lab report.

First

1

⋮ Introduction

2

⋮ Theory

3

⋮ Procedures

4

⋮ Results

5

⋮ Discussion

6

⋮ Conclusion

7

⋮ Lab Appendices (Lab notebook)

Last

2



True or False 1 point Formatting and Style

The lab report in PH201/251 has a style guide which specifies formatting requirements (and can be found on Canvas) and lab reports should be written in accordance with that style guide.

- ☒ True
- ☐ False

3



Multiple Choice 1 point

In introductory physics, as in any written report, the lab report must be tailored to the intended audience. **Select** which of the following is the intended audience for PH201/251 lab reports and thus dictates the level of detail required in the report according to the required reading:

- ☒ Cadets in introductory physics (classmates)
- ☐ A lead experimenter / Principle Investigator
- ☐ Your instructor
- ☐ A technical journal

4



Multiple Answer 2 points Theory Section

Select which of the following are elements included in the **theory** section according to the reading:

- ☒ Lists assumptions which are necessary, valid, and reasonable.
- ☒ Derives model equations using both explanatory text and equations.
- ☐ Explains the measurement procedures with sufficient detail so that it is repeatable
- ☐ Justifies the uncertainty in each measurement.
- ☒ Derives an expression for the objective quantity from the slope equation.
- ☒ Identifies the independent and dependent quantity from the linearized equation.



Stimulus

Propagation of Uncertainty

The area of a piece of paper is calculated using the expression:

$$A = Lw$$

where L is the length and w is the width. The measurements are reported as follows:

$$L = (27.9 \pm 0.2) \text{ cm}$$

$$w = (21.5 \pm 0.2) \text{ cm}$$

5



Numeric 1 point Relative Uncertainty

Calculate the relative uncertainty of the length measurement (L). Report your answer as a percent to two significant figures.

0.70 to 0.73 inclusive

6



Numeric 1 point Percent Difference

The accepted value for the area of the piece of paper (according to the manufacturer) is **605 cm²**. **Calculate** the percent difference between the experimental and accepted value. Report your value as a percent.

7



Num... 2 poi... Propagation of Unce...

Calculate the absolute uncertainty for the experimental value using propagation of uncertainty. Report your answer to 1 significant figure in **cm²**.

8



Essay 0 points Document Assistance and Sources

Take a moment to REFLECT on your efforts when completing this assignment.

Did you receive any assistance in completing this assignment?

Copy and paste which response you choose. If you chose 'Yes' then use the space below to acknowledge assistance and document sources. Reference the question number in your documentation.

NO: I CERTIFY THAT I DID NOT USE ANY SOURCES OR RECEIVE ANY ASSISTANCE REQUIRING DOCUMENTATION WHILE COMPLETING THIS ASSIGNMENT.

YES: I CERTIFY THAT I HAVE COMPLETELY DOCUMENTED ALL SOURCES THAT I USED TO COMPLETE THIS ASSIGNMENT AND THAT I ACKNOWLEDGED ALL ASSISTANCE I RECEIVED IN THE COMPLETION OF THIS ASSIGNMENT.

Edit View Insert Format Tools Table

12pt Paragraph | **B** *I* U **A** ▼ ▼ T^2 ▼ |

▼ ▼ ▼ ▼ | ▼ | ⋮





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LAB#6 HANDOUT

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name)

(OQ variable)

(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

Type equation here.

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

independent variable

IV

units

(Independent Variable Name)

(Indendent Variable)

(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

dependent variable

DV

units

(Dependent Variable Name)

(Dependent Variable)

(units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name)

(Control Variable)

(units)

control variable 2

CV2

units

(Control Variable Name)

(Control Variable)

(units)

control variable 3

CV3

units

(Control Variable Name)

(Control Variable)

(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty)

=

(expression in terms of 'slope')

5. Write a hypothesis statement.

As we (increase/decrease) the (independent quantity), we expect to observe a(n) (linear/exponential/logarithmic/quadratic) (increase/decrease) in the (dependent quantity) based on (Physics Principle).

6. Experiment planning and execution

6a. Review appendix A and design your experiment around your chosen IV and DV

6b. Complete appendix B detailing measurement procedures and uncertainties.

6c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.

6d. Calculate objective quantity and uncertainty using appendix D.

6e. Plot experimental results on a number line to analyze precision and accuracy of findings.

Note: Items in orange are placeholders that you must update.

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>h</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Describe the contribution from each factor of uncertainty and quantify the uncertainty for each:

Factor	Description	Plus/Minus Uncertainty (w. units)
Nature of the quantity		
Quality of the measuring device		
Conditions under which the measurement was performed		
Limitations of humans		

B5. Select the largest of the four factors:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B6. Calculate your relative uncertainty, report as a percent.

	$RU = \frac{B5}{B2} * 100\%$
(relative uncertainty, 2 significant figures)	

B7. Combine B2 and B5 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B5, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

☐ Yes
☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	Objective quantity	<i>OQ</i>	<i>units</i>
	(Objective Quantity Name)	(OQ variable)	(OQ units)

Objective quantity equation (From App A)	<i>OQ = expr in terms of slope (From App A)</i>
---	---

	(Obj qty)	=	(expression in terms of 'slope')
--	-----------	---	----------------------------------

Variable from equation above	Measured Value	Absolute Uncertainty	Units
<i>slope</i>			<i>units</i>
<i>var</i>			

Calculate the objective quantity using the variables and equation above:

List OQ equation with variables here

List OQ equation with values here

<i>OQ</i>	=	use excel to calculate here	<i>units</i>
(Obj qty)		(value)	(units)

Calculate the combined absolute uncertainty of the objective quantity:

$$\delta f(x, slope) = \sqrt{\left(\frac{\partial f}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial f}{\partial x} \delta x\right)^2} \text{ [EXAMPLE]}$$

Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2
<i>slope</i>	$\frac{\partial OQ}{\partial slope} = \text{FILL}$	
<i>var</i>	Type equation here.	
<i>var</i>	Type equation here.	
<i>var</i>	Type equation here.	

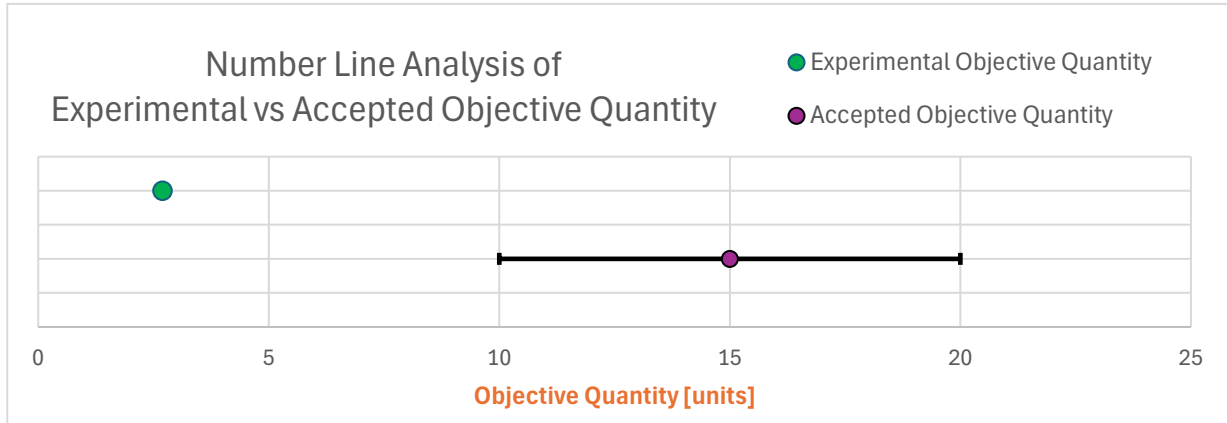
<i>δOQ</i>	=	use excel to calculate here	<i>units</i>
(Obj qty)	=	(value)	(units)

Report the objective quantity as a confidence interval

	<i>OQ</i> =	use excel to calculate here	±	use excel to calculate here	<i>units</i>
	(objective quantity)			(absolute uncertainty, 1SF)	(units)

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	2.7	$\pm$	0.1	units
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	15	$\pm$	5	units
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level
of Precision: 5.0%

(default value for PH2X1 experiments)

Acceptable level of
Accuracy: 5.0%

(default value for PH2X1 experiments)

Calculate Relative Uncertainty (2SF)

$RU =$

$RU =$ 99%

Calculate Percent Difference (2SF)

$PD =$

$PD =$ 99%

Appendix F: Error Analysis Table

Move the **check** to the dominant form of error and follow that flow chart:



Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

var

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

1. Complete the appendices for your experiment
2. For multiple Appendix B's right click the sheet and click 'move or copy', select create a copy and put it in this wc
3. To compile your report click:
 - File
 - Save as adobe pdf OR print to pdf
 - select all appropriate sheets (except this one)
 - Review the pdf to ensure all required information is present.
4. If you need to adjust pdf spacing use the 'page layout' tab on the sheet you need to adjust

orkbook.



UNITED STATES MILITARY ACADEMY
WEST POINT.

LAB#6 SOLUTION

Lab 6 Score Distribution and Appendix Cuts

Deliverable	Description	Points
Entire Report	Formatting and adherence to style guide	3
Results	Content follows guidelines in reading and formatted appropriately	12
Discussion	Content follows guidelines in reading and formatted appropriately	12
Conclusion	Content follows guidelines in reading and formatted appropriately	5
Appendix A	Linearization Table	1
Appendix B	Displacement, mass, speed	3
Appendix C	Data, Plot, and Regression	3
Appendix D	Objective Quantity Calculation	2
Appendix D	Propagation of Uncertainty	2
Appendix E	Number Line	1
Appendix F	Error Analysis	1

Appendix A

0.5

Attempts to complete Appendix.

0.4

Correct OQ eqn

You may also see them sqrt the eqn, still correct just make sure they plot the correct IQ.

0.1

Correct hypothesis statement

Appendix B (3x)

0.5

App B is generally completed.

0.5

Reasonable justification of uncertainty

Appendix C

1.0

Appendix C is generally completed correctly

0.5

Control variable is reported and reasonable

0.5

Table is completely filled including variables and units in headers and absolute uncertainty of velocity changes from trial to trial or is reasonable. Includes accepted value data.

1.0

Near perfect graph: Equation, R-squared, and slope are accurately and correctly reported to include SF and rounding (Slope uncertainty does not need to be reported here) Axes have units and are labelled correctly

Appendix D

1.0

Fills in all App D Data.

1.0

Attempts Objective quantity calculation.

0.5

Attempts propagation of uncertainty

1.0

Correct propagation of uncertainty

0.5

Completely correct confidence interval (rounding, SF, units)

Appendix E

0.5

Appendix E generally complete

0.5

Correct RU and PD with 2SF

Appendix F

0.5

Error type follows from App E.

0.5

Well reasoned / complete responses.

Lab 6 Grading Rubric Distribution

Max Points		Poor		Marginal		Good		Exceptional	
		(Pts)	Description	(Pts)	Description	(Pts)	Description	(Pts)	Description
Formatting	3	(0)	No attempt made to conform to style guide.	(1)	Formatting is generally consistent, though does not adhere to guide.	(2)	Formatting is generally consistent, though does not adhere to guide.	(3)	Formatting completely follows the style guide.
Results	12	(0-6)	Does not report objective quantity results and/or lacks any effort. Unclear results reported. Lacks tables, graphs, figures, or any data.	(7-8)	Does not report objective quantity results though some effort to report results is evident. Results are unclear/not supported by writing and/or provides sparse data.	(9-10)	Reports the objective quantity results and writing supports results. Results are clear and introduced in the written report, though missing some details e.g. one table is missing, a plot isn't well described or incorporated, fails to report the slope, etc.	11-12	Reports the objective quantity full confidence interval completely correctly with all key supporting data. Clear results, well-formatted tables/graphs are error-free and introduced in the report.
Discussion	12	(0-6)	Poor writing that shows no critical thought or effort. Written analysis is almost completely missing or does not demonstrate reasonable thinking to identify sources of error and recommend changes.	(7-8)	Writing is lacking in critical thought or effort. Written analysis which attempts to describe the error. Fails to identify the correct error, does not recommend a reasonable improvement.	(9-10)	Writing attempts critical thought/has some effort. Generally well written analysis. Uses quantitative or qualitative descriptions to report on error, identifies a correct form of error, and recommends changes that may or may not be helpful/reasonable.	11-12	Extremely well written and thoroughly analyzed. Uses quantitative and qualitative descriptions to report on error, identifies a correct form of error, and recommends meaningful changes to improve the experiment.
Conclusions	5	(0-2)	Poor writing with no takeaways from experiment. Bare bones conclusions that is almost completely missing details and has no/incorrect takeaways from the experiment.	(3)	Writing has few takeaways from experiment. Synthesizes some conclusions though missing many details. May report a finding that is in no way related or supported by the rest of the report.	(4)	Well written with some takeaways from experiment. Restates the objective quantity and restates some form of error analysis from the discussion. Synthesizes some findings that are generally correct, though may not be fully supported by the report.	(5)	Extremely well written with synthesized takeaways from experiment. Restates the objective quantity, includes precision and accuracy, restates error and synthesizes findings that follow from the rest of report.

Appendix A: Experiment Overview and Linearization Table			
1. Experiment Goal - Outline your experiment objective			
1a. Describe the goal for this experiment: Determine the initial speed of a cart before it undergoes an inelastic collision with a second cart.			
1b. Objective quantity for this experiment: (objective quantity, OQ)	initial speed of cart A	v_A	kg
(Objective Quantity Name)		(OQ variable)	(OQ units)
2. Physics Model Equations - Create equation which includes OQ			
2a. List the physics equations associated with the OQ		2b. Apply equations from 2a into a model equation:	
$\Sigma \vec{P}_0 = \Sigma \vec{P}_f$$\vec{p} = m \vec{v}$ Just Before $\vec{V}_{A0}$ m_A $+$ cart m_B $\Sigma \vec{P}_0 = \Sigma \vec{P}_f$ Just After $\vec{V}_f$ $m_A + m_B$		$m_A v_{A0} = (m_A + m_B) v_f$$v_{A0} = \frac{(m_A + m_B) v_f}{m_A}$	
3. Determining Independent and dependent variables for your experiment			
3a. The variable changing between trials: (Independent variable, IV)	Mass of cart B	m_A	kg
(Independent Variable Name)		(Indendent Variable)	(units)
3b. Changing the IV will change the following variable: (Dependent variable, DV)	Final velocity of combined carts	v_f	$\frac{m}{s}$
(Dependent Variable Name)		(Dependent Variable)	(units)
3c. The variables in the model equation which will not change between trials: Fill in or blank out as appropriate.	mass of cart A	m_A	kg
(Control Variable Name)		(Control Variable)	(units)
(Control Variable Name)		(Control Variable)	(units)
(Control Variable Name)		(Control Variable)	(units)
4. Linearized Equation and Objective Quantity Calculation			
4a. Rearrange 2b. into a Linearized Equation (y=mx+b)			
$\frac{v_{A0} m_A}{m_A + m_B} = v_f$		The 'm' term must include your objective quantity variable from 1b. 'x' term must include your independent variable from 3b. 'y' term includes DV from 3a.	
$\frac{v_f}{(y)} = \frac{m_A v_{A0}}{(m)} \frac{1}{(m_A + m_B)} + (b)$ (Dependent Qty) (slope) (Independent Qty) Or could be: $v_f = v_{A0} \frac{m_A}{m_A + m_B}$			
4b. Rewrite the slope term below (which includes your objective quantity):			
$slope = m_A v_{A0}$		0.5	Attempts to complete Appendix A.
4c. Rearrange and solve for your objective quantity in terms of slope:			
$\frac{slope}{m_A} = v_{A0}$ (Obj qty) (expression in terms of 'slope')		0.4	Correct OQ eqn You may also see them sqrt the eqn, s correct just make sure they plot the c IQ.
5. Write a hypothesis statement.			
As we _(increase)_ the _(mass of the stationary cart)_, we expect to observe a(n) _(linear)_ (decrease)_ in the _(combined mass velocity)_ based on _(Conservation of Linear Momentum)_.			
6. Experiment planning and execution			
6a. Review appendix A and design your experiment around your chosen IV and DV		0.1 Correct hypothesis statement	
5b. Complete appendix B detailing measurement procedures and uncertainties.			
6c. Complete appendic C for all trials, calculate uncertainties, and conduct linear regression process IAW the SLAM.			
6d. Calculate objective quantity and uncertainty using appendix D.			
6e. Plot experimental results on a number line to analyze precision and accuracy of findings.			

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass of cart B	m_B	digital scale	☒ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.32350	kg	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Describes thoroughly including taring scale. May also include details about how they added masses.

0.5

App B is generally completed.

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☒ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

Justifies thoroughly, most likely quality of the measuring device.

0.5

Reasonable justification of uncertainty

B6. Quantify the absolute uncertainty:

0.00003	kg
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

0.0093%
(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

0.32350	$\pm$	0.00003	kg
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☒ Yes
- ☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Mass of cart A	m_A	digital scale	☐ Independent Variable ☐ Dependent Variable ☒ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.25105	kg	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Describes thoroughly including taring scale. May also include details about how they added masses.

0.5

App B is generally completed.

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☒ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

Justifies thoroughly, most likely quality of the measuring device.

0.5

Reasonable justification of uncertainty

B6. Quantify the absolute uncertainty:

0.00003	kg
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

0.012%
(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

0.25105	$\pm$	0.00003	kg
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☒ Yes
- ☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Speed of Combined carts	v_f	PASCO Smart Cart	☐ Independent Variable ☒ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

0.392	$\frac{m}{s}$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Setup the experiment following worksheet (attached).
 Loaded cart A to two detent spring position and set along end stop then pressed record.
 Used a pencil to tap the spring release mechanism until the collision was complete then stopped recording.
 In PASCO setup speed vs time graph.
 Selected 5 data points right after the collision and averaged them.
 Performed an average and standard deviation in PASCO and recorded the value in our Appendix C.

0.5

App B is generally completed.

B4. Select which factor of uncertainty is dominant

- ☒ 1. Nature of the quantity
☒ 2. Quality of the measuring device
☐ 3. Conditions under which the measurement was performed
☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

The collision has some uncertainty with it as it is an imperfect system and it is not perfectly inelastic so there are some variations in speed. Additionally our sensor may not pick up those minute differences each time. To determine the uncertainty we used a standard deviation.

0.5

Reasonable justification of uncertainty

B6. Quantify the absolute uncertainty:

0.002	$\frac{m}{s}$
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

0.51%
(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

0.392	$\pm$	0.002	$\frac{m}{s}$
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☒ Yes
☐ No

Appendix C: Experimental Results and Linear Regression

Control Variable	Value	Abs Unc	Units
m_A	0.25690	0.00003	kg
0.5	Control variable is reported and is reasonable		

	Independent Variable	IV Absolute Uncertainty	Independent Quantity	Dependent Variable	DV Absolute Uncertainty	Dependent Quantity
Trial number	mass B [kg]	δm_B [kg]	$1/(mA+mB)$ [kg ⁻¹]	v [m/s]	δv [m/s]	v [m/s]
1	0.2498	0.00003	1.973554371	0.392	0.002	0.392
2	0.32355	0.00003	1.722801275	0.355	0.004	0.355
3	0.42365	0.00003	1.46939975	0.309	0.002	0.309
4	0.52375	0.00003	1.280983796	0.268	0.002	0.268
5	0.8218	0.00003	0.92704181	0.237	0.004	0.237
6						
7						
8						
9						
10						

1.0 Appendix C is generally completed correctly

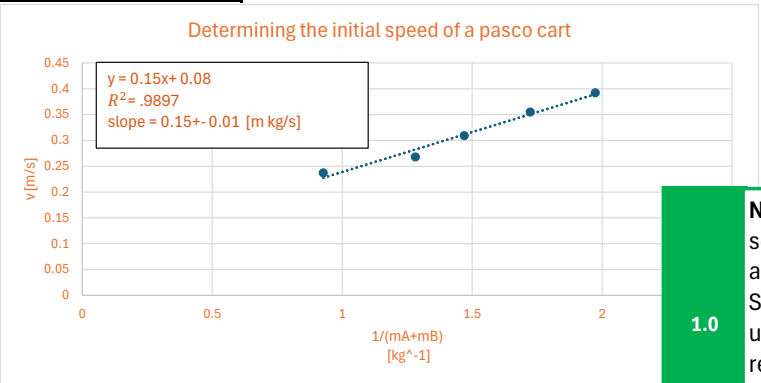
Initial Speed 'Accepted value	
Trial	Speed
1	0.571425
2	0.619545
3	0.607515
4	0.595485
5	0.6015
6	0.6015

0.5 Table is completely filled including variables and units in headers and absolute uncertainty of velocity changes from trial to trial or is reasonable. Includes accepted value data.

0.60 0.02

Note: Items in orange are placeholders that you must update.

Plot



1.0 Near perfect graph: Equation, R-squared, and slope are accurately and correctly reported to include SF and rounding (Slope uncertainty does not need to be reported here) Axes have units and are labelled correctly

Linear Regression results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.989664
R Square	0.979434
Adjusted R Square	0.972579
Standard Error	0.010414
Observations	5

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.015493	0.015493	142.8741	0.00126
Residual	3	0.000325	0.000108		
Total	4	0.015819			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.084143	0.01964	4.284367	0.023366	0.021641	0.146645	0.021641	0.146645
X Variable 1	0.15464	0.012937	11.95299	0.00126	0.113468	0.195813	0.113468	0.195813

0.15464 0.084143
0.012937 0.01964
0.979434 0.010414
142.8741 3
0.015493 0.000325

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	initial speed of cart A	v_A	kg
	(Objective Quantity Name)	(OQ variable)	(OQ units)

Objective quantity equation (From App A)	$\frac{\text{slope}}{m_A} = v_{A0}$
---	-------------------------------------

	(Obj qty)	=	(expression in terms of 'slope')
--	-----------	---	----------------------------------

Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope	0.15	0.01	kg m/s
m_A	0.25690	0.00003	kg
1.0	Fills in all App D Data.		
1.0	Attempts Objective quantity calculation.		
0.5	Attempts propagation of uncertainty		
	on above:		

$$\frac{v_{A0}}{(\text{Obj qty})} = \frac{0.60194737}{(\text{value})} \quad m/s \quad (\text{units})$$

Calculate the combined absolute uncertainty of the objective quantity:

$$\delta v(\text{slope}, m_A) = \sqrt{\left(\frac{\partial v}{\partial \text{slope}} \delta \text{slope}\right)^2 + \left(\frac{\partial v}{\partial m_A} \delta m_A\right)^2}$$

Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2
slope	$\frac{\partial v}{\partial \text{slope}} = \frac{1}{m_A}$	0.001515206
m_A	$\frac{\partial v}{\partial m_A} = -\frac{\text{slope}}{m_A^2}$	-7.02936E-05

1.0 Correct propagation of uncertainty (may differ based on their OQ eqn).

$$\frac{\delta v_{A0}}{(\text{Obj qty})} = \frac{0.038012009}{(\text{value})} \quad m/s \quad (\text{units})$$

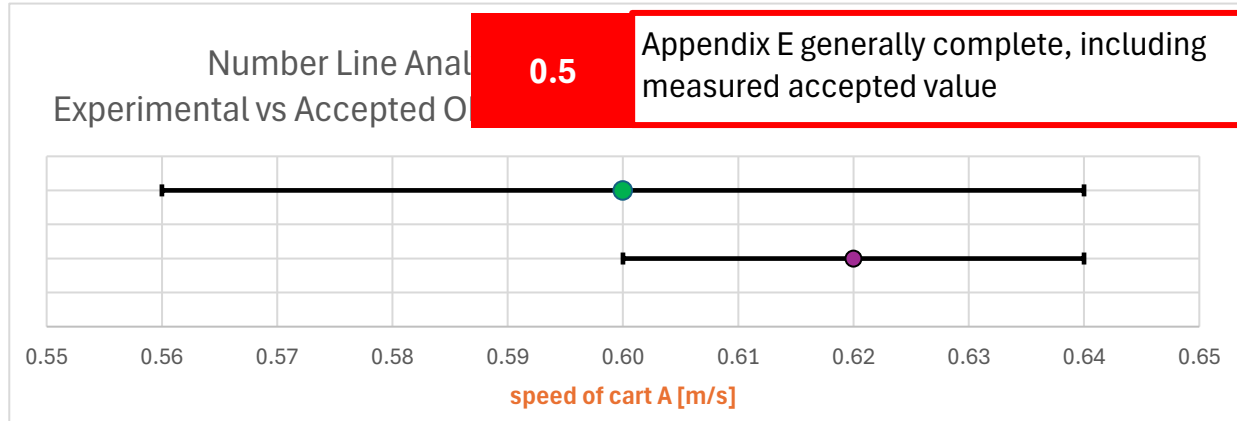
Report the objective quantity as a confidence interval

$v_{A0} =$	0.60	±	0.04	m/s
(objective quantity)			(absolute uncertainty, 1SF)	(units)

0.5 Completely correct confidence interval (rounding, SF, units).

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	0.60	$\pm$	0.04	$\frac{m}{s}$
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	0.62	$\pm$	0.02	$\frac{m}{s}$
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

☐	Systematic	Confidence intervals do not overlap
☒	Random	Confidence intervals do overlap

Acceptable level of Precision: 2.0 %
(default value for PH2X1 experiments)

Acceptable level of Accuracy: 5.0 %
(default value for PH2X1 experiments)

Calculate Relative Uncertainty (2SF)

Calculate Percent Difference (2SF)

0.5 Correct RU and PD with 2SF

RU = 6.7%

PD = 3.2%

Appendix F: Error Analysis Table

Move the **check** to the dominant form of error and follow that flow chart:



Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

slope

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

More measurements, improved speed measuring procedures (sampling closer to launch, using a different sensor)

0.5

Error type follows from App E.

0.5

Well reasoned / complete responses.

Note: If the high quadrature is the slope then discuss more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

Frictionless surface and level surface

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

Spring may not have contacted surface perfectly, release mechanism on cart is prone to friction if not careful

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

Position sensor can be noisy

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

Something from the above is listed.



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LAB# ONLINE POST LAB-N/A



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LAB#7 HANDOUT

Derivation using Conservation of Energy

For this pre-lab you can use either COE-R or N2L-R. Pick one but both are feasible.

1. A rigid body is released from rest at the top of an inclined track and rolls without slipping. The track is raised a distance h above the horizontal. See Figure 1. In this lab the rigid object may be a solid sphere, solid disk, or any other shape listed on the back of your PRC which can roll. The object has a moment of inertia, I , dependent upon its shape. Here we will use $I = cmr^2$ as the general form of the moment of inertia, where m is equal to the mass of the rigid body, r is equal to its outer radius, and c is a coefficient that ranges from 0 to 1 where a solid sphere has coefficient $c = \frac{2}{5}$, a hoop about its center $c = 1$, and a solid cylinder $c = \frac{1}{2}$ (all from the back of the PRC).

Using conservation of energy, **derive** an algebraic expression for the final velocity v of the rigid body once it reaches the bottom of the track. Express your answer in terms of c , h , and the acceleration due to gravity, g .

Note: The next lesson is COE with rotation. The only additional term to consider that we haven't formally introduced is rotational kinetic energy, $K_R = \frac{1}{2}I\omega^2$ where I is the moment of inertia and ω is the angular velocity where we can substitute $\omega = \frac{v}{r}$ to solve for v .



Figure 1: Rigid body capable of rolling being released from rest at the top of an inclined track.

The objective of this lab is to experimentally determine the moment of inertia coefficient, c , for a rigid object. We will do this by varying the height, h , from which the object is released. However, if we were to use the equation derived in problem 1, we would be required to measure the final velocity, v , at the bottom of the track, ensure the object is released perfectly from rest, and accurately measure the height the track is elevated. All of these requirements are extremely difficult and are potential sources of error. Measuring the final velocity is particularly difficult because the object is accelerating down the track so that its velocity is continually changing.

To reduce error in this experiment, we are going to manipulate the equation derived in Problem 1 using a kinematic equation. Our new equation will use the linear acceleration of the object, a , instead of the final velocity, v , and it will use the angle that the ramp makes with the horizontal, θ , instead of the height of the ramp, h . Since the acceleration is a constant, it will be a much more consistent measurement.

2. **Substitute** the expression for the final velocity from Problem 1 into the kinematic equation $v^2 = v_0^2 + 2a(x - x_0)$ to **derive** an expression for the translational acceleration, a , of the object. Express your answer in terms of c , θ , and the acceleration due to gravity, g . Assume the object is released at the origin and from rest ($v_0 = 0$, and $x_0 = 0$). Note: the distance the object rolls down the track, x , is the hypotenuse of a right triangle and $\frac{h}{x} = \sin(\theta)$. See Figure 1.

There are some significant advantages to using this new form as a model equation for this lab. First, once the object is released, its acceleration down the ramp is constant, which allows us to easily measure the acceleration using the PASCO motion sensor, as opposed to trying to measure a final velocity of an object that is continually accelerating. Second, using angle instead of height eliminates the requirement to release the object from the same spot. It won't matter at all where the ball is released as long as the angle of the track doesn't change while it is rolling. Last, we no longer have to ensure the object is released from rest.

3. For Lab 7, the objective is to experimentally determine the moment of inertia constant, c , for a rigid object capable of rolling down a track. We will vary the angle, θ , at which the object is released and measure the resultant translational acceleration, a , of the object. **Complete** appendix A based on the equation you developed in Problem 2.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

In this experiment...

1b. Objective quantity for this experiment:
(objective quantity, OQ)

Objective quantity

OQ

units

(Objective Quantity Name) (OQ variable) (OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

2b. Apply equations from 2a into a model equation:

Type equation here.

Type equation here.

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

independent variable

IV

units

(Independent Variable Name) (Independent Variable) (units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

dependent variable

DV

units

(Dependent Variable Name) (Dependent Variable) (units)

3c. The variables in the model equation which will not
change between trials:
Fill in or blank out as appropriate.

control variable 1

CV1

units

(Control Variable Name) (Control Variable) (units)

control variable 2

CV2

units

(Control Variable Name) (Control Variable) (units)

control variable 3

CV3

units

(Control Variable Name) (Control Variable) (units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

DQ

=

m

IQ

(y)

=

(m)

(x)

+

(b)

(Dependent Qty) (slope) (Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = ?

4c. Rearrange and solve for your objective quantity in terms of slope:

OQ = expr in terms of slope

(Obj qty)

=

(expression in terms of 'slope')

5. Write a hypothesis statement.

Note: Items in orange are to be filled in by the student.

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>y</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☐ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

(alt-enter to go to next line)

B6. Quantify the absolute uncertainty:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☐ Yes
- ☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>y</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☐ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

(alt-enter to go to next line)

B6. Quantify the absolute uncertainty:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☐ Yes
- ☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

	<i>y</i>		☐ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

	<i>units</i>	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

(alt-enter to go to next line)

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☐ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

(alt-enter to go to next line)

B6. Quantify the absolute uncertainty:

	<i>units</i>
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

	$\pm$		<i>units</i>
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☐ Yes
- ☐ No

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	Objective quantity	OQ	units
	(Objective Quantity Name)	(OQ variable)	(OQ units)

Objective quantity equation (From App A)	OQ = expr in terms of slope (From App A)
---	--

	(Obj qty)	=	(expression in terms of 'slope')
--	-----------	---	----------------------------------

Variable from equation above	Measured Value	Absolute Uncertainty	Units
slope			units
var			

Calculate the objective quantity using the variables and equation above:

List OQ equation with variables here

List OQ equation with values here

OQ	=	use excel to calculate here	units
(Obj qty)		(value)	(units)

Calculate the combined absolute uncertainty of the objective quantity:

$$\delta f(x, slope) = \sqrt{\left(\frac{\partial f}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial x}{\partial x} \delta x\right)^2} \text{ [EXAMPLE]}$$

Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2
slope	$\frac{\partial OQ}{\partial slope} = \text{FILL}$	
var	Type equation here.	
var	Type equation here.	
var	Type equation here.	

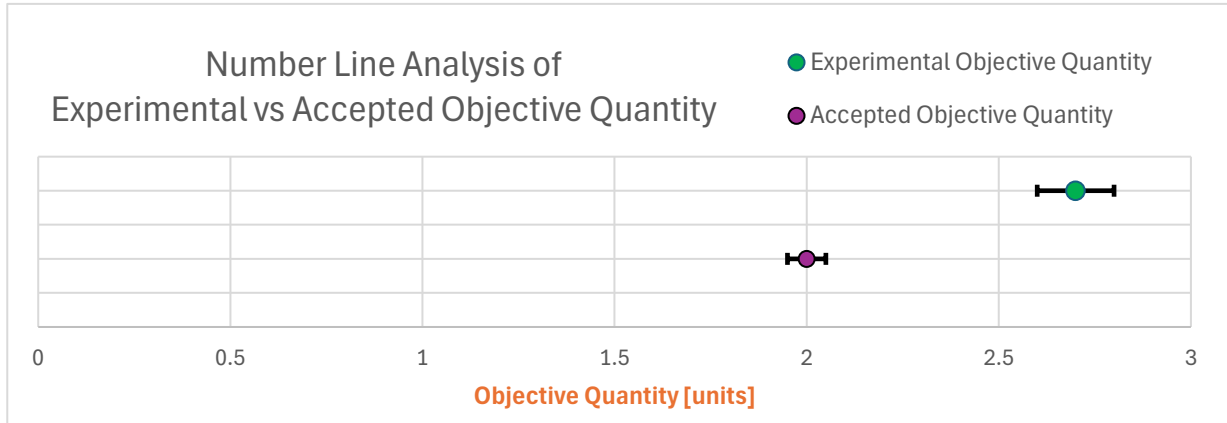
δOQ	=	use excel to calculate here	units
(Obj qty)	=	(value)	(units)

Report the objective quantity as a confidence interval

	OQ =	use excel to calculate here	±	use excel to calculate here	units
	(objective quantity)			(absolute uncertainty, 1SF)	(units)

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	2.7	$\pm$	0.1	units
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	2	$\pm$	0.05	unitless
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

- ☐ Systematic Confidence intervals do not overlap
- ☐ Random Confidence intervals do overlap

Acceptable level
of Precision: 5.0%

(default value for PH2X1 experiments)

Acceptable level of
Accuracy: 5.0%

(default value for PH2X1 experiments)

Calculate Relative Uncertainty (2SF)

$RU =$

RU = 99%

Calculate Percent Difference (2SF)

$PD =$

PD = 99%

Appendix F: Error Analysis Table

Move the **check** to the dominant form of error and follow that flow chart:



Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

var

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

1. Complete the appendices for your experiment
2. For multiple Appendix B's right click the sheet and click 'move or copy', select create a copy and put it in this world
3. To compile your report click:
 - File
 - Save as adobe pdf OR print to pdf
 - select all appropriate sheets (except this one)
 - Review the pdf to ensure all required information is present.
4. If you need to adjust pdf spacing use the 'page layout' tab on the sheet you need to adjust

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LAB#7 SOLUTION

Pre-Lab Tasks

1. A rigid body is released from rest at the top of an inclined track and rolls without slipping. The track is raised a distance h above the horizontal. See Figure 1. The rigid object may be a solid sphere, solid disk, or any other shape listed on the back of your PRC that can roll. The object has a moment of inertia, I , dependent upon its shape. Here we will use $I = cmr^2$ as the general form of the moment of inertia, where m is equal to the mass of the rigid body, r is equal to its outer radius, and c is a coefficient that ranges from 0 to 1 where a solid sphere has coefficient $c = \frac{2}{5}$, a hoop about its center $c = 1$, and a solid cylinder $c = 0.5$ (all from the back of the PRC).

Using conservation of energy, **derive** an algebraic expression for the final velocity v of the rigid body once it reaches the bottom of the track. Express your answer in terms of c , h , and the acceleration due to gravity, g .

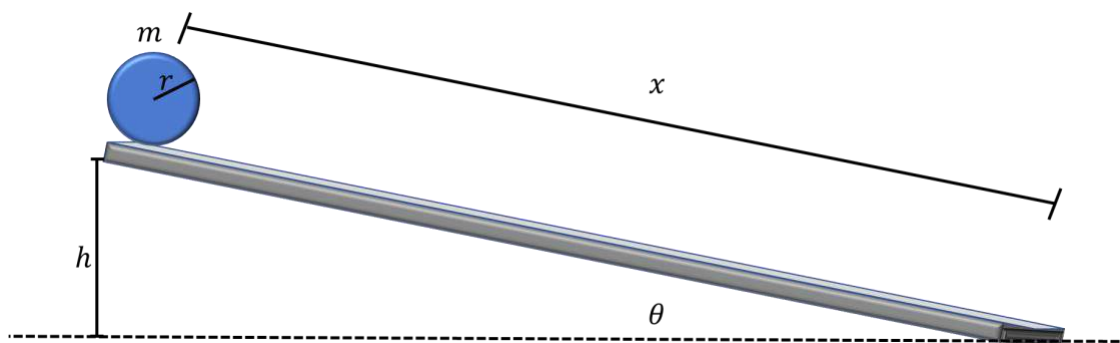


Figure 1: Rigid body capable of rolling being released from rest at the top of an inclined track.

Starting with Conservation of Energy

$$W_{NC} = E_f - E_0$$

Rolling without slipping implies $W_{nc} = 0$ and that we can later use a link equation. Further assuming no initial velocity yields.

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Substituting in $I = cmr^2$ and the link equation that $v = r\omega$ yields.

$$2gh = v^2 + cv^2$$

Solving for v

$$v = \sqrt{\frac{2gh}{(c+1)}}$$

The objective of this lab is to experimentally determine the moment of inertia constant, c , for a rigid object. We will do this by varying the height, h , from which the object is released. However, if we were to use the equation derived in problem 1, we would be required to measure the final velocity, v , at the bottom of the track, ensure the object is released perfectly from rest, and accurately measure the height the track is elevated. All of these requirements are extremely difficult and are potential sources of error. Measuring the final velocity is particularly difficult because the object is accelerating down the track so that its velocity is continually changing. Consequently, we are going to manipulate the equation derived in Problem 1 using a kinematic equation. Our new equation will use the linear acceleration of the object, a , instead of the final velocity, v , and it will use the angle that the ramp makes with the horizontal, θ , instead of the height of the ramp, h . Since the acceleration is a constant, it will be much easier to measure than the final speed.

2. **Substitute** the expression for the final velocity from Problem 1 into the kinematic equation $v^2 = v_0^2 + 2a(x - x_0)$ to **derive** an expression for the translational acceleration, a , of the object. Express your answer in terms of c , θ , and the acceleration due to gravity, g . Assume the object is released at the origin and from rest ($v_0 = 0$, and $x_0 = 0$). Note: the distance the object rolls down the track, x , is the hypotenuse of a right triangle. See Figure 1.

Starting with our kinematic equation

$$v^2 = v_0^2 + 2a(x - x_0)$$

We can substitute in our expression for velocity from Problem 1 along with our initial conditions giving:

$$\frac{2gh}{(c + 1)} = 2a(x)$$

Solving for h/x gives

$$\frac{h}{x} = \frac{a(c + 1)}{g}$$

Substituting in that $\sin \theta = \frac{h}{x}$ and solving for a gives

$$a = \frac{g \sin \theta}{(c + 1)}$$

There are some significant advantages to using this new form as a model equation for this lab. First, once the object is released, its acceleration down the ramp is constant, which allows us to easily measure the acceleration using the PASCO motion sensor, as opposed to trying to measure a final velocity of an object that is continually accelerating. Second, using angle instead of height eliminates the requirement to release the object from the same spot. It won't matter at all where the ball is released as long as the angle of the track doesn't change while it is rolling. Last, we no longer have to ensure the

object is released from rest. You can actually push the object up or down the track when you release it, and it won't matter at all for data collection.

3. For Lab 7, the objective is to experimentally determine the moment of inertia constant, c , for a rigid object capable of rolling down a track. We will vary the angle, θ , at which the object is released and measure the resultant translational acceleration, a , of the object. **Complete** an appendix A based on the equation you developed in Problem 2.

Linearization Table

APPENDIX A

Step 1: Identify fundamental physics Conservation of Energy Moment of Inertia Translational Kinetic Energy Rotational Kinetic Energy Gravitational Potential Energy Kinematic Equations	Step 2: Apply physics in order to build a model equation $a = \frac{g \sin \theta}{(c + 1)}$
Step 3: Identify independent and dependent variables Independent Variable: The angle of the track, θ Dependent Variable: The translational acceleration of the object, a	Step 4: Rearrange equation to $y = mx + b$ form to build the linearized model equation. $a = \frac{g}{(c + 1)} \sin \theta$
Step 5: Identify independent and dependent quantities Independent Quantity: $\sin \theta$ Dependent Quantity: a	Step 6: Identify the objective quantity and control variables Objective Quantity: c Control Variables: g
Step 7: Write the expression for the slope of your equation $\text{slope} = \frac{g}{c + 1}$	Step 8: Rearrange your slope so that you solve for your objective quantity equation $c = \frac{g}{\text{slope}} - 1$

Table 1: Linearization Table for a linearization experiment designed to experimentally determine the moment of inertia constant for a rigid object capable of rolling.

Lab 7 Score Distribution and Appendix Cuts

Deliverable	Description	Points
Entire Report	Formatting and adherence to style guide	3
Introduction	Content follows guidelines in reading and formatted appropriately	4
Theory	Content follows guidelines in reading and formatted appropriately	12
Procedures	Content follows guidelines in reading and formatted appropriately	11
Results	Content follows guidelines in reading and formatted appropriately	11
Discussion	Content follows guidelines in reading and formatted appropriately	11
Conclusion	Content follows guidelines in reading and formatted appropriately	4
Appendices	Check for completeness assign 0-4 pts based on amount completed.	4

Lab 7 Grading Rubric Distribution

Max Points		Poor		Marginal		Good		Exceptional	
		(Pts)	Description	(Pts)	Description	(Pts)	Description	(Pts)	Description
Formatting	3	(0)	No attempt made to conform to style guide.	(1)	Formatting is generally consistent, though does not adhere to guide.	(2)	Formatting is generally consistent, though does not adhere to guide.	(3)	Formatting completely follows the style guide.
Introduction	4	(0-1)	Lacks clarity, no clear objective, minimal or no description of how experiment was conducted	(2)	Objective is unclear or some description of experiment provided but lacks relevance or detail.	(3)	Generally clear objective/description but some elements not fully described.	(4)	Clear, concise, relevant objective with description of how it was accomplished
Theory	12	(0-6)	Minimal or unclear theory that lacks substance or is generally incorrect.	(7-8)	Theory present but vague, lacks sufficient explanation / not effectively communicated.	(9-10)	Theory explained well, with minor omissions of key information from theory checklist.	(11-12)	Thorough, clear, and relevant theory explained with assumptions. Uses correct physics to support theory. Nearly all elements of checklist discussed
Procedures	11	(0-6)	Nearly impossible to repeat experiment. Incomplete or unclear procedures.	(7-8)	Difficult to repeat experiment. Steps taken are included but missing important details (e.g. entire variables not described).	(9-10)	Very possible to repeat experiment. Descriptive steps but limited discussion of uncertainty in measurements/confidence intervals/ other essential elements from the checklist.	11	Extremely easy to repeat experiment. Complete, detailed, and well-organized procedures. Nearly all checklist elements discussed.
Results	11	(0-6)	Does not report objective quantity results and/or lacks any effort. Unclear results reported. Lacks tables, graphs, figures, or any data.	(7-8)	Does not report objective quantity results though some effort to report results is evident. Results are unclear/not supported by writing and/or provides sparse data.	(9-10)	Reports the objective quantity results and writing supports results. Results are clear and introduced in the written report, though missing some details e.g. one table is missing, a plot isn't well described or incorporated, fails to report the slope, etc.	11	Reports the objective quantity full confidence interval completely correctly with all key supporting data. Clear results, well-formatted tables/graphs are error-free and introduced in the report.
Discussion	11	(0-6)	Poor writing that shows no critical thought or effort. Written analysis is almost completely missing or does not demonstrate reasonable thinking to identify sources of error and recommend changes.	(7-8)	Writing is lacking in critical thought or effort. Written analysis which attempts to describe the error. Fails to identify the correct error, does not recommend a reasonable improvement.	(9-10)	Writing attempts critical thought/has some effort. Generally well written analysis. Uses quantitative or qualitative descriptions to report on error, identifies a correct form of error, and recommends changes that may or may not be helpful/reasonable.	11	Extremely well written and thoroughly analyzed. Uses quantitative and qualitative descriptions to report on error, identifies a correct form of error, and recommends meaningful changes to improve the experiment.
Conclusions	4	(0-1)	Poor writing with no takeaways from experiment. Bare bones conclusions that is almost completely missing details and has no/incorrect takeaways from the experiment.	(2)	Writing has few takeaways from experiment. Synthesizes some conclusions though missing many details. May report a finding that is in no way related or supported by the rest of the report.	(3)	Well written with some takeaways from experiment. Restates the objective quantity and restates some form of error analysis from the discussion. Synthesizes some findings that are generally correct, though may not be fully supported by the report.	(4)	Extremely well written with synthesized takeaways from experiment. Restates the objective quantity, includes precision and accuracy, restates error and synthesizes findings that follow from the rest of report.
Appendices	4	(0-1)	No appendices attached.	(2)	Appendices attempted but mostly incomplete/incorrect	(3)	Appendices attached with frequent minor issues in completeness or correctness.	(4)	Appendices attached and nearly fully correct, a few minor issues are ok.

Appendix A: Experiment Overview and Linearization Table

1. Experiment Goal - Outline your experiment objective

1a. Describe the goal for this experiment:

We will determine the moment of inertia coefficient for a rigid body by varying the angle of a ramp and measuring the acceleration.

1b. Objective quantity for this experiment:
(objective quantity, OQ)

moment of inertia constant

c

unitless

(Objective Quantity Name)

(OQ variable)

(OQ units)

2. Physics Model Equations - Create equation which includes OQ

2a. List the physics equations associated with the OQ

$$W_{NC} = E_F - E_0$$
$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

2b. Apply equations from 2a into a model equation:

$$v = \left(\frac{2gn}{c+1}\right)^{.5}$$
$$v^2 = 2ax$$
$$\frac{2gh}{c+1} = 2ax$$
$$\frac{h}{x} = \frac{a(c+1)}{g} \text{ where: } \frac{h}{x} = \sin(\theta)$$
$$a(c+1) = g\sin(\theta)$$

3. Determining Independent and dependent variables for your experiment

3a. The variable changing between trials:
(Independent variable, IV)

angle of the track

θ

degrees

(Independent Variable Name)

(Independent Variable)

(units)

3b. Changing the IV will change the following variable:
(Dependent variable, DV)

acceleraton of object

a

m/s²

(Dependent Variable Name)

(Dependent Variable)

(units)

3c. The variables in the model equation which will not change between trials:
Fill in or blank out as appropriate.

acceleration due to gravity

g

m/s²

(Control Variable Name)

(Control Variable)

(units)

(Control Variable Name)

(Control Variable)

(units)

(Control Variable Name)

(Control Variable)

(units)

4. Linearized Equation and Objective Quantity Calculation

4a. Rearrange 2b. into a Linearized Equation (y=mx+b)

$$a(c+1) = g\sin(\theta)$$
$$a = \frac{g\sin(\theta)}{c+1}$$

The 'm' term must include your objective quantity variable from 1b.
'x' term must include your independent variable from 3b.
'y' term includes DV from 3a.

$$a = \frac{g}{c+1} \sin(\theta)$$

(y)

=

(m)

(x)

+

(b)

(Dependent Qty)

(slope)

(Independent Qty)

4b. Rewrite the slope term below (which includes your objective quantity):

slope = g/(c+1)

4c. Rearrange and solve for your objective quantity in terms of slope:

$$c = \frac{g}{slope} - 1$$

(Obj qty)

=

(expression in terms of 'slope')

5. Write a hypothesis statement.

Note: Items in orange are not to be used in your hypothesis statement.

Appendix B: Measured quantity

B1. Physical quantity measured:

Angle of Track	θ	Phone angle sensor	☒ Independent Variable ☐ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

8.70	degrees	<i>Note: if multiple measurements were made provide one example from appendix C</i>
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

Downloaded an app that read angle data. Placed phone horizontally with the screen facing us along the track and recorded the angle in degrees to the nearest second decimal place.

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☒ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

The phone app does not report uncertainty so 1/2 of the lowest resolution was used as the absolute uncertainty.

B6. Quantify the absolute uncertainty:

0.005	degrees
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

0.057%
(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

8.70	$\pm$	0.005	degrees
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☒ Yes
- ☐ No

Appendix B: Measured quantity

B1. Physical quantity measured:

Acceleration	a	m/s^2	☐ Independent Variable ☒ Dependent Variable ☐ Control Variable
(name)	(variable)	(tool used)	Variable type (Check one)

B2. Measured Value:

1.04	$\frac{m}{s^2}$	Note: if multiple measurements were made provide one example from appendix C
(number)	(units)	

B3. Describe the measurement procedures, be specific so it is repeatable:

We setup a bluetooth motion sensor at the top of a track (upside down) directed down the angle of the track so as to capture the path of the object. We setup the object 10 cm from the motion sensor so that when we release the ball it rolls down the ramp. We setup pasco graph display to measure velocity vs time then hit record in pasco and released the object. We stopped the object near the bottom of the ramp and then stopped recording. We then highlighted a linear section of velocity and conducted a linear regression of the velocity vs time, then the slope of the line with uncertainty is our acceleration.

B4. Select which factor of uncertainty is dominant in this measurement:

- ☐ 1. Nature of the quantity
- ☒ 2. Quality of the measuring device
- ☐ 3. Conditions under which the measurement was performed
- ☐ 4. Limitations of humans

B5. Justify the dominant factor of uncertainty:

The nature of the quantity is well defined, the conditions are ideal, and no limitations of humans are dominant. The quality of the measuring device is the dominant factor and depends on the regression.

B6. Quantify the absolute uncertainty:

0.01	m/s^2
(absolute uncertainty, 1 significant figure)	(units)

B7. Calculate your relative uncertainty, report as a percent.

1.0%
(relative uncertainty, 2 significant figures)

B8. Combine B2 and B6 and write your measurement as a confidence interval below

1.04	$\pm$	0.01	m/s^2
(B2, measured value)		(B6, absolute uncertainty [1 SF])	(B2, units)

B8. Common sense check: Does this measurement seem reasonable?

- ☐ Yes
- ☐ No

ta is for a sphere

Control Variable	Value	Abs Unc	Units

	Independent Variable	IV Absolute Uncertainty	Independent Quantity	SPHERE ACCELERATION	DV Absolute Uncertainty	Dependent Quantity
Trial number	Angle	δIV	sin(theta)	a_x	δDV	DQ
1	8.70	0.005	0.15126082	1.04	0.01	1.04
2	7.15	0.005	0.1244674	0.85	0.02	0.85
3	4.88	0.005	0.08506913	0.573	0.001	0.573
4	2.46	0.005	0.04292191	0.288	0.001	0.288
5	10.50	0.005	0.18223553	1.27	0.01	1.27
6	12.5	0.005	0.21643961	1.51	0.01	1.51

Plot

Angle influence on linear acceleration of a Sphere

acceleration (m/s²)

sin(theta), unitless

$y = mx + b$
 $R^2 = 0.9999$
slope = 6.92 $\pm$ 0.03 m/s²

$y = 7.0692x - 0.0235$

Linear Regression results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.999938
R Square	0.999876
Adjusted R Square	0.799876
Standard Error	0.012324
Observations	6

ANOVA

	df	SS	MS	F	ignificance F
Regression	1	6.107614	6.107614	40216.13	3.71E-09
Residual	5	0.000759	0.000152		
Total	6	6.108373			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
X Variable 1	6.921986	0.034517	200.5396	5.85E-11	6.833258	7.010714	6.833258	7.010714

Note: Items in orange are placeholders that you must update.

7.005

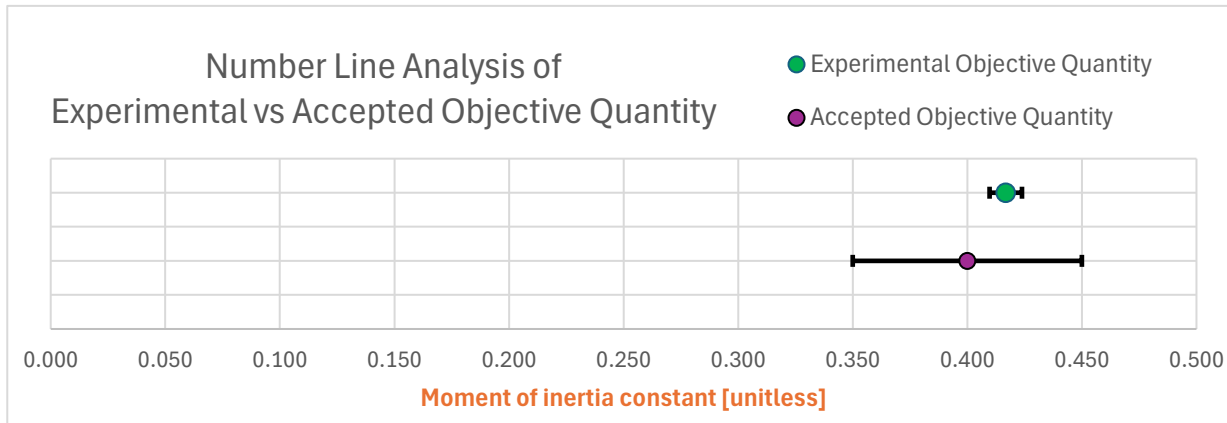
0.41679

Appendix D: Objective Quantity Calculation and Propagation of Uncertainty

Objective quantity (From App A)	moment of inertia constant	c	unitless		
(Objective Quantity Name)		(OQ variable)	(OQ units)		
Objective quantity equation (From App A)	$c = \frac{g}{slope} - 1$				
(Obj qty) = (expression in terms of 'slope')					
Variable from equation above	Measured Value	Absolute Uncertainty	Units		
slope	6.921985799	0.0345	m/s ²		
g	9.807	0.001			
Calculate the objective quantity using the variables and equation above:					
c = 0.416789962 unitless (Obj qty)(value)(units)					
Calculate the combined absolute uncertainty of the objective quantity:					
$\delta c(g, slope) = \sqrt{\left(\frac{\partial c}{\partial slope} \delta slope\right)^2 + \left(\frac{\partial c}{\partial g} \delta g\right)^2}$					
Variable from equation above	Associated Partial Derivative	Quadrature (partial derivative * abs unc)^2			
slope	$\frac{\partial c}{\partial slope} = -g \, slope^{-2}$	4.99127E-05			
g	$\frac{\partial c}{\partial g} = slope^{-1}$	2.08708E-08			
δc = 0.007066366 unitless (Obj qty)(value)(units)					
Report the objective quantity as a confidence interval					
	c =	0.417	±	0.007	unitless
	(objective quantity)			(absolute uncertainty, 1SF)(units)	

Appendix E: Precision and Accuracy Table

Experimental Objective Quantity	0.417	$\pm$	0.007	unitless
(experimental objective quantity)			(absolute uncertainty)	(units)
Accepted Objective Quantity	0.40	$\pm$	0.05	unitless
(accepted objective quantity)			(absolute uncertainty)	(units)



Select the dominant form of error

☐	Systematic	Confidence intervals do not overlap
☒	Random	Confidence intervals do overlap

Acceptable level of Precision:	5.0%
(default value for PH2X1 experiments)	

Acceptable level of Accuracy:	5.0%
(default value for PH2X1 experiments)	

Calculate Relative Uncertainty (2SF)

$RU =$

RU = 1.7%

Calculate Percent Difference (2SF)

$PD =$

PD = 4.2%

Appendix F: Error Analysis Table

Move the **check** to the dominant form of error and follow that flow chart:



Random Error Dominates

R1. Identify term in Appendix D with highest quadrature term:

slope

R2. Discuss ways to reduce the uncertainty in that variable in the space below:

We could take more measurements and also make sure that our track is evenly supported throughout the entire length. Additionally we could try to improve our angle measurements with more accurate tools.

Note: If the high quadrature is the slope then discuss improvement to slope through taking more measurements, improving procedures, improving measuring devices, etc.

Note: If the high quadrature is not the slope then discuss improvement of uncertainty of the variable through minimizing the primary factor of uncertainty listed in its App. B

Systematic Error Dominates

S1. Identify potential flaws in theory or assumptions. Address incorrect assumptions or math errors present in the process.

S2. Identify potential flaws in the procedures. Address any errors that may have occurred during the experiment which contributed to error.

S3. Identify any extraneous conditions which could have impacted your experiment. These should be either environmental lab conditions or equipment limitations.

S4. Suggest improvements to improve subsequent iterations of this experiment. Use your analysis above to guide your suggestions.

1. Complete the appendices for your experiment
2. For multiple Appendix B's right click the sheet and click 'move or copy', select create a copy and put it in this world
3. To compile your report click:
 - File
 - Save as adobe pdf OR print to pdf
 - select all appropriate sheets (except this one)
 - Review the pdf to ensure all required information is present.
4. If you need to adjust pdf spacing use the 'page layout' tab on the sheet you need to adjust

kbook.



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LAB#7 ONLINE POSTLAB -N/A



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LAB#8 HANDOUT



Labs 8,9,10: Design, Conduct, and Report on an Experiment



- Labs 8/9/10 are a sequence
 - Lab 8: Designing an experiment
 - Lab 9: Conducting the experiment + Lab Report
 - Lab 10: Orally present your experiment
- You will be a part of your group for the entire sequence.
- Guiding principles to keep in mind:
 - Be creative
 - Collaborate/be a team player
 - Have a plan (and be adaptable)
 - Communicate effectively (written and oral)
 - Prepare to defend your work



- **For Lab 8 you will complete an experiment design: [5 pts]**
 1. Use this slide deck for some ideas of an experiment you could conduct or produce your own.
 2. You may redo a previous lab, but it must incorporate improved procedures.
 3. Complete a single 'Experiment Concept Slide' for your experiment with your group and an 'Appendix A' for the experiment.
 4. Discuss both documents with your instructor before the end of the lab period.
 5. Submit both documents to your instructor through Canvas.
- **For Lab 9 you will complete the experiment and submit a lab report:**
 1. Conduct your experiment to determine your objective quantity. You may deviate from your original plan as you encounter challenges.
 2. Submit a completed lab report following the due date on Canvas (~1 week).
- **For Lab 10 you will present your findings:**
 1. Prepare a presentation that communicates your experimental goals, theory, procedures, results, and discusses future improvements to the experiment.
 2. Present in the lab 10 period with your group. (presentation details and guidelines follow)

There will be a peer assessment at the end of the sequence.



- **Discuss your experiment concept and Appendix A with your instructor.** Submit Appendix A and Experiment Concept Slide
- **Makeup procedures:** When assigned to a group, back brief your instructor the experiment plan at the start of Lab 9.

<u>Item</u>	<u>5 Points</u>	<u>Description</u>
Appendix A	1	Complete and correct linearized experiment
Experiment Concept slide – Discussion with Instructor	1	Discusses the concept slide with their instructor during the lab.
Experiment Concept slide – Execution steps	1	Descriptive execution steps
Experiment Concept slide – Equipment List	1	Thorough equipment list
Experiment Concept slide – Figure	1	Figure labelled and reasonably formatted



- **Submit a complete lab report after conducting your experiment.** Remember your instructor does not know what you did so you need to be detailed in your reporting.
- **Makeup procedures:** Group member missing a section will coordinate with their group to contribute to the lab report. The instructor has the discretion to schedule a makeup data collection session.

<u>Item</u>	<u>60 Points</u>	<u>Description for Full Points</u>
Introduction	3	Clear, concise, relevant objective with description of how it was accomplished
Theory	12	Thorough, clear, and relevant theory explained with assumptions. Uses correct physics to support theory. Nearly all elements of checklist discussed
Procedures	12	Extremely easy to repeat experiment. Complete, detailed, and well-organized procedures. Nearly all checklist elements
Results	12	Reports the objective quantity full confidence interval completely correctly with all key supporting data. Clear results, well-formatted tables/graphs are error-free and introduced in the report.
Discussion	12	Extremely well written and thoroughly analyzed. Uses quantitative and qualitative descriptions to report on the most significant error in the experiment.
Conclusion	3	Extremely well written with synthesized takeaways from experiment. Restates the objective quantity, includes precision and accuracy, restates error and recommends an improvement to the experiment which follows from the rest of report.
Formatting	2	Follows style guide.
Appendices	4	Fully and correctly completed



- **Present your experiment to the class. Your instructor will ask clarifying questions.**
- **Timing:** 10-12 minutes per group, 2-3 minutes for questions
- **Formatting:** Use MLA citations and the slide template provided on Canvas
- **Makeup procedures:** A group member who misses lab 10 will coordinate with their instructor for a brief oral assessment on their experiment (~3-4 mins but at the instructor's discretion.)

<u>Item</u>	<u>45 Points</u>	<u>Description</u>
Introduction	1	Introduces the group, provides an overview of the experiment, and references how the experiment was conducted
Theory	6	Describes the relevant equations used, the independent/dependent variables/quantities, and an objective quantity equation derived from a linearized form. Discusses assumptions. Explains the fundamental physics concepts and equations in the experiment.
Procedures	6	Describes the procedures to conduct the experiment in a logical sequence. Discusses individual measurement tools and uncertainties. Identifies and explains the greatest factor of uncertainty in their measurements.
Results	6	Presents the linearized plot, objective quantity calculation, and objective quantity uncertainty.
Discussion	6	Discusses precision and accuracy of the results and discusses factors that contributed to error. Identifies the greatest source of error in their experiment.
Conclusion	1	Succinctly concludes by reiterating the outcomes of their experiment. Presents a method to improve the greatest error in subsequent iterations.
Formatting	1	Uses slide template, formatting is consistent and helps to effectively communicate.
Novelty	2	How unique is this experiment? 0-exact reproduction of previous in lab experiment, 1-2 minor variations from previous experiment, 3 major variations from previous or new experiment.
Timing	1	Falls within the 10–12 minute window
Peer Review	15	Points assigned based on instructor's review of peer reviews.



The following experiment ideas are partially completed and only intended to be a starting point.

This list is not exhaustive, you are encouraged to produce your own experiments, using this format to plan the experiment.



Overview

We will drop a ball from varying heights and measure the velocity before impact. Using 1D kinematics we can solve for the acceleration due to gravity. This is an incomplete overview.

Objective Quantity	v_0
Physics Concept	1D Kinematics PRC [3] Height [m]
What will vary? (IV)	Heigh [m]
What will change due to the variation? (DV)	Velocity [m/s]
Control Variables	Acceleration due to gravity
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a PASCO friction box with a felt bottom, determine the coefficient of static friction between the felt and the aluminum track.

Objective Quantity	μ_s felt-aluminum
Physics Concept	N2L, PRC [15,17]
What will vary? (IV)	Mass of the box [kg]
What will change due to the variation? (DV)	Tension on PASCO Force Sensor [N]
Control Variables	
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a PASCO friction box with a cork bottom, determine the coefficient of kinetic friction between the cork and the aluminum track.

Objective Quantity	μ_k cork-aluminum
Physics Concept	N2L, PRC [15,18]
What will vary? (IV)	Mass of the box [kg]
What will change due to the variation? (DV)	Tension on PASCO Force Sensor [N]
Control Variables	
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a PASCO smart cart and PASCO track determine the work by non-conservative forces (friction on track, on axle, imprecise track alignment etc.)

Objective Quantity	Different methods exist
Physics Concept	
What will vary? (IV)	
What will change due to the variation? (DV)	
Control Variables	
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a balloon, determine the acceleration due to gravity based on varying its mass and measuring its terminal velocity.

Objective Quantity	$g \left[\frac{m}{s^2} \right]$
Physics Concept	N2L, PRC [15,19]
What will vary? (IV)	Mass of the balloon [kg]
What will change due to the variation? (DV)	Terminal speed [m/s]
Control Variables	Drag coefficient, density of air, cross sectional area
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a simple pendulum, determine the acceleration due to gravity by varying the height of the pendulum (initial position) and measuring the velocity at the bottom.

Objective Quantity	$g \left[\frac{m}{s^2} \right]$
Physics Concept	COE, PRC [27-30]
What will vary? (IV)	Height at initial position [m]
What will change due to the variation? (DV)	Velocity at lowest position [m/s]
Control Variables	Length of pendulum, mass
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a simple pendulum, determine the acceleration due to gravity by varying the length of the pendulum and measuring the period ([Some help](#) to get started)

Objective Quantity	$g \left[\frac{m}{s^2} \right]$
Physics Concept	Height at initial position [m]
What will vary? (IV)	Mass of the box [kg]
What will change due to the variation? (DV)	Period [s]
Control Variables	
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Determine the spring constant of a ballistic launcher spring by varying the displacement and measuring the speed of the ball when launched.

Objective Quantity	Spring constant, k , of ballistic launcher
Physics Concept	COE, PRC [27-30]
What will vary? (IV)	Spring displacement[m]
What will change due to the variation? (DV)	Launch speed of ball [m/s]
Control Variables	
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a metal arm with a ball catcher on one end and a projectile launcher, calculate the moment of inertia of the arm.

Objective Quantity	$I_{arm} [kg\ m^2]$
Physics Concept	Cons of Ang. Mom., PRC [44,45,46]
What will vary? (IV)	Initial velocity of projectile [m/s]
What will change due to the variation? (DV)	Angular velocity of system [rad/s]
Control Variables	Projectile mass, distance from AOR to catcher
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



Overview

Given a pulley-mass system (Awtod machine) vary one mass measure the angular acceleration (or linear acceleration with link) to determine the moment of inertia of the pulley.

Objective Quantity	$I_{pulley} [kg\ m^2]$
Physics Concept	N2L-R, PRC [22,23]
What will vary? (IV)	Mass 2 hanging from pulley [kg]
What will change due to the variation? (DV)	Angular acceleration of system [rad/s^2]
Control Variables	Mass 1
'Accepted' OQ Value	

INSERT AN EXPERIMENT OVERVIEW
FIGURE HERE

Equipment Required

- ☐ Thing 1
- ☐ Thing 2

Group member responsibilities

Rutherford, Ernest	PASCO, App C, App D
Einstein, Albert	Intro, Theory, Discussion, App F
Bohr, Niels	Setup experiment, Procedures Section
Curie, Marie	App A, App B, App E, and managing final review of report

Experiment Execution Steps

1. Gather equipment from list
2. Setup experiment following figure
3. Measure
4. Measure
5. Record
6. Repeat for X trials
7. Linearize
8. Review results, re-run trials as needed
9. Report



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LAB#8 SOLUTION

Lab 8 Cuts

Deliverable	Description	Points	Cut
Appendix A	Complete and correct linearized experiment	1	1 - Fully complete and correct .75 - Missing minor elements, generally complete .5 - Half completed, missing major elements .25 - Missing most elements 0 - no submission
Experiment Concept slide – Discussion with Instructor	Discusses the concept slide with their instructor during the lab.	1	1 - Fully complete .5 - Partially complete 0 - Incomplete
Experiment Concept slide – Execution steps	Descriptive execution steps	1	1 - Fully complete .5 - Partially complete 0 - Incomplete
Experiment Concept slide – Equipment List	Thorough equipment list	1	1 - Fully complete .5 - Partially complete 0 - Incomplete
Experiment Concept slide – Figure	Figure labelled and reasonably formatted	1	1 - Fully complete .5 - Partially complete 0 - Incomplete
Experiment Concept slide – Responsibilities	Detailed breakdown of Cadets in group and their responsibilities	0	Check for completeness, use to help determine allocation of peer eval points after Lab 10



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LAB#9 HANDOUT-SEE LAB 8 HANDOUT



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LAB#9 SOLUTION

Lab 9 Grading Rubric Distribution

Max Points	Poor			Marginal		Good		Exceptional	
	(Pts)	Description		(Pts)	Description	(Pts)	Description	(Pts)	Description
Formatting	2	0	No attempt made to conform to style guide.	.5	Formatting is generally consistent, though does not adhere to guide.	1	Formatting is generally consistent, though does not adhere to guide.	2	Formatting completely follows the style guide.
Introduction	3	0	Lacks clarity, no clear objective, minimal or no description of how experiment was conducted	1	Objective is unclear or some description of experiment provided but lacks relevance or detail.	2	Generally clear objective/description but some elements not fully described.	3	Clear, concise, relevant objective with description of how it was accomplished
Theory	12	0-6	Minimal or unclear theory that lacks substance or is generally incorrect.	7-8	Theory present but vague, lacks sufficient explanation / not effectively communicated.	9-10	Theory explained well, with minor omissions of key information from theory checklist.	11-12	Thorough, clear, and relevant theory explained with assumptions. Uses correct physics to support theory. Nearly all elements of checklist discussed
Procedures	12	0-6	Nearly impossible to repeat experiment. Incomplete or unclear procedures.	7-8	Difficult to repeat experiment. Steps taken are included but missing important details (e.g. entire variables not described).	9-10	Very possible to repeat experiment. Descriptive steps but limited discussion of uncertainty in measurements/confidence intervals/ other essential elements from the checklist.	11-12	Extremely easy to repeat experiment. Complete, detailed, and well-organized procedures. Nearly all checklist elements discussed.
Results	12	0-6	Does not report objective quantity results and/or lacks any effort. Unclear results reported. Lacks tables, graphs, figures, or any data.	7-8	Does not report objective quantity results though some effort to report results is evident. Results are unclear/not supported by writing and/or provides sparse data.	9-10	Reports the objective quantity results and writing supports results. Results are clear and introduced in the written report, though missing some details e.g. one table is missing, a plot isn't well described or incorporated, fails to report the slope, etc.	11-12	Reports the objective quantity full confidence interval completely correctly with all key supporting data. Clear results, well-formatted tables/graphs are error-free and introduced in the report.
Discussion	12	0-6	Poor writing that shows no critical thought or effort. Written analysis is almost completely missing or does not demonstrate reasonable thinking to identify sources of error and recommend changes.	7-8	Writing is lacking in critical thought or effort. Written analysis which attempts to describe the error. Fails to identify the correct error, does not recommend a reasonable improvement.	9-10	Writing attempts critical thought/has some effort. Generally well written analysis. Uses quantitative or qualitative descriptions to report on error, identifies a correct form of error, and recommends changes that may or may not be helpful/reasonable.	11-12	Extremely well written and thoroughly analyzed. Uses quantitative and qualitative descriptions to report on error, identifies a correct form of error, and recommends meaningful changes to improve the experiment.
Conclusions	3	0	Poor writing with no takeaways from experiment. Bare bones conclusions that is almost completely missing details and has no/incorrect takeaways from the experiment.	1	Writing has few takeaways from experiment. Synthesizes some conclusions though missing many details. May report a finding that is in no way related or supported by the rest of the report.	2	Well written with some takeaways from experiment. Restates the objective quantity and restates some form of error analysis from the discussion. Synthesizes some findings that are generally correct, though may not be fully supported by the report.	3	Extremely well written with synthesized takeaways from experiment. Restates the objective quantity, includes precision and accuracy, restates error and synthesizes findings that follow from the rest of report.



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LAB#10 HANDOUT



An Inspiring and Descriptive Title

Group members



- Outline the key paragraphs of your brief here



- Introduction section goes here
- Use as many slides as needed



- Theory section goes here
- Use as many slides as needed
- Be sure to cite figures



- Procedure section goes here
- Use as many slides as needed
- Be sure to cite figures and provide captions



- Results go here
- Use as many slides as needed
- Be sure to cite figures and provide captions



- Discussion section goes here
- Use as many slides as needed
- Be sure to cite figures and provide captions



- Conclusion goes here
- Use as many slides as needed



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Questions?



- MLA Format
- See the Academic Integrity brief on Canvas:
 - Encl 3 - Core Physics Academic Integrity Brief



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LAB#10 SOLUTION

Lab 10 Grading Rubric Distribution

Max Points	Poor		Marginal		Good		Exceptional		
	(Pts)	Description	(Pts)	Description	(Pts)	Description	(Pts)	Description	
Introduction	1	0	Fails to communicate an introduction	0.5	Introduces the experiment but lacks coherence	.75	Clearly Introduces the experiment	1	Captivates the audience while fully explaining their introduction
Theory	6	0-2	Theory has completely incorrect physics, lacks significant explanation and/or is very poorly explained.	3-4	Theory not clearly communicated, missing assumptions or logical steps.	5	Theory explained well with correct physics.	6	Captivates the audience with a thorough, clear, and relevant theory explained with assumptions.
Procedures	6	0-2	Procedures are lacking major details and it is extremely difficult to understand what the group did.	3-4	Procedures missing many elements that make it difficult to understand the measurements performed.	5	Procedures explained generally well but lacks description of some steps or uncertainties.	6	Procedures clearly explained so that it is easy to understand the groups measurements with the greatest factors of uncertainty mentioned.
Results	6	0-2	Results are poorly communicated and lack significant details. No clear objective quantity value is communicated.	3-4	Results are missing some plots, tables, and do not discuss the absolute uncertainty of the objective.	5	Results include most plots, tables, objective quantity value and uncertainty. Generally well communicated.	6	Results are extremely clear and well articulated with all expected plots and tables. Objective quantity and uncertainty calculations are clearly described.
Discussion	6	0-2	Discussion significantly lacking.	3-4	Discussion mentions type of error but is lacking in details analyzing what contributed to the error.	5	Discussion describes type of error and analyzes contribution of error but has some logical flaws.	6	Discussion is thorough and well articulated, completely describing the dominant experimental error and the source of that error.
Conclusion	1	0	Poor conclusion that lacks significant details and is not coherent.	.5	Conclusion recaps some elements from the experiment but does not make mention of methods to improve in the next iteration.	.75	Conclusion incorporates a recap of all key elements and recommends an improvement but it is not a significant or valid improvement.	1	Conclusion recaps all key elements and includes a well thought out improvement to subsequent experiments.
Formatting	1	0	Poor formatting, no attempts made to have consistent style across presentation.	.5	Inconsistencies in formatting that distract from presentation			1	Nearly perfect formatting consistent across slides.
Novelty	2	0	100% reproduction of a previous lab.	.5	Most of the experiment is repeated from a previous lab.	1	Generally novel experiment but elements are repeated from a previous lab.	2	Group shows novelty / creativity in experiment design
Timing	1	0	Outside of time window (10-12 mins)					1	Within time window (10-12 mins)

	Group 1	Group 2	Group 3	Group 4
Introduction (1)				
Theory (6)				
Procedures (6)				
Results (6)				
Discussion (6)				
Conclusion (1)				
Formatting (1)				
Novelty (2)				
Timing (1)				
Total MAX 30				

5 points of peer assessment applied under separate assignment.

	Group 5	Group 6
Introduction (3)		
Theory (8)		
Procedures (8)		
Results (8)		
Discussion (8)		
Conclusion (3)		
Formatting (2)		
Novelty (3)		
Timing (2)		
Total MAX 45		